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DOCTORAL SCHOOL
BIOMEDICAL SCIENCES

MODEL-BASED EXPLORATION OF DOSE OPTIMIZATION STRATEGIES FOR VANCOMYCIN, MEROPENEM, AND AMIKACIN IN SELECTED PATIENT POPULATIONS

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Dissertation presented in
partial fulfilment of the
requirements for the
degree of Doctor in
Pharmaceutical Sciences

Leuven, 28 April 2026

28 April 2026, 17:00
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Words of thanks

Pursuing a PhD has been one of the most challenging and rewarding experiences of my life. Over these past years, I have grown not only as a researcher, but also as a person. This journey demanded resilience, patience, and determination, and I am grateful for every lesson it brought. Most importantly, I am thankful to the many people whose support made this achievement possible.

My deepest gratitude goes to my promoter, Prof. Dr. Erwin Dreesen. Thank you for your guidance, your trust, and your constant willingness to help. Your support — both scientifically and personally — has been invaluable. You believed in me at moments when I doubted myself, and your advice has shaped the way I think and work. I am grateful for every discussion, every correction, and every opportunity to learn from you. It has been an honor to complete this PhD under your supervision.

I would also like to thank my co-promoter, Prof. Dr. Isabel Spriet, for her encouragement and for providing the clinical context that strengthened this research. Your support, your time, and your commitment throughout this journey are deeply appreciated.

My sincere gratitude extends to my co-promoter, Prof. Dr. Karel Allegaert. Thank you for sharing your expertise, for the thoughtful feedback on my work,

and for the many valuable discussions that helped shape the scientific foundation of this thesis.

I am equally grateful to my co-promoter, Dr. Matthias Gijzen, for his guidance and collaboration. Thank you for your availability, your critical insights, and the practical support that helped drive this work forward.

I would like to thank my internal jury members, Prof. Dr. Yves Debaveye and Dr. Pieter Van Brantegem, for evaluating my work throughout the past years and for their constructive and valuable feedback at every stage of this PhD. I also want to thank the chairs of the committee, Prof. Dr. Pieter Annaert and Prof. Dr. Isabel Huys, for overseeing the evaluation process and for their insightful guidance and support.

My sincere thanks also extend to my external jury members, Prof. Dr. Elke Krekels and Prof. Dr. Claire Roger, for their willingness to evaluate my work and contribute to this important milestone.

I am grateful to my colleagues in the Leuven Pharmacometrics Research Group, both former and current members. Working alongside people who challenge, exchange ideas, and inspire each other has been valuable during this journey. Thank you for the scientific discussions, the encouragement, the interactions, the social events, and the conference memories.

I would also like to thank the colleagues with whom I shared office space outside the Pharmacometrics group. Our daily conversations, coffee

moments, and lunch breaks brought warmth, humor, and balance into the workday. Your presence made the office a welcoming and enjoyable place to return to every morning.

To all collaborators, clinicians, and laboratory teams who contributed data, knowledge, and time to this research — thank you. Your willingness to share expertise made this thesis possible.

Outside of work, I am deeply grateful to my family and friends for their patience, love, and belief in me. I also wish to express my appreciation to those who supported me in practical ways throughout this journey by creating time and space for me to work, particularly during the most demanding periods. Their contribution, though often unseen, made a meaningful and lasting difference.

Finally, this journey has taught me that progress, both professionally and personally, is built step by step, especially during the most challenging times. What matters most is not to stop, but to keep moving forward. I hope this work encourages others who are on their own path. A milestone is rarely quick, never easy, but often achievable with patience, effort, and support. This milestone stands as a testament to resilience, patience, and persistence. And every ending is a new beginning.

List of abbreviations

ADME	Absorption, distribution, metabolism, and excretion
AI	Artificial intelligence
AMR	Antimicrobial resistance
ARC	Augmented renal clearance
AUC	Area under the concentration–time curve
AUC/MIC	Ratio of the area under the concentration–time curve over the minimal inhibitory concentration of the bacteria
AUC _{24h}	24-hour area under the time concentration–time curve
AUC _{24h} /MIC	Ratio of the 24-hour area under the concentration–time curve over the minimal inhibitory concentration of the bacteria
BMI	Body mass index
C _{24h}	24-hour postdose concentration
CE	Conformity European
CI	Confidence interval
CL	Clearance
C _{max}	Peak concentration
C _{max} /MIC	Ratio of the peak concentration over the minimum inhibitory concentration
C _{min}	Minimum concentration
CONSORT	CONsolidated Standards of Reporting Trials
C _{plateau}	Plateau concentration
CrCL	Creatinine clearance
C _{ss,plateau}	Steady state plateau concentration
C _{ss,trough}	Steady state trough concentration
C _{ss,trough/plateau}	Steady state trough or plateau concentration
C _{T+1}	Measured concentrations at relative time +1
C _{T0}	Measured concentrations at relative time 0
C _{T-1}	Measured concentrations at relative time -1
C _{trough}	Trough concentration
eCrCL _{CG}	Estimated creatinine clearance using Cockcroft–Gault equation
ED	Emergency department

eGFR _{CKD-EPI}	Estimated glomerular filtration rate calculated using the Chronic Kidney Disease Epidemiology Collaboration equation
eGFR _{CKD-EPI,24h}	Estimated glomerular filtration rate calculated using the Chronic Kidney Disease Epidemiology Collaboration equation at 24 hours after the start of the infusion
eGFR _{MDRD}	Estimated glomerular filtration rate calculated using the Modification of Diet in Renal Disease equation
EHR	Electronic health record
EMA	European Medicines Agency
EU	European Union
EUCAST	European Committee on Antimicrobial Susceptibility Testing
FDA	Food and Drug Administration
GDPR	General Data Protection Regulation
GFR	Glomerular filtration rate
ICU	Intensive care unit
IIV	Interindividual variability
IOV	Interoccasion variability
IQR	Interquartile range
MAA	Model averaging algorithm
mCrCL _{24h}	Measured creatinine clearance by 24-hour timed urine collection
MDD	Medical Device Directive
MDR	Medical Device Regulation
MIC	Minimum inhibitory concentration
MIC ₉₀	The lowest minimum inhibitory concentration inhibiting ≥90% of bacterial isolates
MIPD	Model-informed precision dosing
ML	Machine learning
MLE	Maximum likelihood estimate
MRSA	Methicillin-resistant <i>Staphylococcus aureus</i>
MSA	Model selection algorithm
NPDE	Normalized prediction distribution error
OFV	Objective function value
PBPK	Physiologically based pharmacokinetics
pcVPC	Prediction-corrected visual predictive check

PD	Pharmacodynamic
PI	Prediction interval
PK	Pharmacokinetic
PKPD	Pharmacokinetic–pharmacodynamic
PopPK	Population pharmacokinetic
PopPK–PD	Population pharmacokinetic–pharmacodynamic
PRISMA	Preferred Reporting Items for Systematic Review and Meta-Analysis
PTA	Probability of target attainment
rBias	Relative bias
RCT	Randomized clinical trial
rRMSE	Relative root mean squared error
RRT	Renal replacement therapy
RSE	Relative standard error
RUV	Residual unexplained variability
SCM	Stepwise covariate modeling
SCr	Serum creatinine
SOFA	Sequential Organ Failure Assessment
TDM	Therapeutic drug monitoring
Time _{3 mg/L}	Time to reach an antibiotic concentration of 3 mg/L
UCr _{24h}	24-hour measured urinary creatinine
V _c	Central volume of distribution
VPC	Visual predictive check
WHO	World Health Organization
X% fT>n×MIC	X percent of time during a dosing interval for which concentrations of the unbound drug is above n-fold the minimum inhibitory concentration throughout a dosing interval

Abstract

Background Optimizing antibiotic therapy in special patient populations (e.g., older adults, critically ill patients with preserved or augmented renal clearance [ARC] admitted to the intensive care unit [ICU], and critically ill adults admitted to the emergency department [ED]), is challenging. These special patient populations experience unique (patho-)physiological changes that may lead to suboptimal drug exposure, making the standard dosing strategies inappropriate for many patients. Their underrepresentation in clinical trials further limits the available evidence and contributes to uncertainty in dose selection.

Pharmacometrics is a discipline that quantifies and predicts pharmacokinetics (PK) and pharmacodynamics (PD) in special patient populations using mathematical models that support data-driven decision-making in clinical research and drug development. Such models enable dose optimization through model-informed precision dosing (MIPD) or dosing simulations.

The aim of this doctoral thesis was to explore the value of population PK (popPK) modeling, simulation, and MIPD in optimizing dosing strategies in special patient populations on intravenously administered antibiotics primarily eliminated by the kidneys and have been on the market for decades. We hypothesized that applying modeling, simulations, and MIPD in these populations would improve probability of target attainment (PTA), minimize toxicity, and ultimately enhance clinical outcomes.

Methods We evaluated popPK modeling, simulation, and MIPD for three antibiotics, each studied in a specific special patient population. We present four chapters. In Chapter 1, we conducted a systematic review summarizing and comparing published popPK models of intravenous vancomycin in adult patients. We also evaluated MIPD software tools participating in our survey based on their clinical applicability. In Chapter 2, we performed MIPD for older adults using nine published vancomycin models, two of which were identified in Chapter 1. In Chapter 3, we developed several meropenem popPK models based on the same structural and statistical model; each having a different kidney biomarker. Using these models, we conducted dosing simulations to optimize dosing strategy while assessing safety in critically ill patients with preserved renal function or ARC admitted to the ICU. In Chapter 4, we created virtual patient datasets using a transformed and truncated multivariate normal distribution. These datasets were used to conduct dosing simulations for amikacin in adults admitted to the ED, using a published model. All pharmacometric analyses were conducted using NONMEM version 7.5.0.

Results The landscape of vancomycin popPK models in adults is heterogeneous, and most MIPD tools lack adequate validation. In older adults, MIPD indicated that the most recent therapeutic drug monitoring (TDM) vancomycin sample provided the best predictive performance, particularly when using the Thomson *et al.* and Ji *et al.* models. For critically ill patients with preserved renal function or ARC, a total daily dose of 4,000 mg meropenem administered as a continuous infusion ($4 \times 1,000$ mg over 6

hours) is recommended instead of the standard dosing strategy of 3,000 mg administered as intermittent infusions ($3 \times 1,000$ mg over 8 hours). For adult patients admitted to the ED, a fixed dose of 1,500 mg amikacin (minimum inhibitory concentration [MIC] 2 mg/L) is recommended over standard weight-based dosing of 15 mg/kg.

Conclusions This thesis illustrates how pharmacometrics can be used to optimize dosing strategies while assessing safety in special patient populations who fail to reach acceptable PTA under standard dosing strategies. However, prospective clinical validation is essential to confirm the results and to establish the real-world value of pharmacometrics in dose optimization.

Samenvatting

Waarom dit belangrijk is Antibiotica redden elke dag levens. Toch werkt een standaarddosisschema niet altijd goed voor kwetsbare patiëntengroepen, zoals oudere mensen, ernstig zieke patiënten op de intensieve zorgafdeling, of patiënten die via de spoedafdeling opgenomen worden. Hun lichaam werkt anders (bijvoorbeeld door een veranderde nierfunctie), waardoor de hoeveelheid geneesmiddel in het bloed te laag (minder werkzaam) of te hoog (meer kans op bijwerkingen) kan zijn.

Doel en aanpak Dit onderzoek had als doel de dosering van drie veelgebruikte antibiotica (vancomycine, meropenem en amikacine) beter af te stemmen op specifieke patiëntengroepen. Daarvoor zijn rekenmodellen, computersimulaties en modelgebaseerde precisiedosering (MIPD) gebruikt. Zo konden we inschatten welke dosering het meest geschikt is en hoe we tijdens de behandeling kunnen bijsturen op basis van gemeten waarden in het bloed.

- Vancomycine: We hebben bestaande modellen en MIPD-software voor dosering op maat geïnventariseerd. We onderzochten welke bloedconcentraties het voorspellen van toekomstige vancomycinespiegels bij oudere patiënten het meest verbeteren.
- Meropenem: We hebben modellen ontwikkeld op basis van verschillende nierfunctiemetingen. Vervolgens werden doseerschema's gesimuleerd

voor patiënten op de intensieve zorgafdeling met een normale of verhoogde nierfunctie.

- Amikacine: We hebben virtuele patiëntengroepen samengesteld voor de spoedafdeling om een vaste startdosis te vergelijken met de gebruikelijke dosering op basis van lichaamsgewicht.

Belangrijkste bevindingen

- Vancomycine bij ouderen: Het landschap van vancomycine-popPK-modellen bij volwassenen is heterogeen, en veel MIPD-hulpmiddelen blijken onvoldoende gevalideerd. Bij oudere patiënten liet MIPD echter zien dat de meest recente bloedconcentratiewaarde doorgaans de nauwkeurigste voorspellingen opleverde, met name bij gebruik van de modellen van Thomson *et al.* en Ji *et al.*
- Meropenem op de intensieve zorgafdeling: Continue toediening leidt vaak tot stabielere, gewenste bloedconcentraties dan meerdere intermitterende infusies per dag, vooral bij verhoogde nierfunctie.
- Amikacine op de spoedafdeling: Een vaste startdosis kan sneller en betrouwbaarder zorgen voor een voldoende werkzame blootstelling dan dosering op basis van lichaamsgewicht.

Betekenis en maatschappelijke impact Rekenmodellen helpen artsen en ziekenhuisapothekers gericht antibiotica te doseren bij patiënten die afwijken van de gemiddelde populatie, wat de kans op goede behandeling vergroot en bijwerkingen verkleint. Voor invoering in de klinische praktijk is

prospectieve klinische bevestiging nodig om de resultaten in de praktijk te valideren.

General Introduction

Dosing challenges in antibiotic therapy

Dose optimization in antibiotic therapy can be challenging in special patient populations whose physiological characteristics differ from those typically represented in clinical trials. Older adults often exhibit age-related changes that may alter drug exposure. Critically ill patients experience rapid and dynamic physiological changes, leading to highly variable pharmacokinetics (PK). These special patient populations are commonly underrepresented in clinical trials or the initial drug development plans, resulting in limited evidence to guide dosing for antibiotics already available on the market. As a result, they are often treated with dosing strategies derived from clinical trials conducted in healthier populations, which may not consistently achieve therapeutic levels in special patient populations. Academic research is therefore essential to identify when standard dosing may be inappropriate for special patient populations, making dose optimization necessary. However, identifying an optimized dosing strategy is only one part of the challenge. For such strategies to improve patient care, they must also be translated into clinically actionable recommendations that can be used at the bedside, within the time constraints and operational realities of hospital care.

Effective antibiotic therapy is shaped by the interplay between the host (special patient populations), the pathogen (bacteria), and the therapy (antibiotic) itself. The first three sections of the introduction discuss these components and their associated challenges. The fourth section examines

how they converge through pharmacokinetic–pharmacodynamic (PKPD) targets and describes the safety metrics. The last section introduces pharmacometrics as a solution to optimize dosing strategies while reducing the risk of toxicity for antibiotic therapy in special patient populations (**Figure 1**), with an emphasis placed on the clinical applicability of model-based dosing strategies, including their feasibility, interpretability, and workflow integration.

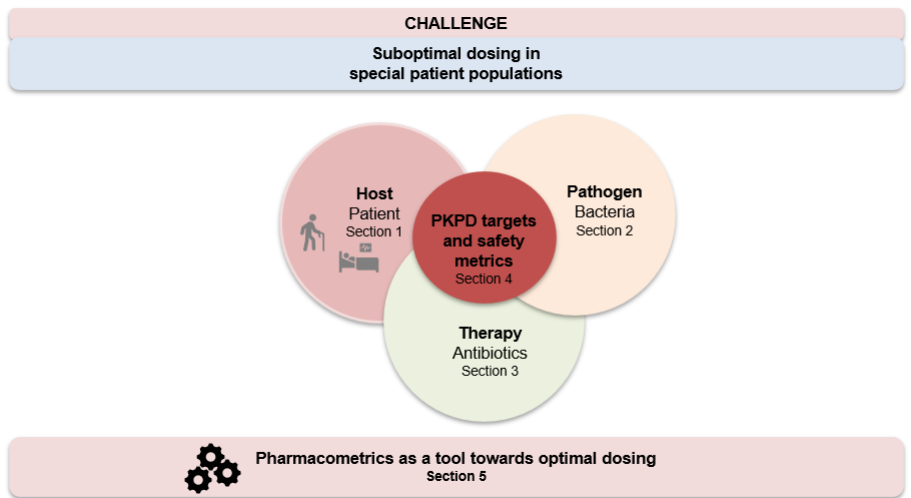


Figure 1 Overview of the introduction. PKPD, pharmacokinetic–pharmacodynamic

1. Specific patient populations

Drug dosing in clinical practice is often more complex than suggested by clinical trials. These trials frequently exclude patients with unstable physiology or organ dysfunction — for example, those with impaired kidney function — resulting in limited evidence for real-world hospital populations. Consequently, dosing recommendations are often extrapolated from data obtained in

healthier adults to patient populations whose physiological profiles differ substantially.

These limitations are particularly relevant for older adults (*i.e.*, 75 years and above) and critically ill patients, who are underrepresented in trials but account for a substantial and clinically relevant proportion of the real-world patient population. Their pathophysiology may differ from that of healthier trial participants, and may also be in a dynamic state, placing these patients at risk of underexposure or overexposure, leading to therapeutic failure or toxicity, respectively.

The following subsections focus on the specific patient populations investigated in this thesis, with emphasis on their distinct and often dynamic pathophysiology characteristics.

1.1 Hospitalized older adults

The global population is experiencing demographic aging. This demographic shift is driven by both an increase in life expectancy and a decrease in fertility rates. Worldwide, the proportion of individuals aged 65 years and older is expected to increase by approximately 1.7-fold between 2020 and 2050 [1]. Similar trends are observed in the European Union (EU), where the share of adults aged 65 years and older is expected to increase 1.5-fold from 2024 to 2100 [2]. The proportion of individuals in Belgium aged 80 years and older has increased from 3.8% in 1995 to 5.6%

in 2025 [3]. National life expectancy is forecasted to increase to 88.3 years for men and 89.9 years for women by 2070 [3].

There is no universally accepted age threshold that defines “older adults”; different fields use cutoffs ranging from 60 to 75 years. In demographic reporting, the age of 65 years is commonly used as a benchmark because it corresponded to the statutory retirement age in Belgium and many European countries. In this thesis, particular attention is given to adults aged 75 years and older. Beyond chronological age, this special patient population shares a constellation of characteristics, often described under the term “geriatric syndrome”, including reduced physiological reserve, declining organ function, cognitive impairment, multimorbidity, and comedications. A decrease in age-related kidney function illustrates these physiological changes. While healthy young adults have an average glomerular filtration rate (GFR) of 112 mL/min, this decreases to 95 mL/min in healthy older adults (≥ 60 years), and to 33 mL/min in older adults with chronic kidney disease [4]. Such a decrease in GFR may prolong drug elimination and increase drug exposure, thereby increasing the risk of toxicity [4–7]. Drug exposure is also influenced by age-related changes in tubular function, including dysfunction in active secretion or reabsorption [4–7]. Both glomerular and tubular changes therefore have the potential to modify drug clearance (CL) and exposure, with possible implications for dosing optimization and minimization of side effects [4–7].

These clinical and physiological characteristics highlight the importance of studying this population. However, despite their healthcare needs and distinct pathophysiology, older adult patients remain underrepresented in clinical trials. In studies of age-related diseases such as Alzheimer's disease — where they constitute nearly three-quarters of the real-world patient population — they account for only about one-quarter of participants [8]. Their representation is even lower in clinical trials for non-age-related diseases where the participation of older adult patients has, in some cases, dropped to as little as 1% [8]. Recognizing this gap, the United States Food and Drug Administration (FDA) published in 2020 the guidance “Enhancing the diversity of clinical trial populations: eligibility criteria, enrolment practices, and trial designs”, encouraging broader inclusion of diverse and clinically complex patient groups, further emphasizing the need for broader participation [9]. Consequently, drug therapy in this special patient population is supported by limited clinical evidence.

1.2 Critically ill patients in the intensive care unit

Critical illness is a state of organ dysfunction, that puts a patient at risk of short-term death, yet for which complete or partial recovery is possible with appropriate therapy [10]. It commonly occurs in the context of trauma, post-surgery, burns, ischemia, or infection [11]. The body initiates a systemic inflammatory response characterized by activation of white blood cells,

complement pathways, and release of cytokines [11]. Hemodynamically, critical illness evolves through two phases: an initial period characterized by a decrease in cardiac output and oxygen consumption, followed by a subsequent phase with an increase in cardiac output and oxygen consumption [11,12].

Critically ill patients may receive care in the intensive care unit (ICU). They are admitted to this ward in cases of imminent organ failure (*e.g.*, acute respiratory failure, severe infection) requiring continuous monitoring, advanced organ support, or intensive medical care that cannot be delivered on another ward [13]. However, ICU admission also depends on the hospital's structure, including the availability of specialized wards, and the number of ICU beds [13]. The risk of ICU admission for adults increases with aging, and multimorbidity [13]. These patient-related characteristics modify drug exposure through changes in body composition, protein binding, and elimination, complicating the identification of an appropriate dosing strategy. Supportive care delivered in the ICU further changes drug exposure. For example, fluid therapy, which is indicated to improve tissue perfusion in cases of hypovolemia, may increase the volume of distribution, particularly for hydrophilic drugs [14].

Kidney function is an important determinant of drug exposure in critically ill patients. Acute kidney injury is common among critically ill patients

admitted to the ICU [15]. Acute kidney injury is defined as an increase in serum creatinine (SCr) of ≥ 0.3 mg/dL within two days, an increase in SCr of ≥ 1.5 times baseline within seven days, or a urine output of < 0.5 mL/kg/h for six hours [15]. It decreases the CL of drugs that rely predominantly on the kidneys. Consequently, drug concentrations may rise above the therapeutic levels, increasing the risk of toxicity and requiring dose reduction.

In contrast, some critically ill patients admitted to the ICU may experience augmented renal CL (ARC), which is characterized by a supraphysiologic GFR, typically defined as a measured creatinine CL (CrCL) by 24-hour timed urine collection ($mCrCL_{24h}$) > 130 mL/min/1.73 m² [16]. It occurs mainly in young adults (*i.e.*, mean or median age range: 34–50 years) as a result of inflammation-induced vasodilation of the afferent glomerular arterioles together with an increase in cardiac output, leading to an increase in renal blood flow [16–18]. ARC accelerates the elimination of drugs that rely predominantly on the kidneys. Consequently, drug concentrations may fall below therapeutic levels, putting patients at risk of therapeutic failure. Under these circumstances, dose optimization should be considered when there is clinical evidence of underexposure.

Historically, clinical trials conducted to obtain regulatory approval have focused on proving efficacy rather than effectiveness [19]. Trial participants are typically selected to fit narrow eligibility criteria, resulting in study

populations that do not reflect the heterogeneity of real-world ICU patients [19]. Consequently, critically ill patients were underrepresented in these trials [19]. Even large trials evaluating drugs central to current ICU practice often enrolled only a few thousand critically ill patients, and none reported participation of patients with ARC, which had not yet been known at the time [16,19].

Given their distinct pathophysiological profile and their historical underrepresentation, drug therapy in critically ill adults — particularly those at the ICU with ARC — follows limited clinical evidence and may require optimization when underexposure is demonstrated.

1.3 Critically ill patients in the emergency department

Other than the ICU, critically ill patients may also receive care in the emergency department (ED). They arrive from outside the hospital with critical illness and are then either admitted to the ICU or to another hospital ward. In the United States, one in five individuals visits the ED once a year [20]. Among the most common reasons for hospital admission after an ED visit are infectious diseases [20]. The likelihood of visiting the ED increases with age [20].

The characteristics of patients in the ED may modify drug exposure differently from those receiving care in the ICU. Patients in the ED may represent an earlier stage of critical illness, sometimes before organ dysfunction has fully developed. In contrast, patients in the ICU may

represent a later and more advanced stage of critical illness, with more profound organ dysfunction. This difference may also apply to supportive care. Patients seen in the ED might be dehydrated on arrival, whereas patients in the ICU have typically already received fluid therapy [21]. ICU patients may require renal replacement therapy (RRT), which accelerates the elimination of drugs that are primarily cleared by the kidneys. RRT also removes inflammatory cytokines and may support recovery of kidney function, a situation that does not apply to most patients in the ED [17].

Critical illness is in a dynamic state, and PK parameters of the same patient may change between ED and ICU admission in ways that can be clinically important. Although such changes are plausible, direct and systematic comparisons of PK between these two wards are largely lacking in the current literature, leaving uncertainty about how PK may differ. Most studies group critically ill patients together rather than separating them by wards, and most available data come from ICU patients [19]. Participation of ED patients in clinical trials is uncommon because of the challenges associated with rapid stabilization, obtaining informed consent, and limited healthcare personnel. Given these limitations, dose optimization may be required when clinical evidence of underexposure or overexposure is present.

2. Bacterial infections

Bacteria are unicellular prokaryotic microorganisms that differ from one

another in several characteristics, including their shape (e.g., cocci, bacilli, spiral), and the structure of their peptidoglycan cell wall, which may be thick or thin. They reproduce asexually by binary fission, enabling rapid population growth. As one of the earliest life forms on Earth, their simple structure, rapid reproduction, and remarkable adaptability have allowed them to survive in nearly all environments. They inhabit soil, water, and other organisms. They may exist as free-living or host-dependent forms. Depending on their interaction with the host, bacteria may be beneficial, commensal, or harmful. In the latter state, pathogenic bacteria can cause infectious diseases. In humans, many bacterial species constitute normal microbiota, which are generally beneficial. However, under certain conditions, some of these species can become pathogenic. In addition to these endogenous sources of infection, other bacterial species – those not part of the normal microbiota – can cause disease when introduced into the body [22].

Bacterial infections remain a major cause of mortality and morbidity worldwide. In 2019, an analysis focusing on 33 bacterial pathogens with sufficient global data estimated that these organisms were responsible for approximately 7.7 million deaths, accounting for 13.6% of all deaths. Nearly three million of these deaths were associated with bloodstream infections [23]. Such infections impose a considerable burden on patients, often requiring prolonged hospitalization and intensive care, due to associated complications. For instance, bloodstream infections may trigger sepsis, a life-threatening condition characterized by systemic inflammation and organ dysfunction

resulting from an excessive host immune response. Sepsis may further progress to septic shock, in which blood pressure drops dangerously low despite adequate fluid resuscitation, necessitating vasopressor therapy [24]. The burden goes beyond individual patients to the healthcare system. The mean total hospital-related cost of sepsis per patient has been reported to range between approximately €1,101 and €91,951 [25]. Enterobacterales, *Pseudomonas aeruginosa*, and *Staphylococcus aureus* are among the most frequent pathogens responsible for infections across older adults and critically ill adult patient [26,27].

Timely diagnosis of bacterial infections remains a major challenge. Gram staining is typically the first diagnostic step, allowing a rapid distinction between gram-positive bacteria, which appear violet because of their thick peptidoglycan cell wall, and gram-negative bacteria, which appear pink or red because of their thin peptidoglycan layer. Definitive identification, however, relies on culturing clinical specimens, a microbiology test that requires 24–96 hours. This process not only identifies the causative pathogen but also characterizes the antimicrobial susceptibility profile by determining the minimum inhibitory concentration (MIC) — the lowest antimicrobial drug concentration required to inhibit bacterial growth. Because this information is not immediately available, initial therapy is usually empirical. In the absence of patient-specific MIC, expected MIC distributions and internationally established susceptibility breakpoints for likely pathogens (e.g., European Committee on Antimicrobial Susceptibility Testing [EUCAST]) are used to

guide empirical antimicrobial selection [28]. Consequently, empirical therapy is commonly initiated based on clinical judgment, gram-staining results, and epidemiological patterns.

3. Antibiotic therapy

Antibiotics are a class of antimicrobial drugs used to prevent and treat bacterial infections. Evidence of antibiotic exposure has been found in human skeletal remains of ancient Nubians dating back to 350–550 Common Era [29]. However, it was not until the 20th century that antibiotics were scientifically discovered, isolated, and developed for clinical use [30]. Penicillin, discovered by Alexander Fleming in 1928 and later massively produced in the 1940s, marked the beginning of the antibiotic era [30]. This breakthrough was followed by the discovery of multiple antibiotic classes — including glycopeptides, β -lactams, and aminoglycosides. As a result, antibiotic therapy made formerly life-threatening bacterial infections treatable, and even preventable. Alongside other major advances in medicine and public health (e.g., vaccination, improved sanitation), the use of antibiotics contributed to a decrease in mortality [31]. For example, in the United States, the mortality rate from infectious diseases decreased by an average of 8.2% per year between 1938 and 1952 [32]. Consequently, the global life expectancy increased — from just over 40 years in 1800 to more than 70 years in 2021 — an improvement to which antibiotic therapy contributed

substantially, together with major advances in medicine, public health, and socioeconomic development [33].

Antibiotics exert their antimicrobial activity by targeting essential bacterial processes. Their mechanisms of action can be broadly classified into: (i) inhibition of cell wall synthesis, (ii) disruption of cell membrane integrity, (iii) inhibition of protein synthesis, (iv) interference with nucleic acid replication or transcription, and (v) inhibition of critical metabolic pathways such as folate synthesis [34]. Their antimicrobial killing effect is either bacteriostatic, meaning they inhibit bacterial growth and allow the host immune system to eliminate the pathogenic bacteria, or bactericidal, meaning they directly kill bacteria [34]. The range of bacterial species targeted by an antibiotic defines its antimicrobial spectrum, which can vary from broad-spectrum drugs with activity against a wide range of species to narrow-spectrum drugs active with activity limited to a number of species [34].

The antimicrobial activity of antibiotics is determined not only by their mechanism of action and antimicrobial spectrum, but also by their pharmaceutical and PK properties. Pharmaceutical properties influence systemic exposure. For example, some antibiotics are chemically unstable and may degrade in acidic environments when stored in solution for extended periods. Lipophilicity influences systemic exposure too: lipophilic antibiotics generally cross cell membranes readily, whereas hydrophilic antibiotics have poorer membrane permeability and thus more limited tissue penetration [34].

These pharmaceutical properties have a direct impact on the PK properties, which describe how the body absorbs, distributes, metabolizes, and excretes a drug, ultimately determining the concentration of antibiotic that reaches the site of infection over time [34]. A standard dosing strategy is usually prescribed with the aim of achieving therapeutic exposure for all patients. However, PK can vary between individuals due to differences in characteristics, such as age, organ function, disease state, or drug–drug interactions. As a result, some patients receiving the same standard dosing may achieve subtherapeutic or supratherapeutic exposure, potentially leading to therapeutic failure or toxicity [35].

Although antibiotics revolutionized modern medicine, the decline in the discovery of new antibiotics and the overuse or misuse of existing antibiotics accelerated the development of antimicrobial resistance (AMR). This occurs when bacteria can survive exposure to antibiotic that would normally directly or indirectly kill them. AMR results from genetic changes acquired through mutation or horizontal gene transfer, leading to resistance mechanisms such as antibiotic-degrading enzymes, altered target sites, or reduced permeability. Today, AMR is recognized as a global public health threat, responsible of 1.27 million deaths in 2019 [36,37]. In Europe, AMR accounts for at least 35,000 deaths annually [38]. These numbers have raised concern about a potential "post antibiotic era", in which the current antibiotics are no longer able to prevent or treat bacterial infections, thus putting in light the urgent need for appropriate antibiotic use [39].

Staphylococcus aureus and *Pseudomonas aeruginosa* are two prominent examples of high-threat bacterial pathogens. Methicillin, a β -lactam antibiotic developed to treat penicillin-resistant *Staphylococcus aureus*, was introduced in 1960; however, methicillin-resistant *Staphylococcus aureus* (MRSA) emerged shortly thereafter, leaving only a few effective therapeutic options (e.g., glycopeptides) [40]. *Pseudomonas aeruginosa* has also developed multidrug resistance (resistant to ≥ 1 drug in ≥ 3 antimicrobial classes) and even extensive drug resistance (resistant to ≥ 1 drug in all but ≤ 2 classes), resulting in difficult-to-treat cases [41,42]. In 2019, an estimated 358,800 deaths were attributable to resistance in *Staphylococcus aureus*, and 169,080 deaths to resistance in *Pseudomonas aeruginosa* [37]. Due to their global public health threat, both organisms are listed by the World Health Organization (WHO) as priority bacterial pathogens [43].

The following subsections focus on antibiotics that have been approved for many years and are investigated in this thesis, with emphasis on their mechanism of action, spectrum of activity, pharmaceutical and PK properties, clinical use, dosing strategies, side effects, and drug–drug interactions.

Vancomycin

Vancomycin is a glycopeptide antibiotic used primarily for gram-positive bacterial infections. It works by binding to the peptidoglycan unit, the D-alanyl-D-alanine, blocking cell-wall synthesis [44]. This leads to bacterial cell death,

making vancomycin bactericidal against susceptible pathogens, except for *Enterococcus* species for which it exhibits a bacteriostatic effect [45].

Vancomycin has a narrow-spectrum activity, limited to gram-positive cocci (e.g., *Staphylococcus*, *Streptococcus*, and *Enterococcus*), and gram-positive bacilli (e.g., *Clostridium difficile*, *Bacillus* species) [46]. It has no activity against gram-negative pathogens because its large molecular size prevents penetration through the bacterial outer membrane [44,46]. This same large molecular size also contributes to its negligible gastrointestinal absorption, which is why oral administration is ineffective for systemic therapy [44]. Oral administration is used only in cases of pseudomembranous colitis, where it eradicates *Clostridium difficile* in the gastrointestinal tract [44]. For systemic infections, vancomycin is administered intravenously [44].

Vancomycin is a hydrophilic molecule with a protein binding capacity ranging between 30% and 55% [47]. Vancomycin is minimally metabolized; at least 75% is eliminated unchanged by the kidneys through glomerular filtration [47]. In adults with normal kidney function, the volume of distribution is approximately 0.9 L/kg (60 L/1.73 m² in a 70 kg, 1.54 m individual), and the elimination half-life varies from 4 to 6 hours [47].

Vancomycin is indicated for severe infections including pneumonia, osteomyelitis, bacteremia, endocarditis, skin and soft-tissue infections, and meningitis [47]. The main target pathogens are MRSA, coagulase-negative *Staphylococci*, and ampicillin-resistant *Enterococcus* species [47]. It is also

indicated for patients allergic to β -lactam antibiotics such as penicillin and cephalosporin [45]. The standard dose recommended on the drug leaflet is 15 to 20 mg/kg every 8 to 12 hours, which can be increased to 25 to 30 mg/kg [46]. The Belgian Society for Infectiology and Clinical Microbiology recommends a loading dose of 25 to 30 mg/kg over 1 to 2 hours, followed by either intermittent infusion of 15 mg/kg every 12 hours or continuous infusion of 30 to 40 mg/kg/day [48]. Because vancomycin exposure must be carefully controlled in clinical practice, dosing is guided by drug concentration monitoring (*i.e.*, therapeutic drug monitoring [TDM]). A detailed explanation of this monitoring approach is provided in Section 5.1.

A common side effect includes hypersensitivity reaction (*e.g.*, urticaria) [49]. Rapid intravenous administration may cause a histamine-mediated “vancomycin infusion reaction”; therefore, vancomycin should not be administered as a bolus [49], and should be administered over at least 60 minutes [46,49]. More serious side effects include nephrotoxicity, neutropenia, and ototoxicity, with the latter potentially becoming irreversible [49]. The risk of serious side effects increases with coadministration of drugs with a similar toxic profile, such as aminoglycosides [46].

Meropenem

Meropenem is a carbapenem, a subgroup of β -lactam. Like other subgroups of β -lactams (penicillin and cephalosporin), it contains a β -lactam ring that binds to penicillin-binding proteins. In particular, meropenem has high affinity

for penicillin-binding proteins 1, 2, 3, and 4 in *Staphylococcus aureus*, *Escherichia coli* and *Pseudomonas aeruginosa* [50,51]. This binding inhibits bacterial cell-wall synthesis and results in cell lysis, giving meropenem a bactericidal effect, except against *Listeria monocytogenes*, where it is bacteriostatic [51]. Unlike many other β -lactams, the β -lactam ring in meropenem contains an amide bond that makes it resistant to degradation by many β -lactamases, thus increasing its spectrum of activity [50].

Meropenem has a broad spectrum of activity against gram-positive and gram-negative aerobic and anaerobic bacteria, including extended-spectrum β -lactamase producing and AmpC β -lactamase producing Enterobacterales [50]. Compared to other carbapenems like imipenem and doripenem, meropenem is less active against gram-positive bacteria but more active against gram-negative bacteria, particularly Enterobacterales. Compared to ertapenem, another carbapenem, meropenem retains its activity against *Pseudomonas aeruginosa* [50].

Meropenem is a relatively small, hydrophilic molecule due to its amide side chain. Although it penetrates most tissues, it cannot be administered orally because it is degraded in the gastrointestinal tract. It has minimal protein binding (~2%), meaning that approximately 98% of the drug is unbound and pharmacologically active — an advantage in patients with hypoalbuminemia [50,51]. Approximately 30% of meropenem is metabolized by hydrolysis of the β -lactam ring, leading to an inactive metabolite [52]. The remaining 70% is

eliminated unchanged by the kidneys through glomerular filtration and tubular secretion [50,51]. In adults with normal kidney function, the mean volume of distribution is 0.25 L/kg, and the mean CL is 17.22 L/h, based on PK studies using a 0.25 g dose [52]. Although this dose is lower than standard doses (*i.e.*, 1–2 g), meropenem displays linear PK, meaning that volume of distribution and CL remain consistent across the clinically used dosing range [53]. The elimination half-life is short (1 hour), requiring multiple daily administrations [52]. Meropenem is supplied as a sterile powder and reconstituted for intravenous administration. Once reconstituted, meropenem undergoes hydrolysis and degrades over time; its stability depends on temperature and the diluent used [51].

Meropenem is indicated for bacterial meningitis, chronic respiratory infections in cystic fibrosis, complicated skin and soft-tissue infections, complicated urinary tract infections, febrile neutropenia, intra-abdominal infections, pneumonia, and sepsis [51,52]. The standard adult dose is 1,000 mg every 8 hours [51]. Higher doses (up to 2,000 mg every 8 hours) are recommended for chronic respiratory infections in cystic fibrosis, or for acute bacterial meningitis [51]. Hence, the total daily dose varies from 1,500 to 6,000 mg depending on the clinical indication. Meropenem is usually administered as an intermittent infusion over 15 to 30 minutes. A 3 to 5 minutes bolus may be administered when the dose is $\leq 1,000$ mg [51]. An extended 3 hours infusion may be used for 2,000 mg doses [48].

Meropenem is well tolerated in general. Common side effects include diarrhea, headache, nausea, vomiting, rash, and pain at the injection site [51]. Rare but serious side effects include nephrotoxicity and neurotoxicity [54]. Laboratory abnormalities may include elevated hepatic enzymes (e.g., alanine aminotransferase) [50].

Meropenem markedly reduces serum valproic acid concentrations, thereby diminishing its anticonvulsant effect and increasing the risk of seizures when both drugs are co-administered [52]. It may also increase the effect of oral anticoagulants (e.g., warfarin) [52]. Meropenem half-life increases when co-administered with probenecid, a drug prescribed for gout, which inhibits meropenem tubular secretion [52].

Amikacin

Amikacin is an aminoglycoside antibiotic. Like other aminoglycosides (e.g., gentamicin), it has a bactericidal effect by binding to the 30S ribosomal subunit, inhibiting protein synthesis and ultimately leading to bacterial cell death [55].

Amikacin has a broad spectrum of activity against most gram-negative bacteria, including multidrug-resistant Enterobacterales (e.g., carbapenem-resistant, gentamicin-resistant) and *Pseudomonas aeruginosa* [55,56]. It has limited activity against gram-positive pathogens such as *Staphylococcus* species and *Mycobacterium* species [55]. It has no activity against anaerobic

pathogens because oxygen-dependent electron transport is required for cellular uptake [55].

Amikacin has minimal protein binding ($\leq 10\%$), meaning that approximately 90% of the drug is unbound, and pharmacologically active [57]. It is also approved as a liposomal inhalation suspension administered through nebulization [58]. Systemic absorption following oral administration is negligible; therefore, amikacin is administered parenterally, typically as an intravenous infusion or intramuscular injection [57]. Amikacin is not metabolized and is eliminated almost entirely unchanged (range: 60%–82%) by the kidneys through glomerular filtration [57]. In adults with normal kidney function, the volume of distribution is approximately 24 L (~28% of body weight), and the elimination half-life ranges from 2.2 and 2.4 hours [57].

Amikacin is indicated for severe infections such as sepsis, pneumonia, intra-abdominal infections, complicated urinary tract infections, and tuberculosis [57]. It is often prescribed in combination with a β -lactam, providing synergistic activity and broadening the coverage against gram-positive pathogens [57]. Standard daily dosing in adults with normal kidney function is 15 mg/kg, administered once daily, with a maximum daily dose of 1,500 mg for up to 10 days [57]. The infusion time ranges from 30 to 60 minutes [57].

Amikacin may cause nephrotoxicity and ototoxicity. These toxicities are more likely in patients with prolonged therapy (*i.e.*, >5 days), dehydration, or co-administered ototoxic or nephrotoxic drugs such as loop diuretics [57]. Rare

adverse effects include apnea, and neuromuscular blockade, particularly in patients with underlying neuromuscular disease [57]. Laboratory abnormalities may include albuminuria, azotemia (*i.e.*, increase in urea and creatinine), oliguria [57].

4. Pharmacokinetic and pharmacodynamic targets and safety metrics

Dosing strategies are determined mainly by PK and pharmacodynamic (PD) properties. PK describes what the body does to a drug in terms of absorption, distribution, metabolism, and excretion (ADME). In contrast, PD describes what the drug does to the pathogen (potency), and ultimately what it does to the host (clinical and toxicological effects) [59].

Historically, antibiotic dosing aimed simply to maintain serum concentrations above the MIC of a pathogen [59]. However, this approach proved inadequate in dose fractionation studies, where the same total dose was divided into different dosing intervals. These studies showed that dosing frequency — not only the total dose — could alter antimicrobial effect [59]. This led to the integration of PK and PD metrics to ensure efficacious drug exposure. Safety, however, is assessed in separate exposure–toxicity analyses.

The PK and PD integration is achieved by linking drug exposure metrics (*e.g.*, area under the concentration–time curve [AUC], peak concentration [$C_{\max}$], time above a threshold concentration [T]) to a potency metric (MIC obtained from a static *in vitro* assay) while considering the antimicrobial killing effect (bacteriostatic or bactericidal), the killing pattern (time- or concentration-

dependent) and, if present, the post-antibiotic effect [59,60]. The post-antibiotic effect represents the period following a brief exposure to an antibiotic – after the drug has been removed – during which bacterial growth remains suppressed even though concentrations have fallen below the minimum inhibitory concentration, before eventually resuming. Its duration can be short or long depending on the antibiotic and pathogen [59].

As a result, three main PKPD indices emerged. The first PKPD index is for time-dependent bacteriostatic antibiotics (*e.g.*, vancomycin). For these antibiotics, total exposure over time is the key determinant of efficacy. Their efficacy is described by the ratio of the AUC over the MIC (AUC/MIC) [59,60]. The second PKPD index is for time-dependent bactericidal antibiotics with no or minimal post-antibiotic effect (*e.g.*, meropenem). These drugs have relatively slow bactericidal action and little bacteriostatic effect gained when concentrations exceed approximately four times the MIC. Their efficacy is described by the percentage of time during a dosing interval for which an unbound antibiotic concentration exceeds the MIC (%*f*T >MIC) [59,60]. The third PKPD index is for concentration-dependent bactericidal antibiotics with a long post-antibiotic effect (*e.g.*, amikacin). For these drugs, higher $C_{\max}$ values lead to faster and more extensive bacterial killing, and the prolonged post-antibiotic effect maintains suppression after concentrations fall. Their efficacy is traditionally described by the ratio of the $C_{\max}$ over the MIC ($C_{\max}/\text{MIC}$) [59,60].

For each antibiotic, a PKPD target value defines the level of exposure required for optimal bacterial killing. Two major developments have shaped the use of PKPD targets. First, there has been a shift towards assessing these targets as early as possible in therapy, in line with early goal-directed antimicrobial therapy, aimed at decreasing mortality and shortening hospital stay [61,62]. Second, target indices and their associated target values have been refined over time. Importantly, PKPD targets define efficacy thresholds only whereas safety is assessed independently using concentrations cutoffs or time-to-toxicity metrics, depending on the drug. Thus, efficacy-based PKPD targets and safety metrics need to be interpreted together when informing dosing decisions.

Some drugs have a narrow therapeutic window, meaning that their exposure must be maintained within a narrow range bounded at the lower end by loss of efficacy and at the upper end by unacceptable toxicity. Definitions of such drugs vary across regulatory agencies, which were historically identified based on clinical experience rather than explicit quantitative criteria [63]. The FDA, for example, attributes narrow therapeutic window label to drugs for which small changes in exposure can result in serious therapeutic failure or toxicity, potentially leading to life-threatening outcomes or causing persistent or significant disability [63]. In contrast, the European Medicines Agency (EMA) does not provide a formal definition for narrow therapeutic window drugs and instead recommends a case-by-case assessment [63].

For antibiotics with a narrow therapeutic window (e.g., vancomycin), PKPD targets alone are therefore not sufficient to determine appropriate dosing at steady state. In these cases, TDM is recommended. TDM involves measuring antibiotic concentrations and comparing them with the boundaries of the therapeutic window. Based on these measurements, dosing is adjusted to avoid underexposure (risk of therapeutic failure), or overexposure (risk of toxicity).

The following subsections focus on the antibiotics investigated in this thesis, with emphasis on their PKPD indices, PKPD target, safety metrics, and the application of TDM, where relevant.

Vancomycin

Due to its narrow therapeutic window, vancomycin is subject to TDM when administered for more than 48 hours. Historically, dosing was guided by trough concentration (C_{trough}), which was used as surrogate marker for 24-hour AUC (AUC_{24h}) [64]. Over the last five years, recommendations have shifted toward AUC_{24h} -based TDM, making the ratio of the AUC_{24h} over the MIC (AUC_{24h}/MIC) the preferred PKPD index [64]. AUC_{24h} -guided TDM can be achieved either by applying first-order PK equations to peak–trough pairs obtained at steady state, or by using Bayesian software that incorporates one or two serum concentrations, with at least one trough measurement [65]. However, AUC_{24h} -based TDM is more complex to implement than steady state trough concentration ($C_{ss, \text{trough}}$)-based TDM. As a result, trough-based TDM is

still commonly applied in clinical practice [64]. As continuous infusion became increasingly adopted, the use of steady state plateau concentrations ($C_{ss,plateau}$) was introduced, and TDM based on steady state trough or plateau concentrations ($C_{ss,trough/plateau}$) became common in practice.

In vivo PKPD studies of vancomycin against MRSA have shown that $AUC_{24h}/MIC \geq 400$ is required to achieve a bactericidal effect [64]. Below this PKPD target, *in vitro* studies have demonstrated an increased risk of development of vancomycin-resistant strains [65,66]. Regarding safety, nephrotoxicity was associated with $AUC_{24h} > 650 \text{ mg}\cdot\text{h/L}$ [64]. To maintain a safety margin below this toxicity threshold while balancing between efficacy and safety, current guidelines recommended AUC_{24h} range of 400 to 600 $\text{mg}\cdot\text{h/L}$ for an MIC of 1 mg/L [65].

Meropenem

The meropenem PKPD target varies according to the severity of illness. In non-critically ill patients, a PKPD target of $40\%fT > 1 \times MIC$ has historically been used [67]. In critically ill patients, higher targets are recommended, such as $100\%fT > 1 \times MIC$, or even $100\%fT > 4 \times MIC$ to maximize antibiotic efficacy [67].

The EUCAST clinical MIC breakpoint for Enterobacterales to meropenem is 2 mg/L [68]. Therefore, achieving $100\%fT > 1 \times MIC$ corresponds to maintaining unbound meropenem concentrations $> 2 \text{ mg/L}$ throughout the dosing interval, whereas $100\%fT > 4 \times MIC$ corresponds to unbound meropenem concentrations $> 8 \text{ mg/L}$.

Regarding safety, the risk of nephrotoxicity and neurotoxicity become more likely when C_{trough} exceeds 44.5 mg/L and 64.2 mg/L, respectively [54].

Amikacin

Historically, the PKPD index used to describe amikacin efficacy was the $C_{\text{max}}/\text{MIC}$ [69]. However, more recent evidence indicates that $\text{AUC}_{24\text{h}}/\text{MIC}$ is a more appropriate PKPD index [69].

The required PKPD target varies according to the severity of illness, immune status, bacterial load, and whether amikacin is used as monotherapy or combination therapy against gram-negative bacteria [69]. In immunocompetent, non-critically ill patients with a low bacterial burden (e.g., urinary tract infections), or in those on combination therapy, an $\text{AUC}_{24\text{h}}/\text{MIC}$ target of 30 to 50 is generally considered sufficient [69]. In contrast, a higher PKPD target is recommended in patients with critical illness and high bacterial load (e.g., pneumonia), or in the infrequent clinical scenarios where amikacin is administered as monotherapy. In such cases, a PKPD target based on $\text{AUC}_{24\text{h}}/\text{MIC} \geq 80$ and PKPD target based on $C_{\text{max}}/\text{MIC} \geq 8$ are recommended [21,69]. Using the EUCAST clinical MIC breakpoint for Enterobacterales to amikacin of 8 mg/L, this translates to an $\text{AUC}_{24\text{h}}$ of 640 mg×h/L and a C_{max} of 64 mg/L [21,69].

Regarding safety, the risk of toxicity increases with elevated and prolonged amikacin exposure. According to UZ Leuven clinical practice, safety depends on two metrics: maintaining a 24-hour postinfusion concentration ($C_{24\text{h}}$)

≤ 3 mg/L and minimizing the time to reach an amikacin concentration exceeding 3 mg/L.

5. Pharmacometric approaches

Pharmacometrics is a quantitative discipline that integrates mathematical, statistical, and pharmacological principles to describe and predict drug PK and PD, and the associated uncertainty [70]. It is widely applied by regulatory authorities, pharmaceutical companies and academic researchers as an evidence-based decision-making tool to optimize dosing strategies while assessing safety in (special) patient populations [70]. In the clinical setting, the value of pharmacometrics lies in its ability to support context-specific dosing decisions, taking into account cost and resource constraints (e.g., availability and timing of laboratory tests), as well as practical considerations related to the clinical workflow.

Pharmacometrics is based on nonlinear mixed-effects modeling, which describes simultaneously the typical behavior of all individuals — the population level — and the variability across those individuals [71]. The term “nonlinear” refers to a nonlinear relationship between the dependent variable (e.g., drug concentrations) and the model parameters or other independent variables (e.g., time) [72]. The term “mixed effects” refers to the parameterization of fixed effects and random effects [71]. Fixed effect parameters take each a single value that represents the typical value in the population, whereas random effect parameters describe variability across its

individuals or any remaining unexplained variability [71]. Parameter estimation aims to identify the set of fixed and random effects that best describe the clinical data, typically by minimizing the objective function value (OFV), calculated as minus twice the log-likelihood. A lower OFV corresponds to a better fit to the clinical dataset under the maximum likelihood estimation [71].

Population PK (popPK) modeling is one of the major applications of nonlinear mixed-effects modeling [71]. PopPK models describe the relationship between drug dose and exposure at the population level while analyzing all individuals simultaneously [71]. They consist of three components [73]. The first is the structural component which describes in mathematical equations the relationship between dose and exposure in the typical patient [73]. It defines for example the typical PK parameters such as CL [73]. The second is the statistical component, which quantifies unexplained variabilities around the structural model at two levels [73]. Level 1 variability reflects between-subject (interindividual variability [IIV]) and, when applicable, within-subject (interoccasion variability [IOV]) variability [73]. The IIV is assumed to follow an exponential (log-normal) distribution to ensure that PK parameters remain positive [73]. Level 2 variability captures the unexplained differences between the individually predicted and observed drug concentrations due to assay variability, timing errors in sample collection, and model misspecification [73]. This residual unexplained variability (RUV) may follow additive, proportional, or combined error models [73]. The third component is the covariate

component, which explains part of the variability through statistically significant patient-, disease-, therapy-, or environment-related characteristics (*i.e.*, covariates such as body weight, kidney function, and comedication) [73]. Identifying covariates that can explain IIV in PK parameters is a key step in popPK model development. Stepwise covariate modeling (SCM) is one of the most used methods [74]. The parameterization depends on the type of covariate data: continuous (*e.g.*, body weight) or categorical (*e.g.*, comedication) [73]. Model performance is evaluated using the OFV together with diagnostic graphics (*e.g.*, goodness-of-fit plots), simulation-based assessments (*e.g.*, visual predictive check [VPC]), and precision of parameter estimates (*e.g.*, bootstrap) [75–77].

In this thesis, the antibiotics under investigation have been approved and used in clinical practice for many years. Consequently, numerous popPK models have already been published, including some developed using clinical datasets from the special patient populations investigated here. In this thesis, these existing models were applied and validated rather than developing new ones. They were applied in model-informed precision dosing (MIPD) and simulation-based dosing optimization. Importantly, the purpose of models was not solely to identify optimized dosing recommendations, but also to ensure their feasibility in real-world clinical settings. This requires consideration of not only model characteristics, but also practical constraints such as the availability of covariates at the time of prescribing, and the complexity of implementation within routine workflows.

All pharmacometric analyses in this thesis were performed using NONMEM (version 7.5.0; ICON Development Solutions, Gaithersburg, MD, USA), with a GNU Fortran 95 compiler and the Perl-speaks-NONMEM (version 4.7.0) toolkit, executed through the Pirana interface (version 2.9.7; Certara, Inc., Princeton, NJ, USA).

Several dosing strategies have been proposed for adults, ranging from TDM, MIPD to simulation-based dosing strategy. The following subsections describe these strategies in detail.

5.1 Therapeutic drug monitoring

TDM was introduced briefly earlier in the thesis. It is used to individualize antibiotic dosing by comparing measured drug concentrations with target concentration. However, TDM has several limitations. First, it relies on one or two drug concentrations obtained at steady state, which provide only a snapshot of drug exposure and often fail to capture the full PK profile [78]. As a result, TDM alone cannot calculate exposure metrics such as the AUC, which require a popPK model. Second, its implementation in clinical practice is hindered by correct drug administration (*e.g.*, complete line flushing), and precise sampling times — conditions that are often difficult to ensure in clinical practice [78]. Third, TDM is resource-intensive, as it requires healthcare personnel, and costs for laboratory analysis. Finally, TDM-guided dose adjustment is not standardized. Different clinicians may adjust dosing in

different ways based on the same TDM result, leading to variability in clinical practice.

Beyond vancomycin, TDM has been proposed for several antibiotics used in adults, including meropenem and amikacin [78]. For meropenem, recent evidence suggests that TDM may be useful in certain special patient populations (e.g., older adult patients, critically ill patients) in whom it exhibits wide PK variability. However, clinical trial evidence is still emerging [78]. For amikacin, TDM has been criticized for sometimes leading to underexposure or overexposure. Hence, the overall clinical benefit of TDM remains uncertain [78]. Additionally, implementation is often hindered by logistical barriers including limited assay availability, regulatory and operational constraints [78]. These limitations underscore that the clinical utility of TDM ultimately depends on its feasibility within the intended clinical setting.

5.2 Model-informed precision dosing

MIPD has emerged as an approach to address several limitations of TDM. MIPD uses Bayesian forecasting, combining a drug-specific popPK model with covariates, with or without measured concentrations, to estimate individual PK parameters and recommend an appropriate dose [78]. In contrast to TDM, which is typically applied once steady state is reached, MIPD provides an *a priori* prediction, with the popPK model with covariates acting as prior information and therefore strongly influencing the parameter estimates [78]. This enables prediction of drug concentrations before steady

state. However, because covariates explain only a small proportion of PK variability, *a priori* predictions are often limited in accuracy [78]. When measured concentrations become available, the influence of the prior decreases as the model is updated (*i.e.*, Bayesian updating), resulting in a *posteriori* prediction that enables more accurate Bayesian forecasting [78]. This update in predictions reduces the impact of operational errors (*e.g.*, correct drug administration, precise sampling times) commonly encountered in clinical practice, which cannot be addressed by TDM. Additionally, because MIPD relies on popPK model, it enables direct model-based estimation of exposure metrics such as the AUC, whereas TDM alone does not directly provide such estimates [78].

The predictive performance of MIPD is typically evaluated through bias and precision [78]. The selection of models used for forecasting depends on both the drug and the special patient population under investigation [78]. Close alignment between the characteristics of the popPK model and the clinical dataset improves the predictive performance [78]. However, validating single popPK models in heterogeneous real-world clinical datasets is challenging, time-consuming and expensive [78]. To overcome this, multi-model approaches such as the model averaging algorithm (MAA) and the model selection algorithm (MSA) have been proposed to outperform MIPD based on a single model [79].

MIPD is increasingly implemented through software tools that incorporate popPK models and Bayesian forecasting algorithms, to support clinical decision-making by estimating the probability that different dosing strategies will achieve PKPD targets [80]. However, regulatory considerations remain. EMA classifies these tools as medical devices, requiring evidence of transparency, safety, and performance before their clinical implementation [81,82]. In addition, the European General Data Protection Regulation (GDPR, Regulation (EU) 2016/679) provides individuals with a right to “meaningful and effective human review”, allowing them to question or reject algorithm-based recommendations, including those generated by MIPD [83]. However, how these requirements apply to MIPD remain unclear, given that MIPD provides dosing suggestions that are reviewed by a pharmacist, and the final prescribing decision remains the responsibility of the treating physician.

5.3 Dosing simulations

PopPK models also enable simulation-based dosing optimization [73]. They allow *in silico* evaluation of an unlimited number of dosing strategies, making it possible to predict drug exposure, assess the efficacy-based PKPD target attainment, and evaluate the drug specific safety metrics in (special) patient populations [73]. The strength of this approach lies in its ability to prioritize the most promising dosing strategies before clinical testing, thereby focusing research efforts, streamlining clinical trials, and accelerating evidence-based implementation [73]. In that sense, dosing simulations becomes a translational

tool that can inform the selection of feasible dosing strategies with potential clinical utility.

Simulation-based dosing optimization typically evaluates numerous dosing strategies globally classified into two categories: fixed dosing and covariate-based dosing [84]. Fixed (population-level) dosing assigns a single dose to all patients within (specific) patient population(s), representing a one-size-fits-all approach. In contrast, covariate-based (group-level) dosing stratifies doses according to (clinically relevant) statistically significant characteristics (*i.e.*, covariate) [84]. These strategies may be derived from literature or developed from incremental changes of the existing standard dosing strategy [85].

Simulation-based dosing optimization requires virtual patient datasets that accurately represent the targeted real-world populations [86]. This involves preserving the distribution of covariates, their frequencies, and their correlations [86]. The first step is to evaluate the clinical impact of each covariate by determining whether variation in that covariate alters the probability of PKPD target attainment for a given dosing strategy [87]. Covariates that meaningfully influence PKPD target attainment are considered clinically relevant and may justify a stratified dosing strategy [87]. When such strategies safely achieve PKPD targets, they can be proposed as optimized dosing strategies. However, if no covariate-based dosing strategy reaches PKPD targets without risking toxicity, more individualized methods — such as TDM or MIPD — become warranted [85,88]. In practice, however, the

selection of a dosing strategy may need to be adapted, to account for feasibility and implementation constraints.

Objectives

Achieving adequate antibiotic exposure remains a challenge in certain special patient populations. This is due to physiological changes associated with aging in older adults and profound hemodynamic changes in critically ill patients, resulting in suboptimal drug exposure under standard dosing strategies.

In this thesis, we hypothesized that the application, and where necessary further development, of popPK models in MIPD and simulation of intravenous antibiotics in selected special patient populations could help identify dosing strategies that improve the efficacy-based PKPD target attainment and reduce risks of side effects, while remaining feasible for implementation in real-world clinical practice.

The main objective was to explore the value of the application, and where necessary further development, of popPK models in MIPD and simulation in optimizing clinically applicable dosing strategies for intravenous antibiotics in clinical settings where patients experience subtherapeutic or supratherapeutic exposure under standard dosing strategies. This PhD project specifically is structured around the following objectives:

Part 1. Hospitalized adults and older patients

Chapter 1. Vancomycin precision dosing: models and tools

To summarize published popPK models of vancomycin in adult patients and provide a comprehensive overview of publicly available MIPD software tools that incorporate these models.

Chapter 2. Predictive performance of vancomycin models in older patients

To evaluate the predictive performance of both single-model and multi-model approaches MIPD using published popPK models to guide vancomycin dosing in older patients.

Part 2. Critically ill patients

Chapter 3. Meropenem dosing in the intensive care unit

To provide an optimized meropenem dosing recommendation for all critically ill adult patients with preserved renal function or ARC admitted to the ICU across a wide range of kidney function markers and underlying patient characteristics.

Chapter 4. Amikacin dose optimization in the emergency department

To provide an amikacin dosing strategy with a clinically acceptable probability of target attainment (PTA) for all critically ill adult patients admitted to the ED.

Research papers

Part 1. Hospitalized adults and older patients

Chapter 1. Vancomycin precision dosing: models and tools

This chapter is based on

Nada Dia *et al.*, Precision Dosing of Vancomycin: A Systematic Review of Population pharmacokinetics. Submitted

Abstract

Background Vancomycin is a widely used antibiotic with a narrow therapeutic window and considerable pharmacokinetic (PK) variability, necessitating accurate and precise dosing. Population PK (popPK) models have become essential for facilitating model-informed precision dosing (MIPD) of vancomycin. We aimed to summarize and compare popPK models of vancomycin and evaluate MIPD software modules incorporating these models.

Methods We systematically searched PubMed, EMBASE, and reference lists of relevant articles from inception through 01 October 2024 to identify articles describing the development of compartmental, one-stage parametric popPK models based on data from adult patients (≥ 18 years) receiving intravenous vancomycin. Studies were excluded if they (i) did not present original research, (ii) were limited to laboratory or animal data, (iii) investigated non-intravenous formulations, or (iv) lacked essential information (*e.g.*, covariates). We extracted and summarized key information on study design, patients, vancomycin dosing strategies, sampling strategies, quantification methods, modeling techniques, and covariates. We registered the protocol in the Open Science Framework registries (osf.io/48sfb). We contacted providers of MIPD software tools and invited them to complete an online questionnaire assessing the features and clinical integration of

their vancomycin module. We evaluated the incorporated models and their clinical applicability. No funding was received.

Results We identified 88 vancomycin popPK models across 86 articles: 44 (50.0%) were one-compartment models, 41 (46.6%) two-compartment models, and 3 (3.4%) three-compartment models. Kidney function estimators and body weight metrics were the most commonly retained covariates on clearance and volume of distribution, respectively. Of 18 identified MIPD software tool providers, 13 (72.2%) completed the questionnaire, confirming the inclusion of vancomycin modules. These tools incorporated a total of 101 vancomycin models (some unpublished), ranging from 1 (NextDose) to 51 (InsightRx Nova) models. Of the 13 tools, four included models exclusively for adult patients (APK, ClinCalc, TDMx, and VancoCalc), while nine supported dosing for both adult and pediatric populations (DoseMeRx, ID-ODS, InsightRx Nova, MWPharm, NextDose, PKJust, PrecisePK, Rx Studio, and Tucuxi). Based on the questionnaire responses, three tools (ID-ODS, InsightRx Nova, and MWPharm) had been evaluated in prospective non-interventional studies, and two (InsightRx Nova, and NextDose) in a prospective interventional trial. Five tools (DoseMeRx, InsightRx Nova, MWPharm, PrecisePK, and Rx Studio) were certified as conforming to EU regulatory standards under the Medical Device

Directive and were in the process of obtaining EU conformity under the Medical Device Regulation.

Conclusion This review presents a comprehensive overview of vancomycin popPK models and MIPD software modules for adult patients. Our findings highlight the diversity among popPK models and the need for prospective evaluation to support clinical implementation of MIPD.

1. Introduction

For over 60 years, the glycopeptide antibiotic vancomycin has been prescribed intravenously to a diverse population of hospitalized patients with serious infections, including central line–associated bloodstream infections, endocarditis, infected thrombi, periprosthetic joint and fracture-related infections, and complicated skin and soft-tissue infections. Vancomycin is effective against gram-positive pathogens such as MRSA, coagulase-negative *Staphylococci*, and ampicillin-resistant *Enterococcus* species. It exhibits a mixed time-dependent, concentration-enhanced antibacterial activity, meaning its efficacy depends on both the duration and amount of exposure.

Vancomycin has a narrow therapeutic window. The lower bound is determined by the PKPD efficacy target, which is the AUC_{24h}/MIC of 400 [65]. The upper bound is constrained by the risk of exposure-dependent nephrotoxicity, which increases with AUC_{24h} values above 650 mg×h/L [89]. For an MIC of 1 mg/L, the target AUC_{24h} is 400–600 mg×h/L [65,90].

Adequate tissue distribution is essential for treating deep-seated infections and may be influenced by factors such as hypoalbuminemia, which reduces protein binding (typically $\leq 50\%$) and increases the free fraction of vancomycin [91]. Because only unbound vancomycin

undergoes glomerular filtration, hypoalbuminemia may also promote renal excretion of vancomycin CL. The net interplay of these effects is not well understood, and the overall clinical significance of hypoalbuminemia remains uncertain. More broadly, the half-life of vancomycin varies widely, from as short as 4.8 hours in patients with ARC to as long as 180 hours in those with impaired kidney function [92,93].

Due to substantial PK variability, most of which is unexplained by covariates such as body weight metrics or kidney function, vancomycin is a prime candidate for TDM. While the AUC_{24h} can be estimated using numerical integration based on multiple vancomycin plasma concentration measurements (trapezoid rule), this method is labor-intensive and burdensome. Therefore, surrogate targets such as C_{trough} of 12.5–17.5 mg/L during intermittent infusion or a plateau concentration ($C_{plateau}$) of 20–25 mg/L during continuous infusion are often used in clinical practice [65].

Over recent decades, numerous popPK models have been developed to describe vancomycin PK in adults. Today, these models have become instrumental in enabling AUC_{24h} -guided MIPD of vancomycin. The first vancomycin dose (*i.e.*, loading dose, if applied) can be individualized based on patient-specific covariates (*a priori* prediction). During the treatment, dosing can be further individualized using

vancomycin plasma concentrations (*a posteriori* prediction). Covariate-based MIPD offers an alternative in settings where plasma concentration measurements are unavailable due to logistical, financial, or technical constraints.

A growing body of evidence supports the clinical and economic benefits of AUC_{24h}-guided MIPD, including reduced nephrotoxicity and shorter hospital stays [94,95]. Despite increasing endorsement in consensus guidelines, MIPD adoption in clinical practice is slow [96]. Barriers include the need for manual data entry, regulatory compliance challenges, and limited pharmacometric expertise in many healthcare settings [97,98]. However, in recent years, MIPD software tools have become increasingly user-friendly, with cloud-based platforms and seamless integration into electronic health record (EHR) systems facilitating efficient integration into clinical workflows [80,88].

In this systematic review, we summarize published popPK models of vancomycin in adult patients and provide a comprehensive overview of publicly available MIPD software tools that incorporate these models.

2. Materials and methods

2.1 Systematic review of vancomycin population pharmacokinetic models

Studies were reported in accordance with the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) Protocols 2020 statement [99]. The PRISMA checklists are provided in **Supplementary Tables 1–2**. The protocol was registered in the Open Science Framework registries (osf.io/48sfb) [100]. Ethics approval was not required for this study.

We conducted a systematic literature search in PubMed and EMBASE from inception through 01 October 2024 using the search strategy detailed in **Supplementary Table 3**. Articles were included if they described the development of a compartmental, one-stage parametric popPK model based on data from adult patients (≥ 18 years) receiving intravenous vancomycin. Articles were excluded if they (i) did not present original research (e.g., reviews, meta-analyses), (ii) were limited to laboratory or animal studies, (iii) investigated non-intravenous formulations, or (iv) lacked essential information (e.g., covariates).

Initial screening of title and abstract was performed in Rayyan by one reviewer (Nada Dia), who also screened the reference lists of included articles to identify additional relevant studies [101]. Potentially eligible articles underwent full-text review. Two independent reviewers (Yannick Hoffert and Angela Elma Edwina) verified the screening process to check its reproducibility. Any disagreements were resolved by consensus.

Data were extracted into Microsoft Excel (Microsoft Corporation, Redmond, WA, USA) using structured templates (**Supplementary Tables 4–8**). Extracted data included characteristics of the clinical study design (authors, year of publication, journal, study location, inclusion and exclusion criteria), patients (number of patients, sex, age, weight, SCr), vancomycin administration (dose, infusion duration, dosing interval), sampling strategies (number of samples, sampling time points), quantification (assay method and instrument, limits of quantification, coefficient of variation), modeling techniques (software, number of compartments, covariate modeling strategies and selection criteria), and covariates (tested covariates, retained covariates, covariates on CL and distribution parameters). Data extraction was filled by one reviewer (Nada Dia) and checked by other reviewers (Yannick Hoffert and Angela Elma Edwina). Any discordance was resolved by consensus.

A descriptive analysis was performed to summarize the extracted data. Categorical variables were presented as frequencies and percentages, whereas continuous variables were presented as medians, with ranges and interquartile ranges (IQR). The data analysis was conducted in R (version 4.1.1; R Core Team, Vienna, Austria), interfaced by RStudio (version 1.4.900; RStudio, Inc., Boston, MA, USA) using the *tidyverse* collection of packages [102].

2.2 Evaluation of model-informed precision dosing software tools

We developed a dynamic web-based questionnaire using Google Forms, consisting of 4 to 13 questions, depending on the respondent's answers, assessing the features of their vancomycin modules (**Supplementary Table 9**). We identified active providers of MIPD software tools through searches for publicly available online information. In July–August 2024, we contacted providers using contact details available on their official websites, inviting them to complete the questionnaire. When contact details were not available, we either filled in the online contact form provided on the website or, in the absence of a website, contacted the last author listed in the published article describing the MIPD software tool. Providers who agreed to participate were invited to complete the questionnaire. Non-respondents received a follow-up reminder.

In February 2025, we conducted a second round of follow-up by email to gather more detailed and updated information on the integrated vancomycin popPK models (references), supported patient populations, and whether the tools allowed users to customize model parameters (yes/no). A disclaimer regarding the temporary storage of collected data was included in all communications.

Responses were manually reviewed and analyzed. We assessed the number, type, and intended populations of the integrated vancomycin models. We also evaluated the user customization options, support of artificial intelligence (AI) or machine learning (ML) components, evidence of clinical validation, and regulatory status of the tools.

3. Results

3.1 Vancomycin population pharmacokinetic models

A total of 631 articles were screened, of which 86 met our selection criteria (**Figure 2**). The articles were published between 1998 and 2024, with slightly more than half appearing in the last six years (45/86, 52.3%).

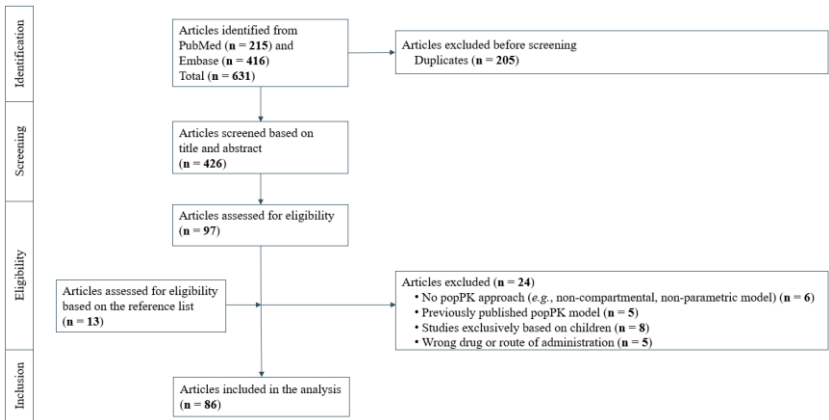


Figure 2 PRISMA flow diagram of the literature search and population pharmacokinetic model selection. PBPK, physiologically based pharmacokinetic; PopPK, population pharmacokinetic

The included articles used clinical datasets originating from 26 countries, predominantly in Asia (42/86, 48.8%) and Europe (22/86, 25.6%), with China contributing the largest share (17/86, 19.8%). Three articles were based on pooled datasets from multiple countries (**Figure 3**) [103–105].

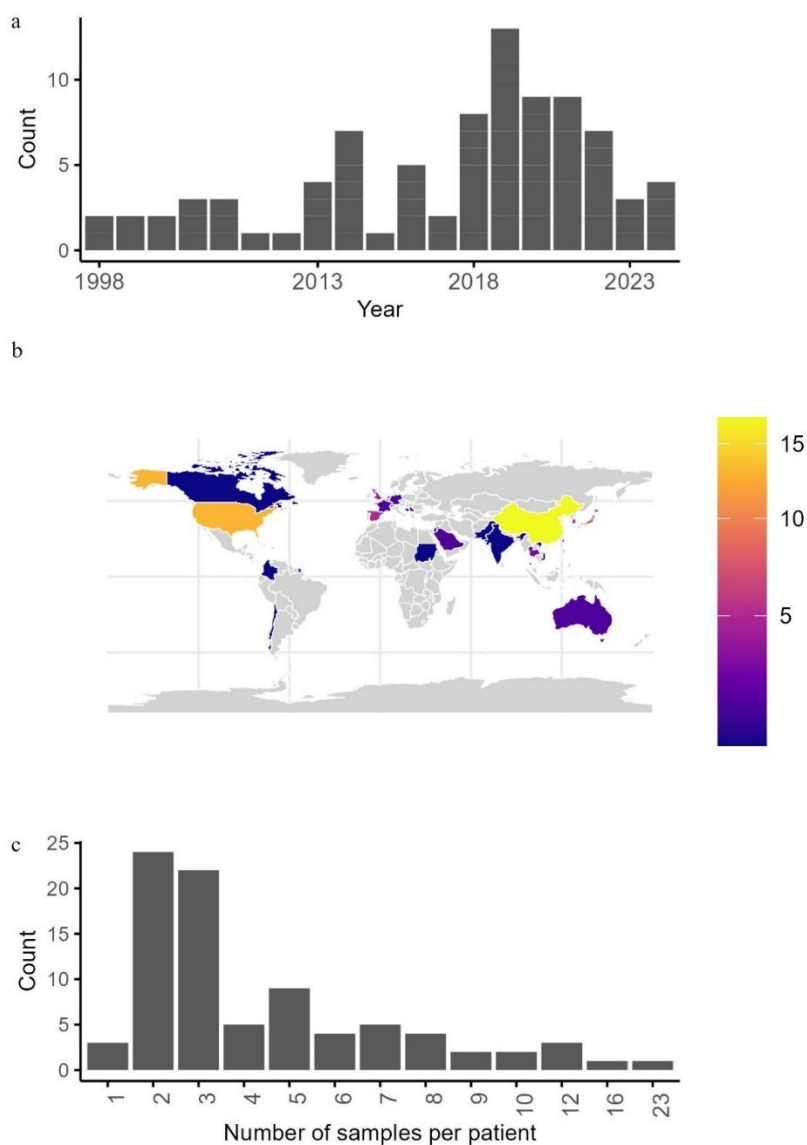


Figure 3 Overview of the (a) yearly published population pharmacokinetic models, (b) the country distribution of the clinical datasets, and (c) the number of samples per patient in the dataset for model development

Most articles (49/86, 57.0%) were based on retrospective data collection. About one-third of the articles (24/86, 27.9%) had datasets based on critically ill patients, with critical illness either explicitly stated as an inclusion criterion [106–123] or inferred from admission to the ICU [124–129]. About one-tenth of the articles (10/86, 11.6%) described datasets based on patients receiving RRT [109,121–123,130–135]. A control group, defined as either healthy individuals or patients without the primary characteristic of interest, was included in 10 articles [124–126,129,136–141] (**Supplementary Table 4**).

Of the 86 articles, 66 (76.7%) focused exclusively on adult patients (≥ 18 years). Among these, 28 included older adults (*i.e.*, ≥ 75 years) [107,112,114–117,119,123,127,130–133,139,140,142–154], although only two specifically targeting older adults (*i.e.*, median or mean age ≥ 75 years) [146,149]. The remaining 20 articles (23.3%) covered broader age ranges [104,105,136,126,106,113,155,134,156–167]: seven focused primarily on pediatric populations (*i.e.*, median or mean age < 18 years) [105,106,113,126,134,155,160], two included both children and adults under 75 years [158,167], and 11 spanned the full age spectrum (*i.e.*, < 18 years to ≥ 75 years) [104,136,156,157,159,161–166] (**Supplementary Table 5**).

Patients received vancomycin via intermittent (46/86, 53.5%), continuous (10/86, 11.6%), or mixed intermittent-continuous (3/86,

3.5%) infusion strategies. The infusion modality was not reported in 27 (31.4%) studies. For model development, the median number of patients was 91 (range: 7–2,554; IQR: 30–191), and the median number of samples was 256 (range: 64–8,877; IQR: 165–602), resulting in a median of three samples per patient (range: 1–23; IQR: 2–6). Rich sampling (defined as >6 samples per patient) was documented in 18 studies [107,109,113,115,116,118,121,125,128,129,132,133,138,154,160,168–170]. Approximately two-thirds of the articles (58/86, 67.4%) included steady state concentrations. Various assay types were used to measure vancomycin concentrations, although nine articles published between 2019 and 2024 did not report the assay type [104,166,107,171,148,144,172,117,135] (**Supplementary Table 6**).

A total of 88 popPK models were developed across the 86 articles, with two articles reporting two models each [141,173]. Half of the models were one-compartment (44/88; 50.0%), while most of the remainder were two-compartment (41/88; 46.6%), and a small number were three-compartment models (3/88; 3.4%). The median number of samples per patient was three for one-compartment models (range 1–23; IQR: 2–3), five for two-compartment models (range 1–12; IQR: 3–7), and nine for three-compartment models (range 9–12; IQR: 9–11) (**Supplementary Table 7**).

IIV was modeled exponentially in 85 of 88 models (96.6%), while RUV was modeled proportionally in 36 of 88 models (40.9%). Four models reported high IIV (>100% for at least one PK parameter) (4.5%) [113,116,136,163]. IOV was included in two models (2.3%) [104,136].

Across all models, 55 patient characteristics were tested as potential covariates. The most frequently tested covariates were body weight metrics (84/88; 95.5%), age (82/88; 93.2%), and kidney function estimators (71/88; 80.7%). In total, 27 characteristics were retained as statistically significant covariates on CL, and 15 on central volume of distribution (V_c). The top three covariates retained on CL were kidney function estimators (58/88; 65.9%), body weight metrics (29/88; 33.0%), and age (14/88; 15.9%). For the V_c , the most retained covariates were body weight metrics (48/88; 54.5%), age (10/88; 11.4%), and fat-free mass (5/88; 5.7%) (**Supplementary Table 8, Figure 4**).

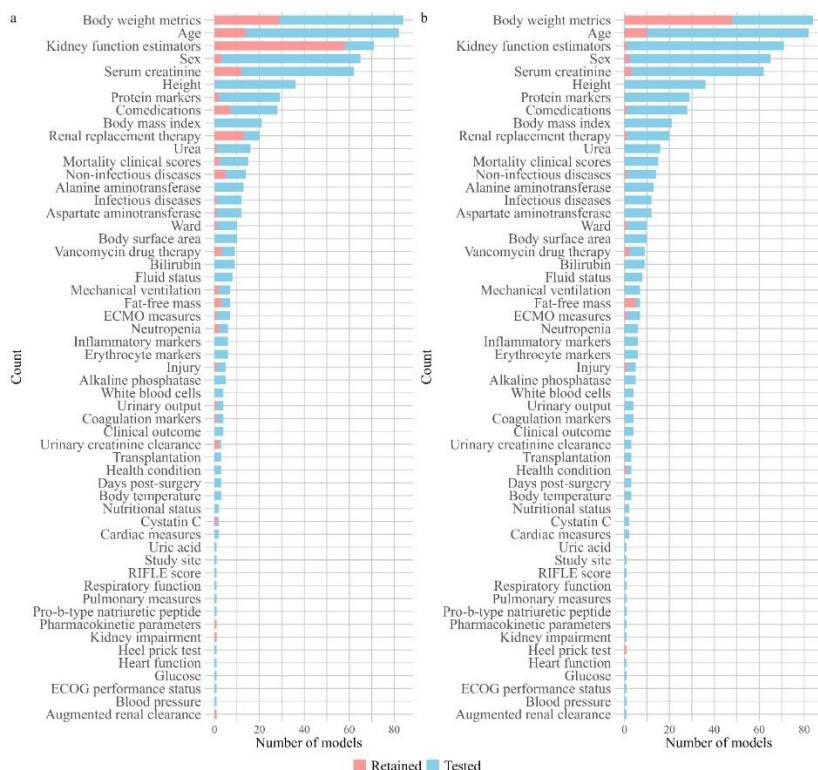


Figure 4 *Tested* and *retained* characteristics in the population pharmacokinetic models on (a) the clearance, and (b) the central volume of distribution. The blue and pink bar represents the total number of models in which characteristics were *tested* and *retained*, respectively. ECMO, extracorporeal membrane oxygenation; ECOG, Eastern Cooperative Oncology Group; RIFLE, risk, injury, failure, loss, end-stage

Among the 24 models developed from datasets of critically ill patients [106–129], none tested ARC using mCrCL_{24h} cutoff of 130 mL/min/1.73 m² [16]. By contrast, SCr (15/24, 62.5%), kidney function estimators (14/24, 58.3%), and clinical mortality scores (11/24, 45.8%) were frequently tested. In the 10 models focusing on patients receiving RRT

[109,121–123,130–135], dialysis-related biomarkers were often tested as covariates, with SCr and kidney function estimators each tested in 4/10 models (40.0%). For the 39 models including older patients [104,107,112,114–117,119,123,127,130–133,136,139,140,142–154,156,157,159,161–166], age-related covariates were infrequently tested: protein markers in 13/39 (33.3%), fluid status in 5/39 (12.8%), and nutritional status in 2/39 (5.1%); no model evaluated a formal geriatric clinical score. Age itself was retained in only a few models (6/39, 15.4% on CL and 5/39, 12.8% on V_c).

3.2 Model-informed precision dosing software tools

We identified 18 active providers of MIPD software tools, of which 13 (72.2%) completed our questionnaire. All respondents confirmed the inclusion of a vancomycin module. Based on publicly available information, four of the five non-responding software tool providers also offer a vancomycin MIPD module, except for iDose [174–176] (**Figure 5**).



Figure 5 Overview of the 18 model-informed precision dosing software tool providers we identified, with 17 having a vancomycin module

A total of 101 vancomycin models were integrated into the MIPD software tools, some of which were unpublished, ranging from 1 (NextDose) to 51 (InsightRx Nova). Out of 101 models, 45 (44.6%) were exclusively recommended for MIPD in adults, 43 (42.6%) in children, 3 (2.9%) across a full age spectrum, and 10 (9.9%) were undisclosed (**Supplementary Table 10**). Approximately half of the models were recommended for hospitalized patients in general

(49/101; 48.5%), with fewer models targeting specific patient populations, such as those with ARC (1/101; 1.0%) [177], cancer (5/101; 5.0%) [103,177–179], cardiac surgery (2/101; 3.0%) [180,181], critical illness (10/101; 9.9%) [106,115–117,119,120,157,182,183], cystic fibrosis (1/101; 1.0%), extracorporeal membrane oxygenation (1/101; 1.0%) [184], injury (*e.g.*, burn) (1/101; 1.0%) [136], kidney impairment (9/101; 8.9%) [115,126,157,161,163,179,185–187], obesity (11/101; 10.9%) [117,138,148,166,188–193], pregnancy (1/101; 1.0%) [158], RRT (11/101; 10.9%) [130,134,144,157,183,194–197]. Of the 13 tools, four included models exclusively for adult patients (APK, ClinCalc, TDMx, and VancoCalc). The remaining nine tools supported dosing for both adult and pediatric populations (DoseMeRx, ID-ODS, InsightRx Nova, MWPharm, NextDose, PKJust, PrecisePK, Rx Studio, and Tucuxi) (**Table 1**).

In five tools (APK, ID-ODS, MWPharm, PrecisePK, and Tucuxi), users could input custom model parameters. Five tools (ID-ODS, InsightRx Nova, MWPharm, PrecisePK, and Rx Studio) reported incorporating AI or ML components [198].

Based on the questionnaire responses, three tools, (ID-ODS, InsightRx Nova, and MWPharm) indicated having been evaluated in prospective non-interventional studies [199–201], while two tools (InsightRx Nova, and NextDose) indicated having been evaluated in

prospective interventional studies [202,203]. All tools reported being in clinical use for vancomycin dosing, except for ClinCalc and VancoCalc. The estimated number of patients whose vancomycin dosing was individualized using these tools in the 12 months prior to the survey (July 2023–August 2024) varied widely, from 255 (PKJust) to 458,693 (InsightRx Nova). Five tools (DoseMeRx, InsightRx Nova, MWPharm, PrecisePK, and Rx Studio) were certified as conforming to EU regulatory standards under the Medical Device Directive (MDD) and were in the process of obtaining EU conformity under the Medical Device Regulation (MDR).

Table 1 Overview of the models per population across the model-informed precision software tools

Population ^a	APK	ClinCalc	DoseMeRx	ID-ODS	InsightRx	MWPharm	NextDose	PKJust	PrecisionPK	RxStudio	TDMx	Tucuxi	VancoCalc
Older adults	–	–	–	–	✓	✓	✓	–	–	–	–	✓	–
Cancer	–	–	–	–	✓	–	–	–	–	–	–	–	–
Adults	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Cancer	–	–	–	–	✓	–	–	–	–	✓	–	–	–
Critical illness	–	✓	✓	✓	✓	✓	–	–	–	–	✓	✓	✓
ECMO	–	–	–	✓	–	–	–	–	–	–	–	–	–
Injury	–	–	–	–	✓	–	–	–	–	–	–	–	–
Kidney impairment	✓	–	–	✓	–	✓	–	–	✓	✓	–	✓	–
Obesity	–	✓	✓	–	✓	✓	–	–	–	–	✓	–	✓
Pregnancy	–	–	–	–	✓	–	–	–	–	–	–	–	–
RRT	–	–	✓	✓	✓	✓	–	–	–	–	–	–	–
Pediatric non-neonates	–	–	✓	✓	✓	✓	✓	✓	✓	✓	–	✓	–
ARC	–	–	–	–	✓	–	–	–	–	–	–	–	–
Cancer	–	–	–	–	✓	–	–	–	–	–	–	–	–
Cardiac surgery	–	–	–	–	✓	–	–	–	–	–	–	–	–
Critical illness	–	–	–	✓	✓	–	–	–	–	–	–	–	–
Kidney impairment	–	–	–	–	✓	–	–	–	–	–	–	–	–

Obesity	–	–	–	–	✓	✓	–	–	–	–	–	–	–
RRT	–	–	–	✓	–	–	–	–	–	–	–	–	–
<i>Pediatric neonates</i>	–	–	✓	✓	✓	✓	✓	✓	✓	–	–	✓	–
ARC	–	–	–	–	✓	–	–	–	–	–	–	–	–
Cancer	–	–	–	–	✓	–	–	–	–	–	–	–	–
Cardiac surgery	–	–	–	–	✓	–	–	–	–	–	–	–	–
Preterm	–	–	✓	–	✓	–	✓	✓	–	–	–	✓	–
RRT	–	–	–	✓	–	–	–	–	–	–	–	–	–

^a Information on the recommended use of one, two, and thirteen models was not fully disclosed by APK, Rx Studio, and PrecisePK, respectively. The model from APK was recommended for model-informed precision dosing in adults, the two models from Rx Studio for pediatric populations, and the thirteen models from PrecisePK for use across adult and pediatric populations, including neonates, in patients with cancer, critical illness, cystic fibrosis, obesity, or undergoing renal replacement therapy (RRT)

ARC, augmented renal clearance; ECMO, extracorporeal membrane oxygenation

4. Discussion

This systematic review provides a comprehensive synthesis of vancomycin popPK models in adults. We critically evaluated their characteristics, clinical applicability, and integration into MIPD software tools.

Our findings underscore the evolution from trough-based to AUC_{24h}-guided vancomycin dosing, which has reshaped the development and utility of popPK models. Historically, one-compartment models sufficed for trough-based monitoring. Yet, the shift toward AUC_{24h} estimation necessitates richer sampling to allow more accurate model structures. Limited sampling strategies (e.g., peak-trough pairs [204]) offer a pragmatic compromise, enabling two-compartment models and accurate AUC_{24h} estimation without overburdening clinical workflows. Furthermore, MIPD using limited sampling strategies has been shown to improve cure rates compared to trough-guided MIPD alone [205]. While the use of AUC_{24h} as a target is the key determinant of clinical performance, a two-compartment model structure alone does not guarantee predictive accuracy [164,206]. For example, using a two-compartment model with only a C_{trough} was found to underestimate AUC_{24h} by 23% thus increasing the risk of nephrotoxicity [195]. This finding indicates that relying solely on a C_{trough} to estimate AUC_{24h} can be inaccurate in two-compartment models, where the distribution phase is not captured. Nevertheless, adding a second concentration generally improves precision for

intermittent-infusion strategies, whereas continuous infusion simplifies AUC estimation.

The diversity among vancomycin popPK models reflects the heterogeneity of patient populations and clinical settings. Most models were originally developed for descriptive purposes rather than MIPD, complicating their direct application in clinical decision-making. Current MIPD software tools for vancomycin incorporate a wide range of popPK models, illustrating the lack of consensus regarding optimal model selection for MIPD. Matching model development populations to target patients for MIPD is widely considered ideal but not always feasible. Therefore, emerging strategies such as pooled data modeling [207], automated multi-model selection [79], and prior flattening [198] offer promising alternatives. Yet, challenges such as dataset restrictions and computational power exist. Furthermore, these approaches require further validation and standardization to confirm their applicability before widespread adoption in MIPD. In this context, Taylor *et al.* provided a tutorial for evaluating and selecting models for MIPD, which might strengthen future MIPD practices [208].

There is substantial variability in vancomycin PK, particularly in critically ill patients, those receiving RRT, and older adults. In critically ill patients, ARC is common and increases the risk of subtherapeutic exposure [16,209]. A similar risk exists in patients receiving RRT, where dialysis markedly enhances vancomycin CL [210]. In older patients, age-related reductions in

kidney function, comorbidities, and malnutrition are common, increasing the risk of supratherapeutic exposure and vancomycin-associated nephrotoxicity [211]. Although some biomarkers such as SCr and kidney function estimators are frequently tested, others such as ARC and age-related covariates are rarely tested. As a result, age often acted as a surrogate, capturing the cumulative effect of several age-related physiological changes rather than directly explaining variability. When more informative covariates such as kidney function estimators are tested as covariates, the independent effect of age often diminishes or disappears, because these variables more directly account for the PK variability in older adults. This is further aggravated by the lack of detailed age distributions in most studies (e.g., proportions of patients aged ≥ 80 years), limiting the ability to explore age-specific variability. In addition, critically ill patients, those on RRT, and older adults are underrepresented in model-development datasets. As a future direction, evaluating the predictive performance of the identified models in these patients will be important.

Inadequate reporting of model development details, such as covariate testing frequency and assay methodology, hampers reproducibility and limits the ability to assess model generalizability. For instance, understanding how often a covariate was tested and retained across studies informs its robustness as a predictor. Similarly, assay compatibility is critical for ensuring consistency between model development and clinical application [212]. While a guideline for reporting popPK analyses in drug development exists, a comparable

guideline for MIPD is lacking [213]. We advocate standardized reporting guidelines for popPK models, akin to CONSolidated Standards of Reporting Trials (CONSORT) or PRISMA, to facilitate model evaluation and selection for MIPD.

Despite the growing availability of MIPD software tools, clinical adoption remains limited. Regulatory certification (e.g., conformity European [CE] marking) is a prerequisite for integration into clinical workflows, particularly in the EU. However, certification alone does not guarantee clinical impact. Prospective validation, ideally through randomized controlled trials (RCTs), provides the highest level of evidence and is essential to demonstrate improved outcomes. Encouragingly, ongoing trials in adults and critically ill children will provide high-quality evidence to support broader implementation [202,203]. Yet, such trials are resource-intensive and expensive, making them challenging to conduct. As an alternative, prospective studies conducted in real-world settings, often without randomization, have shown that MIPD was associated with higher probability of achieving therapeutic vancomycin exposure over conventional dosing strategies, including standard dosing or TDM [214,215].

A CE mark is obligatory for the integration of MIPD software tools into the clinical decision within the EU [81]. Some providers initially obtained CE marking under the MDD regulations. In 2017, the EU adopted the MDR, which replaced the MDD in 2021 [216]. According to the European MDR, MIPD

software tools fall under class II, which includes non-invasive medical devices (software) with medium to high risk to patient health [82]. The CE marking indicates that MIPD software tools comply with EU safety, health, and environmental protection standards, granting them legal access to the EU market [81]. Some providers are currently applying for the MDR certification. In the case of AI or ML-based medical devices, stricter regulatory requirements apply, as they must comply not only with the MDR but also with the EU AI Act. Approved in 2024, the act is the first comprehensive regulation of AI, ensuring safe, transparent, non-discriminatory, and rights-respecting use. However, it is currently being phased in and will only become fully applicable in 2027 [217]. In all cases, software tools handle sensitive data and must therefore comply with the GDPR [218]. Beyond regulatory compliance, one of the primary challenges for MIPD remains in accelerating its integration in hospitals and ensuring appropriate end-user training. Prior research reports clear progress: many MIPD software tools are sufficiently intuitive for clinicians and clinical pharmacists to use without extensive PK training [80,88]. Nevertheless, widespread implementation remains difficult because EHR systems and clinical workflows vary across hospitals. A brief onboarding and clear local standard operating procedures are still required to equip healthcare providers with the necessary training for routine use.

AI and ML are increasingly explored as a complement to and even a substitute for traditional popPK model-based MIPD. Lee *et al.* introduced XGBoost as an ML algorithm to guide popPK model selection [219]. Hughes and Keizer

proposed a hybrid ML/PK approach, where ML acts as a binary classifier to determine whether flattening priors should be applied in popPK models for MIPD [198]. Chan *et al.* used ML to adjust the weight assigned to a model based on covariates [79,220]. Such hybrid approaches, such as ML-guided model selection or prior flattening, enhance flexibility without discarding mechanistic insights. Bououda *et al.* substituted popPK models by XGBoost algorithm in predicting vancomycin AUC_{24h} as good as the automated multi-model approach developed by Uster *et al.* [79,221]. Such fully ML-based models provide several advantages over the conventional vancomycin MIPD. Unlike popPK models, ML is not constrained to a predetermined distribution, allowing it to capture nonlinear patterns. Although it better recognizes complex patterns, making it particularly beneficial for atypical patients, ML tools may face limitations in dose simulation and regulatory acceptance. As ML tools mature, their integration into MIPD workflows will depend on transparent validation, interpretability, and alignment with clinical needs.

Our work has several limitations. First, the lack of reporting proportion of older patients limits our ability to evaluate how well these models represent older patients. As a result, the identification of “older adult” models in this review relied mainly on available summary statistics, such as median or mean age, rather than explicit age-stratified proportions. Second, no validated tool exists for assessing the risk of bias in popPK studies. The absence of such a tool may limit the interpretation of the quality of the included studies. Developing such a tool would require defining popPK-specific bias domains, reaching

expert consensus on the essential assessment criteria, and subsequently pilot-testing and validation prior to widespread adoption. Third, we identified the 18 active providers of MIPD software tools through literature review and internet search. However, it is possible that some tools were missing.

In conclusion, the present work offers a unique contribution by systematically mapping the landscape of vancomycin popPK models and MIPD tools, highlighting their strengths, limitations, and clinical readiness. We provide a structured classification of vancomycin popPK models developed on adult data, as well as MIPD software tools according to their recommended use in various populations. By aligning model characteristics with software capabilities and patient needs, we provide a practical framework for clinicians and developers to advance precision dosing. Our structured classification of models and tools serves as a valuable resource for informed decision-making and future research.

Appendix A

Supplementary Table 1 PRISMA 2020 for abstracts checklist [99]

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g., databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Not applicable
Synthesis of results	6	Specify the methods used to present and synthesize results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarize relevant characteristics of studies.	Yes

Section and Topic	Item #	Checklist item	Reported (Yes/No)
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of studies included and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (<i>i.e.</i> , which group is favored).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (<i>e.g.</i> , study risk of bias, inconsistency and imprecision).	Not applicable
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	Yes
Registration	12	Provide the register name and registration number.	Yes

Supplementary Table 2 PRISMA 2020 checklist [99]

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	L 1-2
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	L 17-51
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	L 131-136
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	L 137-138
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	L 146-150
Information sources	6	Specify all databases, registers, websites, organizations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	L 145-146
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	L 145-146
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved,	L 151-154

Section and Topic	Item #	Checklist item	Location where item is reported
		whether they worked independently, and if applicable, details of automation tools used in the process.	
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	L 155-163
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g., for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	L 155-163
	10b	List and define all other variables for which data were sought (e.g., participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	L 155-163
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Not applicable
Effect measures	12	Specify for each outcome the effect measure(s) (e.g., risk ratio, mean difference) used in the synthesis or presentation of results.	L 164-166

Section and Topic	Item #	Checklist item	Location where item is reported
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g., tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Not applicable
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Not applicable
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	L 155-156
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	L 164-166
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g., subgroup analysis, meta-regression).	Not applicable
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Not applicable
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Not applicable

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Not applicable
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	L 186-190
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	L 186-190
Study characteristics	17	Cite each included study and present its characteristics.	L 155-156
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Not applicable
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g., confidence/credible interval), ideally using structured tables or plots.	L 155-156
Results of syntheses	20a	For each synthesis, briefly summarize the characteristics and risk of bias among contributing studies.	L 191-245

Section and Topic	Item #	Checklist item	Location where item is reported
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g., confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	L 191-245
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	L 191-245
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Not applicable
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Not applicable
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Not applicable
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	L 284-313
	23b	Discuss any limitations of the evidence included in the review.	L 314-320
	23c	Discuss any limitations of the review processes used.	L 359-360
	23d	Discuss implications of the results for practice, policy, and future research.	L 364-369
OTHER INFORMATION			

Section and Topic	Item #	Checklist item	Location where item is reported
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	L 142-144
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	L 142-144
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Not applicable
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	L 61-64
Competing interests	26	Declare any competing interests of review authors.	L 66-69
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	L 155-156

L, lines

Supplementary Table 3 Literature search for population pharmacokinetic models

Databases	Search query
PubMed	(((((vancomycin) OR (vancomycin*))) AND (((pharmacokinetic*)))) AND ((adult*))) AND (model*)) AND (population)
EMBASE	('vancomycin'/exp OR 'vancomycin*') AND ('pharmacokinetics'/exp OR 'pharmacokinetic*') AND ('adult*') AND ('model*') AND ('population')

Supplementary Table 4 Characteristics of the clinical study design

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Adane <i>et al.</i> 2015 Pharmacotherapy [188]	USA	Prospective	Age ≥ 18 years, CrCL ≥ 30 mL/min/1.73 m ² , ≥ 3 days vancomycin therapy, obesity (BMI ≥ 40 kg/m ²), suspected or confirmed <i>Staphylococcus aureus</i> infection	Contraindication to vancomycin, <4 vancomycin doses, meningitis, pregnancy
Ahmed <i>et al.</i> 2024 Drug Des Devel Ther [142]	Sudan	Retrospective	Age ≥ 18 years	Pregnancy, RRT
Aljutayli <i>et al.</i> 2022 Clin Transl Sci [222]	Canada	Retrospective	≥ 4 doses vancomycin therapy	Acute kidney failure, ECMO, impaired kidney function (ESRD), >2 L IV fluids within the last 4 vancomycin doses, RRT
Alqahtani <i>et al.</i> 2018 Antimicrob Agents Chemother [143]	Saudi Arabia	Prospective	Age ≥ 18 years, Cardiac surgical procedures	Vancomycin 3 days prior to the surgery

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Alqahtani <i>et al.</i> 2020 Pharmacotherapy [137]	Saudi Arabia	Retrospective	Cancer, ≥2 days suspected or documented gram-positive bacterial infection A control group of patients without cancer	Impaired kidney or hepatic function, RRT (dialysis)
Avedissian <i>et al.</i> 2017 Pediatr Crit Care Med [106]	USA	Retrospective	Age ≤21 years, critical illness, ≥48 hours vancomycin in the pediatric ICU, normal baseline SCr	Presence of or at risk for kidney impairment at baseline, RRT (hemodialysis)
Bae <i>et al.</i> 2019 Pharmaceutics [144]	South Korea	Retrospective	TDM	[vancomycin] >100 ng/mL, impaired kidney function (eGFR <60 mL/min/1.73 m ²)
Bang <i>et al.</i> 2022 Clin Exp Pharmacol physiol [107]	South Korea	Prospective	Age ≥20 years, BW ≥40 kg, critical illness, surgical ICU admission	Contraindication to vancomycin, Hb <8 g/dL, pregnancy, vancomycin within the last 7 days

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Belabbas <i>et al.</i> 2023 J Infect Chemother [145]	Japan	Retrospective	Age ≥ 20 years, cancer (hematologic malignancies)	Admission to the ICU, < 3.0 mg/L [vancomycin], < 2 days vancomycin therapy, RRT (dialysis)
Belavagi <i>et al.</i> 2022 Germs [108]	India	Prospective	Critical illness, legal consent, sepsis	Comedication (immunosuppressive), contraindication to vancomycin, neutropenia, RRT (hemodialysis)
Buelga <i>et al.</i> 2005 Antimicrob Agents Chemother [179]	Spain	Retrospective	Admission to the hematology unit, age ≥ 15 years, hematological cancer, suspected or documented gram-positive bacterial infection	Admission to the ICU, incomplete data
Bury <i>et al.</i> 2019 Eur J Clin Pharmacol [178]	The Netherlands	Retrospective	Age ≥ 18 years, hematological disease or solid tumor, ≥ 2 days vancomycin, suspected or documented	RRT (dialysis)

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
			gram-positive bacterial infection	
Chu <i>et al.</i> 2020 J Infect Public Health [146]	China	Retrospective	CrCL _{CG} >130 mL/min, suspected or documented gram-positive bacterial infection	Pregnancy, RRT
Chung <i>et al.</i> 2013 J Korean Med Sci [147]	South Korea	Retrospective	Age ≥18 years, SCr ≤1.2 mg/dL	–
Colin <i>et al.</i> 2019 Clin Pharmacokinet [103]	International	Prospective and retrospective	Full age spectrum (premature neonates, children, adults, extremely obese patients, older patients, burns)	ECMO, incomplete data, non-IV vancomycin administration, RRT (continuous hemodialysis, hemofiltration)
De Cock <i>et al.</i> 2014 Pharm Res [105]	International	Prospective and retrospective	Age ≤18 years	Severe impaired kidney function (SCr three times higher than the age-related reference value)

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Deng <i>et al.</i> 2013 Int J Clin Pharmacol Ther [223]	China	Retrospective		–
Dolton <i>et al.</i> 2010 Burns [136]	Australia	Prospective and retrospective	Admission to burns unit, A control group of patients without burns	Incomplete data, <2 mg/L [vancomycin], RRT (dialysis)
Donadello <i>et al.</i> 2014 Critical Care [124]	Belgium	Retrospective	Admission to the ICU, age ≤18 years, continuous IV vancomycin administration, ECMO, serial vancomycin measurement during day 1 of therapy during ECMO A control group of patients without ECMO	Burns, cystic fibrosis, non-simultaneous ECMO and vancomycin, intermittent IV vancomycin administration 2 days prior the continuous administration, pregnancy, prior involvement in a clinical dataset
Dorajoo <i>et al.</i> 2019 Front Pharmacol [172]	Singapore	Retrospective	Impaired kidney function (baseline CrCL _{CG} <60 mL/min), ≥2 doses of IV vancomycin	Incomplete data, RRT, vancomycin within the last 2 weeks

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
			administration over 72 hours	
Escobar <i>et al.</i> 2014 Int J Antimicrob Agents [109]	Chile	Prospective	Admission to the ICU, age ≥ 18 years, critical illness, RRT (high volume hemofiltration), sepsis or severe septic shock	–
Fu <i>et al.</i> 2021 Biopharm Drug Dispos [156]	China	Retrospective	Age ≥ 14 years, hematological diseases, intermittent IV vancomycin administration, neutropenia	Blood purification therapy, incomplete data
Garreau <i>et al.</i> 2021 Antibiotics (Basel) [110]	France	Prospective	Admission to the ICU, age ≥ 18 years, continuous IV vancomycin administration, critical illness, hemodynamic monitoring with PICCO device	Comorbidities influencing vancomycin pharmacokinetics (myeloma, cystic fibrosis, burns), <1 day vancomycin, no TDM

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Gawmahanil <i>et al.</i> 2018 TJPS [111]	Thailand	Retrospective	Admission to the (sub)-ICU >48 hours or ≥1 character related to critical illness, age ≥18 years, critical illness, IV vancomycin due to gram-positive bacterial infection	Incomplete data, kidney transplantation, pregnancy, RRT (hemodialysis, peritoneal dialysis)
Gonzalez <i>et al.</i> 2022 Antibiotics [168]	Colombia	Prospective	Age ≥18 years, comedication (chemotherapy), fever, hematologic neoplasia diagnosis, neutropenia	Acute kidney failure, altered hepatic morphology or architecture, CKD, comorbidities, CHF, DM, fibrosis, hepatic failure, hypothyroidism, necroinflammation, pregnancy
Goti <i>et al.</i> 2018 The Drug Monit [157]	USA	Retrospective	Admission to a unit with high prevalence of ESRD, age >16 years, impaired kidney function (ESRD)	≤5 mg/L [vancomycin], >5 days vancomycin therapy

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Goyal <i>et al.</i> 2022 Front Pharmacol [158]	USA	Retrospective	Admission and discharge during the study period, IV vancomycin, pregnancy	Non-IV vancomycin administration, non-simultaneous extracorporeal RRT with vancomycin, prior vancomycin administration
Hodiamont <i>et al.</i> 2021 J Antimicrob Chemother [128]	The Netherlands	Prospective	Admission to the ICU	–
Holford <i>et al.</i> 2024 Br J Clin Pharmacol [104]	International	Prospective and retrospective	≥1 dose of IV vancomycin	Fat-free mass/TBW >0.95, kidney function >2.5, PNA <197 days, TBW outside the 3–97 th percentile for normal growth
Huang <i>et al.</i> 2021 Intern Emerg Med [112]	China	Retrospective	Critical illness, IV vancomycin, ≥14 days monitor of trough concentration after ICU admission	ECMO, RRT

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Hughes <i>et al.</i> 2024 Ther Drug Monit [148]	USA	Retrospective	Obesity (BMI ≥ 40 kg/m ²), TDM	–
Hui <i>et al.</i> 2019 J Antimicrob Chemother [130]	Australia	Retrospective	Age ≥ 18 years, impaired kidney function, RRT (intermittent high flux hemodialysis)	–
Jaisue <i>et al.</i> 2020 Ther Drug Monit [159]	Thailand	Retrospective	Age ≥ 18 years	Incomplete data, RRT (hemodialysis)
Ji <i>et al.</i> 2018 Acta Pharmacol Sin [149]	China	Retrospective	Continuous IV vancomycin administration	RRT
Jing <i>et al.</i> 2019 J Clin Pharm Ther [173]	China	Retrospective	Admission to neurosurgery, age ≥ 18 years, ≥ 3 days vancomycin therapy, neurologic diagnosis	Cystic fibrosis, hematological malignancies, incomplete data, neutrophil count $< 1,000$ cells/mm, RRT (hemofiltration), thyroid dysfunction
Jung <i>et al.</i> 2021 Antimicrob	South Korea	Prospective	ECMO, respiratory and/or cardiac	RRT due to acute/chronic kidney injury

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Agents Chemother [169]			dysfunction, vancomycin for prophylaxis or therapy	
Kim <i>et al.</i> 2016 Int. J Antimicrob Agents [141]	South Korea	Retrospective	Age ≥ 19 years, confirmed or suspected infections, neurosurgical patients, TDM A control group of non-neurosurgical patients	Incomplete data, RRT (dialysis)
Kloprogge <i>et al.</i> 2019 Antimicrob Agents Chemother [113]	UK	Retrospective	Age < 22 years, critical illness, ≥ 1 hour vancomycin therapy	–
Kovacevic <i>et al.</i> 2020 J Crit Care [114]	Bosnia and Herzegovina	Prospective	Age ≥ 18 years, critical illness, sepsis	Acute or chronic impaired kidney function treated with RRT, pregnancy, termination of vancomycin therapy prior steady state
Le <i>et al.</i> 2013 Pediatr Infect Dis [155]	USA	Prospective and retrospective	Age ≤ 21 years, ≥ 48 hours vancomycin therapy	Comedication 7 days before or during the vancomycin

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
				therapy, neonates in the ICU, RRT (hemodialysis)
Le <i>et al.</i> 2014 J Pharmacol Clin Toxicol [126]	USA	Retrospective	Admission to the pediatric ICU, age ≤ 21 years, ≥ 48 hours vancomycin therapy, A control group of patients without kidney insufficiency	Comedication
Lim <i>et al.</i> 2014 J Clin Pharm Ther [224]	South Korea	Prospective	MRSA infection	–
Lin <i>et al.</i> 2016 Eur J Clin Pharmacol [150]	China	Prospective	Age ≥ 18 years, IV vancomycin after post-craniotomy meningitis diagnosis	RRT, severe cardiopulmonary insufficiency
Lin <i>et al.</i> 2021 Sci Rep [127]	China	Prospective	Admission to the ICU, age ≥ 18 years, suspected or documented infection with MRSA	Hepatic failure, impaired kidney function (ESRD), < 7 days vancomycin therapy, pregnancy, terminal stage of a tumor

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Liu <i>et al.</i> 2019 J Pharm Pharmacol [225]	China	Prospective	Age ≥ 18 years, ≥ 1 steady state non-trough [vancomycin], ≥ 1 steady state [vancomycin]	Abnormal thyroid function, absolute neutrophil count $< 1,000$ cells/mm ³ , DIC, kidney transplantation, ≥ 3 days within 30 days vancomycin prior to the study, multiple organs dysfunction syndromes, RRT (continuous hemodialysis), vascular cystic fibrosis
Liu <i>et al.</i> 2020 J Clin Pharm Ther [129]	China	Prospective	Admission to the ICU after surgery, age ≥ 18 years, lung transplantation recipients, vancomycin for prevention, ECMO support A control group of patients without ECMO	Contraindication to vancomycin, ECMO as a bridge to lung transplantation, hepatic function, impaired kidney function, RRT (continuous), vancomycin prior to surgery
Llopis-Salvia <i>et al.</i> 2006 J Clin Pharm Ther [115]	Spain	Retrospective	Admission to the ICU, critical illness	Impaired kidney function requiring RRT, non-steady

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
				state [vancomycin], SCr fluctuation >0.5 mg/dL
Ma <i>et al.</i> 2020 Front Pharmacol [226]	China	Retrospective	Kidney transplant recipients from brain-dead donors, vancomycin for prevention	–
Mangin <i>et al.</i> 2014 Clin Pharmacokinet [116]	France	Prospective	Admission to the ICU, critical illness, post-sternotomy mediastinitis during a primary closed redon catheter drainage	Incomplete data
Masich <i>et al.</i> 2020 Pharmacotherapy [117]	USA	Prospective	Age ≥18 years, confirmed gram-positive bacteremia, critical illness, obesity (BMI ≥30 kg/m ²), sepsis, vasopressor support	Central nervous system infection, ECMO, history of chronic kidney impairment (CrCL <20 mL/min), pregnancy, RRT, vancomycin for surgical prophylaxis
Medellin-Garibay <i>et al.</i> 2016 J	Spain	Retrospective	Proven or suspected infection, trauma	Burns, cystic fibrosis, hematological malignancies,

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Antimicrob Chemother [151]				pregnancy, RRT (hemofiltration)
Medellin-Garibay <i>et al.</i> 2017 Antimicrob Agents Chemother [118]	Spain	Retrospective	Continuous IV vancomycin administration, critical illness, proven or suspected infections, TDM	Burns, cystic fibrosis, hematological malignancies, pregnancy, RRT (hemofiltration)
Moffett <i>et al.</i> 2019 Eur J Clin Pharmacol [134]	USA	Retrospective	Age ≤ 19 years, ≥ 1 dose vancomycin therapy, RRT (continuous venous-venous hemodiafiltration)	ECMO, RRT (albumin augmented dialysis) due to hepatic failure
Moore <i>et al.</i> 2016 CPT Pharmacometrics Syst Pharmacol [184]	USA	Prospective	Age ≥ 18 years, ECMO	Contraindication to vancomycin, vancomycin-resistant gram-positive culture, vancomycin within 2 days of study entry
Mulla <i>et al.</i> 2005 Br J Clin Pharmacol [160]	UK	Prospective and retrospective	ECMO, full age spectrum (neonates, children, and adults)	Incomplete data

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Munir <i>et al.</i> 2021 Front Pharmacol [152]	Pakistan	Prospective	Age >18 years, surgery	–
Oda <i>et al.</i> 2020 Pharm Res [131]	Japan	Retrospective	Age >18 years, impaired kidney function, RRT (continuous), TDM	–
Oda <i>et al.</i> 2023 Antimicrob Agents Chemother [135]	Japan	Retrospective	Age >18 years, impaired kidney function (ESRD), RRT (intermittent hemodialysis)	–
Okada <i>et al.</i> 2018 J Clin Pharmacol [167]	Japan	Retrospective	Allogeneic hematopoietic stem cell transplantation, vancomycin for prevention	Acute kidney injury (SCr increase by 1.5-fold or by 0.3 mg/dL within 7 days)
Purwonugroho <i>et al.</i> 2012 Scientific World Journal [227]	Thailand	Retrospective	Age ≥18 years	Incomplete data
Revilla <i>et al.</i> 2010 Br J Clin Pharmacol [119]	Spain	Retrospective	Admission to the ICU, age ≥18 years, critical illness	Incomplete data, neoplastic disorders, prior cardiac surgery, RRT

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Roberts <i>et al.</i> 2011 Antimicrob Agents Chemother [120]	Belgium	Retrospective	Admission to the ICU, age ≥ 18 years, continuous IV vancomycin administration, critical illness, sepsis	Burns, cystic fibrosis, <2 days IV vancomycin administration, pregnancy, prior intermittent IV vancomycin administration, RRT
Sanchez <i>et al.</i> 2010 Int J Clin Pharmacol Ther [228]	USA	Retrospective	Age ≥ 18 years, TDM	RRT (hemodialysis)
Schaedeli <i>et al.</i> 1998 Clin Pharmacol Ther [132]	Switzerland	Prospective	Impaired kidney function, RRT (hemodialysis) over long term	–
Scharf <i>et al.</i> 2022 Ann Intensive Care [121]	Germany	Prospective	Critical illness, ≥ 2 [vancomycin] each during and without cytosorb therapy, RRT (continuous veno-venous hemodialysis or hemodiafiltration), sepsis or septic shock	Unavailable TDM

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Shen <i>et al.</i> 2018 Clin Infect Dis [229]	China	Prospective	Age ≥ 18 years, clinical evidence for an infection, ≥ 5 days vancomycin therapy, TDM	Gram-positive bacteria colonization, lactation, pregnancy, prior antibiotics against gram-positive bacterial infection for ≥ 24 h within 72 h prior to the study
Shen <i>et al.</i> 2021 Front Pharmacol [230]	China	Prospective	Clinical evidence for an infection, full age spectrum, ≥ 5 days vancomycin therapy, TDM	Gram-positive bacteria colonization, prior antibiotics against gram-positive bacterial infection for ≥ 24 h within 72 h prior to the study
Smit <i>et al.</i> 2020 Br J Clin Pharmacol [138]	The Netherlands	Prospective	Indication for bariatric surgery, obesity (BMI ≥ 40 kg/m ² or ≥ 35 kg/m ² with comorbidities) A control group of non-obese healthy	Comedication (potential nephrotoxic) 7 days prior the surgery, contraindication to vancomycin, impaired kidney function (eGFR < 60 mL/min/1.73 m ²), lactation, pregnancy
Staatz <i>et al.</i> 2006 Br J Clin Pharmacol [161]	UK	Prospective and retrospective	Admission to the cardiothoracic surgery unit	Incomplete data, RRT

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Tanaka <i>et al.</i> 2010 Antimicrob Agents Chemother [162]	Japan	Prospective	MRSA infection	DIC, multiple organ failure, RRT (hemodialysis, hemofiltration), [vancomycin] collected within 1 h after the completion of the infusion
Thomson <i>et al.</i> 2009 J Antimicrob Chemother [163]	UK	Prospective and retrospective	Age >16 years	Impaired kidney function requiring RRT, incomplete data
Udy <i>et al.</i> 2013 Int J Antimicrob Agents [122]	Belgium	Retrospective	Age ≥18 years, continuous IV vancomycin, critical illness, ≥2 days vancomycin therapy, RRT (continuous), sepsis	Burns, cystic fibrosis, pregnancy, prior intermittent IV vancomycin administration, residual urine output >500 mL/day
Usman <i>et al.</i> 2018 Int J Pharmacol Ther [164]	Germany	Retrospective	–	Incomplete data

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Vu <i>et al.</i> 2019 Int J Antimicrob Agents [231]	Vietnam	Prospective	Age ≥ 18 years	ECMO, lactation, <24 h continuous IV vancomycin administration, pregnancy, prior intermittent IV vancomycin administration prior the continuous administration, RRT
Wang <i>et al.</i> 2021 Antibiotics (Basel) [133]	China	Prospective	Age >16 years, impaired kidney function, ≥ 72 h IV vancomycin simultaneously with RRT (continuous veno-venous hemofiltration), ≥ 16 h RRT daily therapy time, suspected or documented multidrug-resistant gram-positive bacterial infection	Estimated life expectancy <48 h, hematologic diseases, pregnancy
Wei <i>et al.</i> 2022 Front Pharmacol [153]	China	Retrospective	Age ≥ 18 years, postoperative neurosurgery, TDM	Cystic fibrosis, pregnancy

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Wu <i>et al.</i> 2016 J Formos Med Assoc [125]	Taiwan	Prospective	Admission to the ICU, age ≥ 18 years, ECMO A control group of patients without ECMO	AKD, burns (severe 30–40% of the body surface area), plasmapheresis, RRT (continuous hemodialysis)
Yamamoto <i>et al.</i> 2009 J Clin Pharm Ther [139]	Japan	Retrospective	Age ≥ 16 years, suspected or documented gram-positive bacterial infection A control group of healthy volunteers	Incomplete data, RRT (hemodialysis)
Yamazaki <i>et al.</i> 2020 Ther Drug Monitor [123]	Japan	Retrospective	Admission at the ICU, age ≥ 18 years, critical illness, RRT (continuous hemodiafiltration with polymethylmethacrylate membrane hemofilter)	Death on the day of TDM, incomplete data, TDM performed once shortly after being transferred from another hospital
Yasuhara <i>et al.</i> 1998 Ther Drug Monitor [154]	Japan	Retrospective	Age ≥ 18 years, infection with MRSA	–

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Yellepeddi <i>et al.</i> 2024 Antimicrob Agents Chemother [170]	USA	Retrospective	Cystic fibrosis, ≥ 2 [vancomycin]	<24 h vancomycin therapy, pregnancy
Zalloum <i>et al.</i> 2018 TJPR [165]	Jordan	Retrospective	–	–
Zaric <i>et al.</i> 2018 Open Med (Wars) [140]	Serbia	Prospective	Age ≥ 18 years, impaired kidney function (mild-moderate chronic kidney impairment CrCL 30–89 mL/min), ≥ 3 days vancomycin therapy without changes in the daily dose A control group with normal kidney function	Impaired kidney function (severe kidney failure CrCL <30 mL/min), oral vancomycin, RRT (dialysis)
Zhang <i>et al.</i> 2023 Clin Pharmacokinet [166]	USA	Retrospective	Age ≥ 10 years, obesity, overweight	Impaired kidney function (<60 mL/min/1.73 m ²)

Authors year of publication journal [reference]	Study location	Study design	Inclusion criteria	Exclusion criteria
Zhao <i>et al.</i> 2021 Antibiotics (Basel) [232]	China	Retrospective	Age ≥ 18 years, augmented renal clearance, intermittent IV vancomycin administration,	Admission to the hematology department or the surgical department of the ICU, AKD prior or during therapy, CrCL < 15 mL/min, ESRD, incomplete data, prior vancomycin administration in other hospitals, RRT
Zhou <i>et al.</i> 2019 Eur J Drug Metab Pharmacokinet [233]	China	Retrospective	Age ≥ 65 years, ≥ 2 days vancomycin therapy, pulmonary infection	Incomplete data, low volume shock, multiple organ failure, RRT

AKD, acute kidney disease; BMI, body mass index; BW, body weight; CHF, congestive heart failure; CKD, chronic kidney disease; CrCL, creatinine clearance; CrCL_{CG}, creatinine clearance using Cockcroft–Gault equation; DIC, disseminated intravascular coagulation; DM, diabetes mellitus; ECMO, extracorporeal membrane oxygenation; eGFR, estimated glomerular filtration rate; ESRD, end-stage renal disease; Hb, hemoglobin; ICU, intensive care unit; IV, intravenous; MIC, minimum inhibitory concentration; MRSA, methicillin-resistant *Staphylococcus aureus*; PNA, postnatal age; RRT, renal replacement therapy; SCr, serum creatinine; TDM, therapeutic drug monitoring; TBW, total body weight; [vancomycin], vancomycin concentration

Supplementary Table 5 Baseline characteristics of patients

Authors year of publication journal [reference]	Patient number	Female number (%)	Age, mean $\pm$ SD median (range) [IQR], (years)	Weight, mean $\pm$ SD median (range) [IQR], (kg)	SCr, mean $\pm$ SD median (range) [IQR], (mg/dL)
Adane <i>et al.</i> 2015 Pharmacotherapy [188]	31	12 (38.71%)	43 [38.5–53.0]	147.9 [142.8–178.3]	–
Ahmed <i>et al.</i> 2024 Drug Des Devel Ther [142]	99	33 (33.33%)	65 (18–90) [50–75]	–	1.2 (0.2–9.5) [0.7–2.5]
Aljutayli <i>et al.</i> 2022 Clin Transl Sci [222]	116	33 (28.45%)	67.8 $\pm$ 11	72 $\pm$ 8.6	1 $\pm$ 0.5
Alqahtani <i>et al.</i> 2018 Antimicrob Agents Chemother [143]	28	11 (39%)	51.7 $\pm$ 15.9 (18–78)	79.6 $\pm$ 17 (52–111.8)	0.87 $\pm$ 0.25 (0.46–1.52)
Alqahtani <i>et al.</i> 2020 Pharmacotherapy [137]	147 73 patients with cancer 74 patients without cancer	30 (41.1%) patients with cancer 29 (39.19%) patients without cancer	53.8 $\pm$ 15.7 patients with cancer 55.1 $\pm$ 15.9 patients without cancer	72.7 $\pm$ 16.2 patients with cancer 75.5 $\pm$ 19.7 patients without cancer	1.13 $\pm$ 0.4 patients with cancer 1.09 $\pm$ 0.3 patients without cancer

Authors year of publication journal [reference]	Patient number	Female number (%)	Age, mean $\pm$ SD median (range) [IQR], (years)	Weight, mean $\pm$ SD median (range) [IQR], (kg)	SCr, mean $\pm$ SD median (range) [IQR], (mg/dL)
		without cancer			
Avedissian <i>et al.</i> 2017 <i>Pediatr Crit Care Med</i> [106]	250	128 (51.2%)	9.8 [3.2–14]	30 [15–50]	0.4 [0.3–0.54]
Bae <i>et al.</i> 2019 <i>Pharmaceutics</i> [144]	220	81 (36.82%)	63 (21–98)	61.6 60 (30–126.7)	1.7 (0.2–13.3)
Bang <i>et al.</i> 2022 <i>Clin Exp Pharmacol physiol</i> [107]	22	7 (31.82%)	67.5 $\pm$ 13.2 (36–87)	62.9 $\pm$ 10.4 53 (47.1–90.6)	–
Belabbas <i>et al.</i> 2023 <i>J Infect Chemother</i> [145]	148	65 (43.92%)	53.6 $\pm$ 16 59 (20–82)	58.7 $\pm$ 13.1 57.9 (34.5–105)	0.6 $\pm$ 0.4 0.58 (0.2–5.2)
Belavagi <i>et al.</i> 2022 <i>Germes</i> [108]	15	7 (46.67%)	42.4 $\pm$ 17.25 34 (19–70)	69.67 $\pm$ 10.26 70 (50–85)	–
Buelga <i>et al.</i> 2005 <i>Antimicrob</i>	215	96 (44.65%)	51.5 $\pm$ 15.9	64.7 $\pm$ 11.3	0.9 $\pm$ 0.4

Authors year of publication journal [reference]	Patient number	Female number (%)	Age, mean $\pm$ SD median (range) [IQR], (years)	Weight, mean $\pm$ SD median (range) [IQR], (kg)	SCr, mean $\pm$ SD median (range) [IQR], (mg/dL)
Agents Chemother [179]					
Bury <i>et al.</i> 2019 Eur J Clin Pharmacol [178]	116	49 (42.24%)	61.4 $\pm$ 13.4	70	–
Chu <i>et al.</i> 2020 J Infect Public Health [146]	95	24 (25.26%)	45 (11–82) [30–57]	70 [60–80]	0.52 [0.43–0.63]
Chung <i>et al.</i> 2013 J Korean Med Sci [147]	678	278 (41%)	56 57 (18–96)	62.3 60.8 (27–140)	0.8 0.9 (0.39–1.2)
Colin <i>et al.</i> 2019 Clin Pharmacokinet [103]	2,554	1129 (44.21%)	–	(0.42–160)	–
De Cock <i>et al.</i> 2014 Pharm Res [105]	1,760 including 439 patients on vancomycin	–	(0–18)	(0.42–85)	(5–144.1)
Deng <i>et al.</i> 2013 Int J Clin	72	53 (73.61%)	54.07 $\pm$ 18.36	61.12 $\pm$ 10.7	0.9 $\pm$ 0.51 (0.32–3.13)

Authors year of publication journal [reference]	Patient number	Female number (%)	Age, mean $\pm$ SD median (range) [IQR], (years)	Weight, mean $\pm$ SD median (range) [IQR], (kg)	SCr, mean $\pm$ SD median (range) [IQR], (mg/dL)
Pharmacol Ther [223]					
Dolton <i>et al.</i> 2010 Burns [136]	70 37 patients with burns 33 patients without burns	8 (21.62%) patients with burns 7 (21.21%) patients without burns	34 (15–88) patients with burns 72 (38–95) patients without burns	69 (42.5–116) patients with burns 67 (48.9–111) patients without burns	–
Donadello <i>et al.</i> 2014 Critical Care [124]	22 11 patients with ECMO 11 patients without ECMO	7 (63.64%) patients with ECMO 6 (54.55%) patients without ECMO	43 (19–59) patients with ECMO 55 (24–64) patients without ECMO	70 (46–85) patients with ECMO 70 (47–95) patients without ECMO	–
Dorajoo <i>et al.</i> 2019 Front Pharmacol [172]	80	29 (36.25%)	71.7 $\pm$ 13	57.8 $\pm$ 15.7	1.7 $\pm$ 0.83
Escobar <i>et al.</i> 2014 Int J	9	4 (44.44%)	57 $\pm$ 14	70 $\pm$ 18	–

Authors year of publication journal [reference]	Patient number	Female number (%)	Age, mean $\pm$ SD median (range) [IQR], (years)	Weight, mean $\pm$ SD median (range) [IQR], (kg)	SCr, mean $\pm$ SD median (range) [IQR], (mg/dL)
Antimicrob Agents [109]					
Fu <i>et al.</i> 2021 Biopharm Drug Dispos [156]	32	–	43.28 $\pm$ 15.88 (17–83)	56.84 $\pm$ 11.09 (33–83.5)	0.61 $\pm$ 0.21 (0.26–1.41)
Garreau <i>et al.</i> 2021 Antibiotics (Basel) [110]	78	21 (26.92%)	68.9 $\pm$ 12.3	77.9 $\pm$ 20.4	1.58 $\pm$ 0.86
Gawmahanil <i>et al.</i> 2018 TJPS [111]	171	71 (41.52%)	62.2 $\pm$ 17.96	56 (29–125.7)	0.9 (0.19–12.07)
Gonzalez <i>et al.</i> 2022 Antibiotics [168]	14	6 (42.86%)	44	60 [55–67]	0.58
Goti <i>et al.</i> 2018 The Drug Monit [157]	1,812	843 (46.52%)	57 (17–101)	79 (33–255)	1.3 (0.2–16.6)
Goyal <i>et al.</i> 2022 Front Pharmacol [158]	34	34 (100%)	28 (17–38)	74 (43–157)	0.56 (0.27–1.97)

Authors year of publication journal [reference]	Patient number	Female number (%)	Age, mean $\pm$ SD median (range) [IQR], (years)	Weight, mean $\pm$ SD median (range) [IQR], (kg)	SCr, mean $\pm$ SD median (range) [IQR], (mg/dL)
Hodiamont <i>et al.</i> 2021 J Antimicrob Chemother [128]	82	36 (43.3%)	58.8 $\pm$ 14	76.2 $\pm$ 16.6	1.56 $\pm$ 1.36
Holford <i>et al.</i> 2024 Br J Clin Pharmacol [104]	3232	–	(0–80)	(0–200)	(0–5.65)
Huang <i>et al.</i> 2021 Intern Emerg Med [112]	52	13 (25%)	65 [49.5–76.5]	65.5 [59.5–72.5]	1.13 $\pm$ 0.89
Hughes <i>et al.</i> 2024 Ther Drug Monit [148]	83	42 (51%)	56.3 (24.2–89.3)	134 (91.6–218)	0.84 (0.40–2.51)
Hui <i>et al.</i> 2019 J Antimicrob Chemother [130]	37	10 (27.03%)	61.5 (23–86)	78 (40–226)	7.31 (2.53–13.32)
Jaisue <i>et al.</i> 2020 Ther Drug Monit [159]	180	97 (53.89%)	60.8 $\pm$ 17.5 (17–97)	54.2 $\pm$ 11.7 (30–103)	–

Authors year of publication journal [reference]	Patient number	Female number (%)	Age, mean $\pm$ SD median (range) [IQR], (years)	Weight, mean $\pm$ SD median (range) [IQR], (kg)	SCr, mean $\pm$ SD median (range) [IQR], (mg/dL)
Ji <i>et al.</i> 2018 Acta Pharmacol Sin [149]	160	54 (33.75%)	78 (42–95)	65 (38–90)	0.85 (0.27–10.1)
Jing <i>et al.</i> 2019 J Clin Pharm Ther [173]	222	126 (56.76%)	46.95 $\pm$ 12.71 47	60.22 $\pm$ 11.77 60	0.66 $\pm$ 0.2
Jung <i>et al.</i> 2021 Antimicrob Agents Chemother [169]	22	5 (22.73%)	56 [43.3–64.5]	74 [55.5–82.3]	1.33 [0.97–1.77]
Kim <i>et al.</i> 2016 Int. J Antimicrob Agents [141]	132 64 patients with neurosurgery 68 patients without neurosurgery	34 (53.13%) patients with neurosurgery 31 (45.59%) patients without neurosurgery	50.6 $\pm$ 15 patients with neurosurgery 61.6 $\pm$ 15.7 patients without neurosurgery	63.2 $\pm$ 11.6 patients with neurosurgery 61 $\pm$ 12.7 patients without neurosurgery	0.64 $\pm$ 0.24 patients with neurosurgery 0.87 $\pm$ 0.46 patients without neurosurgery
Kloprogge <i>et al.</i> 2019 Antimicrob	616	–	5 (0–21)	19 (0.74–95)	0.39 (0.09–3.29)

Authors year of publication journal [reference]	Patient number	Female number (%)	Age, mean $\pm$ SD median (range) [IQR], (years)	Weight, mean $\pm$ SD median (range) [IQR], (kg)	SCr, mean $\pm$ SD median (range) [IQR], (mg/dL)
Agents Chemother [113]					
Kovacevic <i>et al.</i> 2020 J Crit Care [114]	73	33 (45.21%)	56.9 $\pm$ 17 60 (20–87)	78.2 $\pm$ 14.2 80 (30–120)	1.23 $\pm$ 0.82 1.02 (0.37–1.69)
Le <i>et al.</i> 2013 Pediatr Infect Dis [155]	702	331 (47.15%)	6.6 [2.2–13.4]	22.8 [12.6–46.0]	0.48 $\pm$ 0.33 [0.3–0.6]
Le <i>et al.</i> 2014 J Pharmacol Clin Toxicol [126]	126 63 patients with kidney insufficiency 63 patients without kidney insufficiency	23 (36.51%) patients with kidney insufficiency 32 (50.79%) patients without kidney insufficiency	13 $\pm$ 6 patients with kidney insufficiency 13 $\pm$ 6 patients without kidney insufficiency	52 $\pm$ 26 patients with kidney insufficiency 50 $\pm$ 25 patients without kidney insufficiency	1.3 $\pm$ 0.5 patients with kidney insufficiency 0.6 $\pm$ 0.2 patients without kidney insufficiency
Lim <i>et al.</i> 2014 J Clin Pharm Ther [224]	20	5 (25%)	59.3 $\pm$ 12.9	63.1 $\pm$ 15.7	–

Authors year of publication journal [reference]	Patient number	Female number (%)	Age, mean $\pm$ SD median (range) [IQR], (years)	Weight, mean $\pm$ SD median (range) [IQR], (kg)	SCr, mean $\pm$ SD median (range) [IQR], (mg/dL)
Lin <i>et al.</i> 2016 Eur J Clin Pharmacol [150]	100	34 (34%)	51.6 $\pm$ 16.9 (18–86)	59.1 $\pm$ 10 (38–85)	0.85 $\pm$ 0.71 (0.29–6.15)
Lin <i>et al.</i> 2021 Sci Rep [127]	374	120 (32.09%)	62 (18–93)	65 (40–90.6)	0.8 (0.32–6.57)
Liu <i>et al.</i> 2019 J Pharm Pharmacol [225]	200	72 (36%)	47.4 $\pm$ 15.42 48.5	61.3 $\pm$ 12.06 60	0.68 $\pm$ 0.26
Liu <i>et al.</i> 2020 J Clin Pharm Ther [129]	25 19 patients with ECMO 6 patients without ECMO	3 (15.79%) patients with ECMO 1 (16.67%) patient without ECMO	65 [60–67] patients with ECMO 57 [45.25–61.5] patients without ECMO	63 [60–71] patients with ECMO 60.5 [48.25–65.5] patients without ECMO	0.62 [0.55–0.77] patients with ECMO 0.63 [0.55–0.71] patients without ECMO
Llopis-Salvia <i>et al.</i> 2006 J Clin Pharm Ther [115]	30	11 (36.67%)	67 [18–77]	75 $\pm$ 12.5 (50–130)	–

Authors year of publication journal [reference]	Patient number	Female number (%)	Age, mean $\pm$ SD median (range) [IQR], (years)	Weight, mean $\pm$ SD median (range) [IQR], (kg)	SCr, mean $\pm$ SD median (range) [IQR], (mg/dL)
Ma <i>et al.</i> 2020 Front Pharmacol [226]	56	21 (37.5%)	43.72 $\pm$ 9.92 43.5 (24–70) [36–51.25]	58.27 $\pm$ 8.47 59.95 (37.7–79) [52.2–64.52]	0.29 $\pm$ 2.86 1.85 (0.68–16.85) [1.59–2.97]
Mangin <i>et al.</i> 2014 Clin Pharmacokinet [116]	30	4 (13.33%)	63 (35–81)	82 (62–104)	1.56 (0.36–6.85)
Masich <i>et al.</i> 2020 Pharmacotherapy [117]	16	7 (43.75%)	62 (30–78)	112.7 (72.6–129.1)	–
Medellin-Garibay <i>et al.</i> 2016 J Antimicrob Chemother [151]	118	65 (55.08%)	74.3 $\pm$ 14 77 (37–100)	72 $\pm$ 15	–
Medellin-Garibay <i>et al.</i> 2017 Antimicrob Agents Chemother [118]	54	16 (29.63%)	65 $\pm$ 12.3	75 $\pm$ 20.1	–

Authors year of publication journal [reference]	Patient number	Female number (%)	Age, mean $\pm$ SD median (range) [IQR], (years)	Weight, mean $\pm$ SD median (range) [IQR], (kg)	SCr, mean $\pm$ SD median (range) [IQR], (mg/dL)
Moffett <i>et al.</i> 2019 Eur J Clin Pharmacol [134]	138	75 (54.35%)	4.9 [1–14.5]	31 $\pm$ 25.8	0.72 [0.41–1.29]
Moore <i>et al.</i> 2016 CPT Pharmacometrics Syst Pharmacol [184]	14	3 (21.43%)	47 $\pm$ 16 (19–72)	95 $\pm$ 27	–
Mulla <i>et al.</i> 2005 Br J Clin Pharmacol [160]	45	16 (35.56%)	13.2 (0–61.5)	36.0 (2.5–116)	1.11 (0.30–2.55)
Munir <i>et al.</i> 2021 Front Pharmacol [152]	58	19 (32.76%)	54 (25–86)	75 (53–129)	0.94 (0.4–4.7)
Oda <i>et al.</i> 2020 Pharm Res [131]	17	7 (41.18%)	64 (19–92)	59.7 (43.1–117)	1.7 (0.4–7)
Oda <i>et al.</i> 2023 Antimicrob Agents Chemother [135]	28	20 (71.43%)	61 $\pm$ 14.5	57.8 $\pm$ 13.2	–

Authors year of publication journal [reference]	Patient number	Female number (%)	Age, mean $\pm$ SD median (range) [IQR], (years)	Weight, mean $\pm$ SD median (range) [IQR], (kg)	SCr, mean $\pm$ SD median (range) [IQR], (mg/dL)
Okada <i>et al.</i> 2018 J Clin Pharmacol [167]	75	26 (34.67%)	49 (17–69)	59.4 (39.4–104.5)	0.62 (0.29–1.63)
Purwonugroho <i>et al.</i> 2012 Scientific World Journal [227]	212	100 (47.17%)	66.62 $\pm$ 18.38	57.64 $\pm$ 11.62	3.12 $\pm$ 2.92
Revilla <i>et al.</i> 2010 Br J Clin Pharmacol [119]	191	65 (34.03%)	61.1 $\pm$ 16.3 (18–85)	73 $\pm$ 13.3 (45–150)	1.4 $\pm$ 1 (0.6–5)
Roberts <i>et al.</i> 2011 Antimicrob Agents Chemother [120]	206	81 (39.32%)	58.1 $\pm$ 14.8	74.8 $\pm$ 15.8	–
Sanchez <i>et al.</i> 2010 Int J Clin Pharmacol Ther [228]	141	–	55 $\pm$ 14.58 53.5	73.2 $\pm$ 17.48	1.05 $\pm$ 0.65
Schaedeli <i>et al.</i> 1998 Clin	20	10 (38.46%)	62 $\pm$ 15.2 65 (26–86)	64.7 $\pm$ 13.6 63.6 (42.2–101.3)	–

Authors year of publication journal [reference]	Patient number	Female number (%)	Age, mean $\pm$ SD median (range) [IQR], (years)	Weight, mean $\pm$ SD median (range) [IQR], (kg)	SCr, mean $\pm$ SD median (range) [IQR], (mg/dL)
Pharmacol Ther [132]					
Scharf <i>et al.</i> 2022 Ann Intensive Care [121]	7	1 (14.29%)	52 (20–57)	87 (56–130)	–
Shen <i>et al.</i> 2018 Clin Infect Dis [229]	380	129 (33.95%)	61 [49–74]	60 [55–70]]	0.74 [0.54–1.11]
Shen <i>et al.</i> 2021 Front Pharmacol [230]	488	–	55	58.25	–
Smit <i>et al.</i> 2020 Br J Clin Pharmacol [138]	28 20 obese 8 non-obese healthy	–	38 (23–54) obese 25.5 (20–55) non-obese healthy	139 (110.6–234.6) obese 69.5 (60–84.7) non-obese healthy	0.81 (0.46–1.14) obese 0.79 (0.68–0.97) non-obese healthy
Staatz <i>et al.</i> 2006 Br J Clin Pharmacol [161]	102	31 (30.39%)	66 (17–87)	74 (44–110)	1.14 (0.51–5.96)
Tanaka <i>et al.</i> 2010 Antimicrob	86	28 (32.56%)	73 (17–91)	54 $\pm$ 10 (33–78)	0.7 (0.3–2.2)

Authors year of publication journal [reference]	Patient number	Female number (%)	Age, mean $\pm$ SD median (range) [IQR], (years)	Weight, mean $\pm$ SD median (range) [IQR], (kg)	SCr, mean $\pm$ SD median (range) [IQR], (mg/dL)
Agents Chemother [162]					
Thomson <i>et al.</i> 2009 J Antimicrob Chemother [163]	398	335 (84.17%)	66 (16–97)	72 (40–159)	1.11 (0.34–6.48)
Udy <i>et al.</i> 2013 Int J Antimicrob Agents [122]	81	28 (34.57%)	61 $\pm$ 15.6	83.4 $\pm$ 22.1	–
Usman <i>et al.</i> 2018 Int J Pharmacol Ther [164]	144	51 (35.42%)	62 (16–88)	79.5 (40–177)	0.9 (0.2–11.5)
Vu <i>et al.</i> 2019 Int J Antimicrob Agents [231]	55	19 (34.55%)	55 $\pm$ 18	55.9 $\pm$ 11.1 70	–
Wang <i>et al.</i> 2021 Antibiotics (Basel) [133]	11	5 (45.45%)	63 (20–83)	70 (52–90)	–

Authors year of publication journal [reference]	Patient number	Female number (%)	Age, mean $\pm$ SD median (range) [IQR], (years)	Weight, mean $\pm$ SD median (range) [IQR], (kg)	SCr, mean $\pm$ SD median (range) [IQR], (mg/dL)
Wei <i>et al.</i> 2022 Front Pharmacol [153]	560	190 (33.93%)	52.41 $\pm$ 15.11 (18–89)	69.74 $\pm$ 13.05 (37.05–130)	0.73 $\pm$ 0.87 (0.11–10.83)
Wu <i>et al.</i> 2016 J Formos Med Assoc [125]	22 11 patients with ECMO 11 patients without ECMO	5 (45.45%) patients with ECMO 5 (45.45%) patients without ECMO	47.18 $\pm$ 16.85 patients with ECMO 49 $\pm$ 17.16 patients without ECMO	66.57 $\pm$ 17.53 patients with ECMO 64.07 $\pm$ 12.34 patients without ECMO	1.08 $\pm$ 0.6 patients with ECMO 0.83 $\pm$ 0.39 patients without ECMO
Yamamoto <i>et al.</i> 2009 J Clin Pharm Ther [139]	106 100 patients 6 healthy volunteers	36 (36%) patients 0 (0%) healthy volunteers	65.4 $\pm$ 15.1 (25.8–99.7) patients 21.7 $\pm$ 2 (20–25) healthy volunteers	52.6 $\pm$ 12.7 (28.7–97) patients 60.3 $\pm$ 3.7 (55.2–64.2) healthy volunteers	1.12 $\pm$ 0.08 (1–1.2) patients 0.79 $\pm$ 0.4 (0.2–2.6) healthy volunteers
Yamazaki <i>et al.</i> 2020 Ther Drug Monitor [123]	25	8 (32%)	65 (21–83)	59 (35–92.4)	1.47 (0.23–11.49)
Yasuhara <i>et al.</i> 1998 Ther Drug Monitor [154]	190	59 (31.05%)	64.3 $\pm$ 13.8 (19.3–89.6)	52.3 $\pm$ 9.6 (25.5–75)	1.21 $\pm$ 1.55 (0.2–13.4)

Authors year of publication journal [reference]	Patient number	Female number (%)	Age, mean $\pm$ SD median (range) [IQR], (years)	Weight, mean $\pm$ SD median (range) [IQR], (kg)	SCr, mean $\pm$ SD median (range) [IQR], (mg/dL)
Yellepeddi <i>et al.</i> 2024 Antimicrob Agents Chemother [170]	19	14 (73.68%)	31.2 $\pm$ 12.5 27 (19–60)	63.6 $\pm$ 17.1 57.7 (43.6–105.7)	0.9 $\pm$ 0.33 0.8 (0.4–1.9)
Zalloum <i>et al.</i> 2018 TJPR [165]	110	37 (33.64%)	46.1 $\pm$ 25.6 (0.04–88)	63.5 (1.3–115)	–
Zaric <i>et al.</i> 2018 Open Med (Wars) [140]	110 78 patients with mild-moderate chronic kidney failure 32 patients with normal kidney function	32 (41.03%) patients with mild-moderate chronic kidney failure 11 (34.38%) patients with normal kidney function	67 $\pm$ 10.74 (33–86) patients with mild-moderate chronic kidney failure 59.15 $\pm$ 14.46 (27–86) patients with normal kidney function	78.52 $\pm$ 16.64 (60–180) patients with mild-moderate chronic kidney failure 81.37 $\pm$ 10.11 (60–103) patients with normal kidney function	1.45 $\pm$ 0.53 (0.64–2.83) patients with mild-moderate chronic kidney failure 0.7 $\pm$ 0.19 (0.36–1.12) patients with normal kidney function
Zhang <i>et al.</i> 2023 Clin Pharmakinet [166]	206	88 (42.72%)	40 (10–88)	83.8 (28.3–188)	0.63 (0.21–1.21)

Authors year of publication journal [reference]	Patient number	Female number (%)	Age, mean $\pm$ SD <i>median</i> (range) [IQR], (years)	Weight, mean $\pm$ SD <i>median</i> (range) [IQR], (kg)	SCr, mean $\pm$ SD <i>median</i> (range) [IQR], (mg/dL)
Zhao <i>et al.</i> 2021 Antibiotics (Basel) [232]	209	83 (39.71%)	66 $\pm$ 16.4	63.4 $\pm$ 12.9	–
Zhou <i>et al.</i> 2019 Eur J Drug Metab Pharmacokinet [233]	70	21 (30%)	78.3 $\pm$ 6.96	60.7 $\pm$ 10.2	1.02 $\pm$ 0.35

ECMO, extracorporeal membrane oxygenation; IQR, interquartile range; SCr, serum creatinine; SD, standard deviation

Supplementary Table 6 Characteristics of vancomycin administration, sampling, and quantification

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Adane <i>et al.</i> 2015 Pharmacothe rapy [188]	15–20 mg/kg over 1–2 h q 8–12 h Max 2,500 mg q 12 h Median [IQR] 4,000 [3,625– 4,375] mg/d	II	93	3	Yes	PETINIA	0.8	5
Ahmed <i>et al.</i> 2024 Drug Des Devel Ther [142]	–	II	194	2	Yes	EMIT	–	–

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Aljutayli <i>et al.</i> 2022 Clin Transl Sci [222]	–	–	326	3	Yes	PETINIA	2.5	–
Alqahtani <i>et al.</i> 2018 Antimicrob Agents Chemother [143]	1,000 mg over 0.5 h, 2 h prior skin incision then q 12–48 h	II	168	6	Yes	CMIA (Abbott Laboratories, North Chicago, Illinois, USA) with Architect iVancomycin (Abbott, Wiesbaden, Germany) and Architect i4000SR	0.5	–
Alqahtani <i>et al.</i> 2020 Pharmacotherapy [137]	Mean $\pm$ SD 2,040 $\pm$ 188 and 2,160 $\pm$ 220 mg/d	II	448	3	Yes	Architect iVancomycin (Abbott, Wiesbaden, Germany) and	0.5	–

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
	over 1 h q 12 h					Architect i4000SR immunoassay (Abbott Laboratories, North Chicago, Illinois, USA)		
Avedissian <i>et al.</i> 2017 Pediatr Crit Care Med [106]	Median [IQR] 45 [39.97– 58.61] mg/kg/d	II	658	3	No	CLIA (Advia Centaur, Siemens Medical Solution, Deerfield, IL) FPIA with AxSYM (Abbot Laboratories, Abbot Park, Ill, USA)	0.67	5.5 (2– 100)
Bae <i>et al.</i> 2019	–	CI	1,020	5	Yes	–	–	–

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Pharmaceuti cs [144]								
Bang <i>et al.</i> 2022 Clin Exp Pharmacol physiol [107]	25 mg/kg over 1– 1.5 h	II	197	9	No	–	–	–
Belabbas <i>et al.</i> 2023 J Infect Chemother [145]	–	–	681	5	Yes	CMIA	3	10
Belavagi <i>et al.</i> 2022 Germes [108]	1,000 mg q 12 h	II	83	6	No	HPLC	0.5	5.9
Buelga <i>et al.</i> 2005 Antimicrob Agents	Mean ± SD (range) 1,535 ± 280 (200–	II	1,004	5	Yes	FPIA with TDx (Abbot Laboratories, North Chicago, IL)	0.6	5

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Chemother [179]	3,900) mg/d over 0.5–1 h q 6–48 h							
Bury <i>et al.</i> 2019 Eur J Clin Pharmacol [178]	–	–	742	6	No	Validated spectrophotom etric homogeneous enzyme immunoassay with Viva-E system Emit 2000 (Siemens Healthcare Diagnostics)	–	4.2
Chu <i>et al.</i> 2020 J Infect Public Health [146]	1,000– 4,000 mg/d q 6,8,12 h	II	186	2	Yes	EMIT with Viva-E system (Dade Behring Incorporated, USA)	2	6

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Chung <i>et al.</i> 2013 J Korean Med Sci [147]	2,000 mg/d over 1 h. Median (range) 2,000 (1,000– 2,000) mg/d	II	1,373	2	Yes	FPIA with Cobas Integra 800 Analyzer (Roche, Mannheim, Germany)	–	–
Colin <i>et al.</i> 2019 Clin Pharmacokin et [103]	–	II, CI	8,300	3	Yes	PETINIA, FPIA	–	–
De Cock <i>et al.</i> 2014 Pharm Res [105]	15 mg/kg over 1 h q 12–24 h	II	4,146 including 1,168 for vancomycin	2 with 3 for vancomy cin	Yes	PETINIA	2	–
Deng <i>et al.</i> 2013 Int J Clin	500– 1,000 mg over 1 h	II	167	2	No	FPIA with TDx FLx assay (Abbot	2	4

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Pharmacol Ther [223]						Laboratories, Irving, TX, USA)		
Dolton <i>et al.</i> 2010 Burns [136]	1,000 mg over 1 h q 12 h	II	97	1	No	FPIA	–	–
Donadello <i>et al.</i> 2014 Critical Care [124]	35 mg/kg over 4 h followed by a continuou s infusion dose during the first 24 h of therapy	CI	66	3	Yes	PETINIA with Dimension XPand (Siemens Healthcare Diagnostics, Newark, DE, USA)	0.8	–
Dorajoo <i>et al.</i> 2019 Front	15–20 mg/kg	II	170	2	Yes	–	–	–

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Pharmacol [172]	Max 2,000 mg q 12, 24 h at a rate of 500 mg/h over max 4 h							
Escobar <i>et al.</i> 2014 Int J Antimicrob Agents [109]	1,000 mg/d over 1 h	II	68	8	No	Validated liquid chromatography-tandem mass spectrum assay with Acquity UPLC System (Waters Corporation, Milford, MA)	0.63	5
Fu <i>et al.</i> 2021 Biopharm	Mean $\pm$ SD 2,050 $\pm$ 320	II	152	5	Yes	Viva-E system (Siemens)	–	–

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Drug Dispos [156]	mg/d over 2–8 h					Healthcare Diagnostics)		
Garreau <i>et al.</i> 2021 Antibiotics (Basel) [110]	Mean $\pm$ SD 22.7 $\pm$ 7.5 and 28.6 $\pm$ 9.4 mg/kg/d	CI	335	4	Yes	Immunoturbidi metry with Abbot Architect C8000 analyzer	1.1	–
Gawmahanil <i>et al.</i> 2018 TJPS [111]	Median (range) 30.3 (6.39– 87.53) mg/kg/d over 1–2 h	II	398	2	Yes	CMIA	0.24	10
Gonzalez <i>et al.</i> 2022 Antibiotics [168]	1,000 mg over 2–4 h q 12 h	II	96	7	Yes	Validated chemiluminesc ence assay	3	–

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Goti <i>et al.</i> 2018 The Drug Monit [157]	20–25 mg/kg q 24 h Max 2,000– 2,500 mg	–	2,765	2	Yes	Validated ELISA	–	–
Goyal <i>et al.</i> 2022 Front Pharmacol [158]	–	–	82	2	No	G6PDH enzyme immunoassay with VITROS 5600 Integrated System (Ortho Clinical Diagnostics, Raritan, NJ, USA)	5	10
Hodiamont <i>et al.</i> 2021 J Antimicrob	Loading dose 25 mg/kg or 1,000 mg	II	609	7	Yes	Cobas Integra (Roche Diagnostics,	0.74	–

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Chemother [128]						Basel, Switzerland)		
Holford <i>et al.</i> 2024 Br J Clin Pharmacol [104]	–	–	8,877	3	–	–	–	–
Huang <i>et al.</i> 2021 Intern Emerg Med [112]	Mean ± SD 1,540 ± 788 mg/d over 1 h	–	126	2	Yes	FPIA	3	–
Hughes <i>et al.</i> 2024 Ther Drug Monit [148]	20 mg/kg loading dose Max 4,000 mg followed by a maintena nce dose	II	272	3	Yes	–	–	–

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Hui <i>et al.</i> 2019 J Antimicrob Chemother [130]	1,000– 1,500 mg or 25–30 mg/kg. Median (range) 1,500 (1,000– 4,500) and 1,500 (500– 4,500) mg for the loading and maintena nce dose, respectiv ely	–	180	5	Yes	CLIA with Architect iVancomycin (Abbott Laboratories) and Advia Centaur (Siemens Healthcare)	2	10

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategies	Infusion modality	Total number	Number per patient ^a	At steady state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomycin range (mg/L))
Jaisue <i>et al.</i> 2020 Ther Drug Monit [159]	–	–	474	3	Yes	CMIA with Architect i1000 (Abbott Laboratories, Chicago, Illinois, USA)	2	10
Ji <i>et al.</i> 2018 Acta Pharmacol Sin [149]	1,000 mg q 12 h	CI	251	2	No	FPIA with TDx FLx (Abbott Laboratories, Abbott Park, IL, USA)	2	4
Jing <i>et al.</i> 2019 J Clin Pharm Ther [173]	–	–	436	2	Yes	EMIT with Siemens Viva-E Drug Testing System and Emit 2000 Vancomycin Assay Test Kit (Siemens Healthcare)	2	10

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
						Diagnostic Inc.)		
Jung <i>et al.</i> 2021 Antimicrob Agents Chemother [169]	1,000 mg over 1 h once	II	194	9	No	CMIA with Architect iVancomycin (Abbott Laboratories, Germany)	–	–
Kim <i>et al.</i> 2016 Int. J Antimicrob Agents [141]	Mean ± SD 1,981 ± 219 mg/d with neurosurg ery and 1,810 ± 387 mg/d without neurosurg ery	–	359	3	No	CMIA with Architect iVancomycin Assay Kit (Abbott Laboratories, Wiesbaden, Germany)	–	–

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Kloprogge <i>et al.</i> 2019 Antimicrob Agents Chemother [113]	–	–	4,137	7	No	Microsphere system immunoassay	2	–
Kovacevic <i>et al.</i> 2020 J Crit Care [114]	1,000– 2,000 mg over 1–2 h q 12 h	II	146	2	Yes	Cobas analyzer (Roche/Hitachi Cobas c systems, Cobas c 501)	2	–
Le <i>et al.</i> 2013 Pediatr Infect Dis [155]	Mean ± SD [IQR] 45 ± 12 [39–52] mg/kg/d	II	1,660	2	Yes	CLIA with Advia Centaur (Siemens Medical Solution, Deerfield, IL) FPIA with AxSYM (Abbot	0.67	5.5(2– 100)

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
						Laboratories, Abbot Park, Ill, USA)		
Le <i>et al.</i> 2014 J Pharmacol Clin Toxicol [126]	Mean ± SD 38 ± 14 patients with kidney insufficien cy 42 ± 13 patients without kidney insufficien cy mg/kg/d	II	319	3	Yes	CLIA with Advia Centaur (Siemens Medical Solution, Deerfield, IL) FPIA with AxSYM (Abbot Laboratories, Abbot Park, Ill, USA)	0.67	5.5 (2– 100)

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Lim <i>et al.</i> 2014 J Clin Pharm Ther [224]	1,000 mg over 2 h q 12 h	II	120	6	Yes	FPIA with Cobas Integra (Roche Diagnostics, Rotkreuz, Switzerland)	1.39	–
Lin <i>et al.</i> 2016 Eur J Clin Pharmacol [150]	Mean ± SD (range) 1,910.6 ± 314.2 (1,000– 3,000) mg/d	CI	179	2	Yes	EMIT with SYVA Viva- E/V-Twin (Siemens Laboratories)	–	10
Lin <i>et al.</i> 2021 Sci Rep [127]	500– 1,000 mg q 6, 8, 12, 24 h	–	993	3	Yes	HPLC	–	7
Liu <i>et al.</i> 2019 J	Mean ± SD 916.6	II	514	3	Yes	EMIT with Siemens Viva-	–	10

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategies	Infusion modality	Total number	Number per patient ^a	At steady state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomycin range (mg/L))
Pharm Pharmacol [225]	± 226.56 mg over 1 h q 6, 8, 12, 24 h					E Drug Testing System and Emit 2000 Vancomycin Assay Test Kit (Siemens Healthcare Diagnostic Incorporated, Newark, DE, USA)		
Liu <i>et al.</i> 2020 J Clin Pharm Ther [129]	500 mg over 0.5 h q 8 h	II	175	7	No	CMIA with Architect Vancomycin Reagent Kit (Abbott GmbH and Company KG, BLOKIT) and Abbott Architect	–	–

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
						i1000SR instrument (Abbott Laboratories)		
Llopis-Salvia <i>et al.</i> 2006 J Clin Pharm Ther [115]	18.51 mg/kg/d over 16 days	II	234	8	No	FPIA with TDx (Abbot Irving, TX, USA)	2	6
Ma <i>et al.</i> 2020 Front Pharmacol [226]	500– 2,000 mg q 6/8/12/24 h	–	195	3	Yes	HPLC (Agilent 1260)	1	–
Mangin <i>et al.</i> 2014 Clin Pharmacokin et [116]	15–20 mg/kg over 1 hour q 12 Max 1,000 mg in case of	II	359	12	Yes	PETINIA	2	7

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomycin range (mg/L))
	hemodialysis							
Masich <i>et al.</i> 2020 Pharmacotherapy [117]	Median 15 mg/kg	—	64	4	No	—	—	—
Medellin-Garibay <i>et al.</i> 2016 J Antimicrob Chemother [151]	15–20 mg/kg over 1 hour q 12 h Max 1,000 mg in case of hemodialysis Mean ± SD 5.3 ± 7.8	II	392	3	No	Immunoassay with Roche/Hitachi Cobas c ⁱ assay system (Hoffmann - La Roche, Basel, Switzerland)	1.7	5

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
	mg/kg/day							
Medellin-Garibay <i>et al.</i> 2017 Antimicrob Agents Chemother [118]	Loading dose mean $\pm$ SD 12 $\pm$ 5 mg/kg	CI	874	16	Yes	Immunoassay with Roche/Hitachi Cobas c' assay system (Hoffmann–La Roche, Basel, Switzerland)	1.7	5
Moffett <i>et al.</i> 2019 Eur J Clin Pharmacol [134]	Mean $\pm$ SD 14.3 $\pm$ 1.6 mg/kg/d	II	–	–	No	VITROS 5600 Integrated System	–	6 (5–50)
Moore <i>et al.</i> 2016 CPT Pharmacometrics Syst	10–15 mg/kg q 12 h	–	65	5	No	G6PDH enzyme immunoassay with Roche Cobas C501	1.7	–

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Pharmacol [184]						instrument (Roche Diagnostics, Indianapolis, IN)		
Mulla <i>et al.</i> 2005 Br J Clin Pharmacol [160]	10–15 mg/kg q 6–24 h and 750– 1,000 mg q 12–24 h	–	366	8	Yes	Innofluor FPIA (Opus Diagnostics Incorporated, FortLee, NJ 07024)	1	6
Munir <i>et al.</i> 2021 Front Pharmacol [152]	500– 1,000 mg over 0.5 h	II	176	3	No	HPLC	0.25	–
Oda <i>et al.</i> 2020 Pharm Res [131]	Loading dose <20 mg/kg, maintena nce	–	80	5	Yes	Enzymatic assay	2.5	15

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
	median 8.7 mg/kg/d							
Oda <i>et al.</i> 2023 Antimicrob Agents Chemother [135]	Loading dose median (range) 17.7 (9.3– 31.9)	II	81	3	Yes	–	–	–
Okada <i>et al.</i> 2018 J Clin Pharmacol [167]	Median (range) 2,400 (1,000– 4,500) mg/d Max 1,000 mg over 1 hour q 12 h when	II	227	3	Yes	G6PDH enzyme immunoassay with Emit 2000 (Siemens Healthcare Diagnostics Incorporated, Newark, Delaware)	2	5

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
	CrCL >75 mL/min/1.73 m ²							
Purwonugroho <i>et al.</i> 2012 Scientific World Journal [227]	–	–	391	2	No	FPIA with AxSYM (Abbot Laboratories, Abbot Park, Ill, USA)	2	4.26
Revilla <i>et al.</i> 2010 Br J Clin Pharmacol [119]	1,000 mg over 1 h q 12–24 h or CI	II, CI	569	3	Yes	FPIA with AxSYM (Abbott Laboratories, Abbott Park, EU)	2	7
Roberts <i>et al.</i> 2011 Antimicrob Agents Chemother [120]	15 mg/kg then 30 mg/kg or 750– 1,000 mg over 0.5 h	CI	579	3	Yes	FPIA	0.6	0.6

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
	then 2,000– 3,000 mg							
Sanchez <i>et al.</i> 2010 Int J Clin Pharmacol Ther [228]	1,628 mg/d q 6– 48 h	–	254	2	Yes	EMIT (Syva Corporation, Palo Alto, CA, USA)	3.5	7
Schaedeli <i>et al.</i> 1998 Clin Pharmacol Ther [132]	500– 1,000 mg over 0.3 h	II	162	8	No	HPLC	0.8	–
Scharf <i>et al.</i> 2022 Ann Intensive Care [121]	Loading dose median (range) 1,500 (250– 2,000) mg	CI	160	23	Yes	Cobas 8000 C702 analyzers (Roche Diagnostics, Mannheim, Germany)	–	–

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L))
Shen <i>et al.</i> 2018 Clin Infect Dis [229]	–	–	760	2	Yes	FPIA, CLIA	3	–
Shen <i>et al.</i> 2021 Front Pharmacol [230]	–	–	1011	2	Yes	FPIA, CLIA	3	–
Smit <i>et al.</i> 2020 Br J Clin Pharmacol [138]	1,000 mg over 1.7 h and 12.5 mg/kg once Max 2,500 mg if obese	II	326	12	No	Validated immunoassay	4	(3.7–4.4)
Staatz <i>et al.</i> 2006 Br J Clin	Median (range) 1,000 (120–	–	408	4	No	FPIA with TDx assay system (Abbott Laboratories)	1	4.3

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Pharmacol [161]	2000) mg over 0.2– 4.42 h							
Tanaka <i>et al.</i> 2010 Antimicrob Agents Chemother [162]	Median (range) 20 (5.56– 61) mg/kg/da y over 1– 2 h q 6– 48 h	II	181	2	Yes	FPIA with TDx assay system (Abbott Laboratories, Chicago, IL)	–	–
Thomson <i>et al.</i> 2009 J Antimicrob Chemother [163]	500– 2,500 mg/d q 12–48 hours	II	1,557	4	No	FPIA	2	4.3 TDx and 2.5 FLx
Udy <i>et al.</i> 2013 Int J Antimicrob Agents [122]	Loading dose mean ± SD 16.4 ±	CI	199	2	Yes	PETINIA with Dimension Xpand (Siemens)	0.8	–

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
	5.5 and maintena nce dose mean $\pm$ SD 23.7 $\pm$ 8.1 mg/kg/d					Healthcare Diagnostics, Newark, DE)		
Usman <i>et al.</i> 2018 Int J Pharmacol Ther [164]	Median (range) 1,000 (500– 1,500) mg over 0.5 h	II	256	2	No	HPLC	0.25	(3.03– 17.8)
Vu <i>et al.</i> 2019 Int J Antimicrob Agents [231]	2,261 mg /d Loading dose 20 mg/kg over 1–2 h	CI	274	5	Yes	Homogenous enzyme immunoassay with Cobas C 501 systems (Hoffman–La	1.7	–

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
						Roche, Basel, Switzerland)		
Wang <i>et al.</i> 2021 Antibiotics (Basel) [133]	500 mg q 8, 12, 24 h	–	131	12	Yes	CMIA	–	–
Wei <i>et al.</i> 2022 Front Pharmacol [153]	50–1,500 mg	II	895	2	Yes	CMIA	–	–
Wu <i>et al.</i> 2016 J Formos Med Assoc [125]	15–25 mg/kg over 1 h q 6, 8 h	II	220	10	Yes	FPIA with AxSYM (Abbott Laboratories, Abbott Park, IL, USA)	2	5
Yamamoto <i>et al.</i> 2009 J Clin Pharm Ther [139]	Mean ± SD 1,251 ± 446 mg/day	II	356	3	No	FPIA with TDx (Abbot Japan, Tokyo, Japan)	2	6

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
	over 0.5– 2 h							
Yamazaki <i>et al.</i> 2020 Ther Drug Monitor [123]	–	–	129	5	No	CLIA with Architect analyzer (Abbott Laboratories, Irving, Texas, USA)	3	–
Yasuhara <i>et al.</i> 1998 Ther Drug Monitor [154]	500– 1,000 mg over 1 h q 12 h	II	1,253	7	Yes	FPIA with TDx assay system (Abbott Laboratories, Chicago, Illinois, USA)	–	(2.5–3)
Yellepeddi <i>et al.</i> 2024 Antimicrob Agents	–	II	181	10	Yes	PETINIA	1.1	1.38

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Chemother [170]								
Zalloum <i>et al.</i> 2018 TJPR [165]	–	–	164	1	Yes	FPIA with AxSYM (Abbott Laboratories, Abbott Park, USA)	–	–
Zaric <i>et al.</i> 2018 Open Med (Wars) [140]	Mean ± SD (range) 1,650 ± 540 (500– 3,000) and 1,930 ± 430 (1,000– 3,000) mg/d	–	110	1	Yes	Immunoassay with Cobas e601 analyzers (Roche Diagnostics, Mannheim, Germany)	–	–

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Zhang <i>et al.</i> 2023 Clin Pharmakinet [166]	15–20 mg/kg q 6–12 h over 1 h and 1,000– 1,500 mg/d q 6– 8 h and 3,500 mg/d	II, CI	757	4	Yes	–	–	–
Zhao <i>et al.</i> 2021 Antibiotics (Basel) [232]	1,875 [IQR] 1,46 [1.9– 2,352.0]m g/d	II	424	2	Yes	CMIA	–	–
Zhou <i>et al.</i> 2019 Eur J Drug Metab	500– 1,000 mg over 1.5– 2 h q 6, 8,	II	125	2	Yes	CMIA with Architect i1000 system (Abbott Laboratories,	–	10

Authors year of publication journal [reference]	Vancomycin		Sampling			Quantification method		
	Dosage strategie s	Infusi on modal ity	Total number	Number per patient ^a	At stea dy state	Assay and instrument	LLQ (mg/ L)	CV (%) (vancomy cin range (mg/L)
Pharmacokin et [233]	12, 24, 48 h Mean $\pm$ SD 1,550 $\pm$ 770 mg/d					Chicago, Illinois, USA)		

CI, continuous infusion; CLIA, chemiluminescent immunoassay; CMIA, chemiluminescent microparticle immunoassay; CrCL, creatinine clearance; CV, coefficient of variation; ELISA, enzyme-linked immunosorbent assay; EMIT, enzyme-multiplied immunoassay technique; Flx, fluorescence system; FPIA, fluorescence polarization immunoassay; G6PDH, glucose-6-phosphate dehydrogenase, HPLC, high-performance liquid chromatography; II, intermittent infusion; IQR, interquartile range; LLQ, lower limit of quantification; Max, maximum; PETINIA, particle-enhanced turbidimetric inhibition immunoassay; Q, every; SD, standard deviation; TDx, therapeutic drug monitoring system,(vancomycin range), vancomycin concentration range

^a ratio of the total number of vancomycin concentrations to the total number of patients, rounded to the nearest whole number

Supplementary Table 7 Modeling techniques

Authors year of publication journal [reference]	PK software (version)	Compartment	Covariate modeling	dOFV in FA	dOFV in BE
Adane <i>et al.</i> 2015 Pharmacotherapy [188]	NONMEM 7.3	1	FA	3.84	–
Ahmed <i>et al.</i> 2024 Drug Des Devel Ther [142]	MonolixSuite 2020R1.	1	FA and BE	3.84	6.63
Aljutayli <i>et al.</i> 2022 Clin Transl Sci [222]	NONMEM 7.4.0	1	FA and BE	3.84	7.88
Alqahtani <i>et al.</i> 2018 Antimicrob Agents Chemother [143]	Monolix 4.4	2	FA and BE	3.84	6.63
Alqahtani <i>et al.</i> 2020 Pharmacotherapy [137]	Monolix 4.4	1	FA and BE	3.84	6.63
Avedissian <i>et al.</i> 2017 Pediatr Crit Care Med [106]	NONMEM 7.2.0	1	FA and BE	3.84	7.88
Bae <i>et al.</i> 2019 Pharmaceutics [144]	NONMEM 7.4	2	FA, BE, and GAM	3.84	6.63
Bang <i>et al.</i> 2022 Clin Exp Pharmacol physiol [107]	NONMEM 7.4.0	3	FA	3.84	–
Belabbas <i>et al.</i> 2023 J Infect Chemother [145]	NONMEM 7.4.3	1	FA and BE	3.84	6.63
Belavagi <i>et al.</i> 2022 Germs [108]	Phoenix NLME 8.3	2	FA and BE	3.84	7.88
Buelga <i>et al.</i> 2005 Antimicrob Agents Chemother [179]	NONMEM 5.0.0	1	FA, BE, and GAM	3.84	6.63

Authors year of publication journal [reference]	PK software (version)	Compartment	Covariate modeling	dOFV in FA	dOFV in BE
Bury <i>et al.</i> 2019 Eur J Clin Pharmacol [178]	NONMEM 7.3	2	FA	3.84	–
Chu <i>et al.</i> 2020 J Infect Public Health [146]	NONMEM 7.3.0	1	FA and BE	3.84	7.88
Chung <i>et al.</i> 2013 J Korean Med Sci [147]	NONMEM 7.1.0	1	FA, BE, and GAM	6.63	10.83
Colin <i>et al.</i> 2019 Clin Pharmacokinet [103]	NONMEM 7.3	2	FA and BE	3.84	–
De Cock <i>et al.</i> 2014 Pharm Res [105]	NONMEM 6.0	2	FA and BE	7.88	10.83
Deng <i>et al.</i> 2013 Int J Clin Pharmacol Ther [223]	NONMEM 7.2.0	1	FA and BE	3.84	7.88
Dolton <i>et al.</i> 2010 Burns [136]	NONMEM 6.0	2	FA	3.84	–
Donadello <i>et al.</i> 2014 Critical Care [124]	NONMEM 7.2.0	2	FA	3.84	–
Dorajoo <i>et al.</i> 2019 Front Pharmacol [172]	NONMEM 7.3.0	1	FA and BE	3.84	6.63
Escobar <i>et al.</i> 2014 Int J Antimicrob Agents [109]	NONMEM 7.2.0	2	FA	3.84	–
Fu <i>et al.</i> 2021 Biopharm Drug Dispos [156]	NONMEM 7.3.0	2	FA and BE	3.84	10.83
Garreau <i>et al.</i> 2021 Antibiotics (Basel) [110]	MONOLIX 2020R1	2	FA and BE	3.84	6.63

Authors year of publication journal [reference]	PK software (version)	Compartment	Covariate modeling	dOFV in FA	dOFV in BE
Gawmahanil <i>et al.</i> 2018 TJPS [111]	NONMEM 7.3.1	1	FA and BE	3.84	6.63
Gonzalez <i>et al.</i> 2022 Antibiotics [168]	Monolix 2019	2	FA, BE, and GAM	3.84	6.63
Goti <i>et al.</i> 2018 The Drug Monit [157]	NONMEM 7.2–7.3	2	FA and BE	3.84	10.83
Goyal <i>et al.</i> 2022 Front Pharmacol [158]	Pumas 2.0.0	2	–	–	–
Hodiamont <i>et al.</i> 2021 J Antimicrob Chemother [128]	NONMEM 7.3.0	2	FA and BE	3.84	10.83
Holford <i>et al.</i> 2024 Br J Clin Pharmacol [104]	NONMEM 7.5.1	2	–	–	–
Huang <i>et al.</i> 2021 Intern Emerg Med [112]	Phoenix NLME 1.4.0	1	FA and BE	6.63	10.83
Hughes <i>et al.</i> 2024 Ther Drug Monit [148]	NONMEM 7.4.0 Nlmixr2 2.0.9	2	–	–	–
Hui <i>et al.</i> 2019 J Antimicrob Chemother [130]	NONMEM 7.3.0	2	FA	3.84	–
Jaisue <i>et al.</i> 2020 Ther Drug Monit [159]	ADAPT 5.0;0	2	FA	–	–
Ji <i>et al.</i> 2018 Acta Pharmacol Sin [149]	NONMEM 7.3.0	1	FA and BE	3.84	10.83

Authors year of publication journal [reference]	PK software (version)	Compartment	Covariate modeling	dOFV in FA	dOFV in BE
Jing <i>et al.</i> 2019 J Clin Pharm Ther [173]	NONMEM 7.4.3	1	FA and BE	3.84	10.83
Jung <i>et al.</i> 2021 Antimicrob Agents Chemother [169]	NONMEM 7.4.0	3	FA and BE	3.84	6.63
Kim <i>et al.</i> 2016 Int. J Antimicrob Agents [141]	NONMEM 7.2.0	1	FA and BE	3.84	6.63
Kloprogge <i>et al.</i> 2019 Antimicrob Agents Chemother [113]	NONMEM 7.3.0	2	FA and BE	3.84	10.83
Kovacevic <i>et al.</i> 2020 J Crit Care [114]	NONMEM 7.3.0	1	FA and BE	3.84	6.63
Le <i>et al.</i> 2013 Pediatr Infect Dis [155]	NONMEM 7.2.0	1	FA and BE	3.84	7.88
Le <i>et al.</i> 2014 J Pharmacol Clin Toxicol [126]	NONMEM 7.2.0	1	FA and BE	3.84	7.88
Lim <i>et al.</i> 2014 J Clin Pharm Ther [224]	NONMEM 7.1.2	2	–	–	–
Lin <i>et al.</i> 2016 Eur J Clin Pharmacol [150]	NONMEM 7.2.0	1	FA and BE	3.84	6.63
Lin <i>et al.</i> 2021 Sci Rep [127]	Kinetica 4.4.1	1	FA	–	–
Liu <i>et al.</i> 2019 J Pharm Pharmacol [225]	NONMEM 7.3.0	1	FA and BE	3.84	10.83

Authors year of publication journal [reference]	PK software (version)	Compartment	Covariate modeling	dOFV in FA	dOFV in BE
Liu <i>et al.</i> 2020 J Clin Pharm Ther [129]	NONMEM 7.2.0	2	FA and BE	3.84	10.83
Llopis-Salvia <i>et al.</i> 2006 J Clin Pharm Ther [115]	NONMEM	2	FA	7.88	–
Ma <i>et al.</i> 2020 Front Pharmacol [226]	NONMEM 7.4.0	1	FA and BE	3.84	6.63
Mangin <i>et al.</i> 2014 Clin Pharmacokinet [116]	MONOLIX 4.14s	2	FA	–	–
Masich <i>et al.</i> 2020 Pharmacotherapy [117]	Phoenix NLME 8.1	1	FA	3.84	–
Medellin-Garibay <i>et al.</i> 2016 J Antimicrob Chemother [151]	NONMEM 7.2	2	FA, BE, and GAM	3.84	10.83
Medellin-Garibay <i>et al.</i> 2017 Antimicrob Agents Chemother [118]	NONMEM 7.3.0	1	FA, BE, and GAM	3.84	10.83
Moffett <i>et al.</i> 2019 Eur J Clin Pharmacol [134]	NONMEM 7.4.0	2	FA and BE	3.84	10.83
Moore <i>et al.</i> 2016 CPT Pharmacometrics Syst Pharmacol [184]	NONMEM 7.3.0	2	Full covariate model	6.63	–
Mulla <i>et al.</i> 2005 Br J Clin Pharmacol [160]	WinNonMix 2.0.1	2	FA and BE	7.88	–
Munir <i>et al.</i> 2021 Front Pharmacol [152]	NONMEM 7.4.4	1	FA and BE	3.84	6.63

Authors year of publication journal [reference]	PK software (version)	Compartment	Covariate modeling	dOFV in FA	dOFV in BE
Oda <i>et al.</i> 2020 Pharm Res [131]	NONMEM 7.3.0	2	FA and BE	6.63	10.83
Oda <i>et al.</i> 2023 Antimicrob Agents Chemother [135]	NONMEM 7.3.0	1	FA and BE	6.63	10.83
Okada <i>et al.</i> 2018 J Clin Pharmacol [167]	Phoenix NLME 7.0	2	FA and BE	3.84	6.63
Purwonugroho <i>et al.</i> 2012 Scientific World Journal [227]	NONMEM 7.0.0	2	FA and BE	6.63	9.21
Revilla <i>et al.</i> 2010 Br J Clin Pharmacol [119]	NONMEM 5.1.1	1	FA, BE, and GAM	3.84	6.63
Roberts <i>et al.</i> 2011 Antimicrob Agents Chemother [120]	NONMEM 6.1	1	FA	3.84	–
Sanchez <i>et al.</i> 2010 Int J Clin Pharmacol Ther [228]	NONMEM 6.0.0	2	FA, and GAM	3.84 /6.63 for univariate /multivariate analysis	–
Schaedeli <i>et al.</i> 1998 Clin Pharmacol Ther [132]	NONMEM	2	–	6.63	10.83
Scharf <i>et al.</i> 2022 Ann Intensive Care [121]	NONMEM 7.4.0	1	FA	3.84	–

Authors year of publication journal [reference]	PK software (version)	Compartment	Covariate modeling	dOFV in FA	dOFV in BE
Shen <i>et al.</i> 2018 Clin Infect Dis [229]	NONMEM 7.3.0	1	FA and BE	3.84	10.83
Shen <i>et al.</i> 2021 Front Pharmacol [230]	NONMEM 7.3.0	1	FA and BE	3.84	10.83
Smit <i>et al.</i> 2020 Br J Clin Pharmacol [138]	NONMEM 7.4	3	FA and BE	3.84	10.83
Staatz <i>et al.</i> 2006 Br J Clin Pharmacol [161]	NONMEM 5.1.1	1	FA and GAM	6.63	–
Tanaka <i>et al.</i> 2010 Antimicrob Agents Chemother [162]	NONMEM 5.0.0	1	Linear regression	–	–
Thomson <i>et al.</i> 2009 J Antimicrob Chemother [163]	NONMEM 6.0	2	FA, and GAM	7.88	–
Udy <i>et al.</i> 2013 Int J Antimicrob Agents [122]	NONMEM 6.1.0	1	FA	3.84	–
Usman <i>et al.</i> 2018 Int J Pharmacol Ther [164]	NONMEM 7.2.0	1	FA and BE	3.84	6.63
Vu <i>et al.</i> 2019 Int J Antimicrob Agents [231]	NONMEM 6.0.0	2	FA and BE	6.63	10.83
Wang <i>et al.</i> 2021 Antibiotics (Basel) [133]	NONMEM 7.3.0	2	FA and BE	3.84	10.83
Wei <i>et al.</i> 2022 Front Pharmacol [153]	Phoenix NLME 8.3	1	FA and BE	3.84	10.83

Authors year of publication journal [reference]	PK software (version)	Compartment	Covariate modeling	dOFV in FA	dOFV in BE
Wu <i>et al.</i> 2016 J Formos Med Assoc [125]	NONMEM 6.0.0	1	FA and BE	3.84	10.83
Yamamoto <i>et al.</i> 2009 J Clin Pharm Ther [139]	NONMEM 5.0.0	2	FA and BE	3.84	–
Yamazaki <i>et al.</i> 2020 Ther Drug Monitor [123]	Phoenix NLME 1.4	2	FA	6.63	–
Yasuhara <i>et al.</i> 1998 Ther Drug Monitor [154]	NONMEM 6.0.0	2	Linear regression	-	–
Yellepeddi <i>et al.</i> 2024 Antimicrob Agents Chemother [170]	NONMEM 7.5.0	1	FA and BE	3.84	10.83
Zalloum <i>et al.</i> 2018 TJPR [165]	NONMEM 6.0.0	1	FA	3.84	–
Zaric <i>et al.</i> 2018 Open Med (Wars) [140]	NONMEM 7.3.0	1	FA and BE	6.63	6.63
Zhang <i>et al.</i> 2023 Clin Pharmacokinet [166]	NONMEM 7.4.0	2	FA	3.84	–
Zhao <i>et al.</i> 2021 Antibiotics (Basel) [232]	NONMEM 7.0.0	2	FA and BE	3.84	10.83
Zhou <i>et al.</i> 2019 Eur J Drug Metab Pharmacokinet [233]	NONMEM 7.3.0	1	FA and BE	3.84	6.63

BE, backward elimination; dOFV, drop in objective function value; FA, forward addition; GAM, generalized additive model; OFV, objective function value

Supplementary Table 8 Covariates

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL ^a	Covariate on Vc ^a
Adane <i>et al.</i> 2015 Pharmacotherapy [188]	Age, BMI, BSA, BW metrics, HT, kidney function estimators, sex	BW metrics, kidney function estimators	Kidney function estimators	BW metrics
Ahmed <i>et al.</i> 2024 Drug Des Devel Ther [142]	Age, kidney function estimators, protein markers (albumin), SCr, sex, urea	Kidney function estimators	Kidney function estimators	–
Aljutayli <i>et al.</i> 2022 Clin Transl Sci [222]	Age, BW metrics, HT, infectious diseases, kidney function estimators, neutropenia, non-infectious diseases (liver disease, obesity, renal insufficiency), SCr, sex, type of pathogen, ward	BW metrics, kidney function estimators	Kidney function estimators	BW metrics
Alqahtani <i>et al.</i> 2018 Antimicrob Agents Chemother [143]	Age, BMI, BW metrics, cardiac measures (cardiopulmonary bypass), HT, kidney function estimators, protein markers (albumin), RRT, SCr, sex	BW metrics, kidney function estimators, protein markers (albumin)	Kidney function estimators, protein markers (albumin)	BW metrics
Alqahtani <i>et al.</i> 2020 Pharmacotherapy [137]	Age, BMI, BW metrics, HT, kidney function estimators, non-infectious diseases (cancer), protein markers (albumin), SCr, sex	Kidney function estimators, non-infectious	Kidney function estimators, non-infectious diseases (cancer)	Non-infectious diseases (cancer)

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL ^a	Covariate on Vc ^a
		diseases (cancer)		
Avedissian <i>et al.</i> 2017 Pediatr Crit Care Med [106]	Age, BW metrics, comedication, SCr, sex	BW metrics, SCr	BW metrics, SCr	BW metrics
Bae <i>et al.</i> 2019 Pharmaceutics [144]	Age, BW metrics, HT, infectious diseases, kidney function estimators, protein markers (albumin), RRT, sex, urea	BW metrics, kidney function estimators, RRT	Kidney function estimators, RRT	BW metrics
Bang <i>et al.</i> 2022 Clin Exp Pharmacol physiol [107]	Age, BMI, BSA, BW metrics, FFM, HT, kidney function estimators, protein markers (albumin), sex	Age, BW metrics	Age, BW metrics	BW metrics
Belabbas <i>et al.</i> 2023 J Infect Chemother [145]	Augmented renal clearance, BW metrics, kidney function estimators, neutropenia, non-infectious diseases (acute myeloid leukemia), SCr	Augmented renal clearance, BW metrics, neutropenia, non-infectious diseases (acute myeloid leukemia), SCr	Augmented renal clearance, neutropenia, non-infectious diseases (acute myeloid leukemia), SCr	BW metrics

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL ^a	Covariate on Vc ^a
Belavagi <i>et al.</i> 2022 Germs [108]	BW metrics, kidney function estimators, urinary creatinine clearance	BW metrics, urinary creatinine clearance	Urinary creatinine clearance	–
Buelga <i>et al.</i> 2005 Antimicrob Agents Chemother [179]	Age, BSA, BW metrics, comedication, ECOG performance status, erythrocyte markers, HT, kidney function estimators, neutropenia, non-infectious diseases, protein markers (albumin), SCr, sex, transplantation	BW metrics, kidney function estimators	Kidney function estimators	BW metrics
Bury <i>et al.</i> 2019 Eur J Clin Pharmacol [178]	Age, FFM, kidney function estimators, neutropenia	FFM, kidney function estimators, neutropenia	Kidney function estimators, neutropenia	FFM
Chu <i>et al.</i> 2020 J Infect Public Health [146]	Age, ALP, ALT, AST, BW metrics, infectious diseases (sepsis), injury (trauma), kidney function estimators, non-infectious diseases (malignant tumor, neurological, post-operative, respiratory failure), protein markers (albumin), SCr, sex, urea, vancomycin	Age	Age	–

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL ^a	Covariate on Vc ^a
	drug therapy (dosage, dosing date, dosing time, duration)			
Chung <i>et al.</i> 2013 J Korean Med Sci [147]	Age, BW metrics, comedication, cystatin C, HT, SCr, sex, ward	Age, BW metrics, cystatin C, SCr, sex	Age, BW metrics, cystatin C, SCr, sex	Age, BW metrics, sex
Colin <i>et al.</i> 2019 Clin Pharmacokinet [103]	Age (PMA, PNA), BMI, BW metrics, FFM, health condition, heel prick test, HT, injury, non-infectious diseases (hematological malignancy), pharmacokinetic parameters, SCr, sex	Age (PMA, PNA), BW metrics, heel prick test, non-infectious diseases (hematological malignancy), pharmacokinetic parameters, SCr	Age (PMA, PNA), non-infectious diseases (hematological malignancy), pharmacokinetic parameters, SCr	BW metrics, heel prick test
De Cock <i>et al.</i> 2014 Pharm Res [105]	Age, BW metrics, comedication, kidney function estimators, SCr	BW metrics	BW metrics	BW metrics
Deng <i>et al.</i> 2013 Int J Clin Pharmacol Ther [223]	Age, kidney function estimators, SCr	Kidney function estimators	Kidney function estimators	–
Dolton <i>et al.</i> 2010 Burns [136]	Age, BW metrics, injury, kidney function estimators	BW metrics, injury, kidney	Kidney function estimators	BW metrics, injury

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL ^a	Covariate on Vc ^a
		function estimators		
Donadello <i>et al.</i> 2014 Critical Care [124]	Age, BW metrics, ECMO measures, infectious diseases, kidney function estimators, mortality clinical scores, RRT, sex	RRT	RRT	–
Dorajoo <i>et al.</i> 2019 Front Pharmacol [172]	Age, BMI, BSA, BW metrics, HT, inflammatory markers, kidney function estimators, neutropenia (albumin), protein markers, SCr, sex, WBC	Kidney function estimators	Kidney function estimators	–
Escobar <i>et al.</i> 2014 Int J Antimicrob Agents [109]	Age, BMI, BW metrics, ECMO measures, mortality clinical scores, RRT, SCr, sex	RRT	RRT	–
Fu <i>et al.</i> 2021 Biopharm Drug Dispos [156]	Age, ALT, AST, BW metrics, coagulation markers, erythrocyte markers, kidney function estimators, neutropenia, protein markers, SCr, sex, WBC	BW metrics, kidney function estimators	BW metrics, kidney function estimators	BW metrics
Garreau <i>et al.</i> 2021 Antibiotics (Basel) [110]	Age, BMI, body temperature, BW metrics, cardiac measures, HT, infectious diseases (sepsis), kidney function estimators, mortality clinical	BW metrics, kidney function estimators, RRT	Kidney function estimators, RRT	BW metrics

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL ^a	Covariate on Vc ^a
	scores, protein markers (albumin), pulmonary measures, RRT, SCr, sex			
Gawmahanil <i>et al.</i> 2018 TJPS [111]	Age, BW metrics, comedication (dopamine, dobutamine, norepinephrine, epinephrine, furosemide, gentamicin, ibuprofen, and diclofenac), fluid status, infectious diseases (sepsis), kidney function estimators, mechanical ventilation, mortality clinical scores, non-infectious diseases (diabetes mellitus), protein markers (albumin), SCr, sex	Infectious diseases (sepsis), kidney function estimators, non-infectious diseases (diabetes mellitus), SCr, sex	Infectious diseases (sepsis), kidney function estimators, non-infectious diseases (diabetes mellitus)	SCr, sex
Gonzalez <i>et al.</i> 2022 Antibiotics [168]	Age, BW metrics, HT, kidney function estimators, protein markers, SCr, sex	Kidney function estimators	Kidney function estimators	–
Goti <i>et al.</i> 2018 The Drug Monit [157]	Age, BW metrics, fluid status, HT, kidney function estimators, RRT, SCr, sex, urea, ward	BW metrics, kidney function estimators, RRT	Kidney function estimators, RRT	BW metrics, RRT
Goyal <i>et al.</i> 2022 Front Pharmacol [158]	Age, BMI, BW metrics, FFM, HT, kidney function estimators, SCr	FFM, kidney function estimators	FFM, kidney function estimators	FFM

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL ^a	Covariate on Vc ^a
Hodiamont <i>et al.</i> 2021 J Antimicrob Chemother [128]	Age, BW metrics, kidney function estimators, RRT, RIFLE score, sex	BW metrics, kidney function estimators	BW metrics, kidney function estimators	BW metrics
Holford <i>et al.</i> 2024 Br J Clin Pharmacol [104]	Age, BW metrics, comedication, FFM, HT, kidney function estimators, mechanical ventilation, RRT, SCr, sex	Age, comedication, FFM, kidney function estimators, mechanical ventilation, RRT	Comedication, FFM, kidney function estimators, mechanical ventilation, RRT	Age, FFM
Huang <i>et al.</i> 2021 Intern Emerg Med [112]	Age, BW metrics, cystatin C, kidney function estimators, non-infectious diseases (persistent inflammation, immunosuppression, and catabolism syndrome), SCr	Age, BW metrics, kidney function estimators	Kidney function estimators	Age, BW metrics
Hughes <i>et al.</i> 2024 Ther Drug Monit [148]	Age, BMI, BW metrics, FFM, HT, kidney function estimators, SCr, sex	FFM, kidney function estimators	Kidney function estimators	FFM
Hui <i>et al.</i> 2019 J Antimicrob Chemother [130]	BW metrics, coagulation markers, RRT	BW metrics, RRT	BW metrics, RRT	BW metrics

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL ^a	Covariate on Vc ^a
Jaisue <i>et al.</i> 2020 Ther Drug Monit [159]	Age, BW metrics, comedication, kidney function estimators, non-infectious diseases, nutritional status, sex, ward (ICU)	Kidney function estimators	Kidney function estimators	–
Ji <i>et al.</i> 2018 Acta Pharmacol Sin [149]	Age, BW metrics, kidney function estimators, protein markers (albumin), SCr, sex, urea	Age, kidney function estimators	Age, kidney function estimators	–
Jing <i>et al.</i> 2019 J Clin Pharm Ther [173]	Age, BW metrics, kidney function estimators, sex	Age, BW metrics, kidney function estimators	Age, BW metrics, kidney function estimators	–
	Age, kidney function estimators, sex	Age, kidney function estimators	Age, kidney function estimators	–
Jung <i>et al.</i> 2021 Antimicrob Agents Chemother [169]	Age, BSA, BW metrics, ECMO measures (type, flow rate), HT, kidney function estimators, protein markers, SCr, sex, urea	ECMO measures, Kidney function estimators	Kidney function estimators	–
Kim <i>et al.</i> 2016 Int. J Antimicrob Agents [141]	Age, BW metrics, comedication, days post-surgery, fluid status, kidney function estimators, non-infectious diseases (cirrhosis), protein markers	BW metrics, comedication (nephrotoxic drugs), kidney	BW metrics, comedication (nephrotoxic drugs), kidney	Vancomycin drug therapy (early/late phase treatment)

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL ^a	Covariate on Vc ^a
	(albumin), SCr, sex, vancomycin drug therapy (early/late phase treatment), ward	function estimators, non-infectious diseases (cirrhosis), vancomycin drug therapy (early/late phase treatment), ward (neurosurgery)	function estimators, non-infectious diseases (cirrhosis), vancomycin drug therapy (early/late phase treatment), ward (neurosurgery)	
	Age, BW metrics, comedication, days post-surgery, fluid status, kidney function estimators, non-infectious diseases (cirrhosis), protein markers (albumin), SCr, sex, vancomycin drug therapy (early/late phase treatment), ward	BW metrics, comedication, kidney function estimators, vancomycin drug therapy (early/late phase treatment)	BW metrics, comedication, kidney function estimators, vancomycin drug therapy (early/late phase treatment)	Vancomycin drug therapy (early/late phase treatment)
Kloprogge <i>et al.</i> 2019 Antimicrob Agents Chemother [113]	Age (PMA), BW metrics, SCr	Age (PMA), BW metrics, age-standardized SCr	Age (PMA), BW metrics, age-standardized SCr	BW metrics

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL ^a	Covariate on Vc ^a
Kovacevic <i>et al.</i> 2020 J Crit Care [114]	ALT, AST, BW metrics, comedication (nonsteroidal anti-inflammatory drug), kidney function estimators, protein markers	BW metrics, kidney function estimators	Kidney function estimators	BW metrics
Le <i>et al.</i> 2013 Pediatr Infect Dis [155]	Age, BW metrics, comedication (nephrotoxic drug), neutropenia, SCr, study site, ward	Age, BW metrics, SCr	Age, BW metrics, SCr	BW metrics
Le <i>et al.</i> 2014 J Pharmacol Clin Toxicol [126]	Age, BW metrics, comedication, SCr, sex	Age, BW metrics, SCr	Age, BW metrics, SCr	BW metrics
Lim <i>et al.</i> 2014 J Clin Pharm Ther [224]	Age, BW metrics, HT, kidney function estimators, SCr, sex	Kidney function estimators	Kidney function estimators	–
Lin <i>et al.</i> 2016 Eur J Clin Pharmacol [150]	Age, BW metrics, kidney function estimators, SCr	Kidney function estimators	Kidney function estimators	–
Lin <i>et al.</i> 2021 Sci Rep [127]	Age, BW metrics, comedication (dopamine, noradrenaline, furosemide, dobutamine), injury (burn), RRT, SCr, sex, vancomycin drug therapy (manufacturer)	Age, BW metrics, comedication (dopamine), injury (burn), RRT, SCr	BW metrics, comedication (dopamine), injury (burn), RRT, SCr	Age, BW metrics, SCr

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL ^a	Covariate on Vc ^a
Liu <i>et al.</i> 2019 J Pharm Pharmacol [225]	Age, BW metrics, comedication (nephrotoxic drugs), kidney function estimators, ward	Age, BW metrics, kidney function estimators	Age, BW metrics, kidney function estimators	–
Liu <i>et al.</i> 2020 J Clin Pharm Ther [129]	ECMO measures	ECMO measures	ECMO measures	ECMO measures
Llopis-Salvia <i>et al.</i> 2006 J Clin Pharm Ther [115]	Age, body temperature, BW metrics, clinical outcome, fluid status, infectious diseases, kidney function estimators, non-infectious diseases, SCr, sex, urea, urinary output	BW metrics, kidney function estimators	BW metrics, kidney function estimators	BW metrics
Ma <i>et al.</i> 2020 Front Pharmacol [226]	Age, ALT, AST, BW metrics, clinical outcome, comedication, days post-surgery, erythrocyte markers, kidney function estimators, protein markers, RRT, SCr, sex, transplantation	BW metrics, kidney function estimators	BW metrics, kidney function estimators	BW metrics
Mangin <i>et al.</i> 2014 Clin Pharmacokinet [116]	Age, BW metrics, mortality clinical scores, non-infectious diseases (diabetes mellitus), RRT, SCr, sex	BW metrics, mortality clinical scores, non-infectious diseases	BW metrics, mortality clinical scores, SCr, sex	BW metrics

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL ^a	Covariate on Vc ^a
		(diabetes mellitus), SCr, sex		
Masich <i>et al.</i> 2020 Pharmacotherapy [117]	Age, BW metrics, comedication, health condition (disease status), kidney function estimators, mortality clinical scores, sex	Kidney function estimators	Kidney function estimators	–
Medellin-Garibay <i>et al.</i> 2016 J Antimicrob Chemother [151]	Age, BMI, BSA, BW metrics, comedication (aminoglycosides, furosemide, nonsteroidal anti-inflammatory nonsteroidal anti-inflammatory drugs), HT, kidney function estimators, protein markers, SCr, sex, urea, uric acid	Age, BW metrics, comedication (furosemide), kidney function estimators	Comedication (furosemide), kidney function estimators	Age, BW metrics
Medellin-Garibay <i>et al.</i> 2017 Antimicrob Agents Chemother [118]	Age, BMI, BSA, BW metrics, comedication, glucose, HT, kidney function estimators, mechanical ventilation, SCr, sex	BW metrics, kidney function estimators, mechanical ventilation	Kidney function estimators, mechanical ventilation	BW metrics
Moffett <i>et al.</i> 2019 Eur J Clin Pharmacol [134]	Age, BW metrics, erythrocyte markers, FFM, HT, protein markers (albumin), RRT, SCr, sex, urea, urinary output	FFM, RRT, SCr, urea	FFM, RRT, SCr, urea	FFM

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL _a	Covariate on Vc _a
Moore <i>et al.</i> 2016 CPT Pharmacometrics Syst Pharmacol [184]	Age, BW metrics, kidney function estimators, RRT, SCr, sex	BW metrics, kidney function estimators	Kidney function estimators	BW metrics
Mulla <i>et al.</i> 2005 Br J Clin Pharmacol [160]	Age, BW metrics, SCr	Age, BW metrics, SCr	Age, BW metrics, SCr	BW metrics
Munir <i>et al.</i> 2021 Front Pharmacol [152]	Age, BW metrics, kidney function estimators, SCr, sex	BW metrics, kidney function estimators	BW metrics, kidney function estimators	–
Oda <i>et al.</i> 2020 Pharm Res [131]	Age, BW metrics, kidney function estimators, mortality clinical scores, protein markers, RRT, urinary output	BW metrics, RRT, urinary output	RRT, urinary output	BW metrics
Oda <i>et al.</i> 2023 Antimicrob Agents Chemother [135]	Age, BW metrics, HT, infectious diseases, inflammatory markers, kidney function estimators, mortality clinical scores, protein markers	BW metrics, dialysis, protein markers	BW metrics, dialysis, protein markers	BW metrics
Okada <i>et al.</i> 2018 J Clin Pharmacol [167]	Age, BIL, BSA, BW metrics, comedication (immunosuppressive drugs), erythrocyte markers, inflammatory markers, kidney function estimators, non-infectious diseases (hematologic disease), protein	BW metrics, kidney function estimators	Kidney function estimators	BW metrics

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL _a	Covariate on Vc _a
	markers (albumin), SCr, sex, transplantation, urea, WBC			
Purwonugroho <i>et al.</i> 2012 Scientific World Journal [227]	Age, ALT, AST, BIL, BW metrics, comedication, kidney function estimators, protein markers (albumin), SCr, sex, urea	Age, kidney function estimators	Kidney function estimators	Age
Revilla <i>et al.</i> 2010 Br J Clin Pharmacol [119]	Age, BW metrics, clinical outcome, comedication, fluid status, HT, kidney function estimators, mechanical ventilation, mortality clinical scores, non-infectious diseases, nutritional status, protein markers (albumin), SCr, sex	Age, BW metrics, kidney function estimators, SCr	Age, BW metrics, kidney function estimators	BW metrics, SCr
Roberts <i>et al.</i> 2011 Antimicrob Agents Chemother [120]	Age, BMI, BW metrics, kidney function estimators, mortality clinical scores, sex, urinary creatinine clearance	BW metrics, urinary creatinine clearance	Urinary creatinine clearance	BW metrics
Sanchez <i>et al.</i> 2010 Int J Clin Pharmacol Ther [228]	Age, BMI, BW metrics, kidney function estimators, sex, vancomycin drug therapy (duration)	Age, BW metrics, kidney function estimators	Kidney function estimators	BW metrics

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL ^a	Covariate on Vc ^a
Schaedeli <i>et al.</i> 1998 Clin Pharmacol Ther [132]	BW metrics, fluid status, kidney function estimators, RRT, urinary creatinine clearance	BW metrics, kidney function estimators, RRT	Kidney function estimators, RRT	BW metrics
Scharf <i>et al.</i> 2022 Ann Intensive Care [121]	Adsorber for extracorporeal cytokine, age, BIL, BW metrics, inflammatory markers, mortality clinical scores, protein markers (albumin), RRT	Adsorber for extracorporeal cytokine, RRT	Adsorber for extracorporeal cytokine, RRT	–
Shen <i>et al.</i> 2018 Clin Infect Dis [229]	Age, ALP, ALT, AST, BMI, BW metrics, HT, injury, kidney function estimators, non-infectious diseases (chronic kidney disease, diabetes, cancer), protein markers (albumin), SCr, sex	Age, kidney function estimators	Kidney function estimators	Age
Shen <i>et al.</i> 2021 Front Pharmacol [230]	Age, ALP, ALT, AST, BMI, BW metrics, HT, injury, kidney function estimators, non-infectious diseases (chronic kidney disease, diabetes, cancer), protein markers (albumin), SCr, sex	Age, BW metrics, kidney function estimators	BW metrics, kidney function estimators	Age, BW metrics
Smit <i>et al.</i> 2020 Br J Clin Pharmacol [138]	Age, BMI, BW metrics, HT, kidney function estimators, SCr, sex	Age, BW metrics	BW metrics	Age

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL _a	Covariate on Vc _a
Staatz <i>et al.</i> 2006 Br J Clin Pharmacol [161]	Age, BMI, BSA, BW metrics, HT, kidney function estimators, SCr, sex, vancomycin drug therapy (day of therapy)	kidney function estimators, BW metrics	kidney function estimators	BW metrics
Tanaka <i>et al.</i> 2010 Antimicrob Agents Chemother [162]	Age, BW metrics, HT, kidney function estimators, SCr, sex	BW metrics, kidney function estimators	Kidney function estimators	BW metrics
Thomson <i>et al.</i> 2009 J Antimicrob Chemother [163]	Age, BW metrics, BSA, HT, kidney function estimators, SCr, sex, vancomycin drug therapy (day of therapy)	BW metrics, kidney function estimators	Kidney function estimators	BW metrics
Udy <i>et al.</i> 2013 Int J Antimicrob Agents [122]	Age, BW metrics, clinical outcome, comedication (vasopressor), HT, kidney function estimators, mechanical ventilation, mortality clinical scores, RRT, SCr, sex, ward (ICU)	BW metrics	-	BW metrics
Usman <i>et al.</i> 2018 Int J Pharmacol Ther [164]	Age, BW metrics, kidney function estimators, sex, SCr	Kidney function estimators	kidney function estimators	–
Vu <i>et al.</i> 2019 Int J Antimicrob Agents [231]	Age, BW metrics, kidney function estimators, mechanical ventilation, mortality clinical scores, SCr, sex	BW metrics, kidney function estimators	Kidney function estimators	BW metrics

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL ^a	Covariate on Vc ^a
Wang <i>et al.</i> 2021 Antibiotics (Basel) [133]	Age, ALP, ALT, BIL, BMI, body temperature, BW metrics, coagulation marker, RRT, erythrocyte markers, heart function, kidney function estimators, mortality clinical scores, protein markers (albumin), respiratory function, sex, urea, urinary output, WBC	BW metrics, protein markers (albumin), RRT	Protein markers (albumin), RRT	BW metrics
Wei <i>et al.</i> 2022 Front Pharmacol [153]	Age, BMI, BW metrics, comedication (furosemide, mannitol, torsemide), HT, kidney function estimators, RRT, SCr, sex,	BW metrics, comedication (mannitol), kidney function estimators	BW metrics, comedication (mannitol), kidney function estimators	–
Wu <i>et al.</i> 2016 J Formos Med Assoc [125]	Age, ALT, AST, BIL, BW metrics, ECMO measures (indication, type of pump, mode), HT, kidney function estimators, SCr, sex, vancomycin drug therapy (dosing)	Age, kidney function estimators	Kidney function estimators	Age
Yamamoto <i>et al.</i> 2009 J Clin Pharm Ther [139]	Age, BW metrics, health condition, infectious diseases (bacteremia, pneumonia, other), kidney function estimators	BW metrics, health condition, kidney function estimators	Kidney function estimators	BW metrics, health condition

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL ^a	Covariate on Vc ^a
Yamazaki <i>et al.</i> 2020 Ther Drug Monitor [123]	Age, BW metrics, HT, inflammatory markers, mortality clinical scores, protein markers (albumin), SCr, sex, urea	BW metrics, mortality clinical scores	Mortality clinical scores	BW metrics
Yasuhara <i>et al.</i> 1998 Ther Drug Monitor [154]	Age, ALT, ALP, AST, BIL, BW metrics, kidney function estimators, SCr, urea	Kidney function estimators	Kidney function estimators	–
Yellepeddi <i>et al.</i> 2024 Antimicrob Agents Chemother [170]	Age, BW metrics, comedication (aztreonam, tobramycin), kidney function estimators, SCr	BW metrics	BW metrics	–
Zalloum <i>et al.</i> 2018 TJPR [165]	Age, ALT, AST, BIL, BW metrics, comedication (n-acetylcysteine), fluid status, HT, infectious diseases, kidney function estimators, kidney impairment, mechanical ventilation, non-infectious diseases, protein markers, SCr, sex, urea	BW metrics, comedication (n-acetylcysteine), kidney impairment, SCr, sex	BW metrics, kidney impairment, SCr, sex	BW metrics, comedication (n-acetylcysteine)
Zaric <i>et al.</i> 2018 Open Med (Wars) [140]	Age, ALT, AST, BIL, BNP, BW metrics, coagulation markers, comedication (angiotensin converting enzyme inhibitor, colistin, furosemide, heparin, nonsteroidal anti-	AST, coagulation markers, kidney function estimators,	AST, coagulation markers, kidney function estimators,	Kidney function estimators

Authors year of publication journal [reference]	Tested characteristics ^a	Retained covariates ^a	Covariate on CL ^a	Covariate on Vc ^a
	inflammatory drug, piperacillin tazobactam), infectious diseases (sepsis), inflammatory markers, injury, kidney function estimators, protein markers (albumin), sex, vancomycin drug therapy	vancomycin drug therapy (dosing)	vancomycin drug therapy (dosing)	
Zhang <i>et al.</i> 2023 Clin Pharmakinet [166]	Age, BMI, BW metrics, HT, kidney function estimators, SCr, sex	Age, BW metrics, sex	Age, BW metrics, sex	BW metrics
Zhao <i>et al.</i> 2021 Antibiotics (Basel) [232]	Age, BW metrics, infectious diseases (shock, multiple organ failure), kidney function estimators, sex, ward (ICU)	Kidney function estimators, ward (ICU)	Kidney function estimators	Ward (ICU)
Zhou <i>et al.</i> 2019 Eur J Drug Metab Pharmacokinet [233]	Age, ALT, AST, BIL, BW metrics, comedication, infectious diseases (sepsis), kidney function estimators, protein markers (albumin), SCr, sex, urea	Kidney function estimators	Kidney function estimators	–

^a Body weight metrics (total body weight, lean body weight, adjusted body weight), cardiac measure (cardiac index, cardiopulmonary pass), clinical outcome (complication, death, hospitalization), coagulation marker (interleukin, platelet count), fluid status (fluid, osmolality, total body water, volume status), health condition (disease status, healthy, immunosuppression), heart function (diastolic blood pressure, heart rate, systolic blood pressure), inflammatory marker (c-reactive protein markers, interleukin, procalcitonin), injury (burn, trauma), mortality clinical scores (acute physiology and chronic health evaluation [APACHE] score, simplified acute physiology score II [SAPSII] score, sequential organ failure assessment [SOFA] score),

pulmonary measure (extravascular lung water index, pulmonary vascular permeability index), respiratory function (respiratory rate)

ALT, alanine aminotransferase; ALP, alkaline phosphatase; AST, aspartate aminotransferase; BIL, bilirubin; BNP, b-type natriuretic peptide; BMI, body mass index; BSA, body surface area; BW metrics, body weight metrics; V_c , central volume of distribution; CL, clearance; ECOG, eastern cooperative oncology group; ECMO, extracorporeal membrane oxygenation; FFM, fat-free mass; HT, height; SCr, serum creatinine; RIFLE, risk, injury, failure, loss, end-stage renal disease; RRT, renal replacement therapy; WBC, white blood cells

Supplementary Table 9 Questionnaire on the model-informed precision dosing software tools

Question	Answer
Email ^a	
Name of the software tool ^a	
Name of the company	
Does your software tool include a module for model-informed dose optimization of vancomycin (e.g., model-informed precision dosing, target concentration intervention, model-based therapeutic drug monitoring, etc.)?	- No - Yes
Your software tool does not include a module for model-informed dose optimization of vancomycin. Please elaborate why?	
Information about vancomycin population pharmacokinetic model(s).	
What approach is your vancomycin module based on? If the approach you apply not listed, please fill in the 'Other' option.	- A single population pharmacokinetic model approach - A multiple population pharmacokinetic model approach - Artificial intelligence or machine learning - Other
How many vancomycin population pharmacokinetic model(s) is/are implemented in your software tool? ^a	
Is there a traceable publication identifier (e.g., PMID, DOI) for this/these model(s)?	- No - Yes

You indicated that traceable publication identifier(s) is/are available.	
Please provide the traceable publication identifier(s) (e.g., PMID, DOI). ^a	
You indicated that traceable publication identifier(s) is/are not available.	
Please select the population(s) that are covered by your vancomycin module? If the population(s) is/are not listed, use the 'Other' option to fill in any additional population(s).	- Adults - Adolescents - Children - Neonates - Older patients - Pregnant women - Critically ill patients - Other
Please provide how many of the included model(s) is/are 1-compartment, 2-compartment, and 3-compartment model(s). ^a	
Please list the covariates in the model(s). ^a	
Medical use of the vancomycin module.	
Did you perform external validation(s) of the vancomycin module? In case of a prospective interventional trial, please use the 'Other' option to provide a clinical trial identifier.	- No - Yes, retrospectively - Yes, prospectively, non-interventional - Yes, prospectively, interventional - Other
Is your software tool with the vancomycin module currently in clinical use to guide vancomycin dosing? If yes, please use the 'Other' option to provide an estimated range of the number of patients in the last 12 months.	- No - Yes - Other

Is your vancomycin module intended for medical use?	- No - Yes
Is your vancomycin module regulated by any competent authority (e.g., Medical Device Regulation in the European Union: Regulation (2017/745)? If yes, please use the 'Other' option to list the competent authorities which registered the vancomycin module?	- No - Yes - Other

^a A response is required to proceed to the next question

Supplementary Table 10 Models integrated into model-informed precision dosing software tools

Population Model [reference]^a	APK	ClinCalc	DoseMeRx	ID-ODS	InsightRx Nova	MWPharm	NextDose	PKJust	PrecisePK	Rx Studio	TDMx	Tucuxi	VancoCalc
Full age spectrum													
Colin <i>et al.</i> 2019 Clin Pharmacokinet [103]	–	–	–	–	✓	✓	–	–	–	–	–	✓	–
De Cock <i>et al.</i> 2014 Pharm Res [105]	–	–	–	–	✓	–	–	–	–	–	–	✓	–
Holford <i>et al.</i> 2024 Br J Clin Pharmacol [104]	–	–	–	–	–	–	✓	–	–	–	–	–	–
Adults													
#vancomycin_adult_C2 ^b [185]	–	–	–	–	–	✓	–	–	–	–	–	–	–
#vancomycin_adult_k_C2 ^b [185]	–	–	–	–	–	✓	–	–	–	–	–	–	–
Adane <i>et al.</i> 2015 Pharmacotherapy [188]	–	✓	–	–	✓	–	–	–	–	–	✓	–	✓
Ariano <i>et al.</i> 2005 Am J Kidney Dis ^b [196]	–	–	–	✓	–	–	–	–	–	–	–	–	–
Bae <i>et al.</i> 2019 Pharmaceutics [144]	–	–	–	–	✓	✓	–	–	–	–	–	–	–
Buelga <i>et al.</i> 2005 Antimicrob Agents Chemother [179]	–	✓	✓	–	✓	–	–	–	–	✓	–	–	✓
Bury <i>et al.</i> 2019 Eur J Clin Pharmacol [178]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Carreno <i>et al.</i> 2017 Antimicrob Agents Chemother ^b [191]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Claisse <i>et al.</i> 2019 Br J Clin Pharmacol ^b [183]	–	–	–	✓	–	–	–	–	–	–	–	–	–

Crass <i>et al.</i> 2018 J Antimicrob Chemother ^b [190]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Dolton <i>et al.</i> 2010 Burns [136]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Dutch therapeutic drug monitoring monograph ^b [234]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Forrest <i>et al.</i> 2007 annual meeting of the Infectious Diseases Society of America ^b [235]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Goti <i>et al.</i> 2018 The Drug Monit [157]	–	–	✓	✓	✓	–	–	–	–	✓	✓	✓	–
Goyal <i>et al.</i> 2022 Front Pharmacol [158]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Hughes <i>et al.</i> 2024 Ther Drug Monit [148]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Hughes <i>et al.</i> 2024 unpublished ^b	–	–	–	–	✓	–	–	–	–	–	–	–	–
Hui <i>et al.</i> 2019 J Antimicrob Chemother [130]	–	–	✓	–	✓	–	–	–	–	–	–	–	–
Liu <i>et al.</i> 2019 J Pharm Pharmacol [225]	–	–	–	–	✓	–	–	–	–	–	–	✓	–
Llopis-Salvia <i>et al.</i> 2006 J Clin Pharm Ther [115]	–	–	–	–	✓	–	–	–	–	–	–	✓	–
Van Maarseveen <i>et al.</i> 2014 Eur J Clin Pharmacol ^b [199]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Mangin <i>et al.</i> 2014 Clin Pharmacokinet [116]	–	–	–	–	–	–	–	–	–	–	✓	–	–
Masich <i>et al.</i> 2020 Pharmacotherapy [117]	–	✓	–	–	–	–	–	–	–	–	–	–	✓

Matzke <i>et al.</i> 1984 Antimicrob Agents Chemother ^b [186]	✓	–	–	✓	–	–	–	–	–	–	–	–	–
Moore <i>et al.</i> 2016 CPT Pharmacometrics Syst Pharmacol [184]	–	–	–	✓	–	–	–	–	–	–	–	–	–
Neely <i>et al.</i> 2014 Antimicrob Agents Chemother ^b [236]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Oda <i>et al.</i> 2020 Pharm Res [131]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Revilla <i>et al.</i> 2010 Br J Clin Pharmacol [119]	–	–	–	–	✓	–	–	–	–	–	✓	–	–
Roberts <i>et al.</i> 2011 Antimicrob Agents Chemother [120]	–	✓	–	–	✓	✓	–	–	–	–	✓	–	✓
Rodvold <i>et al.</i> 1988 Antimicrob Agents Chemother ^b [187]	–	–	–	–	✓	–	–	–	✓	–	–	–	–
Sabourenkou <i>et al.</i> 2019 Open Forum Infect Dis ^b [189]	–	–	✓	–	–	–	–	–	–	–	–	–	–
Smit <i>et al.</i> 2020 Br J Clin Pharmacol [138]	–	–	–	–	–	✓	–	–	–	–	–	–	–
Staatz <i>et al.</i> 2006 Br J Clin Pharmacol [161]	–	–	–	–	–	–	–	–	–	–	–	✓	–
Tanaka <i>et al.</i> 2010 Antimicrob Agents Chemother [162]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Thomson <i>et al.</i> 2009 J Antimicrob Chemother [163]	–	–	–	–	✓	–	–	–	–	–	✓	✓	–
Vancomycin_C1 ^b [237]	–	–	–	–	–	✓	–	–	–	–	–	–	–
Vancomycin__adult_C2 ^b [194]	–	–	–	–	–	✓	–	–	–	–	–	–	–

Vancomycin__capd_C2 ^b [194]	–	–	–	–	–	✓	–	–	–	–	–	–	–
Vancomycin__dialysis_C2 ^b [194]	–	–	–	–	–	✓	–	–	–	–	–	–	–
Yamamoto <i>et al.</i> 2009 J Clin Pharm Ther [139]	–	–	–	–	–	–	–	–	–	–	–	✓	–
Yasuhara <i>et al.</i> 1998 Ther Drug Monitor [154]	–	–	–	–	–	✓	–	–	–	–	–	–	–
Zdovc <i>et al.</i> 2016 Population Approach Group Europe meeting ^b [182]	–	–	–	–	✓	–	–	–	–	–	–	–	–
<i>Pediatric non-neonates and neonates</i>													
Jacqz-Aigrain <i>et al.</i> 2019 J Antimicrob Chemother [238]	–	–	–	–	✓	✓	–	–	–	–	–	–	–
Chen <i>et al.</i> 2018 Eur J Clin Pharmacol [239]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Chung <i>et al.</i> 2021 Kidney Int Rep [197]	–	–	–	✓	–	–	–	–	–	–	–	–	–
Grimsley <i>et al.</i> 1999 Arch Dis Child Fetal Neonatal Ed [240]	–	–	–	–	–	–	–	–	–	–	–	✓	–
Gwee <i>et al.</i> 2019 JAMA Pediatr [241]	–	–	✓	–	–	–	–	–	–	–	–	–	–
Kloprogge <i>et al.</i> 2019 Antimicrob Agents Chemother [113]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Moffett <i>et al.</i> 2019 J Pediatr Pharmacol Ther [180]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Shimamoto <i>et al.</i> 2021 J Antimicrob Chemother [177]	–	–	–	–	✓	–	–	–	–	–	–	–	–

Shimamoto <i>et al.</i> 2023 Clin Pharmacol Ther [181]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Song <i>et al.</i> 2017 Oncotarget [242]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Pediatric non-neonates													
#vancomycin_child_C2 [185]	–	–	–	–	–	✓	–	–	–	–	–	–	–
Avedissian <i>et al.</i> 2017 Pediatr Crit Care Med [106]	–	–	–	✓	✓	–	–	–	–	–	–	–	–
Lamarre <i>et al.</i> 2000 Antimicrob Agents Chemother [243]	–	–	✓	–	✓	–	–	–	–	–	–	–	–
Le <i>et al.</i> 2014 J Pharmacol Clin Toxicol [126]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Le <i>et al.</i> 2014 Ther Drug Monit [244]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Le <i>et al.</i> 2015 Clin Ther [192]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Moffett <i>et al.</i> 2019 Eur J Clin Pharmacol [134]	–	–	–	✓	–	–	–	–	–	–	–	–	–
Smit <i>et al.</i> 2021 AAPS J [193]	–	–	✓	–	✓	–	–	–	–	✓	–	–	–
Zhang <i>et al.</i> 2023 Clin Pharmacokinet [166]	–	–	–	–	–	✓	–	–	–	–	–	–	–
Pediatric neonates													
#vancomycin_neonate_0.5_1m_C2 [185]	–	–	–	–	–	✓	–	–	–	–	–	–	–
#vancomycin_neonate_0_0.5m_C1 [185]	–	–	–	–	–	✓	–	–	–	–	–	–	–
Anderson <i>et al.</i> 2007 Br J Clin Pharmacol [245]	–	–	–	–	✓	–	–	–	–	–	–	–	–

Bhongsatiern <i>et al.</i> 2015 Ther Drug Monitor [246]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Capparelli <i>et al.</i> 2001 J Clin Pharmacol [247]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Dao <i>et al.</i> 2020 Pharmacol Resp [248]	–	–	–	–	✓	✓	–	–	–	–	–	✓	–
Frymoyer <i>et al.</i> 2014 Antimicrob Agents Chemother [249]	–	–	✓	✓	✓	–	–	–	–	–	–	✓	–
Germovsek <i>et al.</i> 2019 J Antimicrob Chemother [250]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Kimura <i>et al.</i> 2004 Antimicrob Agents Chemother [251]	–	–	–	–	–	–	–	–	–	–	–	✓	–
Lee <i>et al.</i> 2021 Sci Rep [252]	–	–	–	–	–	–	–	–	–	–	–	✓	–
Li <i>et al.</i> 2018 Front Pharmacol [253]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Lo <i>et al.</i> 2010 Antimicrob Agents and Chemother [254]	–	–	–	–	✓	–	–	–	–	–	–	✓	–
Mehrotra <i>et al.</i> 2012 Pharmacotherapy [255]	–	–	–	–	–	–	–	–	–	–	–	✓	–
Mulubwa <i>et al.</i> 2020 Drugs R D [256]	–	–	–	–	–	–	–	–	–	–	–	✓	–
Sheng <i>et al.</i> 2017 Int J Clin Pharmacol Therap [257]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Tseng <i>et al.</i> 2018 Antimicrob Agents and Chemother [257]	–	–	–	–	✓	–	–	–	–	–	–	–	–
Vancomycin_neonate_C2 [194]	–	–	–	–	–	✓	–	–	–	–	–	–	–
Zhao <i>et al.</i> 2013 Arch Dis Child [258]	–	–	–	–	✓	–	–	–	–	–	–	–	–

^a The model references of one, two, three and thirteen models were not disclosed by APK, Rx Studio, PKJust and PrecisePK, respectively. The model from APK was recommended for model-informed precision dosing in adults; one model from PKJust was intended for use in adults and two others for pediatric populations, including neonates; the two models from Rx Studio for pediatric populations; and the thirteen models from PrecisePK for use across adult and pediatric populations, including neonates

^b The model developed for adults was not included in the systematic review as it did not meet the inclusion criteria

Chapter 2. Predictive performance of vancomycin models in older patients

This chapter is based on

Angela E Edwina*, Nada Dia* *et al.*, Optimizing vancomycin dosing in older patients: predictive performance of pharmacometric models. In preparation

[*Shared co-first authorship]

Abstract

Background Model-informed precision dosing (MIPD) is increasingly used to guide vancomycin dosing. However, the suitability of published population pharmacokinetic (popPK) models for older patients has not been evaluated. We aimed to assess the predictive performance of these models in the older patient population.

Methods We encoded nine published popPK models of vancomycin in NONMEM 7.5.0. We used retrospectively collected data from 255 hospitalized older patients aged 75 years and above who were on vancomycin treatment and contributed three therapeutic drug monitoring (TDM) concentrations. We evaluated the *a priori* (solely based on covariates) and *a posteriori* (based on covariates and vancomycin concentrations) predictive performance (accuracy and precision) of single-model and multi-model selection/averaging algorithms.

Results *A priori* predictions varied largely (relative bias; rBias -37%–45%, relative root mean squared error; rRMSE 47%–78%). *A posteriori* prediction using the most recent TDM sample (time between TDM sample and predicted occasion: median 36.0 hours, interquartile range 24.0–48.6 hours) provided the best predictive performance (rBias -18%–24%, rRMSE 25%–45%) with the models of Thomson *et al.* and Ji *et al.* (age median 78 years) performing best. The multi-model algorithm did not improve performance. Adding an earlier sample resulted in only slight improvement or even worsening

performance. Models with less accurate and precise predictive performance produced highly variable and skewed 24-hour area under the time concentration–time curve (AUC_{24h}) predictions.

Conclusion PopPK models developed specifically for older populations and with high interindividual variability, were clinically suitable for guiding vancomycin dosing in older patients, especially when paired with Bayesian forecasting using recent concentration data. The Thomson model showed the best performance, followed by the Ji model, both of which provided more reliable AUC_{24h} predictions.

1. Introduction

Infectious diseases are one of the leading causes of hospitalization among older adults [259–263]. A particularly concerning pathogen in this regard is MRSA, given its higher prevalence and lethality among older adults [264–266]. The glycopeptide antibiotic vancomycin is often the preferred agent for treating MRSA and other severe gram-positive infections [267]. Due to its narrow therapeutic window, guidelines recommend TDM of vancomycin [65]. The AUC_{24h}/MIC of 400–600 mg×h/L has been shown to be the surrogate marker for vancomycin efficacy and safety. However, determining AUCs is laborious and financially demanding [268]. As a result, many clinical settings continue to use single-point concentration measurements as a standard alternative, aiming for a target C_{trough} of 15–20 mg/L for intermittent dosing and $C_{plateau}$ of 20–25 mg/L for continuous infusion [269,270].

MIPD has emerged as a promising strategy to anticipate individualized dosing strategies [88]. Typically, MIPD uses popPK models to streamline TDM-based dose optimization with fewer TDM samples [78]. This means that vancomycin concentrations measured in TDM samples are used for Bayesian forecasting to deduce the maximum *a posteriori* PK parameter estimates of the patient. However, in the absence of TDM samples, patient covariates can be used to deduce *a priori* PK parameter estimates of the patient, making MIPD suitable for first-dosing determination [78].

We previously performed a systematic search and identified 86 popPK models of vancomycin developed in adult patient populations, but their predictive performance and thus suitability for MIPD of vancomycin in older patients remains unclear [271]. Selecting the right popPK model for MIPD of vancomycin to older patients is challenging, yet besides manual selection, automated MSAs exist, as well as MAAs [79,272]. These multi-model algorithms may offer a potential advantage by automating vancomycin MIPD for more accurate and precise predictions [79,272].

With this study, we aimed to evaluate the predictive performance of both single-model and multi-model approaches using published popPK models to guide vancomycin dosing in older patients.

2. Materials and methods

2.1 Clinical dataset

Vancomycin TDM data were collected retrospectively from hospitalized older patients aged 75 years and above admitted to any ward at the University Hospitals Leuven between October 2019 and May 2022 and treated with vancomycin infusions. Patients were excluded if they did not provide a minimum of three vancomycin plasma concentrations measured consecutively during one treatment course. Patients without complete vancomycin dosing records within 48 hours prior to the concentration measurement, and patients with vancomycin concentrations above the upper

limit of quantification of 50 mg/L were excluded from analysis. The study was approved by the Ethics Committee Research of UZ/KU Leuven (S65213).

2.2 Candidate models

Nine popPK models were selected. Seven of these models were developed in distinct patient populations covering a broad range of settings, including critically ill, extremely obese, dialysis, sepsis, trauma, and post-cardiac surgery patients [116,119,120,151,157,163,188]. Therefore, these vancomycin models were considered representative of the hospital-wide adult patient population investigated in our study. Additionally, two popPK models developed using data from older patients, with a mean or median age of 75 years or more, were selected based on a systematic literature search to ensure adequate representation of older patients [149,233,271].

2.3 Model evaluations

The global fit of the clinical dataset to the candidate models was visually inspected with goodness-of-fit plots, including measured versus individual predicted concentrations and conditional weighted residual versus time plots. In addition, simulation-based evaluations were performed, including prediction-corrected VPC (pcVPCs) and normalized prediction distribution errors (NPDEs). A normal distribution of NPDEs $\mathcal{N}(0, 1)$ was assessed using three tests: the Wilcoxon signed-rank test to compare the NPDE median to zero, the Fisher variance test to compare its variance to one, and the Shapiro–Wilk test to evaluate whether the NPDEs followed a normal distribution. An

adjusted p-value of all three tests (a global test) was calculated to identify the best-fitting model [273].

2.4 Multi-model approaches

Two multi-model approaches were assessed using the nine candidate models: a MSA and a MAA [79]. Both approaches aimed to predict the vancomycin concentration at relative time +1 (T_{+1}) using earlier vancomycin concentration at relative times -1 (T_{-1}) and 0 (T_0). First, the individual PK parameters were estimated with each of the nine popPK models based on observed covariate data alone (*a priori* prediction) and covariate data in combination with vancomycin concentrations at T_{-1} and/or T_0 (*a posteriori* prediction). Next, the individually predicted concentration–time data were automatically weighted based on the maximum likelihood estimate (MLE) of each candidate model i in relation to the sum of MLEs of all n candidate models (**Equation 1**).

$$W_{MLE_i} = \frac{MLE_i}{\sum_{i=1}^n MLE_i} = \frac{e^{(-0.5 \times OFV_i)}}{\sum_{i=1}^n e^{(-0.5 \times OFV_i)}} \quad (1)$$

In the MSA, the model with the highest weight was selected, while the others were excluded. In the MAA, a combined prediction was generated from all models using **Equation 2**.

$$\text{Prediction} = \sum_{i=1}^n (\text{Prediction}_{\text{model}_i} \times W_{\text{model}_i}) \quad (2)$$

The theory behind these multi-model approaches was previously published by Uster *et al.* [79].

2.5 Predictive performance evaluations

For both *a priori* and *a posteriori* predictions, the predictive performance was evaluated from the differences between the predicted and the measured concentrations at T_{+1} (C_{T+1}). Three *a posteriori* prediction settings were compared: Prediction with measured concentrations at T_0 (C_{T0}) only, measured concentrations at T_{-1} (C_{T-1}) only, and both C_{T0} and C_{T-1} . The model-predicted versus measured C_{T+1} in the different single-/multi-model approaches, *a priori/a posteriori* prediction modalities, and evaluation settings were compared by calculating the relative bias (rBias; **Equation 3**) and the relative root mean square error (rRMSE; **Equation 4**) to determine accuracy and precision, respectively. A rBias between $\pm 20\%$ with a 95% confidence interval (CI) including zero was deemed clinically acceptable [79]. No rRMSE threshold for clinical acceptability was prespecified. Lower rRMSE values indicated more precise predictions.

$$\text{rBias} = \frac{1}{n} \times \sum_{i=1}^n \left(\frac{\text{predicted}_i - \text{observed}_i}{\text{observed}_i} \right) \times 100\% \quad (3)$$

$$\text{rRMSE} = \sqrt{\frac{1}{n} \times \sum_{i=1}^n \frac{(\text{predicted}_i - \text{observed}_i)^2}{(\text{observed}_i)^2}} \times 100\% \quad (4)$$

with n the total number of patients, and i each individual patient.

2.6 Robustness analysis, sensitivity analysis, classification accuracy, and Bland–Altman analysis

The predictive performance and the computation time of the multi-model approaches were assessed with a subset of popPK models compared to the nine popPK models in a robustness analysis. The predictive performance of the single and the multi-model approaches in the sensitivity analysis was assessed by 1) changing the covariate data using a single imputation – using the model’s median covariate value when centered, and otherwise the clinical dataset’s median – to assess the impact of each covariate, and 2) modifying the models by using flattened priors for all the PK parameters [198]. The classification accuracy was also evaluated; it was defined as no change in exposure category, *i.e.*, subtherapeutic (<12.5 mg/L, supratherapeutic (>17.5 mg/L), or therapeutic (12.5–17.5 mg/L) between the predicted vancomycin concentration and the observed concentration. The agreement between the predicted and measured C_{T+1} values over the entire range of measured C_{T+1} was evaluated in a Bland–Altman analysis. In addition, the analysis was conducted over the time intervals from C_{T-1} or C_{T0} to C_{T+1} to determine if the difference between predicted and measured C_{T+1} values changed over time after TDM [274].

2.7 Software

All models were coded in NONMEM (version 7.5.0; Icon plc) and are available in Appendix B. Predictions were executed using NONMEM with a GNU Fortran 95 compiler. Dataset formatting, data analysis, and visualization were

conducted in R (version 4.3.1, R Core Team, Vienna, Austria) in conjunction with RStudio (2023.06.1+524).

3. Results

3.1 Clinical dataset

Data were collected from 255 older patients, each contributing three consecutive TDM samples, resulting in a total of 765 vancomycin concentrations. The median age was 79 years, with an IQR of 77 to 84 years. Detailed patient characteristics are presented in **Table 2**. The median (IQR) vancomycin concentration C_{T-1} , C_{T0} , and C_{T+1} were 17.0 mg/L (13.0–21.0), 19.9 mg/L (15.7–23.9), and 19.4 mg/L (15.9–23.6), respectively.

Table 2 Patient characteristics

Characteristic	Values
Number of patients	255
<i>Demographics</i>	
Age, years, median [IQR]	79 [77–84]
Sex, male, n (%)	147 (57.6)
Body weight, kg, median [IQR]	68.0 [60.8–80.9]
Height, cm, median [IQR]	167 [161–174]
Body mass index, kg/m ² , median [IQR]	24.7 [22.4–28.6]
<i>Serology</i>	
Serum creatinine, mg/dL, median [IQR]	1.00 [0.78–1.44]
eCrCL _{CG} , mL/min, median [IQR]	54.0 [36.5–73.8]
Blood urea nitrogen, mg/dL, median [IQR]	49.0 [31.5–67.5]
Serum albumin, g/L, median [IQR]	31.0 [27.2–34.8]
Total protein, g/L, median [IQR]	62.0 [54.5–68.0]
<i>Comorbidities, comedication, dialysis, admission to the intensive care unit</i>	
Patients with diabetes ^a , n (%)	65 (25.5)
Comedication with furosemide, n (%)	166 (65.1)
Intensive care unit patients, n (%)	72 (28.2)
Patients with CVVH, n (%)	7 (2.7)

CVVH, continuous veno-venous hemofiltration; eCrCL_{CG}, estimated creatinine clearance using

Cockcroft–Gault equation; IQR, interquartile range

^a Diabetes status was missing for 12 patients

3.2 Population pharmacokinetic models of vancomycin

The nine popPK models, including two elderly models by Ji *et al.* (2018) and Zhou *et al.* (2019), varied in structure as five are one-compartment models and four are two-compartment models (**Supplementary Table 11**). Differences were also observed in the covariates included, as detailed in **Supplementary Table 12**. Eight models included estimated CrCL using Cockcroft–Gault equation (eCrCL_{CG}) to estimate the vancomycin CL, and one

model incorporated SCr, body weight, and sex as covariates on vancomycin CL. Seven models included body weight for estimating the volume of distribution. Age was included as a covariate in three models, including Ji model, affecting CL in two models and the volume of distribution in one.

3.3 Model evaluations

The individual predicted concentrations from each model generally aligned closely with the observed concentrations, although not all models capture the full variability of the data well (**Supplementary Figures 1–3**). From visual inspections of these plots, we concluded that the models by Goti *et al.* and Ji *et al.* fitted the clinical dataset best. The Goti *et al.*, Ji *et al.*, and Thomson *et al.* models were identified as the best-fitting models in terms of the NPDEs (**Supplementary Figures 4–5, Supplementary Table 13**).

3.4 Influence of the models in the multi-model approaches

When calculating model weights for multi-model approaches using single sample C_{T0} or C_{T-1} , the Zhou *et al.* model was the most frequently selected in both intermittent and continuous infusion administration. When both C_{T-1} and C_{T0} were included following intermittent infusion administration, the distribution among models was more balanced, with the Zhou model still holding the highest weight at 26% ($n=66/255$) and the Thomson model close behind at 25% ($n=63/255$). In continuous infusion administration, the

Thomson model was selected slightly more often than the Zhou model (28% vs. 26%, respectively) (**Figure 6**).

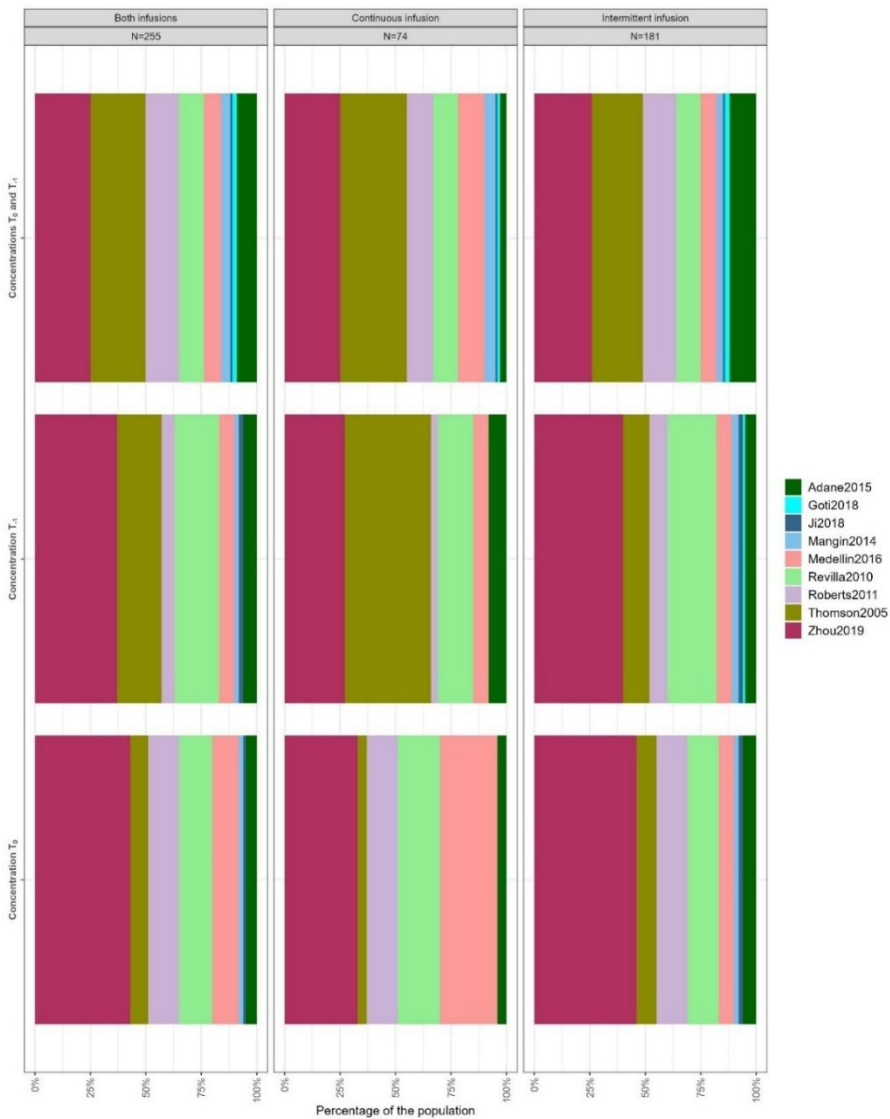
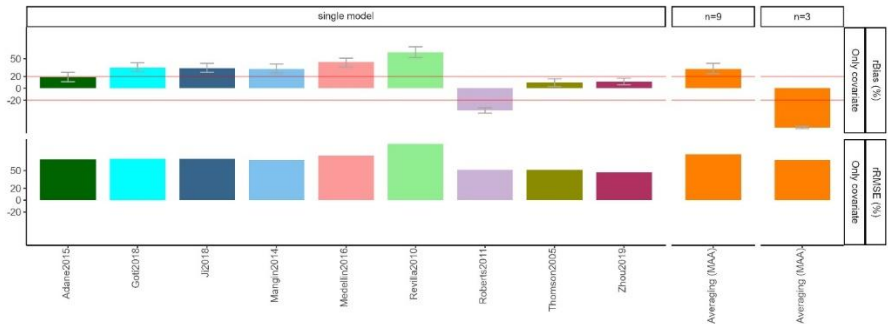


Figure 6 Weight distribution of the population pharmacokinetic models in three *a posteriori* prediction settings. Zhou model was selected the most whereas Goti model was selected the least. T, relative time; N, total number of patients in the population

3.5 Predictive performance evaluations

A priori prediction of C_{T+1} was clinically unacceptable with the single and the multi-model approaches (rBias -37% to +45%, rRMSE 47% to 78%; **Figure 7a**). The predictive performance improved when at least one concentration was accounted for in all the popPK models (**Figure 7b**). The greatest improvement in predictive performance was observed when the most recent concentration (C_{T0}) was included (rBias -18% to +24%, rRMSE 25% to 45%). The least improvement in predictive performance was observed when an earlier concentration (C_{T-1}) was included (rBias -25% to +33%, rRMSE 39% to 60%). When the two concentrations were considered (C_{T-1} and C_{T0}), the predictive performance improved slightly or worsened depending on the model/approach (rBias -16% to +21%, rRMSE 27% to 42%). Overall, the Thomson *et al.* model had the best predictive performance, followed by the Ji *et al.* model, while for continuous infusion Adane and Ji models showed the best predictive performance (**Supplementary Figures 6–7**). The multi-model approaches did not perform better than the single-model approach (**Figure 7b**).

a *A priori* predictions



b *A posteriori* predictions

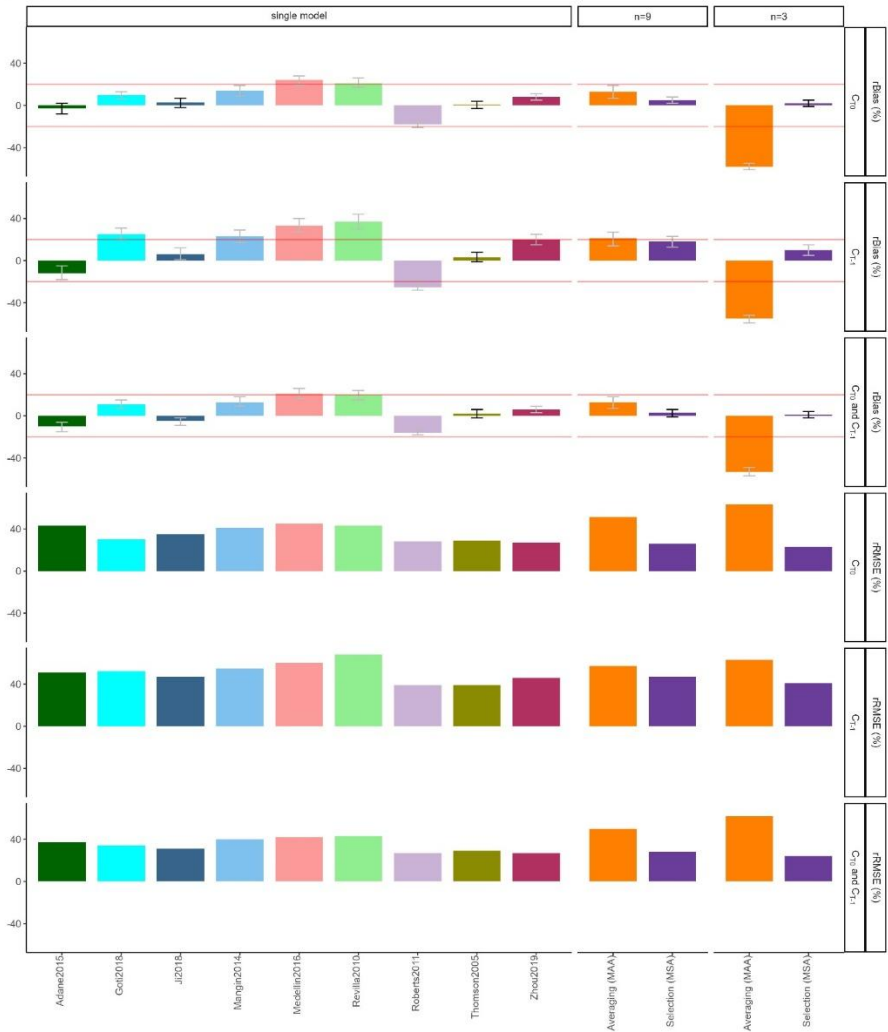


Figure 7 The predictive performance of the single and multi-model approaches using all the nine models versus the three models to predict vancomycin concentration (C) at a time +1 (T+1). (a) *A priori* predictions using the clinical dataset with covariate data without previous concentrations, and (b) *a posteriori* prediction settings using the clinical dataset with covariate data and previous concentrations (CT₋₁, CT₀, CT₋₁ and CT₀). Whiskers cover the 95% confidence interval of the relative bias calculated via the standard error (black whisker indicated 95% CI including 0). Model weighing in *a priori* predictions correspond to 1/number of models, disabling model selection. The horizontal lines in relative bias panels indicate $\pm 20\%$ of the relative bias. rBias, relative bias; rRMSE, relative root mean square error; C, concentration; T, relative time

3.6 Robustness analysis, sensitivity analysis, classification

accuracy, and Bland–Altman analysis

Three popPK models, Roberts, Thomson, and Zhou models, were selected based on their predictive performance in the *a posteriori* prediction settings. The predictive performance of the multi-model approaches with the subset models showed more bias compared to approaches with all nine models, especially for MAA (**Figure 7**). The average computation time with the subset versus all the models decreased by 68% from 0.52 to 0.16 seconds per patient. Removing the effect of covariates makes the *a posteriori* prediction of all single models not clinically acceptable, except for MAA and MSA. Flattening priors resulted in rBias closer to 0 and smaller rRMSE. Despite an overall misclassification of 28%, Thomson *et al.* model as well as the MSA based on five models had the lowest misclassification among all the single and multi-model approaches (**Supplementary Table 14**). The prediction bias was not different across time after TDM (**Figure 8**) and was the least across

a measured concentration range of 12.5 mg/L to 17.5 mg/L when C_{T0} was included in the *a posteriori* predictions (**Supplementary Figure 8**).

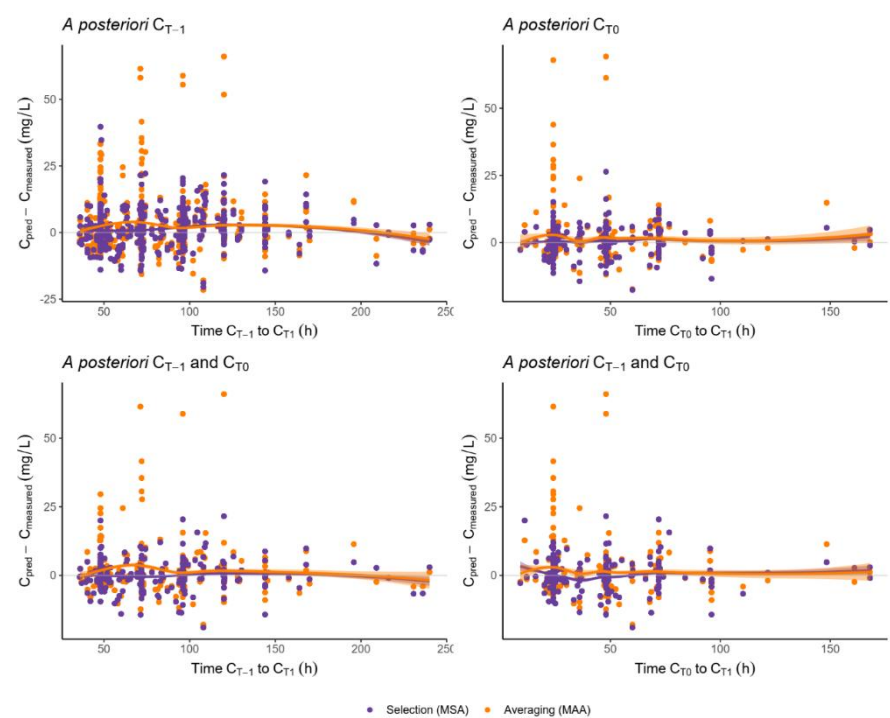


Figure 8 Bland–Altman plots where an agreement between the measured and the predicted vancomycin concentrations was observed in the *a posteriori* predictions using the model selection algorithm (MSA; purple) and the model averaging algorithm (MAA; orange) over time after therapeutic drug monitoring. C, concentration; C_{pred} , predicted concentration; T, relative time

3.7 Model-informed precision dosing software tools

The AUC_{24h} of a course of therapy was predicted *a priori* (covariates only) and estimated from the observed vancomycin concentrations (Bayesian) with each model. The distribution of the predicted vancomycin drug exposures was heterogeneous (**Figure 9**). The *a priori* predicted AUC_{24h} ranged from 279–

757 mg×h/L showing a wider range compared to the *a posteriori* predictions, which ranged from 348 to 656 mg×h/L. Models such as that of Thomson *et al.* which displayed a comparably low bias and precision in the previous analyses demonstrated similar ranges for the *a priori* and Bayesian estimated AUC_{24h}.

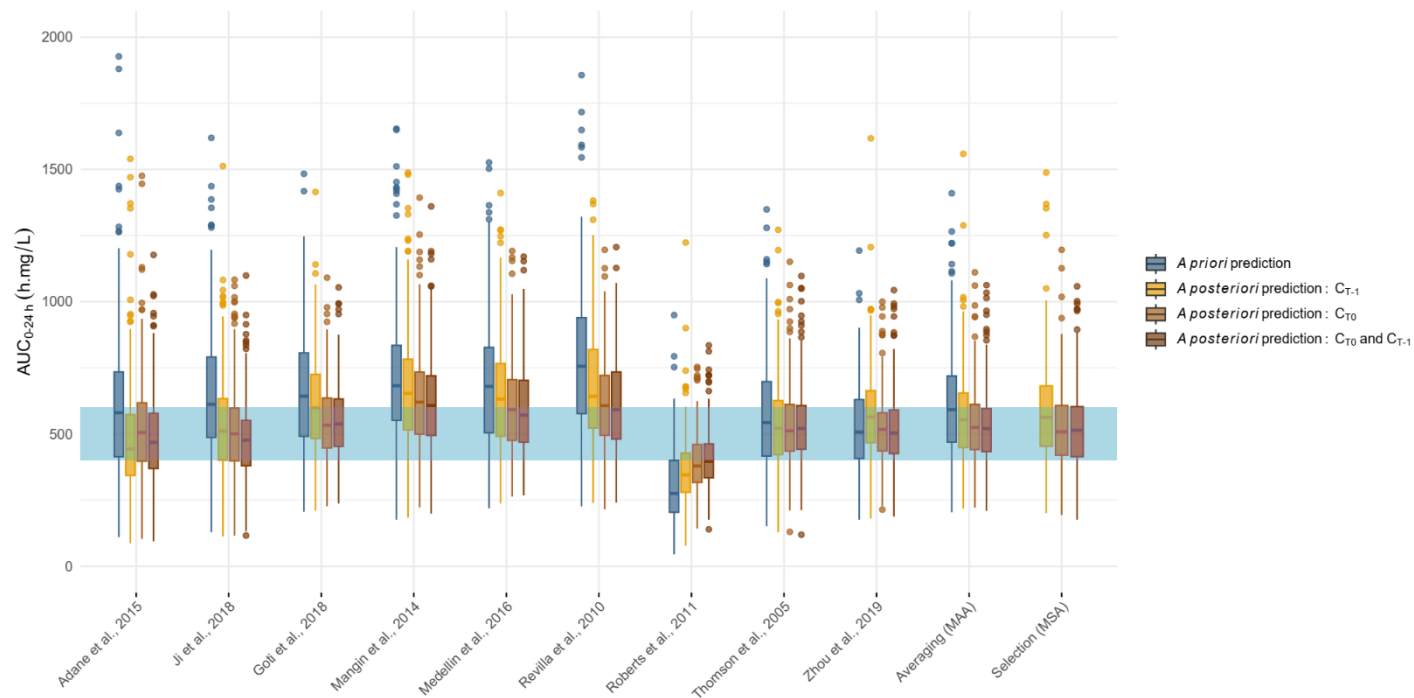


Figure 9 *A priori* predicted and Bayesian estimated vancomycin AUC_{24h} in 255 older inpatients. Box plots represent the 25th, 50th and 75th percentiles; the whiskers cover the 1.5-fold interquartile range and outliers are marked as dots. The blue area indicates the targeted AUC_{24h} therapeutic window of 400–600 mg×h/L

4. Discussion

When applying MIPD in special patient populations, such as older hospitalized patients, careful selection or development of the algorithm is crucial to ensure accurate guidance. In our external validation of nine vancomycin popPK models, we found that the superior predictive performance of the Thomson model is likely due to its weaker priors, characterized by higher IIV compared to other models. Our analysis showed that when the priors were flattened, predictive performance improved, suggesting that models with less restrictive assumptions about variability can better capture the individual differences in this population. Additionally, models developed using data from older patients, such as the Ji and Zhou models (median age 78 years), also resulted in superior predictions for our older patient cohort. This demonstrates the importance of using models with suitable variability assumptions and aligning them with the target population, as models that accounted for high IIV and were developed for similar age groups tend to predict outcomes more accurately in older adults.

These single models were more accurate and precise compared to multi-model algorithms, which contrasts with previous findings by Uster *et al.*, where MAA and MSA were reported to result in unbiased and precise predictions [79]. The observed underperformance of the multi-model algorithms in our study is likely due to potentially PK alterations in older patients. In this population, rapid or unpredictable changes in CL, volume of distribution, or

both can occur, meaning that the observed concentration does not consistently map onto a specific part of the concentration–time profile. This variability may have complicated the weight calculation process in the multi-model approach. In contrast, single-model predictions do not rely on comparative weighting across models and are therefore less sensitive to this variability.

The top-performing models identified in previous studies evaluating the predictive performance in adults are different from that we identified. In the analysis by Broeker *et al.*, the Goti model demonstrated the best predictive performance among the models evaluated [157,206]. Broeker *et al.* included a broad hospital-wide population without restrictions, and the Goti model was developed using the largest dataset of all the investigated models (2,765 vancomycin concentrations from 1,812 hospitalized patients). In contrast, our study focused solely on older patients. Although the Goti model showed good overall performance in our study as well, it was surpassed by models built on older population. On the other hand, the Roberts model did not perform as well in our study, particularly in terms of general fit. This contrasts with the findings of Cunio *et al.*, who analyzed critically ill patients and found that the Roberts model, which was developed for this population, provided clinically acceptable bias in both overall model evaluation and Bayesian forecasting analyses [275]. The difference in results can be explained by the fact that our dataset included a smaller proportion of critically ill patients.

Removing the effect of covariates from a *a posteriori* prediction resulted in lower predictive performance, indicating that including covariates is likely crucial, particularly for older hospitalized patients. Importantly, this does not contradict with how flattening the priors improved performance. Flattening the priors relaxes the model's assumptions about variability. In contrast, removing covariates eliminates patient-specific information altogether. These modifications therefore affect different components of the model and are not expected to produce similar effects. Our finding on the impact of covariates aligns with the argument by Broeker *et al.*, who emphasized the importance of body weight and CrCL in vancomycin dosing [206]. Our analysis suggests that, besides weight and CrCL, age is also an important covariate, as models developed from older patient populations performed better for the older patients in our dataset, especially the Ji model, which includes age as a covariate. Age may serve as a proxy for various (patho-)physiological changes that accompany aging, such as changes in body composition and comorbidities, which influence both drug distribution and CL.

The predicted AUC_{24h} varied significantly between models, which has important clinical implications since dosing decisions based on these predictions would differ. For example, the supratherapeutic AUCs predicted by the Revilla, Medellin, and Mangin models would suggest a dose reduction, while the subtherapeutic AUCs from the Roberts model would indicate a need for a dose increase. The latter is particularly concerning due to the notable negative bias in the Roberts model, which could lead to overdosing. Although

even the best-performing models misclassified 28% of patients when classifying based on observed C_{trough} , this level of misclassification is expected given the sparse sampling and the lack of rich data to compare predicted and true AUCs directly. Importantly, an integrated evaluation of predictive performance metrics and models evaluations using VPC, NPDE and goodness-of-fit plots provides evidence that models such as those from Thomson and Ji are more reliable for unbiased AUC predictions.

This study has several limitations that should be considered. First, the analysis was restricted to a limited number of popPK models, which may have excluded potentially relevant models that could offer further insights or improved predictions. Second, as a single-center retrospective study, the generalizability of the findings may be constrained. Third, while all models use accessible covariates, there is one model that included the Sequential Organ Failure Assessment (SOFA) score, restricting its use to only ICU patients. Furthermore, none of the models accounted for geriatric patients with a geriatric syndrome profile, instead focusing on older adults without explicitly addressing this syndrome. While our dataset represents the largest cohort of older patients analyzed to date, addressing the notable underrepresentation of this population in pharmacometric studies, we acknowledge that larger datasets alone do not necessarily enhance generalizability if they primarily consist of similar patient profiles. However, our cohort includes diverse patient characteristics as it is a hospital-wide study, which we believe enhances the applicability of our findings to a broader geriatric population and strengthens

the relevance of our conclusions. In addition, much can be learned from data in other populations. Age alone is not the primary determinant of PK alterations; instead, it is the (patho-)physiological changes associated with aging. For instance, studies in non-geriatric populations with a wide range of renal function can provide valuable insights into PK changes, which are particularly relevant for geriatric patients who often experience age-related declines in renal function, a key factor in vancomycin CL. Moreover, in the case of either observed or predicted concentration is uncertain, future evaluations of vancomycin models could benefit from incorporating modified error structures, such as the one demonstrated by Schatz *et al.* [274].

In conclusion, popPK models developed from older populations and with high IIV were found to be clinically acceptable for guiding vancomycin dosing in older patients, particularly when using Bayesian forecasting with the most recent concentration. The Thomson model demonstrated the best performance, followed by the Ji model.

Appendix B

Supplementary Table 11 Patient characteristics in the included studies

Authors. Year of publicati on Journal	Patient population	Age (years), mean $\pm$ SD, median (range) [IQR]	Body weight (kg), mean $\pm$ SD, median (range) [IQR]	BMI (kg/m ²), mean $\pm$ SD, median (range) [IQR]	Serum creatinine, mean $\pm$ SD, median (range) [IQR]	CrCL, mean $\pm$ SD, median (range) [IQR]
Adane <i>et al.</i> , 2015	Extremely obese (BMI ≥ 40 kg/m ²) adults with suspected or confirmed <i>Staphylococcus aureus</i> infection requiring treatment for ≥ 3 days in USA	43 [38.5–53.0]	147.9 [142.8–178.3]	49.5 [44.3–54.8]	–	124.8 [106.0–133.9] mL/min/1.73 m ²
Goti <i>et al.</i> , 2018	Adult patients hospitalized at a unit with high prevalence of end-stage renal diseases	57 (17–101)	79 [33–255]	–	1.3 [0.2–16.6] mg/dL	62 (4–150) mL/min
Ji <i>et al.</i> , 2018	Hospitalized adult patients on continuous	78 (42–95)	65 (38–90)	22.31 (12.85–36.89)	75 (24–893) μ mol/L	58.02 (5.45–224.0) mL/min

	infusion of vancomycin, and not in need for renal replacement therapy					
Mangin <i>et al.</i> , 2014	Intensive care unit adult patients with post-sternotomy mediastinitis during their primary closed Redon catheter drainage	63 (35–81)	82 (62–104)	28 (22–36)	138 (32–606) $\mu\text{mol/L}$	–
Medellin-Garibay <i>et al.</i> , 2016	Traumatic adult patients with proven or suspected infection	74.3 ± 14 , 77 (37–100)	72.0 ± 15	27.5 ± 5	–	90.5 ± 51.67 mL/min
Revilla <i>et al.</i> , 2010	Intensive care unit adult patients	61.1 ± 16.3 (18–85)	73.0 ± 13.3	26.2 ± 4.1	–	86.1 ± 55.1 mL/min
Roberts <i>et al.</i> , 2011	Septic critically ill adults in the intensive care unit	58.1 ± 14.8	74.8 ± 15.8	25.9 ± 5.4	–	90.7 ± 60.4 mL/min/ 1.73 m^2
Thomson <i>et al.</i> , 2009	Hospitalized adult patients during routine clinical care at the hospital	66 (16–97)	72 (40–159)	–	98 (30–573) $\mu\text{mol/L}$	64 (12–216) mL/min

Zhou <i>et al.</i> , 2019	Hospitalized geriatric patients (≥65 years) with a pulmonary infection (a hospital- or a community-acquired pneumonia)	78.3 ± 6.96	60.7 ± 10.2	–	90.6 ± 31 μmol/L	56.3 ± 22.1 mL/min
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BMI, body mass index; CrCL, creatinine clearance; SD, standard deviation; IQR, interquartile range

Supplementary Table 12 Properties and standardized pharmacokinetic parameters (standard patient, 50 years old, male, body weight: 75 kg, body height: 1.7 m, serum creatinine of 85 $\mu\text{mol/L}$) of the evaluated pharmacometric models

Authors. Year of publication Journal	Vancomycin administration	No. patients	Nr. samples	Covariates	CL	Vc	Vp	Q	IIV	RU V
Adane <i>et al.</i> , 2015	Intermittent infusion	29	93	TVCL = $\text{CL} * (\text{CrCL}/125)$ TVV = $\text{Vc} * \text{Weight}$	4.74	38.3	–	–	CL: 26.7 % Vp: 23.9 %	P: 18.9 % A: –
Goti <i>et al.</i> , 2018	n.r.	1812	2765	TVCL = $\text{CL} * (\text{CrCL}/120) **0.8)$ *0.7 (if dialysis) TVVc = $\text{Vc} * (\text{Weight}/70) *0.5$ (if dialysis) TVVp = $\text{Vp} * (\text{Weight}/70)$	3.82	62.6	38.4	6.5	CL: 38.4 % Vc: 81.9 %, Vp: 57.1%	P: 22.7 % A: 3.4 mg/L
Ji <i>et al.</i> , 2018	Continuous infusion	160	251	TVCL = $\text{CL} * (1 + 0.0084 * (\text{CrCL} - 80)) * (75/\text{Age}) **0.8143$	4.52	52.1	–	–	CL: 32.4 %	P: 26.8 %

									Vc: 28.9 %	A: 2.7 mg /L
Mangin <i>et al.</i> , 2014	Intermittent infusion	30	359	$TVCL = CL * 0.66$ (if female) $*((Weight/70)$ $**0.75)$ $*((SAPSII/50) **(-0.5)$ $*((SCr/88.42)/100) **(-$ $0.9)$ $TVVc = Vc * (Weight/70)$ $TVVp = Vc * (Weight/70)$ $TVQ = Q * 0.3$ (if diabetes) $*((Weight/$ $70) **0.75)$	2. 6 0	23 .5	72 .9	6. 0 1	CL: 29% Vc: 53% Q: 101 %	P: – A: 7.3 mg /L
Medellín-Garibay <i>et al.</i> , 2016	Intermittent infusion	118	392	$TVCL = CL * CrCL$ $TVCL = CL_{furosemide} * CrCL$ $TVVc = Vc * Weight$	2. 8 7	55 .5	44 2. 5	0. 8 1	CL: 36.7 % Vc: 40.0 %	P: 19. 2 % A: 3.5

				$TVVp = Vp * Weight$						mg /L
Revilla <i>et al.</i> , 2010	Intermittent and continuous infusion	191	569	$TVCL = CL * CrCL + (Age^{**}(-0.24)) * Weight$ $TVVc = Vc * 2.49$ (if $SCr > 1$) * Weight	4.84	61.5	–	–	CL: 30.1 % Vc: 22.8 %	P: – A: 4.2 mg /L
Roberts <i>et al.</i> , 2011	Continuous infusion	206	579	$TVCL = CL * (UCrCL / 100)$ $TVVc = Vc * Weight$	4.15	114.8	–	–	CL: 38.9 % Vc: 37.4 %	P: 19.9 % A: 2.4 mg /L
Thomson <i>et al.</i> , 2009	Intermittent infusion	398	1557	$TVCL = CL * CrCL * 0.0154$ $TVVc = Vc * Weight$ $TVVp = Vp * Weight$	4.45	50.6	54.9	2.28	CL: 27%, Q: 49% Vc: 15%, Vp:	P: 15 % A: 1.6 mg /L

									120 %	
Zhou <i>et al.</i> , 2019	Intermittent infusion	70	125	TVCL = CL *(CrCL/56.28) **0.542	3. 3 0	15 4	–	–	CL: 17.4 % Vc: 33.9 %	P: 6.6 %

A, additive; Vc, central volume of distribution; CrCL, creatinine clearance; CL, clearance; Q, intercompartmental clearance; IIV, interindividual variability; n.r., not reported; Vp, peripheral volume of distribution; P, proportional; TBW, total body weight; RUV, residual unexplained variability; SCr, serum creatinine; UCrCL, urinary creatinine clearance

Supplementary Table 13 Statistical tests to assess normality of the normalized prediction distribution errors (NPDE) of the population pharmacokinetic models.

Model	Wilcoxon test ¹	Fisher test ²	Shapiro–Wilk test ³	Global test ⁴	Ranking ⁵
Adane <i>et al.</i> , 2015	1.5e-15	0e+00	2.8e-01 (.)	0	2
Goti <i>et al.</i> , 2018	9.3e-01 (.)	0e+00	8.2e-11	0	3
Ji <i>et al.</i> , 2018	0e+00	2.9e-12	8.6e-12	0	4
Mangin <i>et al.</i> , 2014	9e-09	0e+00	3e-20	0	5
Medellín-Garibay <i>et al.</i> , 2016	0e+00	5.2e-06	1.3e-12	0	6
Revilla <i>et al.</i> , 2010	0e+00	0e+00	3.7e-06	0	7
Roberts <i>et al.</i> , 2011	0e+00	3.1e-05	9.9e-07	0	8
Thomson <i>et al.</i> , 2009	1e-02 (**)	9.3e-11	2.1e-08	2.8 e-10	1
Zhou <i>et al.</i> , 2019	9e-04	0e+00	2.8e-10	0	9

¹ Wilcoxon test compared the mean of the NPDE to 0

² Fisher test compared the variance of the NPDE to 1

³ Shapiro–Wilk test compared the NPDE to the normal distribution

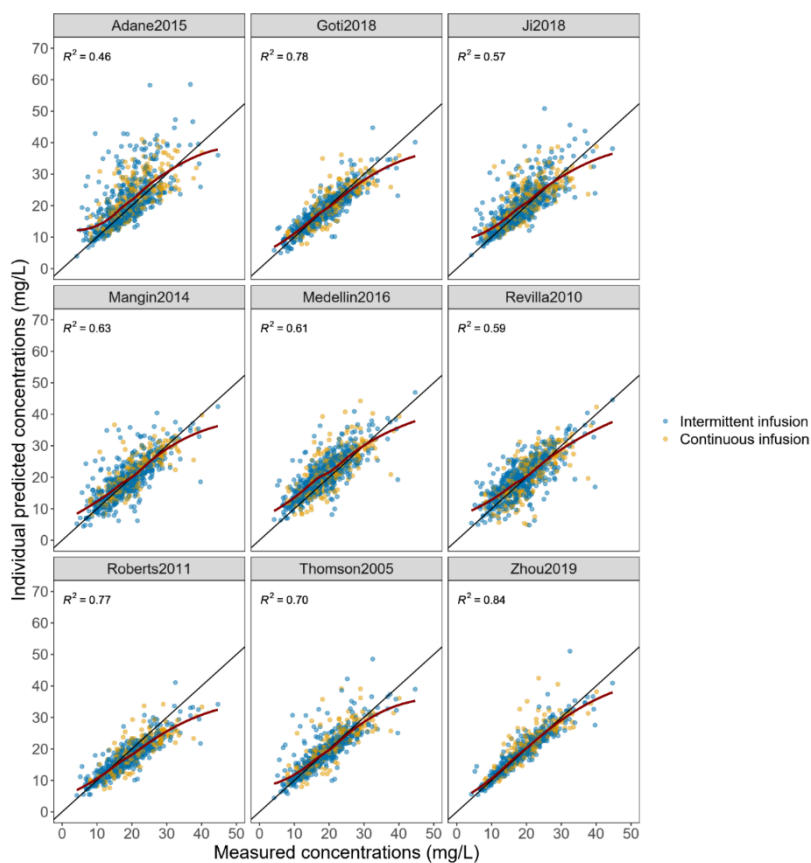
⁴ P-value of the global test was equal to the minimum of the adjusted p-values. The adjusted p-values are the raw p-value of the three individual tests multiplied by 3

⁵ Ranking based on p-value of the global test (highest value to lowest value)

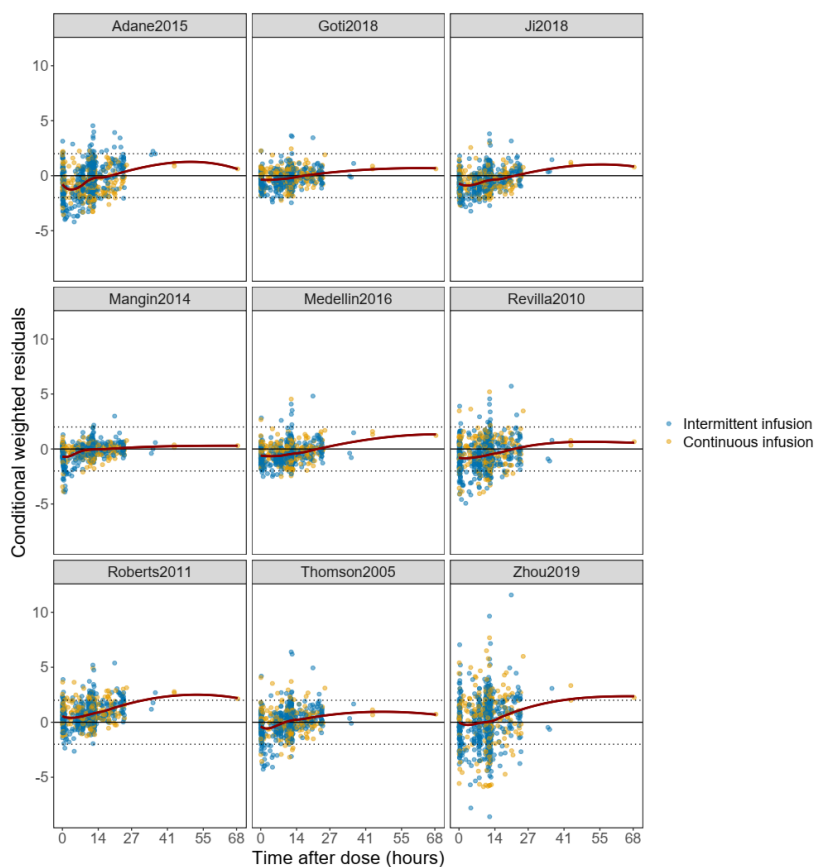
'.' >0.1; '*' >0.05; '**' >0.01; '***' >0.001, Significant levels indicate that the NPDEs are different from $\mathcal{N}(0,1)$ distribution for the specified tests

Supplementary Table 14 Classification accuracy is defined as no change in exposure category (*i.e.*, subtherapeutic, supratherapeutic or therapeutic) according to the predicted vancomycin concentration compared to the observed concentration

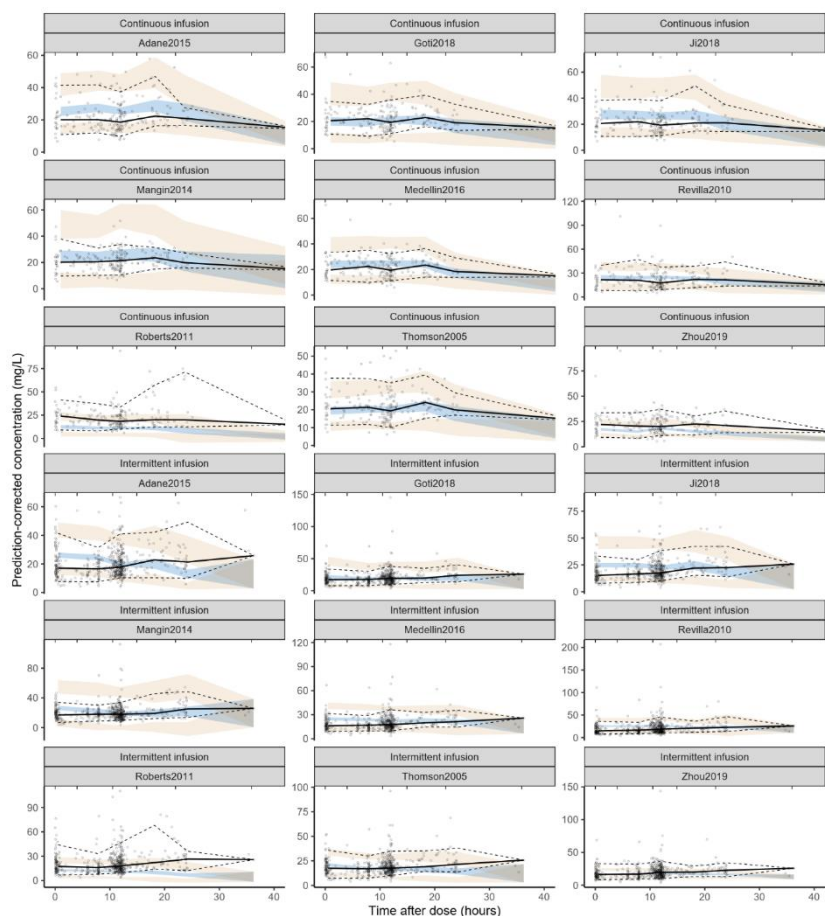
Model	Classification accuracy, n (%)
Adane <i>et al.</i> , 2015	150 (59%)
Goti <i>et al.</i> , 2018	177 (69%)
Ji <i>et al.</i> , 2018	168 (66%)
Mangin <i>et al.</i> , 2014	166 (65%)
Medellin-Garibay <i>et al.</i> , 2016	167 (65%)
Revilla <i>et al.</i> , 2010	168 (66%)
Roberts <i>et al.</i> , 2011	134 (53%)
Thomson <i>et al.</i> , 2009	183 (72%)
Zhou <i>et al.</i> , 2019	182 (71%)
multi-model averaging algorithm based on nine models	177 (69%)
multi-model selection algorithm based on nine models	171 (67%)
multi-model averaging algorithm based on five models	105 (59%)
multi-model selection algorithm based on five models	183 (72%)



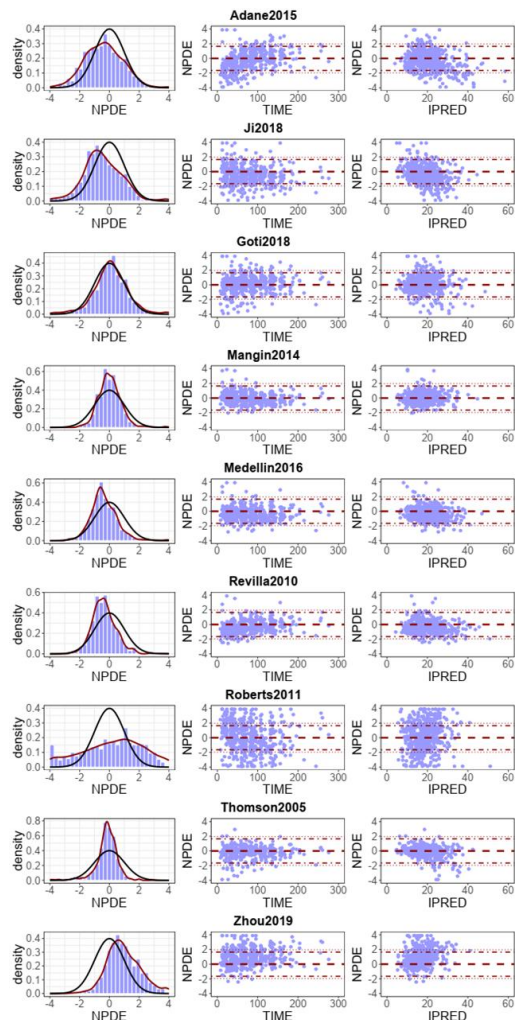
Supplementary Figure 1 Goodness-of-fit plots of the measured concentration versus individual predicted concentration. The solid black and red lines represent the line of identity, and a locally weighted smoother, respectively



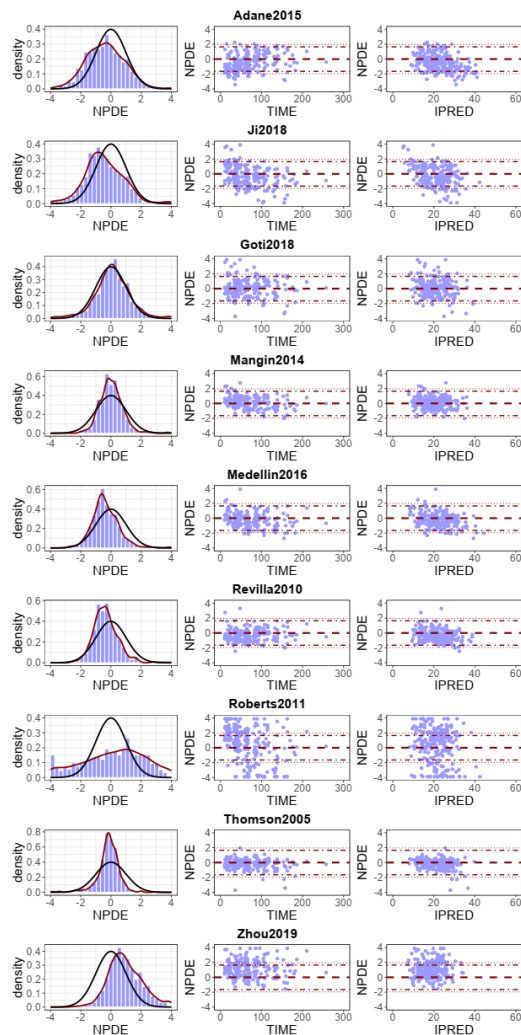
Supplementary Figure 2 Goodness-of-fit plots of the time after dose (hours) versus conditional weighted residuals. The solid black and red lines represent the line of identity, and a locally weighted smoother, respectively. The dashed lines represent a conditional weighted residual of ± 3



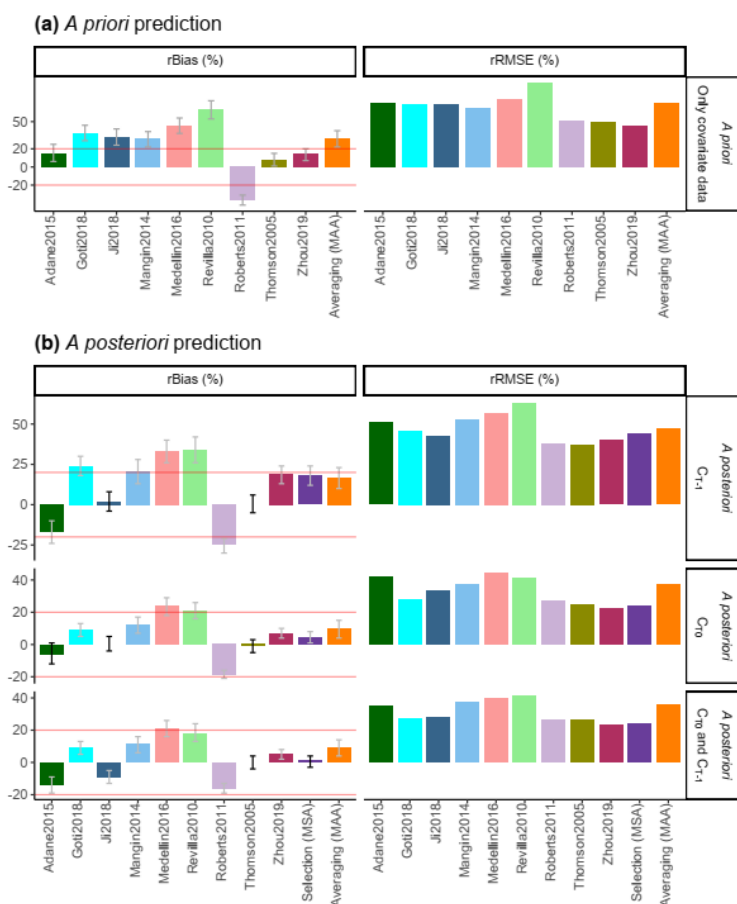
Supplementary Figure 3 Prediction-corrected visual predictive checks of the prediction-corrected vancomycin concentration versus time after dose (hours) in 255 patients foreach of the nine population pharmacokinetic models. The solid black lines represent the median of the data. The dashed lines represent the 5th and 95th percentile of the data. The shaded areas represent the 90% confidence intervals of the predictions. The vertical dashes at the top of each plot represent the binning intervals



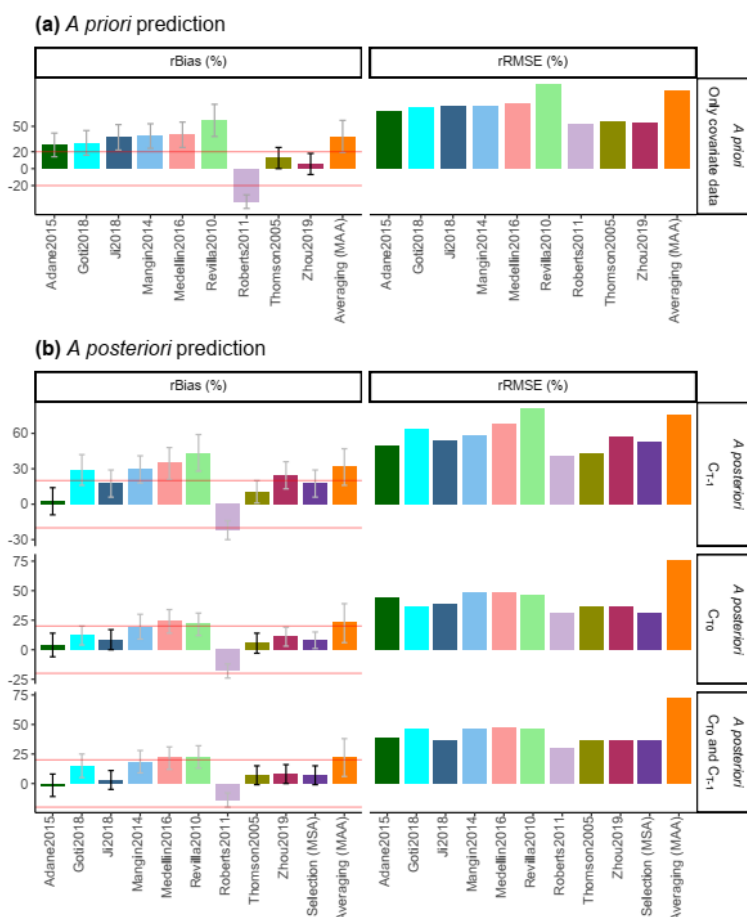
Supplementary Figure 4 Plots illustrating normalized prediction distribution errors (NPDE) of the nine population pharmacokinetic models following **intermittent infusion** administration. *The first plot:* Histogram and density (solid red line) of the distribution of NPDE overlayed with theoretical $\mathcal{N}(0,1)$ distribution (solid black line). *The second plot:* Scatterplot of NPDE versus Time (hours). *The third plot:* Scatterplot of NPDE versus individual predicted concentrations (IPRED). The dashed horizontal lines are the null line (NPDE = 0). The dot dashed horizontal lines indicate the 90% range (NPDE = -1.645, 1.645). The dotted lines indicate the 95% range (NPDE = -1.96, 1.96). ESAMPLE was set at 10,000



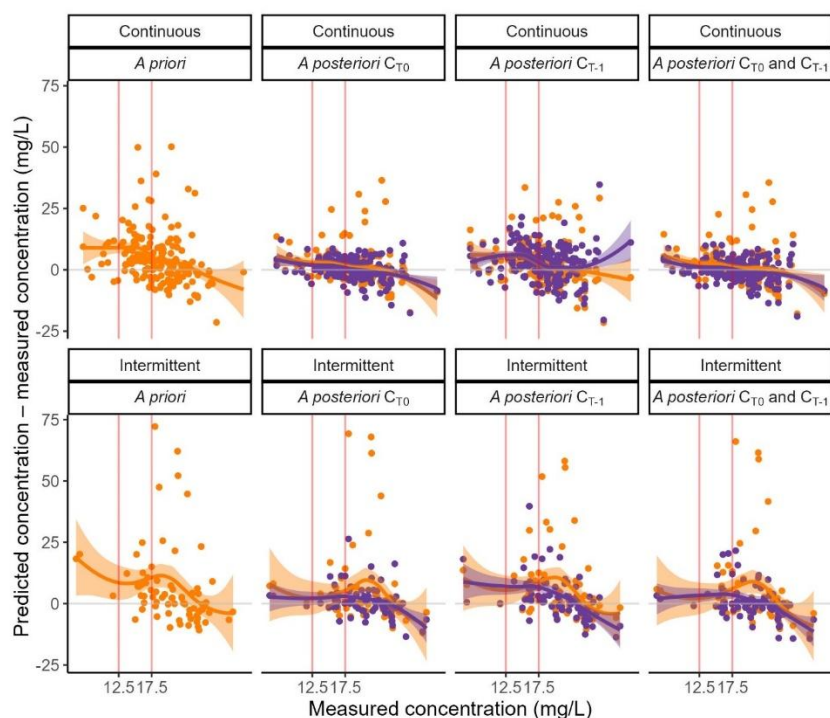
Supplementary Figure 5 Plots illustrating normalized prediction distribution errors (NPDE) of the nine population pharmacokinetic models following **continuous infusion** administration. *The first plot:* Histogram and density (solid red line) of the distribution of NPDE overlaid with theoretical $\mathcal{N}(0,1)$ distribution (solid black line). *The second plot:* Scatterplot of NPDE versus Time (hours). *The third plot:* Scatterplot of NPDE versus individual predicted concentrations (IPRED). The dashed horizontal lines are the null line (NPDE = 0). The dot-dashed horizontal lines indicate the 90% range (NPDE = -1.645, 1.645). The dotted lines indicate the 95% range (NPDE = -1.96, 1.96). ESAMPLE was set at 10,000



Supplementary Figure 6 The predictive performance of the single and multi-model approaches using all the nine models versus the three models to predict vancomycin concentration (C) at a time +1 (T+1) following **intermittent infusion** administration. (a) *A priori* predictions using the clinical dataset with covariate data without previous concentrations, and (b) *a posteriori* prediction settings using the clinical dataset with covariate data and previous concentrations (CT₋₁, CT₀, CT₋₁ and CT₀). Whiskers cover the 95% confidence interval of the relative bias calculated via the standard error (black whisker indicated 95% CI including 0). Model weighing in *a priori* predictions correspond to 1/number of models, disabling model selection. The horizontal lines in relative bias panels indicate $\pm 20\%$ of the relative bias. rBias, relative bias; rRMSE, relative root mean square error; C, concentration; T, relative time



Supplementary Figure 7 The predictive performance of the single and multi-model approaches using all the nine models versus the three models to predict vancomycin concentration (C) at a time +1 (T+1) following **continuous infusion** administration. (a) *A priori* predictions using the clinical dataset with covariate data without previous concentrations, and (b) *a posteriori* prediction settings using the clinical dataset with covariate data and previous concentrations (CT-1, CT₀, CT₋₁ and CT₀). Whiskers cover the 95% confidence interval of the relative bias calculated via the standard error (black whisker indicated 95% CI including 0). Model weighing in *a priori* predictions correspond to 1/number of models, disabling model selection. The horizontal lines in relative bias panels indicate $\pm 20\%$ of the relative bias. rBias, relative bias; rRMSE, relative root mean square error; C, concentration; T, relative time



Supplementary Figure 8 Bland–Altman plots stratified by type of infusion where an agreement between the measured and the predicted vancomycin concentrations was observed in the *a priori* and *a posteriori* predictions using the model selection algorithm (MSA; purple) and the model averaging algorithm (MAA; orange). The vertical red lines indicate the 12.5 and 17.5 mg/L concentration target range. The solid line with shaded area is a locally smoother with 95% confidence interval based on the data (MSA in purple and MAA in orange). C, concentration; T, relative time

NONMEM control streams

```

$PROBLEM      Adane et al. 2015 Pharmacotherapy – PMID: 25644478
$INPUT        ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE
              DIABETES WEIGHT HEIGHT BMI ICU SCr EGFR BUN
              ALBUMIN  TOTAL_PROTEIN  DIALYSIS  CVVH
              FUROSEMIDE DOSE ETH
$DATA         data.csv IGNORE=C
$SUBROUTINE   ADVAN13 TOL=9
$MODEL        NCOMPARTMENTS=1
              COMP=(CENTRAL)
$PK           BSA          =(WEIGHT**0.425)*(HEIGHT**0.725)*0.0
              07184
              IF(SCr.LT.1) SCT=1
              )
              IF(SCr.GE.1) SCT=SCr
              IF(SEX.EQ.1) CrCL =((1.00*(WEIGHT*(140-
              AGE)))/(72*SCT))*(1.73/BSA) ; Males
              IF(SEX.EQ.0) CrCL =((0.85*(WEIGHT*(140-
              AGE)))/(72*SCT))*(1.73/BSA) ; Females
              CLBSA        =CrCL
              TVCL         =THETA(1)*(CLBSA/125)
              CL           =TVCL*EXP(ETA(1))
              TVV1         = THETA(2)*WEIGHT
              V1           =TVV1*EXP(ETA(2))
              S1           =V1
              K10          =CL/V1
$DES          DADT(1)      = - K10*A(1)
$error        IPRED       =F
              IRES        =DV-IPRED
              DEL         =0
              IF (F.EQ.0) =1
              DEL
              W           =IPRED+DEL
              Y           =IPRED+W*ERR(1)
$THETA        6.54        ; TVCL
              0.51        ; TVV1
$OMEGA        0.071      ; CL
              0.057      ; V1
$SIGMA        0.035      ; Proportional error
$SIMULATION   METHOD=1 INTER MAXEVAL=0 PRINT=5 NOABORT
$TABLE        POSTHOC
              FILE=Output_data.csv

```

\$PROBLE M Goti *et al.* 2018 *The Drug Monit* – PMID: 29470227

\$INPUT ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE
DIABETES WEIGHT HEIGHT BMI ICU SCr EGFR BUN
ALBUMIN TOTAL_PROTEIN DIALYSIS CVVH
FUROSEMIDE DOSE ETH

\$DATA data.csv IGNORE=C

\$SUBRO ADVAN13 TOL=9

UTINE

\$MODEL NCOMPARTMENTS=2
COMP=(CENTRAL)
COMP=(PERIPH)

\$PK IF(SCr.LT.1.AND.AGE.GT.60) SCT = 1.0
IF (SCr.GE.1.OR.AGE.LE.60) SCT = SCr
IF(SEX.EQ.1) CrCL = (1.00*WEIGHT*(140-AGE))/(72*SCT); Male
IF(SEX.EQ.0) CrCL = (0.85*WEIGHT*(140-AGE))/(72*SCT); Female
IF (CrCL.GT.150) CCT = 150
IF (CrCL.LE.150) CCT = CrCL
TVCL =
THETA(1)*((CCT/120)**THETA(2))*(THETA(3)**DIALYSIS)
CL = TVCL*EXP(ETA(1))
TVV1 =
THETA(4)*(WEIGHT/70)*(THETA(5)**DIALYSIS)
V1 = TVV1*EXP(ETA(2))
TVQ = THETA(6)
Q = TVQ
TVV2 = THETA(7)*(WEIGHT/70)
V2 = TVV2*EXP(ETA(3))
S1 = V1
S2 = V2
K10 = CL/V1
K12 = Q/V1
K21 = Q/V2

\$DES DADT(1) = K21*A(2) - K12*A(1) - K10*A(1)
DADT(2) = K12*A(1) - K21*A(2)

\$ERROR	IPRED	= F
	IRES	= DV-IPRED
	Y	= IPRED*(1+ERR(1))+ ERR(2)
\$THETA	4.5	; TVCL
	0.8	; CrCL on CL
	0.7	; Dialysis on CL
	58.4	; TVV1
	0.5	; Dialysis on V1
	6.5	; TVQ
	38.4	; TVV2
\$OMEGA	0.158	; CL
	0.666	; V1
	0.326	; V2
\$SIGMA	11.56	; Proportional error
	0.515	; Additive error
\$SIMULATION	METHOD=1 INTER	MAXEVAL=0 PRINT=5 NOABORT
	POSTHOC	

\$PROBLEM	Ji et al. 2018 <i>Acta Pharmacol Sin</i> PMID: 28836582
\$INPUT	ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE DIABETES WEIGHT HEIGHT BMI ICU SCr EGFR BUN ALBUMIN TOTAL_PROTEIN DIALYSIS CVVH FUROSEMIDE DOSE ETH
\$DATA	data.csv IGNORE=C
\$SUBROUTINE	ADVAN13 TOL=9
\$MODEL	NCOMPARTMENTS=2 COMP=(CENTRAL) COMP=(PERIPH)
\$PK	IF(SEX.EQ.1) CrCL = (1.00*WEIGHT*(140-AGE))/(72*SCr) ; Male IF(SEX.EQ.0) CrCL = (0.85*WEIGHT*(140-AGE))/(72*SCr) ; Female IF (FUROSEMIDE.EQ.0) TVCL = THETA(1)*CrCL*(60/1000) IF (FUROSEMIDE.EQ.1) TVCL = THETA(2)*CrCL*(60/1000) CL = TVCL*EXP(ETA(1)) (AGE.LE.65) TVV1 = THETA(3)*WEIGHT (AGE.GT.65) TVV1 = THETA(4)*WEIGHT V1 = TVV1*EXP(ETA(2)) TVQ = THETA(4) Q = TVQ TVV2 = THETA(5)*(WEIGHT) V2 = TVV2 S1 = V1 S2 = V2 K10 = CL/V1 K12 = Q/V1 K21 = Q/V2 DADT(1) = K21*A(2) - K12*A(1) - K10*A(1) DADT(2) = K12*A(1) - K21*A(2) IPRED = F IRES = DV-IPRED Y = IPRED*(1+ERR(1))+ ERR(2) \$THETA 0.49 ; TVCL without Furosemide 0.34 ; TVCL with Furosemide 0.7 ; Dialysis on CL 0.74 ; TVV1 in young patients 1.07 ; TVV1 in older patients 0.81 ; TVQ

```
$OMEGA      5.86      ; TVV2
             0.137     ; CL
             0.16      ; V1
$SIGMA      0.037     ; Proportional error
             12.25     ; Additive error
$SIMULATION  METHOD=1 INTER MAXEVAL=0 PRINT=5 NOABORT
             POSTHOC
$TABLE      FILE=Output_data.csv
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$PROB Mangin et al. 2014 Clin Pharmacokinet – PMID: 25117184
LEM
$INPUT ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE
DIABETES WEIGHT HEIGHT BMI ICU SCr EGFR BUN
ALBUMIN TOTAL_PROTEIN DIALYSIS CVVH FUROSEMIDE
DOSE ETH
$DATA data.csv IGNORE=C
$SUBR ADVAN13 TOL=9
OUTLINE
$MODE NCOMPARTMENTS=2
L COMP=(CENTRAL)
COMP=(PERIPH)
$PK IF(SEX.E = (1.00*WEIGHT*(140-AGE))/(72*SCr); Males
Q.1)
CrCL
IF(SEX.E = (0.85*WEIGHT*(140-AGE))/(72*SCr); Females
Q.0)
CrCL
IF(SEX.E = THETA(1)
Q.1)
F_FEM
IF(SEX.E = 1
Q.0)
F_FEM
SCT = SCr*88.4
IF(DIABE = THETA(2)
TES.EQ.1
) F_DM
IF(DIABE = 1
TES.EQ.0
) F_DM
SAPSII = 50
TVCL =THETA(3)*F_FEM**((WEIGHT/70)**0.75)*((SAP
SII/50)**THETA(4))*((SCT/100)**THETA(5))
CL =TVCL*EXP(ETA(1))
TVV1 = THETA(6)*(WEIGHT/70)
V1 =TVV1
TVQ =THETA(7)*F_DM*((WEIGHT/70)**0.75)
Q =TVQ*EXP(ETA(2))
TVV2 THETA(8)*(WEIGHT/70)
V2 =TVV2*EXP(ETA(3))
S1 =V1
S2 =V2
K10 = CL/V1
K12 = Q/V1

```

	K21	= $Q/V2$
\$DES	DADT(1)	= $K21*A(2) - K12*A(1) - K10*A(1)$
	DADT(2)	= $K12*A(1) - K21*A(2)$
\$ERRO	IPRED	= F
R		
	IRES	= DV-IPRED
	Y	= IPRED + ERR(1)
\$THET	0.66	; F_FEM on CL
A		
	0.30	; F_DM on Q
	1.91	; TVCL
	-0.5	; SAPSII on CL
	-0.9	; SCT on CL
	21.9	; TVV1
	5.71	; TVQ
	68.0	; TVV2
\$OMEG	0.084	; CL
A	1.020	; Q
	0.271	; V2
\$SIGMA	53.58	; Additive error
\$SIMUL	METHOD=1	INTER MAXEVAL=0 PRINT=5 NOABORT
ATION	POSTHOC	
\$TABLE	FILE=Output_data.csv	

\$PROBLEM	Medellin-Garibay <i>et al.</i> 2016 <i>J Antimicrob Chemother</i> – PMID: 26568565	
\$INPUT	ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE DIABETES WEIGHT HEIGHT BMI ICU SCr EGFR BUN ALBUMIN TOTAL_PROTEIN DIALYSIS CVVH FUROSEMIDE DOSE ETH	
\$DATA	data.csv IGNORE=C	
\$SUBROUTINE	ADVAN13 TOL=9	
\$MODEL	NCOMPARTMENTS=2 COMP=(CENTRAL) COMP=(PERIPH)	
\$PK	IF(SEX.EQ.1) CrCL	= (1.00*WEIGHT*(140-AGE))/(72*SCr) ; Male
	IF(SEX.EQ.0) CrCL	= (0.85*WEIGHT*(140-AGE))/(72*SCr) ; Female
	IF (FUROSEMIDE.EQ.0)	= THETA(1)*CrCL*(60/1000)
	TVCL	
	IF (FUROSEMIDE.EQ.1)	= THETA(2)*CrCL*(60/1000)
	TVCL	
	CL	= TVCL*EXP(ETA(1))
	(AGE.LE.65) TVV1	= THETA(3)*WEIGHT
	(AGE.GT.65) TVV1	= THETA(4)*WEIGHT
	V1	= TVV1*EXP(ETA(2))
	TVQ	= THETA(4)
	Q	= TVQ
	TVV2	= THETA(5)*(WEIGHT)
	V2	= TVV2
	S1	= V1
	S2	= V2
	K10	= CL/V1
	K12	= Q/V1
	K21	= Q/V2
\$DES	DADT(1)	= K21*A(2) - K12*A(1) - K10*A(1)
	DADT(2)	= K12*A(1) - K21*A(2)
\$ERROR	IPRED	= F
	IRES	= DV-IPRED
	Y	= IPRED*(1+ERR(1))+ ERR(2)
\$THETA	0.49	; TVCL without Furosemide
	0.34	; TVCL with Furosemide
	0.7	; Dialysis on CL
	0.74	; TVV1 in young patients
	1.07	; TVV1 in older patients

	0.81		; TVQ
	5.86		; TVV2
\$OMEGA	0.137		; CL
	0.16		; V1
\$SIGMA	0.037		; Proportional error
	12.25		; Additive error
\$SIMULATION	METHOD=1	INTER	MAXEVAL=0 PRINT=5 NOABORT
	POSTHOC		
\$TABLE	FILE=Output_data.csv		

\$PROBL Revilla *et al.* 2010 *Br J Clin Pharmacol* – PMID: 20653673
 EM
 \$INPUT ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE
 DIABETES WEIGHT HEIGHT BMI ICU SCr EGFR BUN
 ALBUMIN TOTAL PROTEIN DIALYSIS CVVH
 FUROSEMIDE DOSE ETH
 \$DATA data.csv IGNORE=C
 \$SUBRO ADVAN13 TOL=9
 UTINE
 \$MODEL NCOMPARTMENTS=1
 COMP=(CENTRAL)
 \$PK BSA =
 (WEIGHT**0.425)*(HEIGHT**0.725)*0.007184
 IF(SEX.EQ.1.AND.ETH.EQ.0) CrCL = 170*(SCr**(-0.999))*(AGE**(-0.176))*(BUN**(-0.17))*((ALBUMIN**0.318)*1.000*1.00)*((BSA/1.73)/WEIGHT) ; Male Caucasian
 IF(SEX.EQ.1.AND.ETH.EQ.1) CrCL = 170*(SCr**(-0.999))*(AGE**(-0.176))*(BUN**(-0.17))*((ALBUMIN**0.318)*1.000*1.18)*((BSA/1.73)/WEIGHT) ; Male Black
 IF(SEX.EQ.0.AND.ETH.EQ.0) CrCL = 170*(SCr**(-0.999))*(AGE**(-0.176))*(BUN**(-0.17))*((ALBUMIN**0.318)*0.762*1.00)*((BSA/1.73)/WEIGHT) ; Female Caucasian
 IF(SEX.EQ.0.AND.ETH.EQ.1) CrCL = 170*(SCr**(-0.999))*(AGE**(-0.176))*(BUN**(-0.17))*((ALBUMIN**0.318)*0.762*1.18)*((BSA/1.73)/WEIGHT) ; Female Black
 IF (SCr.GE.1) AB = 1
 IF (SCr.LT.1) AB = 0
 TVCL =(THETA(1)*CrCL+(AGE**THETA(2)))*((60/1000)*WEIGHT)
 CL =TVCL*EXP(ETA(1))
 TVV1 = THETA(3)*(THETA(4)**AB)*WEIGHT
 V1 =TVV1*EXP(ETA(2))
 S1 =V1
 K10 =CL/V1
 \$DES DADT(1) =- K10*A(1)
 \$ERROR IPRED =F
 IRES =DV-IPRED
 Y =IPRED+ERR(1)

\$THETA	0.67		; TVCL
	0.82		; TVV1
	-0.24		; AGE on CL
	2.49		; Serum creatinine categorical variable on V
\$OMEGA	0.087		; CL
	0.051		; V1
\$SIGMA	17.89		; Additive error
\$SIMULA TION	METHOD=1	INTER	MAXEVAL=0 PRINT=5 NOABORT
\$TABLE	POSTHOC		
	FILE=Output_data.csv		


```

$PROBLEM      Roberts et al. 2011 Antimicrob Agents Chemother –
               PMID: 21402850
$INPUT        ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE
               DIABETES WEIGHT HEIGHT BMI ICU SCr EGFR BUN
               ALBUMIN TOTAL_PROTEIN DIALYSIS CVVH
               FUROSEMIDE DOSE ETH
$DATA         data.csv IGNORE=C
$SUBROUTINE   ADVAN13 TOL=9
$MODEL        NCOMPARTMENTS=1
               COMP=(CENTRAL)
$PK           UCrCL      =100
               TVCL      =THETA(1)*(UCrCL/100)
               CL        =TVCL*EXP(ETA(1))
               TVV1      = THETA(2)*WEIGHT
               V1        =TVV1*EXP(ETA(2))
               S1        =V1
               K10       =CL/V1
$DES          DADT(1)    =- K10*A(1)
$ERROR        IPRED     =F
               IRES      =DV-IPRED
               Y         =IPRED*(1+ERR(1))+ ERR(2)
$THETA        4.58      ; TVCL
               1.53      ; TVV1
$OMEGA        0.151     ; CL
               0.140     ; V1
$SIGMA        0.039     ; Proportional error
               5.76      ; Additive error
$SIMULATION   METHOD=1 INTER MAXEVAL=0 PRINT=5 NOABORT
               POSTHOC
$TABLE        FILE=Output_data.csv

```

\$PROBLEM	Thomson <i>et al.</i> 2009 <i>J Antimicrob Chemother</i> – PMID: 19299472
\$INPUT	ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE DIABETES WEIGHT HEIGHT BMI ICU SCr EGFR BUN ALBUMIN TOTAL_PROTEIN DIALYSIS CVVH FUROSEMIDE DOSE ETH
\$DATA	data.csv IGNORE=C
\$SUBROUTINE	ADVAN13 TOL=9
\$MODEL	NCOMPARTMENTS=2 COMP=(CENTRAL) COMP=(PERIPH)
\$PK	IF (SCr.LT.0.68) = 0.68 SCT IF = SCr (SCr.GE.0.68) SCT IF(SEX.EQ.1) = (1.00*WEIGHT*(140- CrCL AGE))/(72*SCT)) ; Male IF(SEX.EQ.0) = (0.85*WEIGHT*(140- CrCL AGE))/(72*SCT)) ; Female TVCL = THETA(1)*THETA(2)*CrCL CL = TVCL*EXP(ETA(1)) TVV1 = THETA(3)*WEIGHT V1 = TVV1*EXP(ETA(2)) TVQ = THETA(4) Q = TVQ*EXP(ETA(4)) TVV2 = THETA(5)*WEIGHT V2 = TVV2*EXP(ETA(3)) S1 = V1 S2 = V2 K10 = CL/V1 K12 = Q/V1 K21 = Q/V2 \$DES DADT(1) = K21*A(2) - K12*A(1) - K10*A(1) DADT(2) = K12*A(1) - K21*A(2) \$ERROR IPRED = F IRES = DV-IPRED Y = IPRED*(1+ERR(1))+ ERR(2) \$THETA 2.99 ; TVCL 0.0154 ; CrCL on CL 0.675 ; TVV1 2.28 ; TVQ 0.732 ; TVV2 \$OMEGA 0.073 ; CL 0.023 ; V1

	1.690	; V2
	0.240	; Q
\$SIGMA	0.023	; Proportional error
	2.56	; Additive error
\$SIMULATION	METHOD=1 INTER MAXEVAL=0 PRINT=5 NOABORT	
	POSTHOC	
\$TABLE	FILE=Output_data.csv	

```

$PROBLEM      Zhou et al. 2019 Eur J Drug Metab Pharmacokinet –
               PMID: 30506225
$INPUT        ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE
               DIABETES WEIGHT HEIGHT BMI ICU SCr EGFR BUN
               ALBUMIN TOTAL_PROTEIN DIALYSIS CVVH
               FUROSEMIDE DOSE ETH
$DATA         data.csv IGNORE=C
$SUBROUTINE   ADVAN13 TOL=9
$MODEL        NCOMPARTMENTS=1
               COMP=(CENTRAL)
$PK           IF(SEX.EQ.1) = (1.00*(WEIGHT*(140-
               CrCL AGE))/((72*SCr); Males
               IF(SEX.EQ.0) = (0.85*(WEIGHT*(140-
               CrCL AGE))/((72*SCr); Females
               TVCL          =THETA(1)*
               ((CrCL/56.28)**THETA(2))
               CL            =TVCL*EXP(ETA(1))
               TVV1          = THETA(3)
               V1            =TVV1*EXP(ETA(2))
               S1            =V1
               K10           =CL/V1
$DES          DADT(1)      =- K10*A(1)
$ERROR        IPRED       =F
               IRES        =DV-IPRED
               Y            =IPRED*(1+ERR(1))
$THETA        2.45        ; TVCL
               0.54        ; CrCL on CL
               154         ; TVV1
$OMEGA        0.030       ; CL
               0.115       ; V1
$SIGMA        0.004       ; Proportional error
$SIMULATION   METHOD=1 INTER MAXEVAL=0 PRINT=5 NOABORT
               POSTHOC
$TABLE        FILE=Output_data.csv

```

Part 2. Critically ill patients

Chapter 3. Meropenem dosing in the intensive care unit

This chapter is based on

Nada Dia *et al.*, Model-based evaluation of an optimized meropenem dosing strategy for critically ill patients with preserved or augmented renal clearance. In preparation

Abstract

Background Standard eight-hourly intermittent intravenous infusions of 1,000 mg meropenem provide inadequate pharmacokinetic–pharmacodynamic (PKPD) target attainment (100 percent of time during a dosing interval for which concentration of the unbound drug is above 1-fold the minimum inhibitory concentration throughout a dosing interval [$100\% fT > 1 \times \text{MIC}$]) in critically ill septic patients with preserved or augmented renal clearance (ARC) at the European Committee on Antimicrobial Susceptibility Testing (EUCAST) susceptibility breakpoint of 2 mg/L for Enterobacterales. Our aims were to compare the predictive performance of different kidney function markers and the patient characteristics required to calculate these markers on meropenem pharmacokinetics (PK) and to identify an optimized dosing strategy with $\geq 90\%$ probability of target attainment (PTA), while considering safety metrics.

Methods Using data from 58 adult patients with preserved renal function or ARC admitted to the intensive care unit, we developed five meropenem population PK models including measured serum creatinine (SCr) or estimated creatinine-based kidney function markers (estimated creatinine clearance using Cockcroft–Gault equation [$e\text{CrCL}_{\text{CG}}$], estimated glomerular filtration rate (eGFR) calculated using the Modification of Diet in Renal Disease equation, eGFR calculated using the Chronic Kidney Disease Epidemiology Collaboration equation, measured creatinine clearance by

24-hour timed urine collection. We compiled all observed patient characteristics required to calculate these markers and, for each marker, evaluated the impact of any observed characteristics not already included in the marker. We identified a model based solely on observed characteristics that provided a fit comparable to the best model kidney function-based model. We then performed stochastic simulations with each of the six models using a virtual patient dataset under standard and optimized dosing strategies. We defined PKPD target attainment as concentration of unbound meropenem >2 mg/L throughout the dosing interval ($100\% fT > 1 \times \text{MIC}$) on the end of day 1 after the start of treatment onwards. We compared PTA and predicted population rate of target attainment between the standard and optimized dosing strategies, while evaluating safety metrics separately. We developed an interactive tool to allow flexible evaluation of meropenem dosing under user-defined scenarios.

Results The eCrCL_{CG} model resulted in the biggest drop in objective function value compared with the other four kidney function markers models. The identified model included measured SCr and total body weight as observed characteristics, and resulted in a model fit similar to the eCrCL_{CG} model. The PTA for efficacy was $\geq 90\%$ across the range of each kidney function marker and body weight under continuous infusion of a total daily dose of 4,000 mg administered at a rate of 1,000 mg every 6 hours. Across the six models, the predicted population rate of target attainment was $\leq 46.7\%$ (95% prediction

interval 45.4%–47.3%) under standard dosing strategy and $\leq 99.9\%$ (99.9%–100%) under the optimized dosing strategy.

Conclusion SCr and total body weight drive the meropenem PK in critically ill septic patients with preserved renal function or ARC who need a total daily dose of 4,000 mg administered as a continuous infusion of $4 \times 1,000$ mg over 6 hours. An interactive simulation tool is accessible at <https://lpmx.shinyapps.io/meropenem/>.

1. Introduction

Meropenem, a broad-spectrum β -lactam antibiotic, is commonly used in the ICU to treat patients with sepsis and septic shock [276]. Optimal treatment implies timely and continuous efficacy-based PKPD target attainment, defined as concentrations of unbound meropenem above 1- to 4-fold the MIC throughout the dosing interval ($100\% fT > 1-4 \times \text{MIC}$) [277]. While meropenem is generally well tolerated, nephrotoxicity and neurotoxicity may occur at C_{trough} above 44.5 mg/L and 64.2 mg/L, respectively [54].

Meropenem is primarily excreted by the kidney through glomerular filtration and tubular secretion. This is consistent with the vast majority of popPK studies where a statistically significant relation between meropenem CL and kidney function was identified [278–282]. Standard eight-hourly intermittent intravenous infusions of 1,000 mg meropenem were shown to result in inadequate PKPD target attainment ($100\% fT > 1 \times \text{MIC}$) in critically ill septic patients with preserved renal function or ARC (below 50%) [278]. Although kidney function has frequently been a covariate in meropenem popPK models, the clinical relevance of a kidney function on the PTA remains insufficiently characterized. In particular, the impact of different kidney function markers on the PKPD target attainment of meropenem in the ICU has not been comprehensively evaluated.

Numerous popPK models of meropenem in critically ill patients with sepsis have been developed [278,280,283–294]. In these models, the $eCrCL_{CG}$

emerged as the primary kidney function marker retained as a covariate on meropenem CL. The popPK model of Jaruratanasirikul *et al.* stands out as the sole exception, as it includes the estimated GFR calculated using the Modification of Diet in Renal Disease equation ($eGFR_{MDRD}$) [278,280,283–293]. While other kidney function markers, such as the estimated GFR calculated using the Chronic Kidney Disease Epidemiology Collaboration equation ($eGFR_{CKD-EPI}$), the $mCrCL_{24h}$, or measured SCr have been suggested, they have been rarely evaluated as potential covariates. Calculation of these kidney function markers relies on SCr in combination with a number of patient characteristics, making certain markers more suitable for specific patient populations [295]. For instance, the $eCrCL_{CG}$ and $mCrCL_{24h}$ depend on total body weight, which more accurately reflects the individual's muscle mass contributing to SCr production, and consequently kidney function [296]. The $eGFR_{MDRD}$ and $eGFR_{CKD-EPI}$ equations include race as a factor to correct for the relatively higher SCr levels measured in African-American versus white individuals [297]. However, race cannot be considered an accurate classifier as racial groups are genetically and socially highly heterogeneous [142,298,299].

Investigating opportunities for meropenem dose optimization based on popPK models including single kidney function marker fails to reliably ensure equitable meropenem therapy for all critically ill patients with preserved renal function or ARC admitted to the ICU. Therefore, the objective of this popPK modeling and simulation study was to provide an optimized meropenem

dosing recommendation for these patients across a wide range of kidney function markers and underlying patient characteristics while considering efficacy targets and safety metrics.

2. Materials and methods

2.1 Ethics statement

The study was approved by the Ethics Committee Research UZ/KU Leuven (S54511) and registered at ClinicalTrials.gov (NCT03560557). Written informed consent was obtained from each patient or the closest relative before inclusion in this study.

2.2 Clinical dataset

Data from a prospective PK study including 58 adult patients with preserved (*i.e.*, $\text{eGFR}_{\text{CKD-EPI}} \geq 70 \text{ mL/min/1.73 m}^2$ on the day of PK sampling) or ARC (*i.e.*, $\text{mCrCl}_{24\text{h}} \geq 130 \text{ mL/min/1.73 m}^2$) admitted to the ICU were used. All participants had severe sepsis (*i.e.*, [suspected] infection causing organ dysfunction) or septic shock (*i.e.*, severe sepsis associated with hypotension despite appropriate fluid therapy) at the moment meropenem therapy was started [278]. A total of 345 meropenem plasma concentrations were measured. A detailed description of the clinical dataset was reported by Gijsen *et al.* [278]. Baseline data on the kidney function markers (*i.e.*, eCrCL_{CG} , $\text{eGFR}_{\text{MDRD}}$, $\text{eGFR}_{\text{CKD-EPI}}$, and $\text{mCrCl}_{24\text{h}}$) and the patient characteristics needed to calculate these kidney function markers (*i.e.*, age, total body weight, height, SCr,

24-hour measured urinary creatinine [UCr_{24h}], sex assigned at birth, and race) were retrieved (**Supplementary Table 15**).

2.3 Population pharmacokinetic modeling and model comparison

Gijssen *et al.* developed a two-compartment popPK model of meropenem based on the clinical dataset, assuming negligible protein binding [278]. The model included eCrCL_{CG} at the start of meropenem therapy as a covariate on CL:

$$CL_i = CL_{pop} \text{ (L/h)} \times \left(\frac{eCrCL_{CG} \text{ (mL/min)}}{111.7 \text{ (mL/min)}} \right)^{0.638} \times e^{\eta_i} \text{ with } \eta_i \sim \mathcal{N}(0, 0.288) \quad (5)$$

with CL_{*i*} the individual clearance, CL_{*pop*} the typical estimate of the clearance, 111.7 mL/min the median value of the eCrCL_{CG} in the clinical dataset, 0.638 the estimated exponent of the eCrCL_{CG} effect on meropenem clearance, and η_i the difference between the typical and individual clearance.

Using the clinical dataset, we ran four additional popPK models including eGFR_{MDRD}, eGFR_{CKD-EPI}, mCrCL_{24h}, and SCr as a covariate on meropenem CL. We compared all popPK models in terms of (i) drop in OFV and IIV with respect to the base model, (ii) parameter precision and physiological plausibility, and (iii) goodness-of-fit plots and pcVPC model calibration plots (based on 1,000 simulations). Next, we performed stepwise forward addition (OFV drop ≥3.84 points; p-value ≤0.05) and backward elimination (OFV increase <7.88 points; p-value <0.005) of patient characteristics that were not already used to calculate the kidney function marker in the respective popPK

model (**Supplementary Table 16**). We compared and internally validated the final popPK models using the same tools as described above. When selecting a model among the final models, we relied on two criteria: drop in OFV and type of covariates (*i.e.*, measured or estimated).

2.4 Dosing strategies

A total of eight dosing strategies were evaluated over seven treatment days. Four of these strategies were recommended by established guidelines, three of which were prescribed at the ICU of UZ Leuven [68,278,300,301]. Two dosing strategies followed the protocol applied in the BLING III trial (**Table 3**) [302]. The remaining two strategies were included because they represent clinically feasible alternatives. For dosing strategies involving a loading dose followed by continuous infusion, the timing between the loading dose and the start of the continuous infusion aims to allow the loading dose to distribute fully, ensuring that the subsequent infusion maintains therapeutic concentration without toxicity. The efficacy-based PKPD target was defined as $100\% fT > 1 \times \text{MIC}$, corresponding to a concentration of unbound meropenem throughout a dosing interval above 2 mg/L, at the EUCAST breakpoint for susceptibility of 2 mg/L for Enterobacterales [68,278].

Table 3 Dosing strategies

Infusion strategy	Dosing strategy	References
Intermittent	1,000 mg over 30 minutes q8h ^a	[68,278,300]
	3,000 mg over 30 minutes LD followed 8 hours later by 1,000 mg over 30 minutes q8h	–
	2,000 mg over 30 minutes q8h ^b	[68,278,300]
	40 mg/kg over 30 minutes q8h	[301]
Extended	2,000 mg over 3 hours q8h	[68,300]
Continuous	1,000 mg over 30 minutes LD followed 4 hours later by 1,000 mg over 8 hours q8h	[302]
	1,000 mg over 30 minutes LD followed 4 hours later by 1,000 mg over 12 hours q12h	[302]
	1,000 mg over 6 hours q6h	–

^a Standard dosing strategy

^b Main dosing strategy used in the intensive care unit of UZ Leuven
LD, loading dose; qxh, every x hours

2.5 Clinical relevance assessment and comparison of kidney function markers

We performed simulations with each of the five popPK models, each having a kidney function marker, to calculate the PTA along predefined ranges of covariate values (*i.e.*, patient characteristics and kidney function markers) (**Supplementary Table 17**). Therefore, we created virtual patient datasets, each with a patient characteristic or kidney function marker of interest being uniformly distributed. For the remaining patient characteristics, we set the values to the clinical dataset's median to assess the pure impact of each characteristic without confounding effects. For each virtual patient, we

performed 1,000 simulations of the standard dosing strategy with each popPK model. The impact of a patient characteristic or kidney function marker on the PTA was considered clinically relevant when the PTA gradually increased/decreased along the investigated range at the end of day 1 of treatment [303]. Following covariate evaluation, the model including patient characteristics and achieving a fit comparable to the best-performing model with kidney function marker was retained as a sixth model.

2.6 Identification of an optimized dosing strategy

Using a total of six models (*i.e.*, five popPK models, each with a kidney function marker, and the selected popPK model with patient characteristics), we now performed 1,000 simulations per virtual patient for each model under various optimized dosing strategies. A meropenem dosing strategy resulting in $\geq 90\%$ PTA at the end of day 1 and day 3 of treatment, in all virtual patients along the studied covariate ranges with each of the six popPK models, was considered optimized.

To predict the impact of the optimized meropenem dosing strategy on the real-world patient population, we created a virtual patient dataset ($n=200$) with the covariates sampled from the covariance matrix of the clinical dataset (**Supplementary Figure 9**). For each virtual patient, we performed 1,000 simulations under the standard and optimized dosing strategies with each of the six popPK models. The population-level PTA, the $C_{ss, trough/plateau}$, the prescribed daily meropenem dose, and the dispensed daily amount at the end

of day 1, and day 3 of treatment were compared between the standard and optimized dosing strategies. The safety limit for neurotoxicity was set to $C_{ss, trough/plateau}$ 64.2 mg/L [54].

2.7 Software

PopPK modeling and simulation were performed with NONMEM (version 7.5.0; Icon Development Solutions, Gaithersburg, MD, USA) and the Perl-speaks-NONMEM (version 4.7.0) toolkit on the Pirana modeling workbench (version 2.9.7; Certara, Inc., Princeton, NJ, USA). Virtual patients were created, and data was analyzed using R (version 4.2.3; R Core Team, Vienna, Austria) in the RStudio integrated development environment (version 3.1.446; RStudio, Inc., Boston, MA, USA) with the *tidyverse* collection of packages [304].

2.8 Interactive tool

PopPK modeling and simulation were performed with NONMEM (version 7.5; Icon Development Solutions). We developed an interactive tool based on the selected popPK model with patient characteristics to allow flexible investigation of user-defined dosing strategies. The *Shiny* package (version 1.7.4), *mrgsolve* package (version 1.3.0), and *plotly* package (version 4.10.3) in R were used for developing the web application [305–307]. We cross-validated the performance of the tool against NONMEM to ensure consistency.

3. Results

3.1 Population pharmacokinetic modeling and model comparison

The inclusion of each kidney function marker on meropenem CL significantly improved the goodness-of-fit of the popPK model to the study data (p-value <0.0001) (**Table 4**). The $eCrCL_{CG}$ resulted in the biggest drop in OFV (34.38 points; p-value <0.0001), explained most IIV (13%), and had the most precise parameter estimate (relative standard error [RSE] = 20.7%). However, visual inspection of goodness-of-fit plots and model calibration did not show clear differences between kidney function markers (**Supplementary Figures 10–15**).

Accounting for total body weight on top of $eGFR_{MDRD}$, $eGFR_{CKD-EPI}$, and SCr models resulted in a further significant drop in OFV (p-value <0.01), resulting in an OFV close to that of the model with only $eCrCL_{CG}$ (**Supplementary Table 18**). The popPK model with the effects of measured SCr and total body weight on meropenem CL – using traditional power functions – resulted in a model fit and calibration similar to the popPK model based on $eCrCL_{CG}$ (**Supplementary Figures 10, 15–16**).

Table 4 Population pharmacokinetic model parameter estimates

Parameter estimate (%RSE)	No covariate	eCrCL _{CG}	eGFR _{MDRD}	eGFR _{CKD-EPI}	mCrCL _{24h}	SCr
<i>dOFV</i>	–	-34.38	-24.25	-22.93	-21.47	-21.18
<i>Condition number</i>		98.26	126.9	71.45	93.5	93.59
<i>Typical values</i>						
CL (L/h)	14.20 (8.4)	13.70 (8.4)	14.10 (7.6)	14.70 (8.8)	14.40 (7.3)	13.80 (8.3)
Kidney function on CL	–	0.638 (20.7)	0.505 (24.0)	0.950 (33.4)	0.369 (42.3)	-0.572 (27.6)
V _c (L)	21.00 (18.3)	25.50 (9.1)	25.00 (6.4)	25.10 (35.9)	21.5 (14.8)	24.90 (7.8)
Q (L/h)	12.9 (34.4)	8.13 (26.4)	8.68 (27.1)	8.89 (160.9)	12.2 (30.2)	8.68 (34.3)
V _p (L)	15.9 (21.6)	12.40 (9.8)	12.90 (15.4)	12.90 (119.4)	15.7 (21.9)	12.90 (19.4)
<i>Interindividual variability</i>						
IIV on CL (%CV)	70.8 (16.6)	57.80 (12.3)	64.10 (10.5)	63.20 (11.2)	58.2 (14.9)	65.50 (10.3)
IIV on V _c (%CV)	45.9 (44.3)	57.00 (16.7)	55.90 (16.6)	56.30 (21.1)	47.3 (27.5)	55.70 (16.7)
IIV on V _p (%CV)	80.5 (20.6)	74.40 (24.1)	80.00 (21.1)	77.70 (28.4)	87.2 (16.1)	80.50 (20.5)

Correlation between CL and V_c (%CV)	95 (22.5)	65 (20.7)	70 (16.3)	67 (19.7)	94 (16.3)	69 (16.6)
<i>Residual variability</i>						
Proportional (%CV)	34.6 (22.3)	31.10 (17.3)	30.90 (16.5)	31.20 (19.1)	34.4 (19.9)	30.80 (16.9)
Additive (mg/L)	0.137 (57.8)	0.147 (45.6)	0.156 (40.1)	0.141 (49.3)	0.130 (50)	0.160 (40.6)

CL, clearance; CV, coefficient of variation calculated as $\sqrt{\exp(\omega^2) - 1} \times 100\%$; dOFV, drop in objective function value with respect to the base population pharmacokinetic model; eCrCL_{CG}, estimated creatinine clearance calculated using the Cockcroft–Gault equation; eGFR_{CKD-EPI}, estimated glomerular filtration rate calculated using the Chronic Kidney Disease Epidemiology Collaboration equation; eGFR_{MDRD}, estimated glomerular filtration rate calculated using Modification of Diet in Renal Disease equation; IIV, interindividual variability; mCrCL_{24h}, measured creatinine clearance by 24-hour timed urine collection; OFV, objective function value; Q, intercompartmental clearance; RSE, relative standard error; SCr, serum creatinine; V_c , central volume of distribution; V_p , volume of distribution in the peripheral compartment

3.2 Clinical relevance assessment and comparison of kidney function markers

Under the standard dosing strategy of 1,000 mg administered over 30 minutes every eight hours, SCr had a clinically relevant impact on PTA in all five popPK models, with higher SCr predicting higher PTA (**Figure 10**). Total body weight is accounted for in the $eCrCL_{CG}$ and $mCrCL_{24h}$ models, but in the $eCrCL_{CG}$ model PTA decreased with increasing total body weight, while in the $mCrCL_{24h}$ model PTA increased marginally with increasing total body weight. The UCr_{24h} is only accounted for in the $mCrCL_{24h}$ model where a decrease in PTA was observed with increasing UCr_{24h} . Age is accounted for in three models ($eCrCL_{CG}$, $eGFR_{MDRD}$ and $eGFR_{CKD-EPI}$ models), but it had a clinically relevant impact only in two ($eCrCL_{CG}$, and $eGFR_{CKD-EPI}$ models). The other patient characteristics had minor effects on PTA. PTAs calculated from $eGFR_{MDRD}$, $eGFR_{CKD-EPI}$, and SCr were highly correlated (correlation coefficient ≥ 0.86) (**Supplementary Figure 17**). This is in line with the observations in **Figure 10**, where PTAs of different popPK models aligned most as the correlation between the included kidney markers was higher.

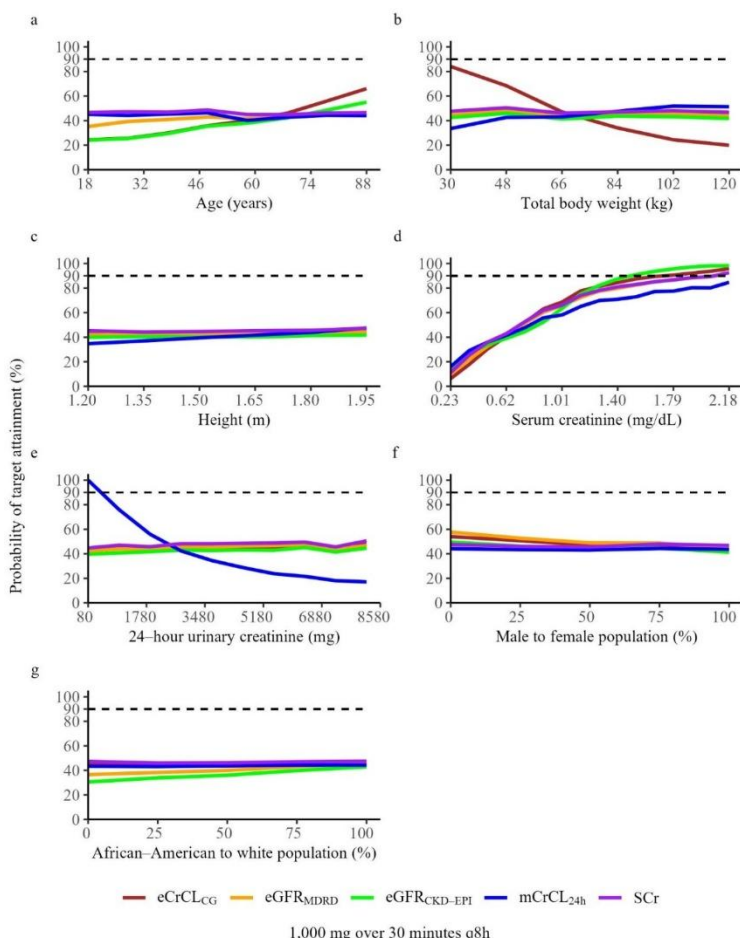


Figure 10 Probability of target attainment on the end of day 1 of treatment in five population pharmacokinetic models with different kidney function markers across the range of (a) age, (b) total body weight, (c) height, (d) serum creatinine, (e) 24-hour measured urinary creatinine, (f) male to female population, and (g) African-American to white population at a susceptibility breakpoint of 2 mg/L for Enterobacteriales, targeting a concentration of unbound meropenem throughout a dosing interval above 2 mg/L, with 1,000 mg standard dose administered over 30 minutes every 8 hours. The dashed horizontal line marks 90% probability of target attainment. eCrCl_{CG}, estimated creatinine clearance using the Cockcroft–Gault equation; eGFR_{CKD-EPI}, estimated glomerular filtration rate using the Chronic Kidney Disease Epidemiology Collaboration equation; eGFR_{MDRD}, estimated glomerular filtration rate using the Modification of Diet in Renal Disease equation; mCrCl_{24h}, measured creatinine clearance by 24-hour timed urine collection; q_{xh}, every x hours; SCr, serum creatinine

3.3 Identification of an optimized dosing strategy

The PKPD target attainment at the end of day 1 and day 3 of treatment was $\geq 90\%$ across the range of the kidney function markers and total body weight only under a continuous infusion at a rate of 1,000 mg every 6 hours (**Supplementary Figures 18–19**). The $C_{ss, trough/plateau}$ was highly variable but slightly higher under the optimized dosing strategy (continuous infusion) compared to the 1,000 mg standard dosing strategy over 30 min every 8 hours (**Supplementary Figures 20–21**). However, meropenem concentrations remained below the neurotoxicity threshold of 64.2 mg/L over 72 hours of treatment with continuous infusion, while the standard intermittent dosing strategy resulted in concentrations above this threshold in 7–9% of the population (**Supplementary Figure 22**). Under the proposed optimized dosing strategy, the PKPD target attainment at the end of day 1, and day 3 of treatment was $\geq 90\%$ across all the combinations of patient characteristics in the selected popPK model (SCr and total body weight) (**Figure 11**). Additionally, none of the patient characteristics had a clinically relevant impact on PKPD target attainment as PTA was similar along the entire range of each patient characteristic (**Supplementary Figure 23**).

At the end of day 1 and day 3 of treatment, the predicted rate of target attainment in the ICU population of patients with preserved renal function or ARC, as indicated by the ranges of patient characteristics and kidney function markers listed in **Supplementary Table 15**, was below 50% following

standard dosing strategy administered over 30 min every 8 hours and increased to 99.6% or higher under the proposed optimized dosing strategy using any of the popPK models (**Table 5**). The predicted $C_{ss, trough/plateau}$ varied from 3.3 mg/L to 41.6 mg/L under the optimized dosing strategy. Compared to standard dosing strategy, the optimized dosing strategy required an additional daily dose of 1,000 mg, resulting in a maximum additional daily cost of €13.92 in Belgium.

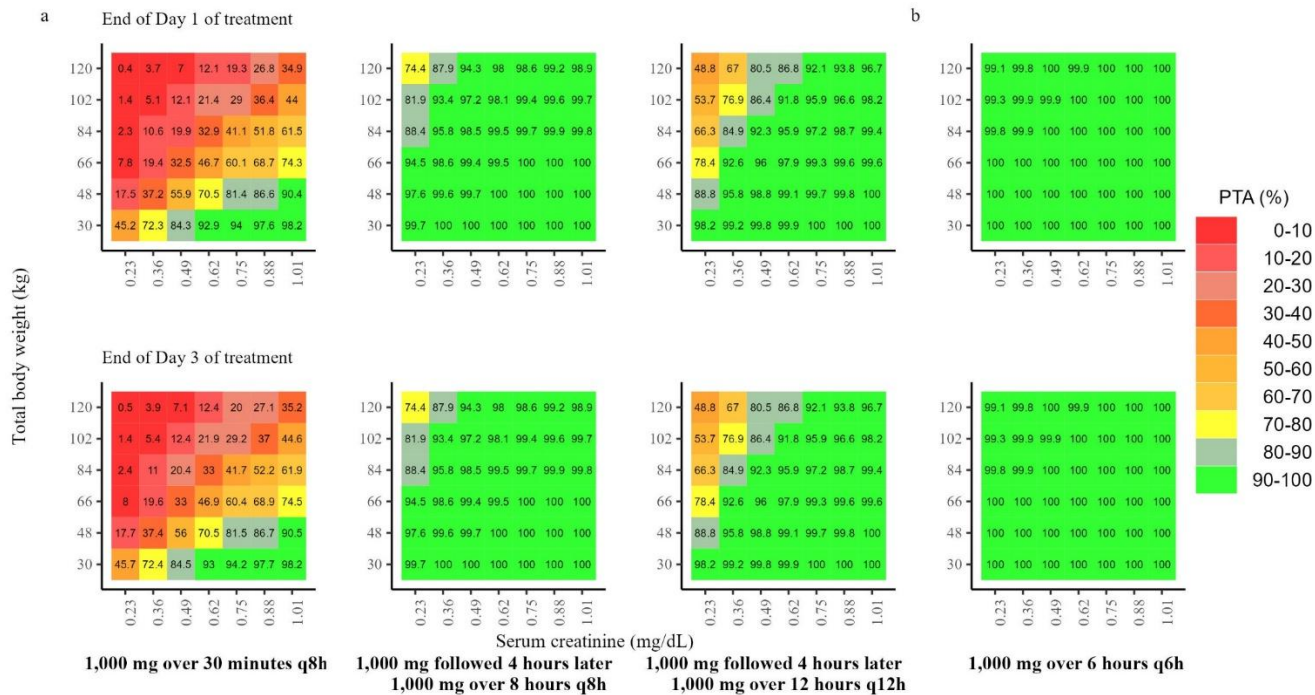


Figure 11 Heat map showing the probability of target attainment in the selected population pharmacokinetic model across a serum creatinine range of 0.23–1.01 mg/dL with stepwise of 0.13 mg/dL and total body weight of 30–120 kg with stepwise of 18 kg at

the end of (a) day 1, and (b) day 3 of treatment at a susceptibility breakpoint of 2 mg/L for Enterobacterales, targeting a concentration of unbound meropenem throughout a dosing interval above 2 mg/L, with four dosing strategies including the 1,000 mg standard dose administered over 30 minutes every 8 hours (left), two dosing strategies administered in the BLING III trial (middle), and the 1,000 mg optimized dose administered over six hours every six hours (right). PTA, probability of target attainment; q_{xh}, every x hours

Table 5 Comparison between standard and optimized dosing strategies of meropenem at the population level^a

	day 1		day 3	
	Standard dosing	Optimized dosing ^e	Standard dosing	Optimized dosing ^e
Predicted population rate of target attainment (%), mean (95% PI) ^b				
eCrCl _{CG}	46.0 (44.6–46.9)	99.8 (99.8–100)	46.3 (44.9–47.3)	99.8 (99.8–100)
eGFR _{MDRD}	46.4 (45.0–46.4)	99.7 (99.7–100)	46.7 (45.4–46.7)	99.7 (99.7–100)
eGFR _{CKD-EPI}	42.7 (41.2–42.7)	99.7 (99.7–100)	43.1 (41.6–43.1)	99.7 (99.7–100)
mCrCl _{24h}	43.6 (42.2–44.1)	99.9 (99.9–100)	44.1 (42.7–44.5)	99.9 (99.9–100)
Standalone SCr	43.6 (42.2–43.6)	99.6 (99.6–100)	44.0 (42.5–44.0)	99.6 (99.6–100)
SCr + TBW	44.0 (42.6–44.0)	99.7 (99.7–100)	44.3 (42.9–44.3)	99.7 (99.7–100)
Predicted C _{ss, trough/plateau} (mg/L), mean (95% PI), at the population level				
eCrCl _{CG}	3.0 (0.1–12.9)	13.8 (3.8–35.7)	3.1 (0.1–14.0)	14.1 (3.8–37.5)
eGFR _{MDRD}	3.2 (0.0–14.7)	14.2 (3.4–39.2)	3.4 (0.0–16.2)	14.6 (3.4–41.6)

eGFR _{CKD-EPI}	2.8 (0.0–12.9)	13.4 (3.4–35.9)	2.9 (0.0–14.0)	13.7 (3.4–37.7)
mCrCl _{24h}	2.7 (0.1–11.6)	13.6 (4.0–34.4)	2.8 (0.1–12.7)	13.9 (4.0–36.0)
Standalone SCr	3.0 (0.0–14.3)	13.8 (3.3–38.2)	3.2 (0.0–15.8)	14.1 (3.3–40.5)
SCr + TBW	3.0 (0.0–14.1)	13.7 (3.4–37.1)	3.2 (0.0–15.5)	14.0 (3.4–39.2)
Prescribed daily dose (g) ^c	3	4	9	12
Cumulative cost (€) ^d				
Meropenem, 1 g (Pfizer)	26.85	35.80	80.55	107.40
Meropenem Eugia, 1 g (Aurobindo)	37.30	49.72	111.87	149.16

^a Patients with a median (range) estimated creatinine clearance calculated using the Cockcroft–Gault equation (eCrCL_{CG}) of 111.70 (34.70–463.00) mL/min; estimated glomerular filtration rate calculated using the Chronic Kidney Disease Epidemiology Collaboration equation (eGFR_{CKD-EPI}) of 104.50 (25.00–200.00) mL/min/1.73 m²; estimated glomerular filtration rate calculated using Modification of Diet in Renal Disease equation (eGFR_{MDRD}) of 126.45 (23.90–509.00) mL/min/1.73 m²; measured creatinine clearance by 24-hour timed urine collection (mCrCl_{24h}) of 88.00 (23.00–217.00) mL/min/1.73 m²; serum creatinine (SCr) of 0.63 (0.23–2.14) mg/dL; and total body weight (TBW) of 69.5 (35–116) kg. ^b PKPD target defined as a 100% *fT*>1×MIC corresponding to a concentration of unbound meropenem throughout a dosing interval above 2 mg at the EUCAST breakpoint for susceptibility of 2 mg/L for Enterobacterales [68,278]. ^c The dispensed daily amount is equal to the prescribed daily dose. ^d Cost analysis according to the Belgian Information Center on Pharmacotherapy limited to meropenem with sustained stability during continuous infusion [308,309]. ^e P-values were statistically significant. C_{ss,trough/plateau}, steady state trough or plateau concentration; PI, prediction interval

3.4 Interactive simulation tool

We developed an interactive tool that allows end-users to explore more meropenem dosing strategies within the range of covariate values observed in the patient dataset. The tool is accessible at <https://lpmx.shinyapps.io/meropenem/>. In the sidebar panel on the left, the user can choose (i) the popPK model with the kidney function marker of interest, (ii) covariates (iii) dosing details, (iv) exposure and safety targets, and (v) the simulation settings (**Figure 12**). After clicking the “*Simulate*” button, the corresponding output is generated and visualized in the main panel on the right. This includes plots of simulated meropenem concentrations and PTA, which are displayed against the dosing interval.

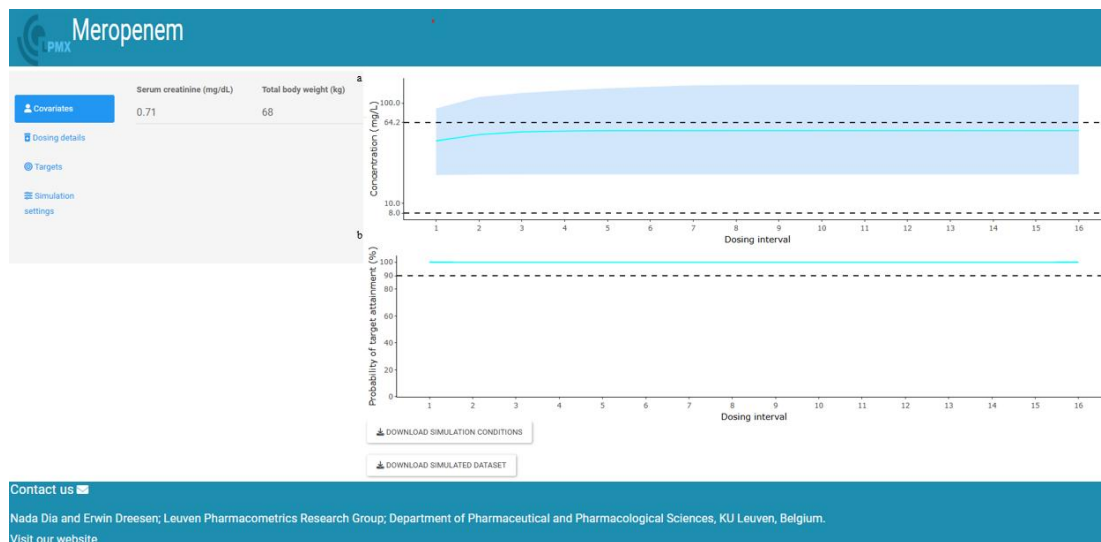


Figure 12 Snapshot of the user interface of the simulation software tool. On the left side, the user gives value for the covariates, dosing details, targets, and simulation settings. An output of simulated dataset is generated based on the user input. By default, the output is visualized as (a) simulated meropenem concentration, and (b) probability of target attainment over a dosing interval. In (a), the blue line within the blue-shaded area represents the median concentration within the 95% prediction interval. The black dashed lines indicate safety and exposure targets. In (b), the blue and black lines represent the calculated probability of target attainment and the user-selected probability of target attainment, respectively. LPMX, KU Leuven Pharmacometrics research group

The default example case in the software tool considers a PKPD target defined as $100\% fT > 4 \times \text{MIC}$ at the EUCAST susceptibility breakpoint of 2 mg/L for Enterobacterales, which corresponds to a concentration of unbound meropenem throughout a dosing interval above 8 mg/L in a patient with SCr of 0.71 mg/dL and total body weight of 68 kg (**Supplementary Table 19**). Under these definitions and through incremental increases from the proposed dose targeting a concentration of unbound meropenem throughout a dosing interval above 2 mg/L, a PTA $\geq 90\%$ was achieved with a continuous meropenem infusion of 4,000 mg over 6 hours every 6 h (**Supplementary Figures 24–26**). This dosing strategy has therefore been set as default setting strategy (**Supplementary Table 19**). Additionally, the tool was cross validated against NONMEM (**Supplementary Figure 27**).

4. Discussion

Our popPK simulation study indicates that the SCr and total body weight in power function were the main drivers of meropenem CL. Additionally, it indicates that, in critically ill patients with preserved (*i.e.*, $\text{eGFR}_{\text{CKD-EPI}} \geq 70 \text{ mL/min/1.73 m}^2$ on the day of PK sampling) or ARC (*i.e.*, $\text{mCrCl}_{24\text{h}} \geq 130 \text{ mL/min/1.73 m}^2$), a continuous intravenous infusion of meropenem with a total daily dose of 4,000 mg administered at a rate of 1,000 mg over six hours provides at least a 90% probability of

maintaining meropenem concentrations one time above 2 mg/L ($100\% fT > 1 \times \text{MIC}$) at the EUCAST breakpoint for susceptibility of 2 mg/L for Enterobacterales [310].

Numerous popPK models of meropenem in critically ill patients have been published. These models mainly focused on patients with renal impairment and mechanical/respiratory support [278,280,283–293]. Very few were specific to patients with preserved renal function or ARC [294,310]. Seventeen out of 58 patients (29%) in our clinical dataset had ARC [310]. Although we only relied on data from intermittent meropenem infusions, previous work has shown that meropenem popPK models parameterized using intermittent-infusion data can reliably reproduce concentration–time profiles under continuous infusion. This is because continuous-infusion profiles are determined by PK parameters such as CL, which are fully identifiable from intermittent-infusion data, thereby supporting the use of our model to simulate continuous infusion [280].

TDM of meropenem has been suggested [311]. However, it is not widely available in clinical routine at the ICU (*i.e.*, 39%) [269]. Additionally, it can't help reaching a rapid and timely target attainment within the first 24 hours of sepsis, jeopardizing the survival rate of patients [312]. Therefore, we provided an optimized dosing strategy that

ensures PKPD target attainment within the first 24 hours while minimizing toxicity risks based on safety metrics.

Relying on an estimated or measured kidney function marker to guide dosing strategies is cumbersome as it is based on a set of patient characteristics that may not always be readily available (*e.g.*, height), accurately reported (*e.g.*, race), or quickly obtained (*e.g.*, UCr_{24h}). Our study showed that two easily measured patient characteristics drive the meropenem CL: the SCr and total body weight. Therefore, we call for a shift in focus away from using estimated kidney function markers. Instead, relying solely on the measured values of SCr and total body weight is both sufficient and straightforward.

Several clinical trials, and more recently also the meta-analyses including the results of the BLING III RCT, indicated the clinical benefits of continuous infusions compared to intermittent dosing of meropenem. A decrease in mortality and an increase in clinical cure was reported [313–315]. However, findings are inconsistent [285,302,309,316]. This inconsistency may be due to differences in patient populations (ICU versus non-ICU, need for RRT versus exclusion of RRT) and the timing of measuring clinical outcomes (short-term versus long-term benefits). Although our study did not investigate clinical outcomes, we conducted a simulation study that demonstrated the superiority of continuous infusion in terms of PTA. Chemical stability remains a major concern

with the continuous infusion. In the United Kingdom, stability is maintained for up to 12 hours [302]. However, more studies demonstrate meropenem stability for only six hours [317,318]. Based on these findings, a 1,000 mg dose of meropenem administered over six hours every six hours via continuous infusion is feasible. Additionally, it is unlikely to result in nephrotoxicity. While the recommended daily dose (4,000 mg) may be more costly than the standard daily dose (3,000 mg), overall savings are expected due to shorter hospital stays and the lack of need for TDM. However, these savings may vary between countries.

Several dosing strategies were evaluated in our simulation study, including dosing strategies administered in the BLING III trial for which the PTA analysis from the PK substudy has not yet been published [302]. BLING III demonstrated a clinical benefit – a reduction in the 90-day mortality rate – for the continuous dosing strategy over the intermittent standard dosing strategy only in the subsequent meta-analysis, but not in the original trial, despite the inclusion of over 7,000 patients. One explanation for this smaller-than-anticipated difference is the lack of meaningful differences in target attainment (*i.e.*, $\geq 90\%$) between the dosing strategies. We are therefore particularly interested in the forthcoming PK substudy results, especially in patients with preserved renal function or ARC. The recommended body weight–

based dosing strategy in children was also investigated because some popPK models identified a body weight–related parameter without the kidney function marker as a covariate [281,301,310].

An interactive tool was developed to allow users to simulate what if scenarios. For meropenem, several definitions of efficacy target attainment exist 100% $fT>1\times MIC$ and 100% $fT>4\times MIC$ as well as 40% $fT>1\times MIC$. Moreover, the EUCAST clinical breakpoints vary by pathogen and are set to reflect the most challenging clinical situations [68]. Specifically in the ICU, high PKPD targets and MIC values were considered to improve the clinical outcome [319]. In our simulation study, the EUCAST breakpoint for susceptibility of common bacteria was considered given that the MIC values are not available before the third day of culture.

Our study has several limitations. First, an external validation was not performed for the selected model, which includes SCr and total body weight as covariates. However, it is a model based on rich sampling of patients with preserved renal function or ARC rarely represented in clinical trials. Furthermore, we have provided the code for the selected model in the appendix to allow external evaluation. Second, in this manuscript, we only investigated eight dosing strategies. However, we developed an interactive tool to allow users to explore meropenem dosing for any efficacy-based PKPD target. Third, prospective

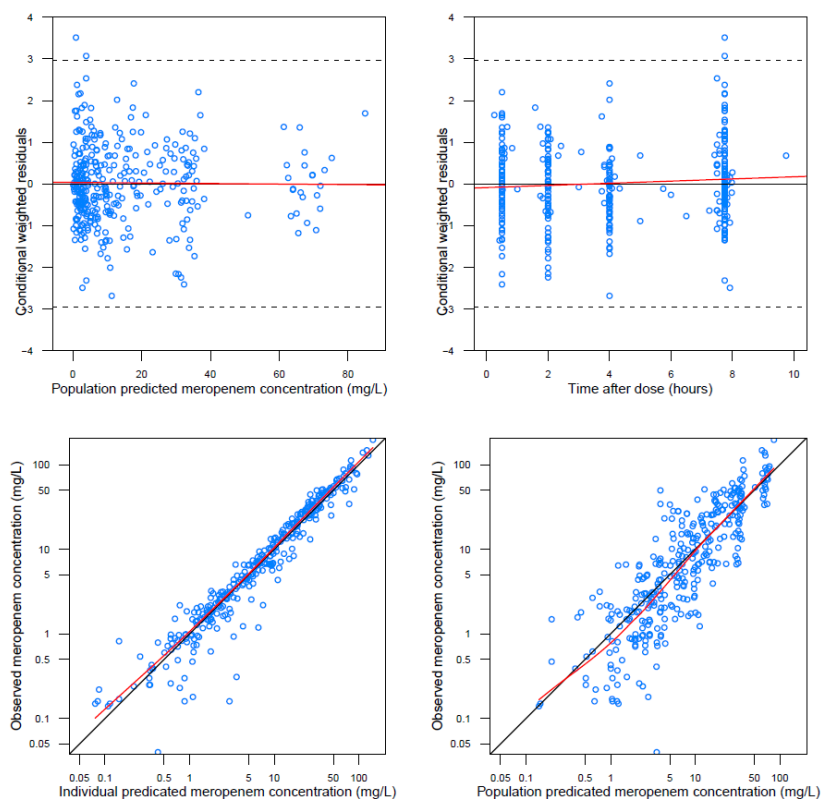
evaluation of our suggested dosing strategy in a clinical trial is needed. This is planned to be conducted at our hospital and can be of added value to the ongoing EMPRESS trial [320].

In conclusion, our simulation study advocates for the use of SCr and total body weight to predict meropenem CL rather than the estimation of kidney function markers. Additionally, our study supports an optimized six-hourly continuous intravenous infusion of 1,000 mg meropenem (total daily dose of 4,000 mg) in critically ill septic patients with preserved renal function or ARC. The recommended dosing strategy ensures a 100% $fT>1\times\text{MIC}$ target attainment at a EUCAST breakpoint for susceptibility of 2 mg/L for Enterobacterales from day 1 of treatment with a minimal risk of toxicity.

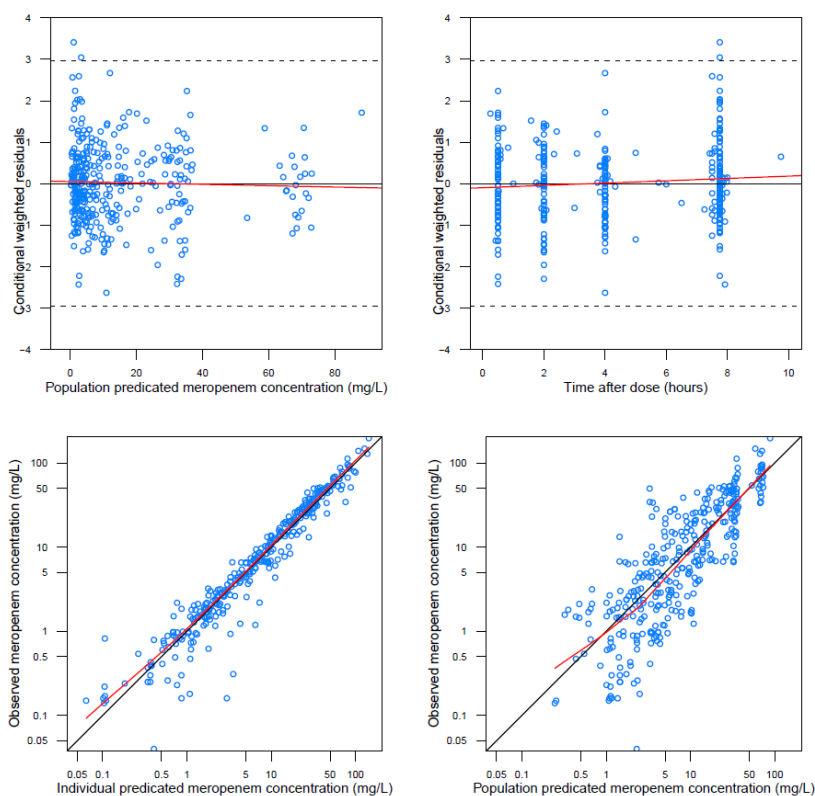
Appendix C

	Age	TBW	Height	SCr	UCr _{24h}	Sex	Race	eCrCL _{CG}	eGFR _{MDRD}	eGFR _{CKD-EPI}	mCrCL _{24h}
Age		0.01	0.14	0.09	-0.04	0.08	-0.19	-0.44	-0.29	-0.50	-0.29
TBW	0.01		0.38	0.28	0.47	0.15	-0.01	0.24	-0.16	-0.24	0.09
Height	0.14	0.38		-0.05	0.43	0.55	-0.08	0.34	0.23	0.15	0.42
SCr	0.09	0.28	-0.05		0.32	0.05	-0.02	-0.58	-0.71	-0.84	-0.40
UCr _{24h}	-0.04	0.47	0.43	0.32		0.44	-0.03	-0.07	-0.29	-0.19	0.59
Sex	0.08	0.15	0.55	0.05	0.44		-0.08	0.11	0.12	0.15	0.30
Race	-0.19	-0.01	-0.08	-0.02	-0.03	-0.08		-0.03	0.03	0.04	-0.01
eCrCL _{CG}	-0.44	0.24	0.34	-0.58	-0.07	0.11	-0.03		0.87	0.79	0.53
eGFR _{MDRD}	-0.29	-0.16	0.23	-0.71	-0.29	0.12	0.03	0.87		0.86	0.47
eGFR _{CKD-EPI}	-0.50	-0.24	0.15	0.84	-0.19	0.15	0.04	0.79	0.86		0.56
mCrCL _{24h}	-0.29	0.09	0.42	-0.40	0.59	0.30	-0.01	0.53	0.47	0.56	

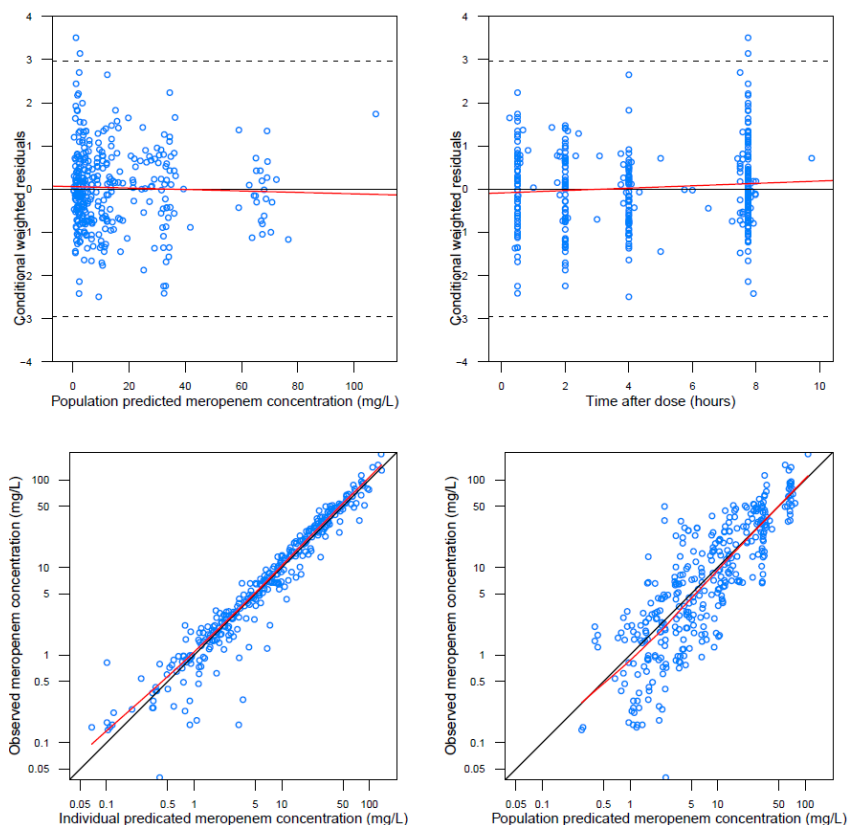
Supplementary Figure 9 Covariance matrix of the clinical dataset. eCrCL_{CG}, estimated creatinine clearance calculated using the Cockcroft–Gault equation; eGFR_{CKD-EPI}, estimated glomerular filtration rate calculated using the Chronic Kidney Disease Epidemiology Collaboration equation; eGFR_{MDRD}, estimated glomerular filtration rate calculated using the Modification of Diet in Renal Disease equation; mCrCL_{24h}, measured creatinine clearance by 24-hour timed urine collection; SCr, serum creatinine; TBW, total body weight; UCr_{24h}, 24-hour measured urinary creatinine



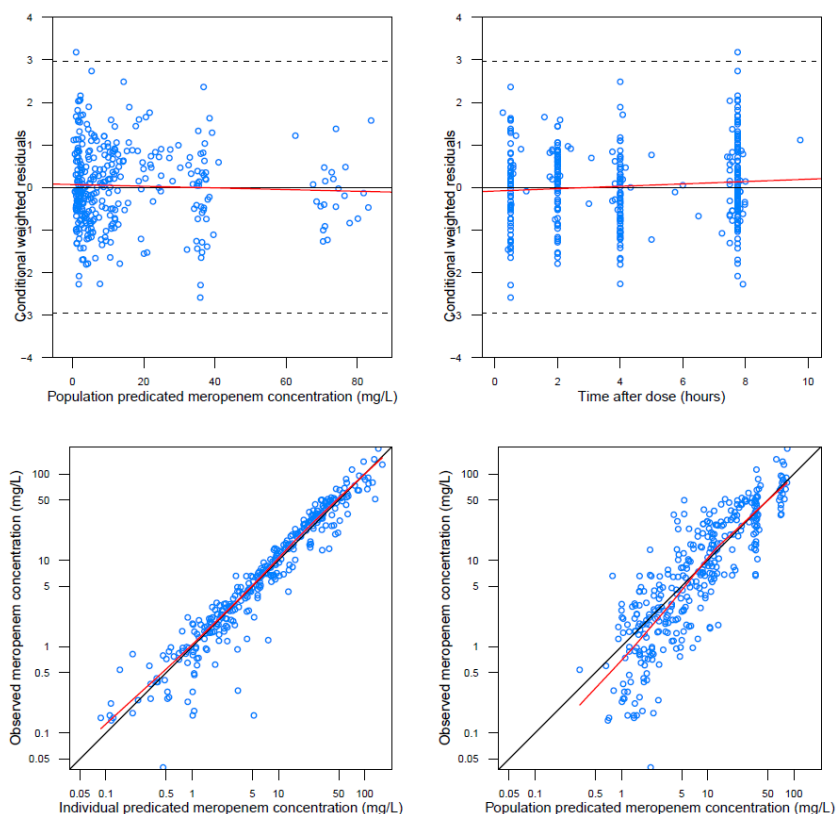
Supplementary Figure 10 Goodness-of-fit plots of the model with estimated creatinine clearance calculated using the Cockcroft–Gault equation (a) Conditional weighted residuals (CWRES) versus population predicted (PRED) meropenem concentration, (b) CWRES versus time after dose, (c) observed versus individual predicted meropenem concentration, and (d) observed versus PRED meropenem concentration. Observed-versus-predicted concentration plots are presented on a logarithmic scale. Solid black lines and solid red lines represent lines of identity and linear regression lines, respectively



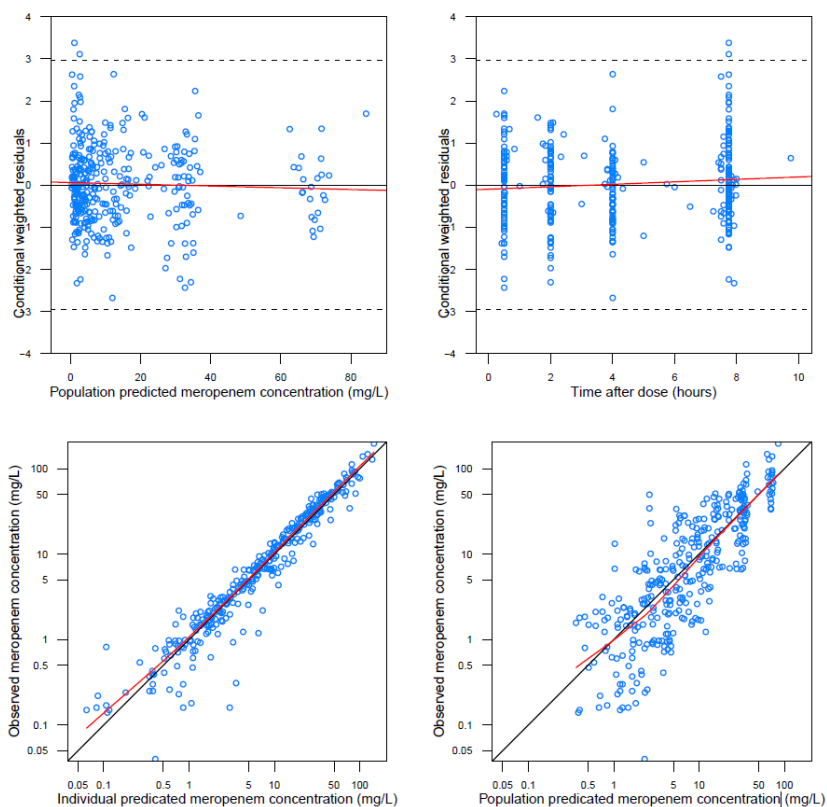
Supplementary Figure 11 Goodness-of-fit plots of the model with estimated glomerular filtration rate calculated using the Modification of Diet in Renal Disease equation (a) Conditional weighted residuals (CWRES) versus population predicted (PRED) meropenem concentration, (b) CWRES versus time after dose, (c) observed versus individual predicted meropenem concentration, and (d) observed versus PRED meropenem concentration. Observed-versus-predicted concentration plots are presented on a logarithmic scale. Solid black lines and solid red lines represent lines of identity and linear regression lines, respectively



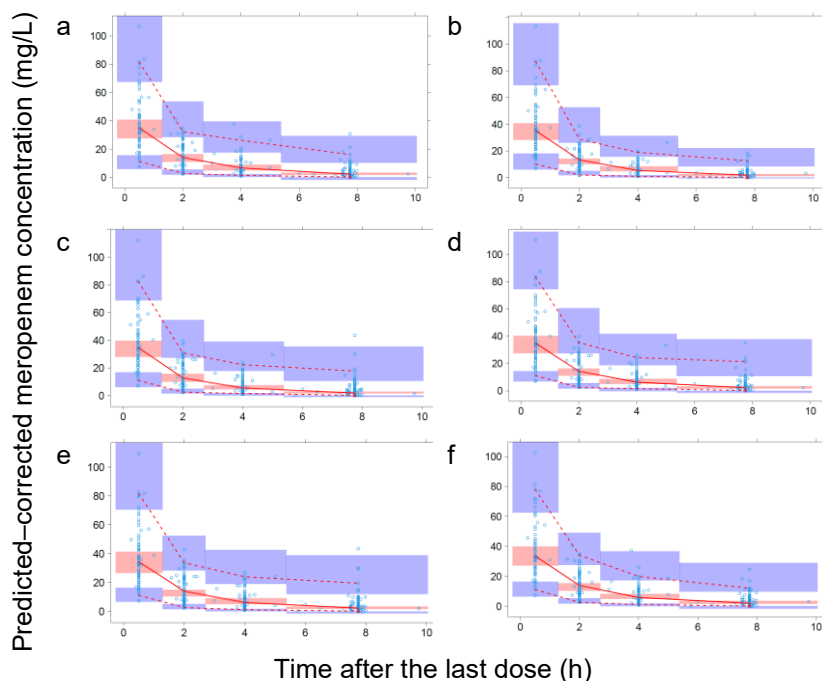
Supplementary Figure 12 Goodness-of-fit plots of the model with estimated glomerular filtration rate calculated using the Chronic Kidney Disease Epidemiology Collaboration equation (a) Conditional weighted residuals (CWRES) versus population predicted (PRED) meropenem concentration, (b) CWRES versus time after dose, (c) observed versus individual predicted meropenem concentration, and (d) observed versus PRED meropenem concentration. Observed-versus-predicted concentration plots are presented on a logarithmic scale. Solid black lines and solid red lines represent lines of identity and linear regression lines, respectively



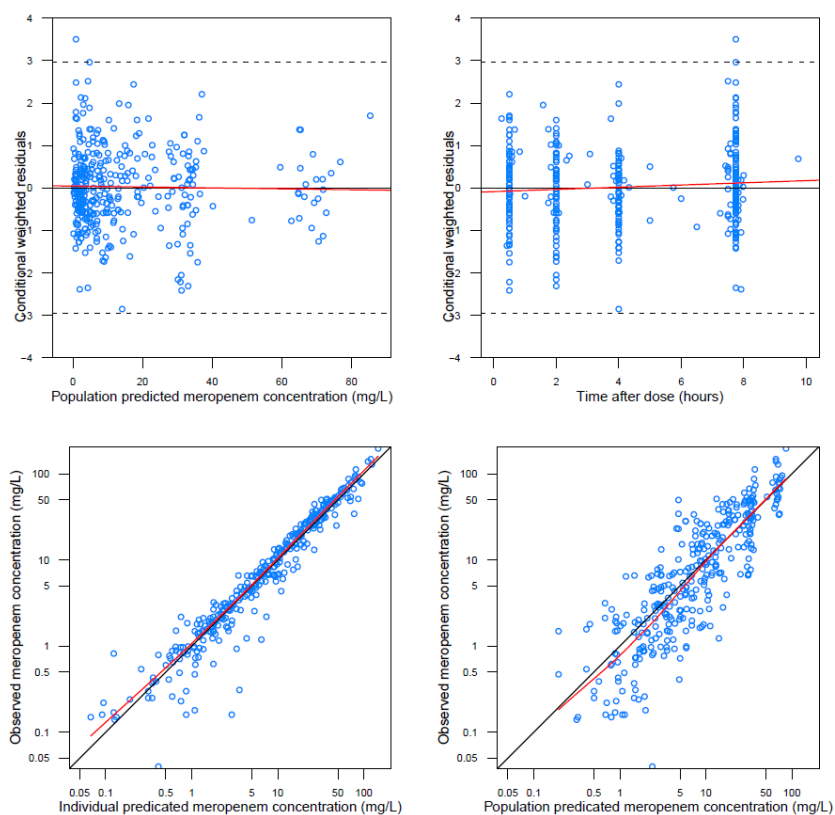
Supplementary Figure 13 Goodness-of-fit plots of the model with measured creatinine clearance by 24-hour timed urine collection (a) Conditional weighted residuals (CWRES) versus population predicted (PRED) meropenem concentration, (b) CWRES versus time after dose, (c) observed versus individual predicted meropenem concentration, and (d) observed versus PRED meropenem concentration. Observed-versus-predicted concentration plots are presented on a logarithmic scale. Solid black lines and solid red lines represent lines of identity and linear regression lines, respectively



Supplementary Figure 14 Goodness-of-fit plots of the model with standalone serum creatinine (a) Conditional weighted residuals (CWRES) versus population predicted (PRED) meropenem concentration, (b) CWRES versus time after dose, (c) observed versus individual predicted meropenem concentration, and (d) observed versus PRED meropenem concentration. Observed-versus-predicted concentration plots are presented on a logarithmic scale. Solid black lines and solid red lines represent lines of identity and linear regression lines, respectively



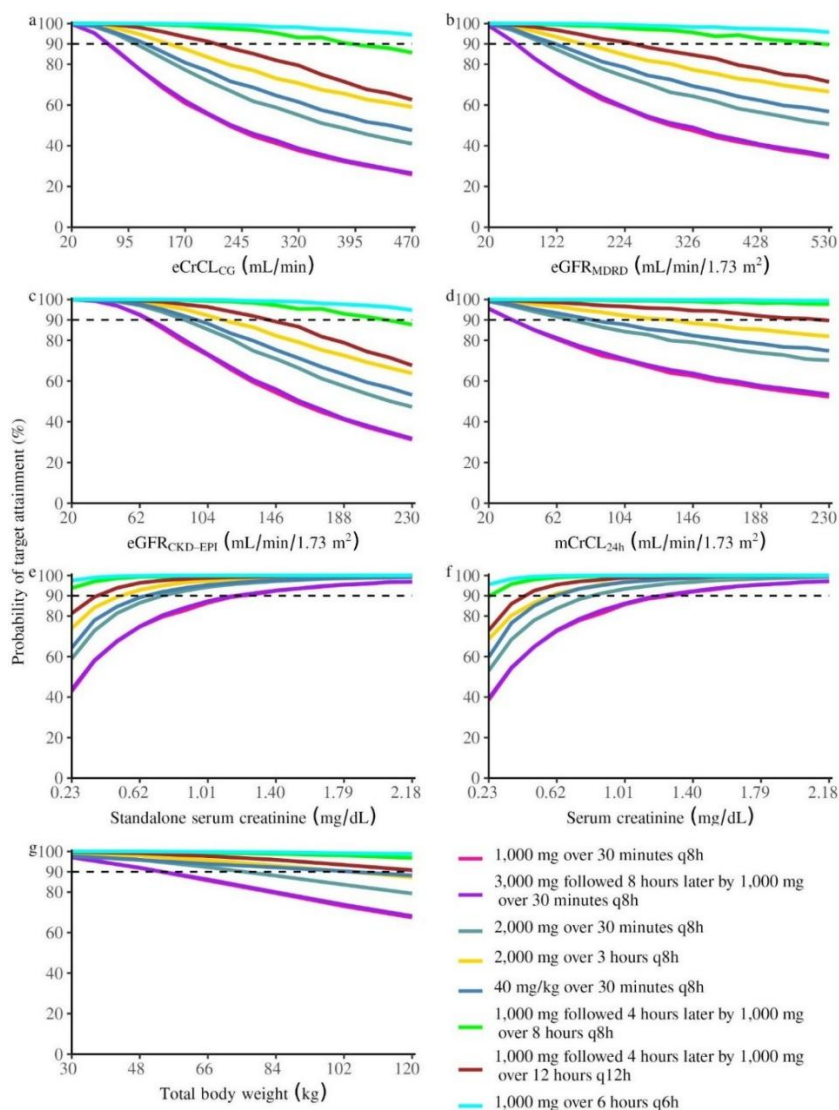
Supplementary Figure 15 Prediction-corrected visual predictive check of the model with (a) estimated creatinine clearance calculated using the Cockcroft–Gault equation, (b) estimated glomerular filtration rate calculated using the Modification of Diet in Renal Disease equation, (c) estimated glomerular filtration rate calculated using the Chronic Kidney Disease Epidemiology Collaboration equation, (d) measured creatinine clearance by 24-hour timed urine collection, (e) standalone serum creatinine, and (f) serum creatinine and total body weight. The observed prediction-corrected meropenem concentrations are represented by circles. The red solid and the two dashed lines represent the median, 2.5th and 97.5th percentile of the prediction-corrected observations, respectively. The shaded areas indicate 95% prediction intervals of the median and 2.5th and 97.5th percentile of the simulated values



Supplementary Figure 16 Goodness-of-fit plots of the model with serum creatinine and total body weight (a) Conditional weighted residuals (CWRES) versus population predicted (PRED) meropenem concentration, (b) CWRES versus time after dose, (c) observed versus individual predicted meropenem concentration, and (d) observed versus PRED meropenem concentration. Observed-versus-predicted concentration plots are presented on a logarithmic scale. Solid black lines and solid red lines represent lines of identity and linear regression lines, respectively

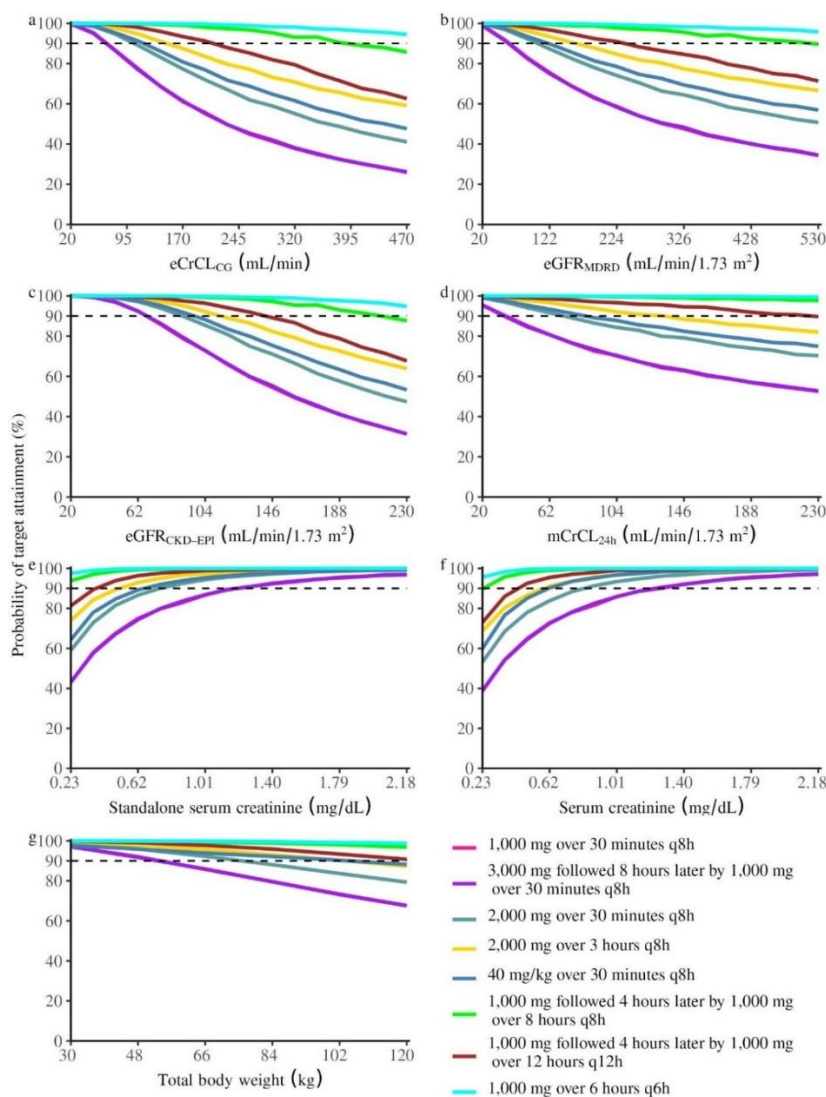
	eCrCL _{CG}	eGFR _{MDRD}	eGFR _{CKD-EPI}	mCrCL _{24h}	SCr
eCrCL _{CG}		0.79	0.76	0.15	0.73
eGFR _{MDRD}	0.79		0.94	0.23	0.86
eGFR _{CKD-EPI}	0.78	0.94		0.24	0.94
mCrCL _{24h}	0.15	0.23	0.24		0.24
SCr	0.73	0.86	0.94	0.24	

Supplementary Figure 17 Correlation matrix of the probabilities of target attainment at the end of day 1 of treatment, based on the kidney function markers estimated from the virtual patient datasets containing all the patient characteristics. eCrCL_{CG}, estimated creatinine clearance calculated using the Cockcroft–Gault equation; eGFR_{CKD-EPI}, estimated glomerular filtration rate calculated using the Chronic Kidney Disease Epidemiology Collaboration equation; eGFR_{MDRD}, estimated glomerular filtration rate calculated using the Modification of Diet in Renal Disease equation; mCrCL_{24h}, measured creatinine clearance by 24-hour timed urine collection; SCr, serum creatinine



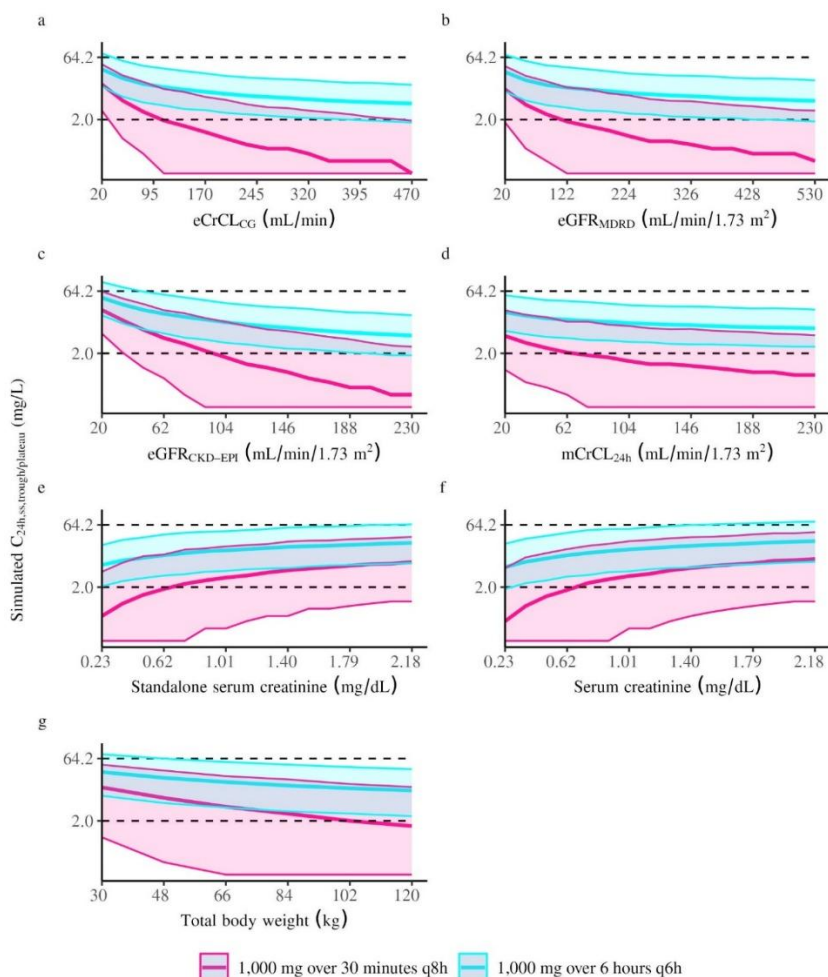
Supplementary Figure 18 Probability of target attainment on the end of day 1 of treatment across the range of (a) the estimated creatinine clearance using the Cockcroft–Gault equation (eCrCl_{CG}), (b) the estimated glomerular filtration rate using the Modification of Diet in Renal Disease equation (eGFR_{MDRD}), (c) the estimated glomerular filtration rate using the Chronic Kidney Disease Epidemiology Collaboration equation (eGFR_{CKD-EPI}), (d) the measured creatinine clearance by 24-hour timed urine collection (mCrCl_{24h}), (e) standalone

serum creatinine, (f) serum creatinine, and (g) total body weight at a susceptibility breakpoint of 2 mg/L for Enterobacterales, targeting a concentration of unbound meropenem throughout a dosing interval above 2 mg/L for eight dosing strategies. The dashed horizontal line marks a 90% probability of target attainment. q_xh, every x hours



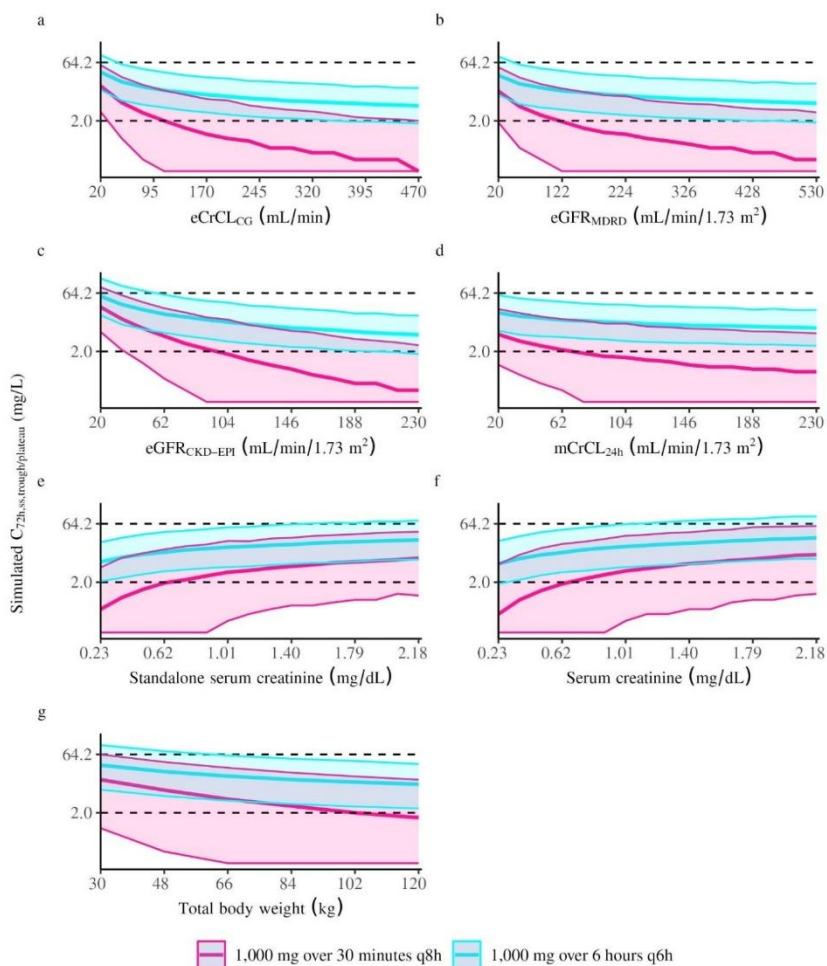
Supplementary Figure 19 Probability of target attainment on the end of day 3 of treatment across the range of (a) the estimated creatinine clearance according to the Cockcroft–Gault equation (eCrCL_{CG}), (b) the estimated glomerular filtration rate using the Modification of Diet in Renal Disease equation (eGFR_{MDRD}), (c) the estimated glomerular filtration rate using the Chronic Kidney Disease Epidemiology Collaboration equation (eGFR_{CKD-EPI}), (d) the measured creatinine clearance by 24-hour timed urine collection (mCrCL_{24h}), (e) standalone

serum creatinine, (f) serum creatinine, and (g) total body weight at a susceptibility breakpoint of 2 mg/L for Enterobacterales, targeting a concentration of unbound meropenem throughout a dosing interval above 2 mg/L for eight dosing strategies including the 1,000 mg standard dose administered over 30 minutes every 8 hours. The dashed horizontal line marks a 90% probability of target attainment. q_{xh}, every x hours



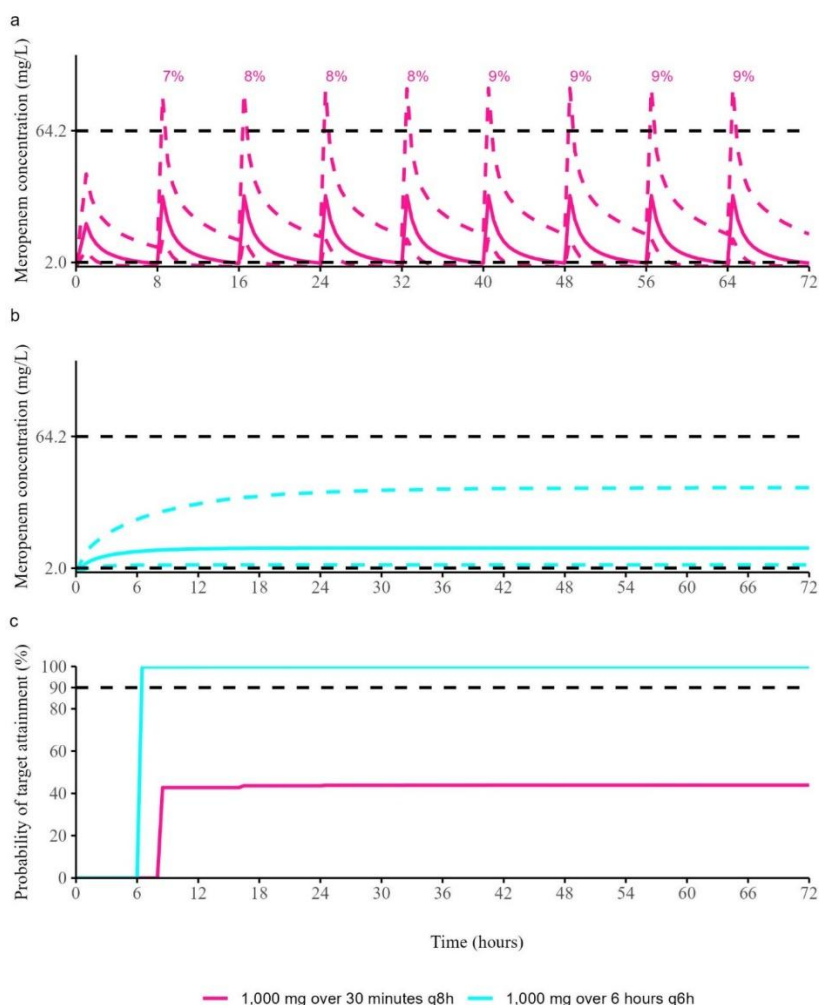
Supplementary Figure 20 Simulated 24-hour steady state concentration across the range of (a) the estimated creatinine clearance according to the Cockcroft–Gault equation ($eCrCl_{CG}$), (b) the estimated glomerular filtration rate using the Modification of Diet in Renal Disease equation ($eGFR_{MDRD}$), (c) the estimated glomerular filtration rate using the Chronic Kidney Disease Epidemiology Collaboration equation ($eGFR_{CKD-EPI}$), (d) the measured creatinine clearance by 24-hour timed urine collection ($mCrCl_{24h}$), (e) standalone serum creatinine, (f) serum creatinine, and (g) total body weight at a susceptibility breakpoint of 2 mg/L for Enterobacterales, targeting a concentration of unbound meropenem throughout a dosing interval above 2 mg/L for the 1,000 mg standard dose administered over 30

minutes every eight hours and the 1,000 mg optimized dose administered over six hours every six hours. The dashed horizontal lines represent a concentration of 2 mg/L, and 64.2 mg/L. The solid lines within the shaded area represent the median concentration and the 95% prediction interval. $C_{24h,ss,trough/plateau}$, 24 hours steady state trough or plateau concentration; q_{xh}, every x hours



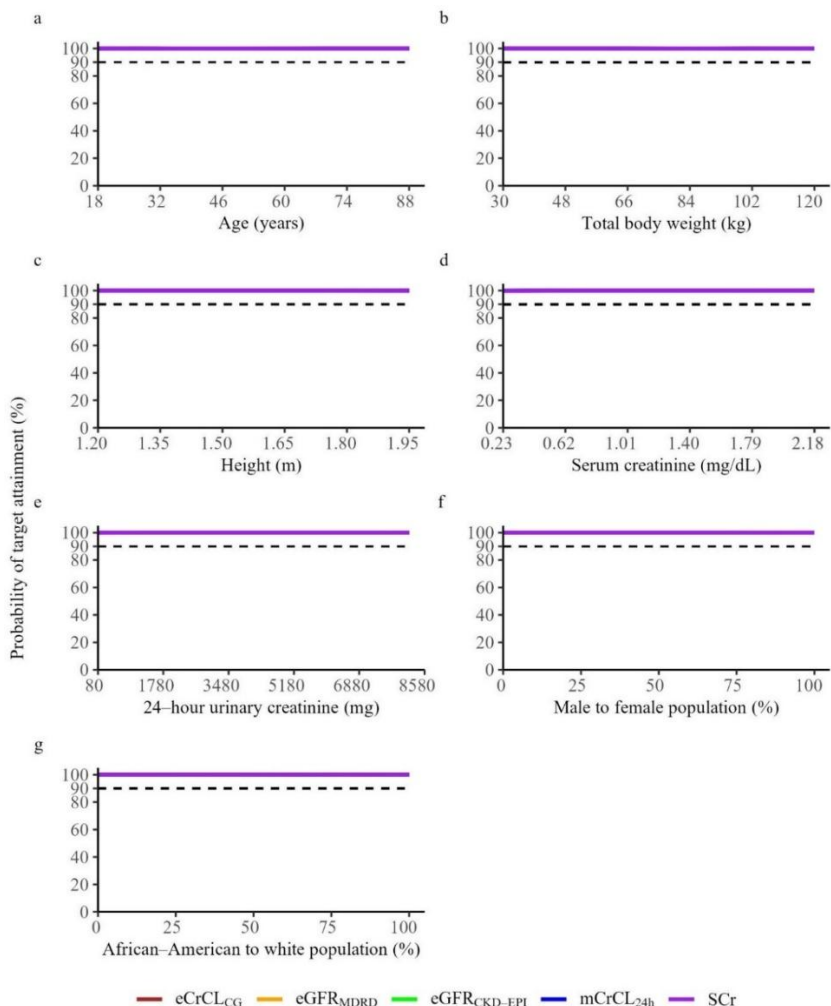
Supplementary Figure 21 Simulated 72-hour steady state concentration across the range of (a) the estimated creatinine clearance according to the Cockcroft–Gault equation ($eCrCl_{CG}$), (b) the estimated glomerular filtration rate using the Modification of Diet in Renal Disease equation ($eGFR_{MDRD}$), (c) the estimated glomerular filtration rate using the Chronic Kidney Disease Epidemiology Collaboration equation ($eGFR_{CKD-EPI}$), (d) the measured creatinine clearance by 24-hour timed urine collection ($mCrCl_{24h}$), (e) standalone serum creatinine, (f) serum creatinine, and (g) total body weight at a susceptibility breakpoint of 2 mg/L for Enterobacteriales, targeting a concentration of unbound meropenem throughout a dosing interval above 2 mg/L for the 1,000 mg standard dose administered over 30

minutes every eight hours and the 1,000 mg optimized dose administered over six hours every six hours. The dashed horizontal lines represent a concentration of 2 mg/L, and 64.2 mg/L. The solid non-black lines represent the median concentration for dosing strategies. $C_{72h,ss,trough/plateau}$, 72 hours steady state trough or plateau concentration; q_{xh}, every x hours



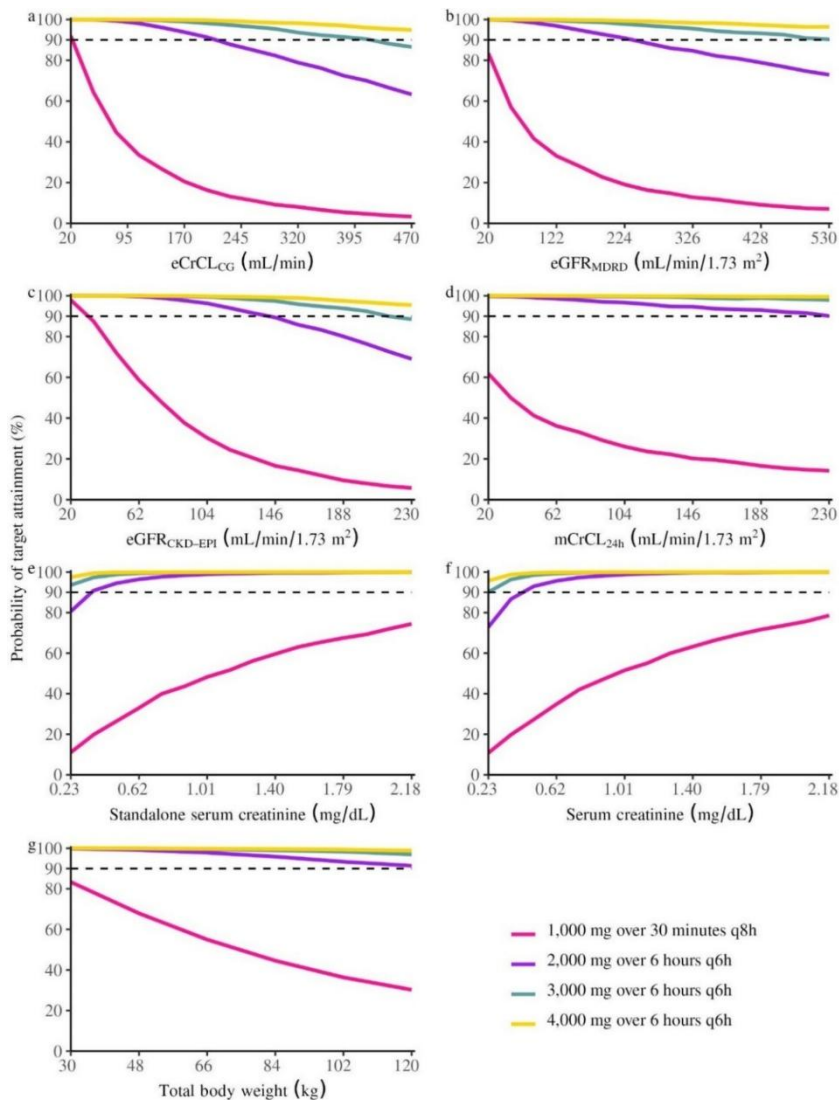
Supplementary Figure 22 Comparison of (a–b) simulated meropenem concentrations and (c) probability of target attainment over 72 hours of treatment using the population pharmacokinetic model with serum creatinine, and total body weight at a susceptibility breakpoint of 2 mg/L for Enterobacterales, targeting a concentration of unbound meropenem throughout a dosing interval above 2 mg/L for the 1,000 mg standard dose administered over 30 minutes every eight hours and the 1,000 mg optimized dose administered over six hours every six hours. The percentages in pink represent the percentages of patients with meropenem concentration ≥ 64.2 mg/L. The dashed black horizontal lines represent a concentration of 2 mg/L, a concentration of 64.2 mg/L,

and a PTA of 90%. The non-black solid and the two dashed lines represent the median, 2.5th and 97.5th percentile of the meropenem concentrations. q_{xh}, every x hours



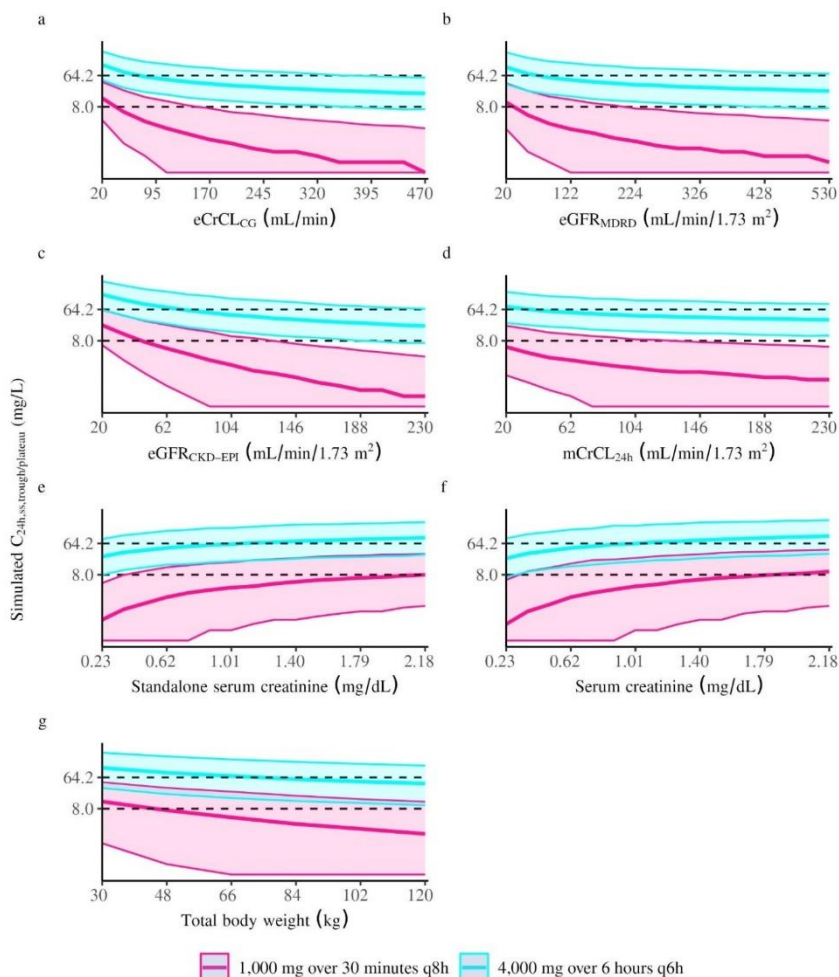
Supplementary Figure 23 Probability of target attainment (PTA) on the end of day 1 of treatment in five population pharmacokinetic models with different kidney function markers across the range of (a) age, (b) total body weight, (c) height, (d) serum creatinine, (e) 24-hour measured urinary creatinine, (f) male to female population, and (g) African-American to white population at a susceptibility breakpoint of 2 mg/L for Enterobacterales, targeting a concentration of unbound meropenem throughout a dosing interval above 2 mg/L with 1,000 mg optimized dose administered over six hours every six hours. The PTA was $\geq 90\%$ for all models, resulting in substantial overlap of the curves.

The dashed horizontal line marks 90% PTA. $eCrCl_{CG}$, estimated creatinine clearance according to the Cockcroft–Gault equation; $eGFR_{CKD-EPI}$, estimated glomerular filtration rate using the Chronic Kidney Disease Epidemiology Collaboration equation; $eGFR_{MDRD}$, estimated glomerular filtration rate using the Modification of Diet in Renal Disease equation; $mCrCl_{24h}$, measured creatinine clearance by 24-hour timed urine collection; SCr , serum creatinine



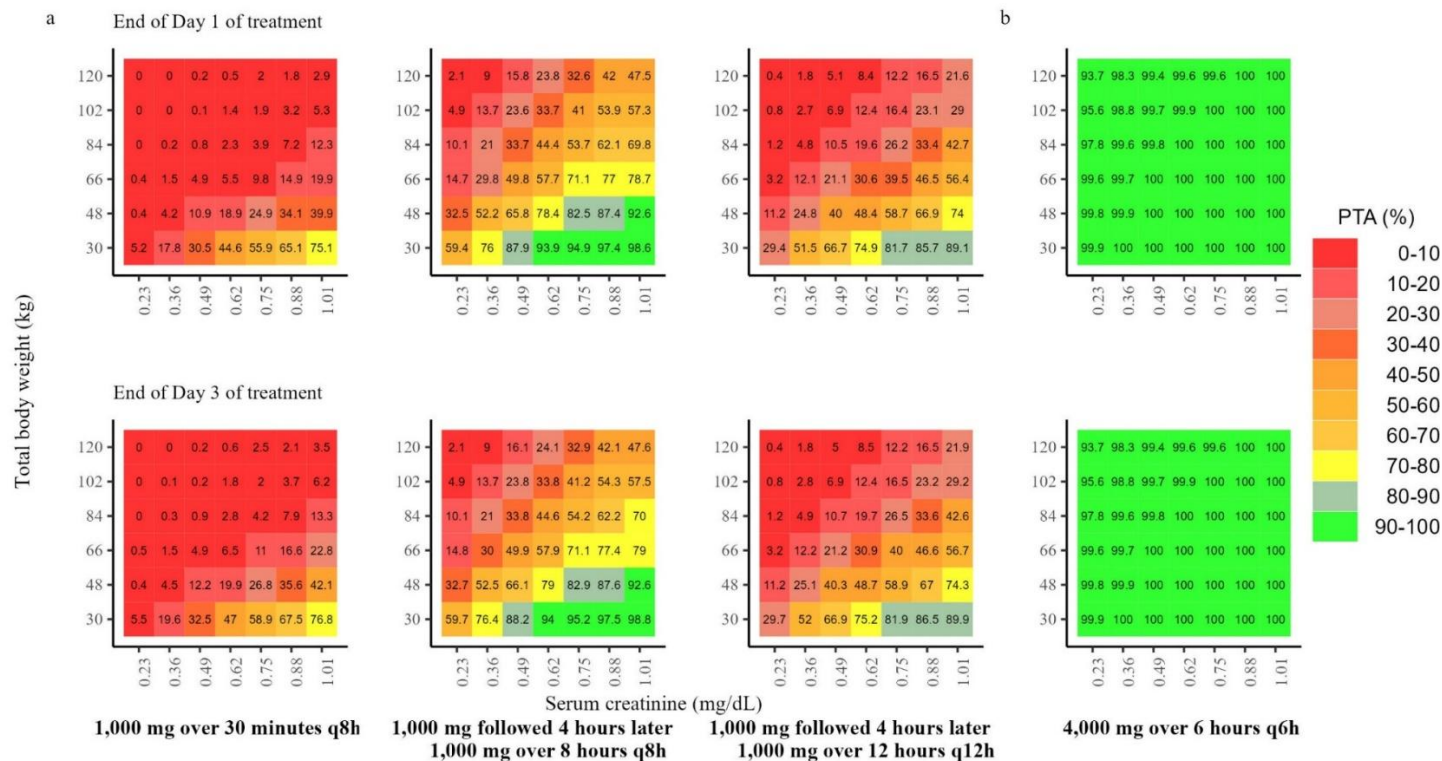
Supplementary Figure 24 Probability of target attainment on the end of day 1 of treatment across the range of (a) the estimated creatinine clearance according to the Cockcroft–Gault equation (eCrCl_{CG}), (b) the estimated glomerular filtration rate using the Modification of Diet in Renal Disease equation (eGFR_{MDRD}), (c) the estimated glomerular filtration rate using the Chronic Kidney Disease Epidemiology Collaboration equation (eGFR_{CKD-EPI}), (d) the measured creatinine clearance by 24-hour timed urine collection (mCrCl_{24h}), (e) standalone

serum creatinine, (f) serum creatinine, and (g) total body weight at a susceptibility breakpoint of 2 mg/L for Enterobacterales, targeting a concentration of unbound meropenem throughout a dosing interval above 8 mg/L for four dosing strategies including the 1,000 mg standard dose administered over 30 minutes every 8 hours. The dashed horizontal line marks a 90% probability of target attainment. q_{xh}, every x hours

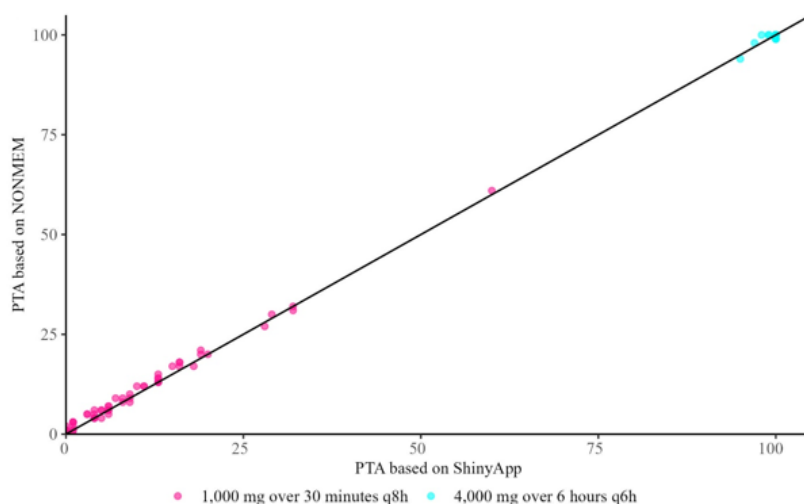


Supplementary Figure 25 Simulated 24-hour steady state concentration across the range of (a) the estimated creatinine clearance according to the Cockcroft–Gault equation ($eCrCL_{CG}$), (b) the estimated glomerular filtration rate using the Modification of Diet in Renal Disease equation ($eGFR_{MDRD}$), (c) the estimated glomerular filtration rate using the Chronic Kidney Disease Epidemiology Collaboration equation ($eGFR_{CKD-EPI}$), (d) the measured creatinine clearance by 24-hour timed urine collection ($mCrCL_{24h}$), (e) standalone serum creatinine, (f) serum creatinine, and (g) total body weight at a susceptibility breakpoint of 2 mg/L for Enterobacteriales, targeting a concentration of unbound meropenem throughout a dosing interval above 8 mg/L for the 1,000 mg standard dose administered over 30

minutes every eight hours and the 4,000 mg optimized dose administered over six hours every six hours. The dashed horizontal lines represent a concentration of 8 mg/L, and 64.2 mg/L. The solid lines within the shaded area represent the median concentration and the 95% prediction interval. $C_{24h,ss,trough/plateau}$, 24 hours steady state trough or plateau concentration; q_{xh}, every x hours



Supplementary Figure 26 Heat map showing the probability of target attainment in the selected population pharmacokinetic model with two patients characteristics across a serum creatinine range of 0.23–1.01 mg/dL with stepwise of 0.13 mg/dL and total body weight of 30–120 kg with stepwise of 18 kg at the end of (a) day 1, and (b) day 3 of treatment at a susceptibility breakpoint of 2 mg/L for Enterobacterales, targeting a concentration of unbound meropenem throughout a dosing interval above 8 mg/L, with four dosing strategies including the 1,000 mg standard dose administered over 30 minutes every 8 hours (left), two dosing strategies administered in the BLING III trial (middle), and the 4,000 mg optimized dose administered over six hours every six hours (right). PTA, probability of target attainment; q_{xh}, every x hours



Supplementary Figure 27 Agreement of the probability of target attainment (PTA) between NONMEM and the interactive ShinyApp tool in scatter plot for the real-world clinical dataset of 58 patients with preserved or augmented renal clearance, using the model with serum creatinine and total body weight, for the 1,000 mg standard dose administered over 30 minutes every eight hours and the 4,000 mg optimized dose administered over six hours every six hours, targeting a 100% concentration of unbound meropenem four times above the EUCAST breakpoint for susceptibility of 2 mg/L for Enterobacterales, which corresponds to a concentration of unbound meropenem throughout a dosing interval above 8 mg/L, on the end of day 1 of treatment. The solid diagonal line represents the identity line

Supplementary Table 15 Baseline patient characteristics and kidney function markers, unless otherwise stated. Eligibility was based on kidney function at the time of pharmacokinetic sampling

	Units	Median (minimum–maximum) or n (%)
Patient characteristics		
Age	year	63 (23–83)
Total body weight	kg	69.5 (35–116)
Body mass index	kg/m ²	23.45 (15.80–45.30)
Body mass index category		
Underweight		8 (13.79)
Normal weight		32 (55.17)
Pre-obese		9 (15.52)
Obese		9 (15.52)
Height ^a	cm	170.05 (130.00–187.00)
Serum creatinine	mg/dL	0.63 (0.23–2.14)
24-hour measured urinary creatinine ^b	g	2.67 (0.97–8.05)
Sex assigned at birth		
Males		40 (68.97)
Females		18 (31.03)
Race ^c		
African		2 (3.45)
American		56 (96.55)
White		
Meropenem dose		
500 mg		1 (1.72)
1,000 mg		38 (65.52)
2,000 mg		19 (32.76)
Kidney function markers^d		
eCrCL _{CG}	mL/min	111.70 (34.70–463.00)
eGFR _{MDRD}	mL/min/1.73 m ²	126.45 (23.90–509.00)
eGFR _{CKD-EPI}	mL/min/1.73 m ²	104.50 (25.00–200.00)
mCrCL _{24h} ^e	mL/min/1.73 m ²	88.00 (23.00–217.00)

^a Height calculated as $\sqrt{\text{total body weight (kg)}/\text{body mass index (kg/m}^2\text{)}} \times 100$. ^b 24-hour measured urinary creatinine calculated as (10,34496 × measured urinary creatinine clearance by 24-hour timed

urine collection $\times$ serum creatinine (mg/dL) $\times$ height (m)^{0.725} $\times$
total body weight (kg)^{0.425})/1.73.

^c Race designates a white patient if the estimated glomerular filtration rate is calculated using the Modification of Diet in Renal Disease equation / (175 $\times$ serum creatinine (mg/dL)^{-1.154} $\times$

age (year)^{-0.203} $\times$ 0.742 if female = 1. ^d A total of 12 patients exhibited a change in kidney function marker after 72-hour of meropenem administration. ^e A total of 4 patients had missing values

eCrCL_{CG}, estimated creatinine clearance calculated using the Cockcroft–Gault equation; eGFR_{CKD-EPI}, estimated glomerular filtration rate calculated using the Chronic Kidney Disease Epidemiology Collaboration equation; eGFR_{MDRD}, estimated glomerular filtration rate calculated using the Modification of Diet in Renal Disease equation; mCrCL_{24h}, measured creatinine clearance by 24-hour timed urine collection; n (%), number and percentage of patients

Supplementary Table 16 Kidney function markers equations

Kidney function markers	Equation	Age	Total body weight	Height	Serum creatinine	24-hour measured urinary creatinine	Sex	Race
eCrCL _G	$\frac{\text{Total body weight (kg)} \times (140 - \text{age (years)})}{72 \times \text{SCr (mg/dL)}} \times 0.85 \text{ if female}$	✓	✓	–	✓	–	✓	–
eGFR _M DRD	$175 \times \text{SCr}^{-1.154} \text{ (mg/dL)} \times \text{age}^{-0.203} \text{ (years)}$ $\times 0.742 \text{ if female} \times 1.21 \text{ if African-American}$	✓	–	–	✓	–	✓	✓
eGFR _C KD-EPI	$141 \times \text{minimum} \left(\frac{\text{SCr (mg/dL)}}{K}, 1 \right)^{\alpha} \times \text{maximum} \left(\frac{\text{SCr (mg/dL)}}{K}, 1 \right)^{\alpha}$ $\times 0.993^{\text{age (years)}}$ $\times 1.018 \text{ if female} \times 1.159 \text{ if African-American}$ with $K = 0.7$ for females and 0.9 for males, and $\alpha = -0.329$ for females and -0.411 for males	✓	–	–	✓	–	✓	✓

	<u>24-hour measured urinary creatinine (mg)</u>							
mCrCL _{24h}	$\frac{1,440 \times \text{SCr (mg/dL)}}{1.73}$	–	✓	✓	✓	✓	–	–
	$\times \frac{0.007184 \times \text{height}^{0.725} \text{ (m)} \times \text{total body weight}^{0.425} \text{ (kg)}}{1.73}$							

eCrCL_{CG}, estimated creatinine clearance calculated using the Cockcroft–Gault equation; eGFR_{CKD-EPI}, estimated glomerular filtration rate calculated using the Chronic Kidney Disease Epidemiology Collaboration equation; eGFR_{MDRD}, estimated glomerular filtration rate calculated using the Modification of Diet in Renal Disease equation; mCrCL_{24h}, measured creatinine clearance by 24-hour timed urine collection; SCr, serum creatinine

Supplementary Table 17 Patient characteristics and kidney function markers of the virtual patients' dataset

	Unit	Range	Stepsize	Number of scenarios
Patient characteristics				
Age	year	18–88	10	8
Total body weight	kg	30–120	18	6
Height	cm	120–220	10	5
Serum creatinine ^a	mg/dL	0.23–2.18	0.13	15
24-hour measured urinary creatinine	g	0.08–8.58	0.9	10
Male/female ratio	%	0–100	25	5
African American/white ratio	%	0–100	25	5
Kidney function markers				
eCrCL _{CG}	mL/min	20–470	30	15
eGFR _{MDRD}	mL/min/1.73 m ²	20–530	34	15
eGFR _{CKD-EPI}	mL/min/1.73 m ²	20–230	14	15
mCrCL _{24h}	mL/min/1.73 m ²	20–230	14	15

^a Serum creatinine is both a patient characteristic and a kidney function marker.

eCrCL_{CG}, estimated creatinine clearance calculated using the Cockcroft–Gault equation; eGFR_{CKD-EPI}, estimated glomerular filtration rate calculated using the Chronic Kidney Disease Epidemiology Collaboration equation; eGFR_{MDRD}, estimated glomerular filtration rate calculated using the Modification of Diet in Renal Disease equation; mCrCL_{24h}, measured creatinine clearance by 24-hour timed urine collection

Supplementary Table 18 Drop in objective function with the stepwise covariate modeling in six population pharmacokinetic models

Model	No covariate	eCRCL _{CG}	eGFR _{MDRD}	eGFR _{CKD-EPI}	mCrCL _{24h}	SCr
OFV (dOFV)	1300.84	1266.46 (-34.38)	1276.59 (-24.25)	1277.91 (-22.93)	1279.37 (-21.47)	1279.66 (-21.18)
Forward addition^a						
Testing round 1						
Age	1298.93 (-1.91)	—	—	—	1278.73 (-0.65)	1279.15 (-0.51)
TBW	1298.97 (-1.87)	—	1266.59 (-10.00)	1266.85 (-11.06)	—	1269.17 (-10.49)
Height	1299.07 (-1.77)	1266.04 (-0.42)	1275.62 (-0.97)	1276.18 (-1.73)	—	1277.44 (-2.22)
UCr _{24h}	1296.92 (-3.92)	1262.94 (-3.52)	1271.17 (-5.42)	1273.07 (-4.84)	—	1272.04 (-7.62)
Sex assigned at birth	1298.90 (-1.94)	1265.73 (-0.73)	—	—	1279.37 (0.00)	1276.22 (-3.44)
Race	1300.83 (-0.01)	—	—	—	1279.36 (0.02)	1279.60 (-0.06)
Testing round 2						
	UCr _{24h}	eCRCL _{CG}	eGFR _{MDRD} + TBW	eGFR _{CKD-EPI} + TBW	mCrCL _{24h}	SCr + TBW
Age	1294.85 (-2.07)	—	—	—	—	1268.37 (-0.80)

TBW	1296.42 0.50)	(- -	-	-	-	-		
Height	1296.45 0.47)	(- -	1266.33 0.26)	(- 1266.82 (-0.03)	-		1269.16 (-0.01)	
UCr _{24h}	-	-	1263.35 3.24)	(- 1265.44 (-1.41)	-		1263.91 (-5.26)	
Sex assigned at birth	1294.33 1.53)	(- -	-	-	-		1266.85 (-2.32)	
Race	1296.92 0.00)	(- -	-	-	-		1269.06 (-0.11)	
Testing round 3	UCr _{24h}	eCRCL _{CG}	eGFR _{MDRD} TBW	+	eGFR _{CKD-EPI} TBW	+	mCrCL _{24h}	SCr + TBW + UCr _{24h}
Age	-	-	-	-	-	-		1263.18 (-0.73)
Height	-	-	-	-	-	-		1263.57 (-0.34)
Sex assigned at birth	-	-	-	-	-	-		1263.33 (-0.58)
Race	-	-	-	-	-	-		1263.79 (-0.12)
Backward elimination	UCr _{24h}	eCRCL _{CG}	eGFR _{MDRD} TBW	+	eGFR _{CKD-EPI} TBW	+	mCrCL _{24h}	SCr + TBW + UCr _{24h}
TBW	-	-	1276.59 (10.00)		1277.91 (11.06)		-	1272.40 (8.49)
UCr _{24h}	1300.84 (3.92)	-	-	-	-	-		1296.17 (5.26)

Final model	No covariate	eCrCL _{CG}	eGFR _{MDRD} TBW	+	eGFR _{CKD-EPI} TBW	+	mCrCL _{24h}	SCr + TBW
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^a Continuous patient characteristics were introduced as $CL_i = CL_{pop} (L/h) \times \left(\frac{\text{patient characteristics}}{\text{median patient characteristics}} \right)^{THETA (1)}$ with CL_i and CL_{pop} being individual clearance, and typical estimate of the clearance, respectively. The categorical patient characteristics were introduced as $CL_i = CL_{pop} (L/h) \times THETA (1)$ otherwise $CL_i = CL_{pop} (L/h) \times THETA (1) \times (1 + THETA (2))$. Red and violet highlights indicate $|dOFV| \geq 7.88$ points and $3.84 \leq |dOFV| < 7.88$ points, respectively.

dOFV, drop in objective function value; eCrCL_{CG}, estimated creatinine clearance calculated using the Cockcroft–Gault equation; eGFR_{CKD-EPI}, estimated glomerular filtration rate calculated using the Chronic Kidney Disease Epidemiology Collaboration equation; eGFR_{MDRD}, estimated glomerular filtration rate calculated using the Modification of Diet in Renal Disease equation; mCrCL_{24h}, measured creatinine clearance by 24-hour timed urine collection; OFV, objective function value; SCr, serum creatinine; TBW, total body weight; UCr_{24h}, 24-hour measured urinary creatinine

Supplementary Table 19 Default simulation conditions in the interactive software tool

Section	Simulation condition	Value
Covariates	Serum creatinine, mg/dL	0.71
	Total body weight, kg	68
Dosing details	Loading dose	
	Amount, mg	0
	Infusion duration, h	0
	Dosing interval, h	0
	Additional doses, n	0
	Maintenance dose	
	Amount, mg	4,000
	Infusion duration, h	6
	Dosing interval, h	6
	Additional doses, n	15
	Time between the start of the first loading dosing strategy to the first maintenance dosing strategy, h	0
Targets	Protein binding, %	0
	Minimum inhibitory concentration, mg/L	2
	Percentage of time above minimum inhibitory concentration, %	100
	Multiple of minimum inhibitory concentration	4
	Toxic concentration, mg/L	64.2
Simulation settings ^a	Stochastic simulation number	100
	Prediction interval, %	95
	Interindividual variability	Yes
	Intraindividual variability	Yes

^aSimulation time is limited to the longest time point reached under the selected dosing strategy

NONMEM control stream

```

$PROBLEM      MEROPENEM SIMULATION
$INPUT        ID OCC TIME DV EVID MDV AMT RATE NDOSE
              CG MDRD CKDEPI MCRCL SCR TBW EVENT
$DATA         SIM_DS.csv IGNORE=@
$SUBROUTINE   ADVAN13 TOL=9
$MODEL        NCOMPARTMENTS=2
              COMP=(CENTRAL DEFOBS)
              COMP=(PERIPH)
$PK           TVCL      =THETA(1) * (SCR/0.63)**THETA(5) *
                      (TBW/69.5)**THETA(6)
              ;TVCL     =THETA(1) * (CG/111.7)**THETA(5)
              ;TVCL     =THETA(1) * (MDRD/126.45)**THETA(5)
              ;TVCL     =THETA(1) * (CKDEPI/104.5)**THETA(5)
              ;TVCL     =THETA(1) * (MCRCL/88.0)**THETA(5)
              ;TVCL     =THETA(1) * (SCR/0.63)**THETA(5)
              CL        =TVCL*EXP(ETA(1))
              TVV1      =THETA(2)
              V1        =TVV1*EXP(ETA(2))
              TVQ       =THETA(3)
              Q         =TVQ
              TVV2      =THETA(4)
              V2        =TVV2*EXP(ETA(3))
              S1        =V1
              S2        =V2
              KE        =CL/V1
              K12       =Q/V1
              K21       =Q/V2
$DES          DADT(1)   =K21*A(2) - K12*A(1) - KE*A(1)
              DADT(2)   =K12*A(1) - K21*A(2)
$ERROR        IPRED     =F
              W         = SQRT(SIGMA(1,1)*IPRED**2 +
                      SIGMA(2,2))
              Y         = IPRED * (1+EPS(1))+ EPS(2)
              IRES      = DV-IPRED
              IWRES     =IRES/W
$THETA        14.0      ; TVCL

```

```

25.5      ; TVV1
7.99      ; TVQ
12.4      ; TVV2
-0.678    ; SCR
0.817     ; TBW
; 0.638   ; CG
; 0.505   ; MDRD
; 0.950   ; CKDEPI
; 0.369   ; MCRCL
; -0.572  ; Standalone SCR

$OMEGA    BLOCK(2)
0.282     ; CL
0.174     ; V1
0.275
$OMEGA    0.428 ; V2
$SIGMA    0.0919 ; Proportional error
0.152     ; Additive error

$SIMULATION (9784275) ONLYSIM SUBPROBLEMS=1000
$TABLE     ID OCC TIME EVENT MDV DV EVID AMT RATE
           PRED IPRED NDOSE
           CG MDRD CKDEPI MCRCL SCR TBW ONEHEADER
           NOPRINT
           FILE=Output_data_SCRTBW.csv
           ;FILE=Output_data_CG.csv
           ;FILE=Output_data_MDRD.csv
           ;FILE=Output_data_CKDEPI.csv
           ;FILE=Output_data_MCRCL.csv
           ;FILE=Output_data_SCR.csv

```

Chapter 4. Amikacin dose optimization in the emergency department

This chapter is based on

Nada Dia *et al.*, Dose optimization of amikacin in the emergency department: a population pharmacokinetic simulation study. *Ther Drug Monit* 47(3), 353–362

Abstract

Background In adult patients with sepsis or septic shock admitted to the emergency department (ED), a single intravenous 15 mg/kg amikacin dose provides inadequate pharmacokinetic–pharmacodynamic (PKPD) target attainment at the locally reported minimum inhibitory concentration (MIC) of 2 mg/L and the European Committee on Antimicrobial Susceptibility Testing (EUCAST) clinical breakpoint for Enterobacterales of 8 mg/L. The aim of the study was to provide an amikacin dosing strategy with a clinically acceptable probability of target attainment (PTA) for all patients.

Methods Stochastic simulations were performed using a two-compartment population pharmacokinetic (popPK) model of amikacin (NONMEM 7.5.0). PTA was evaluated for various dosing strategies across a range of virtual patients' body weight, body mass index, serum total protein, serum sodium, fluid balance, and estimated glomerular filtration rate using the Chronic Kidney Disease Epidemiology Collaboration equation ($eGFR_{CKD-EPI}$), at the locally reported MIC of 2 mg/L and the clinical breakpoint of 8 mg/L. The ratio of the 24-hour area under the concentration–time curve over the MIC (AUC_{24h}/MIC) of ≥ 80 and a 24-hour postdose concentration (C_{24h}) of < 3 mg/L were evaluated as efficacy and safety targets, respectively.

Results The PTA for the clinical breakpoint of 8 mg/L was $< 90\%$ with standard 15 mg/kg dosing, across all patient characteristics. A flat 1,500 mg dose achieved $\geq 90\%$ PTA for the entire population at an MIC of 2 mg/L. However,

at the clinical breakpoint of 8 mg/L, a flat 3,500 mg dose provided $\geq 90\%$ PTA only when the $\text{eGFR}_{\text{CKD-EPI}}$ was $< 96 \text{ mL/min/1.73 m}^2$. The $\text{C}_{24\text{h}}$ was similar for 1,500 mg and 15 mg/kg dosing (median range across patient characteristics: 0.9–40 mg/L versus 0.6–35 mg/L, respectively), whereas 3,500 mg resulted in a higher $\text{C}_{24\text{h}}$ (2–100 mg/L).

Conclusions A flat dose is recommended over weight-based dosing. However, selecting a 1,500 mg or 3,500 mg dose may compromise either efficacy (MIC 2 mg/L) or safety (clinical breakpoint 8 mg/L), posing a dilemma. Clinical validation is warranted.

1. Introduction

Sepsis is a dysregulated host response to an infection, which can progress to organ dysfunction and mortality. According to the WHO, sepsis is a global health priority, responsible for the death of 11 million individuals annually [321,322]. Management includes antimicrobial therapy, fluid replacement, and organ support [61].

Patients with sepsis or septic shock are often initiated on empiric antimicrobial combination therapy, such as a broad-spectrum β -lactam with an aminoglycoside, within the first hour of hospitalization [61]. Aminoglycosides are favored because of their low resistance rates and efficacy against multidrug-resistant gram-negative bacteria, including Enterobacterales species [323]. However, aminoglycosides have a narrow therapeutic window and are ototoxic and nephrotoxic [324,325].

Early and adequate achievement of the PKPD target is pivotal for patient survival [312]. Historically, the PKPD target of aminoglycosides was defined as a $C_{\max}/MIC \geq 8$ [21]. However, the AUC_{24h}/MIC is now believed to be a more appropriate PKPD index [69]. An $AUC_{24h}/MIC \geq 80$ has been proposed for critically ill patients [69]. However, the antimicrobial susceptibility of Enterobacterales species may vary between hospitals. Therefore, locally reported MICs or the lowest MIC inhibiting $\geq 90\%$ of bacterial isolates (MIC_{90}) are commonly used to calculate the PKPD target. To avoid errors due to

regional and technical variation, the EUCAST recommended using the Enterobacterales clinical breakpoint of 8 mg/L [326].

Amikacin is generally given as a single dose within the first hour of admission to the ED. It is administered intravenously as a weight-based dose, implying clinical relevance of body weight on PKPD target attainment. Although a 15 mg/kg amikacin dose is commonly prescribed, EUCAST breakpoints demand a 25–30 mg/kg dose. Yet, out of safety concerns, the latter is not widely adopted [327]. The 15 mg/kg and 25 mg/kg doses have resulted in 94% and 95% target attainment, respectively, considering the $C_{\max}/\text{MIC}$ target of 8 using an MIC_{90} of 2 mg/L [328]. However, when considering the EUCAST breakpoint of 8 mg/L, only 40% and 76% of patients achieved the $C_{\max}/\text{MIC}$ target [328].

De Winter *et al.* suggested that patients with a high body mass index (BMI), elevated CrCL, low serum sodium concentration, low total protein concentration, or a high fluid balance may need a higher amikacin dose to reach adequate target attainment [21]. This conclusion was based on a population popPK modeling study where these covariates were identified based on goodness-of-fit criteria. However, it remains unclear to what extent these statistically significant covariates impact PKPD target attainment. Therefore, there is a medical need to investigate the clinical relevance of these covariate effects, to identify an optimized yet feasible amikacin dosing strategy

with improved PKPD target attainment and minimal toxicity profiles in all patients.

The objectives of this simulation study are 2-fold: (1) to identify patient characteristics with a clinically relevant impact on PKPD target attainment and (2) to determine an amikacin dosing strategy with an optimized probability ($\geq 90\%$) of attaining the AUC_{24h} and C_{max} PKPD targets, while minimizing toxicity risk by targeting a $C_{24h} \leq 3$ mg/L as a safety objective while monitoring the time to reach an amikacin concentration of 3 mg/L ($Time_{3\text{ mg/L}}$) when using a locally reported MIC_{90} and the EUCAST breakpoint for Enterobacterales species.

2. Materials and methods

2.1 Population pharmacokinetic model

De Winter *et al.* described a two-compartment popPK model of amikacin in adult patients admitted to the ED with sepsis or septic shock [21]. The model was built on a total of 279 plasma samples from 97 patients. Five covariates were identified to have a statistically significant impact on the amikacin PK CL and V_c of patient i , which is given by the equation with the $eGFR_{CKD-EPI}$ at 24 hours after the start of the infusion ($eGFR_{CKD-EPI,24h}$), BMI at inclusion, serum total protein at inclusion, serum sodium at inclusion, and fluid balance for the first 24 hours after the start of the infusion. The NONMEM control stream is provided in Appendix D.

$$CL_i = 1.8 \text{ L/h} \times \left(\frac{eGFR_{CKD-EPI,24h,i}}{52.0 \text{ mL/min/m}^2} \right)^{0.86} \times \left(\frac{BMI_i}{24.0 \text{ kg/m}^2} \right)^{0.86} \times e^{\eta_i} \text{ with } \eta_i \sim N(0, 0.14)$$

(6)

$$V_{c,i} = 16 \text{ L} \times \left(\frac{BMI_i}{24.0 \text{ kg/m}^2} \right)^{0.42} \times \left(\frac{\text{serum total protein}_i}{66.0 \text{ g/L}} \right)^{-0.53} \times \left(\frac{\text{serum sodium}_i}{138.25 \text{ mmol/L}} \right)^{-2.1}$$

$$\times \left(1 + \frac{5}{10^5} \times (\text{Fluid Balance}_{24h,i} - 2796.5 \text{ mL}) \right) \times e^{\eta_i} \text{ with } \eta_i \sim N(0, 0.03)$$

(7)

2. 2 Clinical dataset

Data on patients' body weight and popPK covariate values were retrieved from the real-world clinical dataset set (n = 97) [21]. Information on body weight was required to simulate weight-based dosing strategies. The study was approved by the Ethics Committee Research UZ/KU Leuven (S52801). Written informed consent was obtained from the patient or the closest relative. This study was registered at ClinicalTrials.gov (NCT02365272).

2. 3 Virtual populations

The normality of the body weight and covariate data distributions was first tested using the Shapiro–Wilk test. Where necessary, transformations to normality were applied based on the skewness coefficient to make all covariates data follow a normal distribution. Next, a covariance matrix was created from the (transformed) clinical dataset. Virtual patients were created by sampling (transformed) body weight and the covariate values from a

truncated multivariate normal distribution, constrained within the range of the clinical dataset [21,329,330]. These sampled transformed covariate values were converted back to the original scale.

First, six virtual patient datasets were created, each having either body weight or one covariate normally distributed, whereas values for the other covariates/body weight were sampled from the covariance matrix. These datasets were used to evaluate PKPD target attainment across a wide range of covariate/body weight values (*cf.* Identification of an optimized dosing strategy).

Next, a virtual patient data set ($n = 200$) representing the real-world patient population was created by randomly sampling body weight and covariates from the covariance matrix. This data set was used to evaluate PKPD target attainment at the population level (*cf.* Evaluation of the dosing strategy at the population level).

2. 4 Dosing strategies and simulations

Eleven dosing strategies were evaluated (**Table 6**). Stochastic simulations (with IIV and RUV) were performed with each dosing strategy in each virtual patient population. A total of 1,000 and 100 simulations were carried out per virtual patient in the uniformly distributed covariate populations and the recreated real-world populations, respectively.

Table 6 Dosing strategies under investigation

Type	Quantity
Weight-based dosing	15 mg/kg
	25 mg/kg
	30 mg/kg
	35 mg/kg
Weight-based dosing with a dose cap at 80 kg	15 mg/kg limited to 1 200 mg
	25 mg/kg limited to 2 000 mg
Flat dosing	1,000 mg
	1,500 mg
	2,000 mg
	3,000 mg
	3,500 mg

2. 5 Determination of MIC

Two MICs were considered. The first was the MIC₉₀ of the Enterobacterales species in blood, urine, lung, or wound cultures obtained from patients on amikacin admitted to UZ Leuven from January 1, 2018, until November 15, 2022. The second was the EUCAST clinical breakpoint for susceptibility given a wild-type distribution for Enterobacterales species (MIC 8 mg/L) [326].

2. 6 Identification of an optimized dosing strategy

An optimized amikacin dosing strategy was determined using virtual patient datasets with uniformly distributed body weight and covariate data. For each dosing strategy, four exposure metrics were calculated, comprising two safety metrics (C_{24h} and Time_{3 mg/L}) and 2 effectiveness metrics (AUC_{24h} and C_{max}). These metrics were evaluated over the entire range of body weight and

covariate values in all 11 dosing strategies. The safety targets were $C_{24h} \leq 3$ mg/L and $Time_3 \text{ mg/L}$, with the latter investigated for four days following the administration of amikacin, mirroring hospital monitoring practices. There, amikacin is routinely assessed once a patient has been on amikacin for more than three days, expecting it to exacerbate or resolve on the fourth day [331,332]. PKPD target attainment was evaluated based on an $AUC_{24h}/MIC \geq 80$ and a $C_{max}/MIC \geq 8$, with a PTA $\geq 90\%$ considered clinically acceptable [303]. The clinical relevance of body weight and covariates was established if resulted in clinically unacceptable target attainment across the studied ranges. Because amikacin is minimally bound to protein ($\leq 10\%$), the simulated plasma concentration of unbound amikacin was considered equal to the plasma concentration of total amikacin [333]. Finally, a dosing strategy with clinically acceptable PTA and minimal toxicity profile in the entire virtual population was identified.

2. 7 Evaluation of the dosing strategy at the population level

The impact of the proposed dosing strategy on the population-level rate of AUC_{24h} and C_{max} target attainment was evaluated in comparison to the 15 mg/kg dosing strategy using the virtual patient data set representing the real-world patient population (*cf.* Virtual populations).

2. 8 Software

R (version 4.1.1; R Core Team, Vienna, Austria) in the RStudio integrated development environment (version 1.4.900; RStudio, Boston, MA) with the

tmvtnorm package and tidyverse collection of packages, NONMEM (version 7.5; Icon Development Solutions, Gaithersburg, MD), and the Perl-speaks-NONMEM (version 4.7.0) toolkit on the Pirana modeling workbench (version 2.9.7; Certara, Princeton, NJ) were used [102,334]. The ordinary differential equation solver ADVAN13 and evaluation tolerance TOL9 were integrated into the \$DES block to calculate the exposure metrics.

3. Results

3.1. Virtual populations

The baseline characteristics of patients and the covariance matrix of body weight and covariates are presented in Appendix D (**Supplementary Tables 20–21**). The median (range) of body weight and $\text{eGFR}_{\text{CKD-EPI},24\text{h}}$ were 70.0 (36.0–115.0) kg and 53.0 (6.0–200.0) mL/min/1.73 m², respectively. Correlations between body weight and covariates are visualized in Appendix D (**Supplementary Figure 28**), whereas a summary of all patient characteristics is provided by De Winter *et al.* [21].

3.2. Determination of MIC

Over the 5-year study period, a total of 61 903 Enterobacterales species isolates were cultured. MIC values varied from 2 mg/L to 64 mg/L, with the MIC₉₀ being 2 mg/L corresponding to 55 713 isolates (**Supplementary Figure 29**). The AUC_{24h} and C_{max} targets based on an MIC₉₀ of 2 mg/L were 160 mg×h/L and 16 mg/L, respectively. The AUC_{24h} and C_{max} targets

corresponding to the EUCAST breakpoint for Enterobacterales of 8 mg/L were 640 mg×h/L and 64 mg/L, respectively.

3.3. Identification of an optimized dosing strategy

Considering an MIC₉₀ of 2 mg/L, the 15 mg/kg dose resulted in high IIV in AUC_{24h} and C_{max} (**Supplementary Figures 30–31**). eGFR_{CKD-EPI,24h} was identified as a clinically relevant covariate under the 15 mg/kg dosing strategy because the probability of attaining the AUC_{24h} target was compromised when eGFR_{CKD-EPI,24h} was above 126 mL/min/1.73 m² (**Figure 13**). However, C_{max} target attainment remained clinically acceptable across all patients.

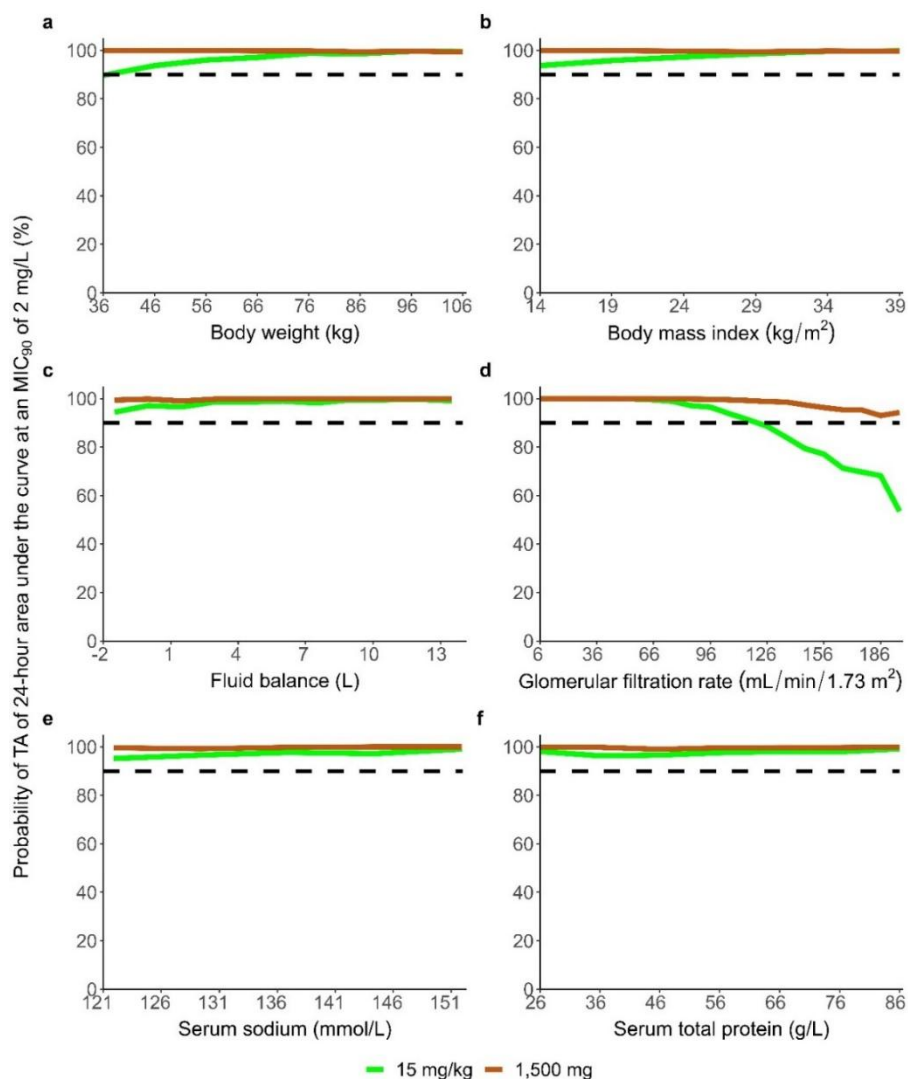


Figure 13 Probability of target attainment in terms of 24-hour area under the concentration–time curve over the range of (a) body weight, (b) body mass index, (c) fluid balance, (d) glomerular filtration rate, (e) serum sodium, and (f) serum total protein at an MIC₉₀ of 2 mg/L for the dosing strategies of 15 mg/kg and 1,500 mg. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient’s other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line indicates 90% probability of target attainment. TA, target attainment

A flat 1,500 mg dose resulted in a clinically acceptable probability of AUC_{24h} target attainment across the entire range of $eGFR_{CKD-EPI,24h}$ values, as well as body weight and other covariates (**Figure 13**). Although C_{24h} was highly variable, it was only marginally higher with 1,500 mg amikacin compared with 15 mg/kg, except for patients with low body weight/BMI (**Figure 14**). The $Time_{3\text{ mg/L}}$ postadministration was highly variable, being longest in patients with low $eGFR_{CKD-EPI,24h}$ (**Supplementary Figure 32**). Major differences between flat 1,500 mg and 15 mg/kg of amikacin dosing were only observed in patients with low body weight/BMI.

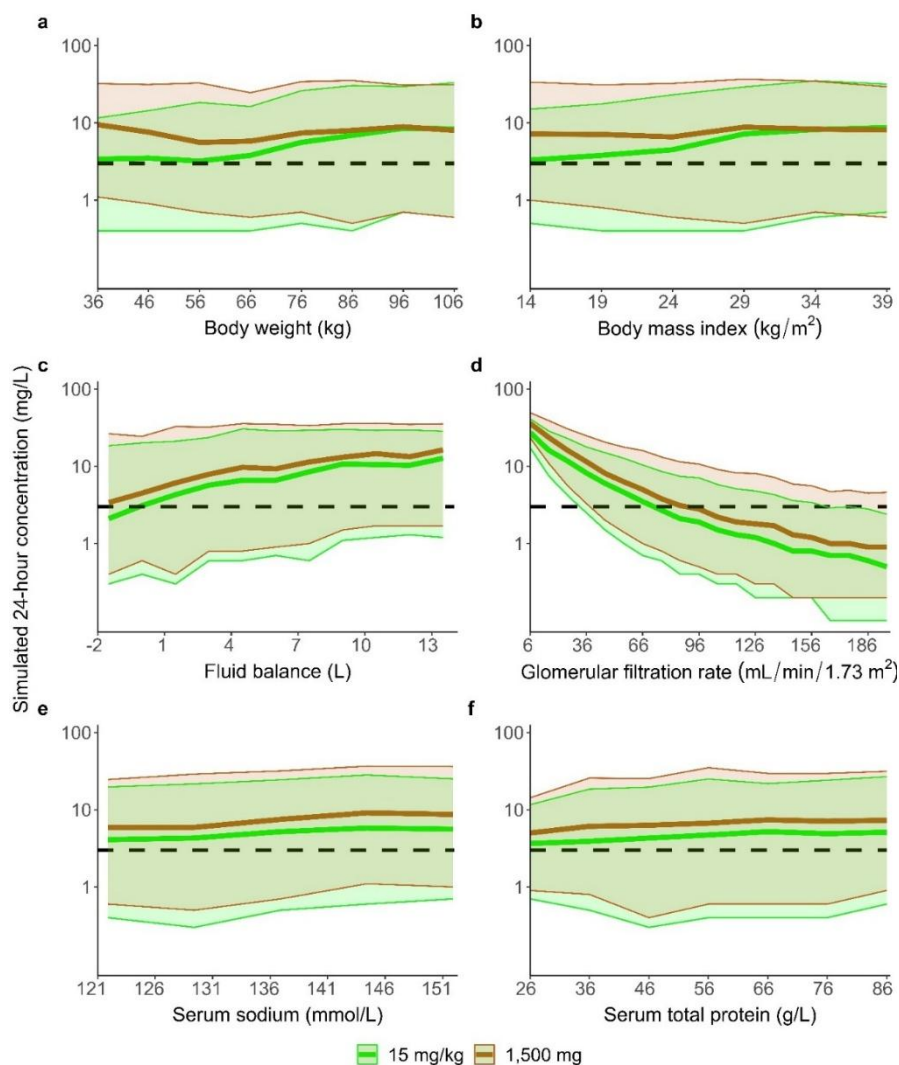


Figure 14 Simulated 24-hour concentration (C_{24h}) over the range of (a) body weight, (b) body mass index, (c) fluid balance, (d) glomerular filtration rate, (e) serum sodium, and (f) serum total protein at an MIC_{90} of 2 mg/L for the dosing strategies of 15 mg/kg and 1,500 mg. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line, the solid lines, and the shaded areas represent a C_{24h} of 3 mg/L, the median C_{24h} , and 95% prediction interval, respectively

Results of simulated AUC_{24h} and C_{max} with other dosing strategies are detailed in Appendix D (**Supplementary Figures 33–34**). $eGFR_{CKD-EPI,24h}$ was identified as a clinically relevant covariate under the 15 mg/kg, 15 mg/kg limited to 1,200 mg, and flat 1,000 mg dosing strategies, as compromised AUC_{24h} target attainment was evident when $eGFR_{CKD-EPI,24h}$ exceeded 126 mL/min/1.73 m² (**Supplementary Figure 35**). The probability of C_{max} target attainment was 90% or higher with all the dosing strategies, including the EUCAST-recommended 25–30 mg/kg dose. Low $eGFR_{CKD-EPI,24h}$ was associated with the highest C_{24h} and the longest Time_{3 mg/L} (**Supplementary Figures 36–37**).

Considering the EUCAST breakpoint for Enterobacterales of 8 mg/L, the probability of attaining the AUC_{24h} and C_{max} targets under the 15 mg/kg dose was compromised across the entire range of body weight and covariate values (**Figure 15 and Supplementary Figure 38**). Even at the highest tested dose of 3,500 mg, the $eGFR_{CKD-EPI,24h}$ had a clinically relevant impact on AUC_{24h} target attainment, as PTA dropped below 90% when $eGFR_{CKD-EPI,24h}$ was ≥ 96 mL/min/1.73 m² (**Figure 15**). A flat 3,500 mg dose showed higher IIV in AUC_{24h} , C_{max} , and C_{24h} compared with the 15 mg/kg dose, with the maximum simulated values consistently higher across all body weight and covariate values (**Figure 16, Supplementary Figures 39–40**). The time to achieve a concentration of 3 mg/L after the administration varied with the shortest in patients with high $eGFR_{CKD-EPI,24h}$ (**Supplementary Figure 41**). Results of AUC_{24h} and C_{max} simulations with other dosing strategies are shown in

Supplementary Figures 42–43. The probability of AUC_{24h} target attainment was never $\geq 90\%$ with a dosing strategy other than 3,500 mg when eGFR_{CKD-EPI,24h} was < 96 mL/min/1.73 m² (**Supplementary Figure 44**). Conversely, the probability of C_{max} target attainment was 90% and above with flat dosing $\geq 3,500$ mg across the entire population (**Supplementary Figure 45**).

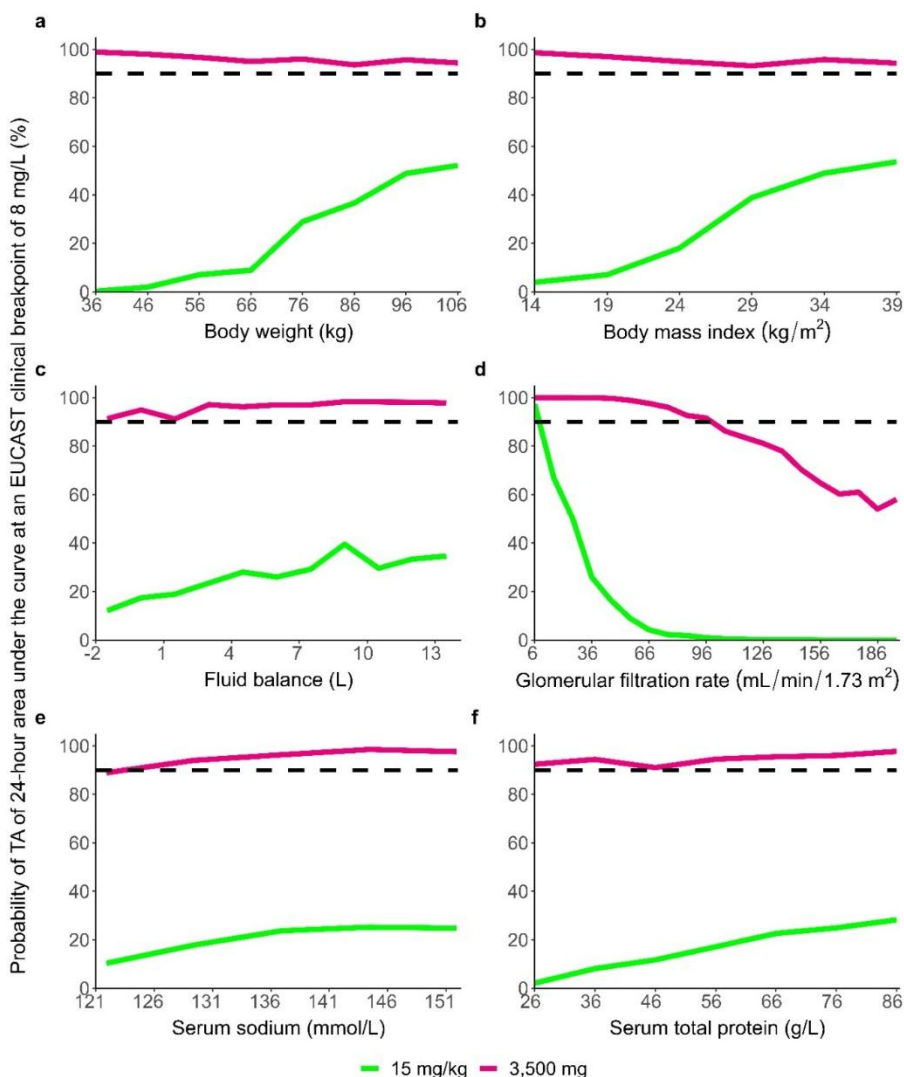


Figure 15 Probability of target attainment in terms of 24-hour area under the concentration–time curve over the range of (a) body weight, (b) body mass index, (c) fluid balance, (d) glomerular filtration rate, (e) serum sodium, and (f) serum total protein at the EUCAST clinical breakpoint of 8 mg/L for the dosing strategies of 15 mg/kg and 3,500 mg. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient’s other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line indicates 90% probability of target attainment. TA, target attainment

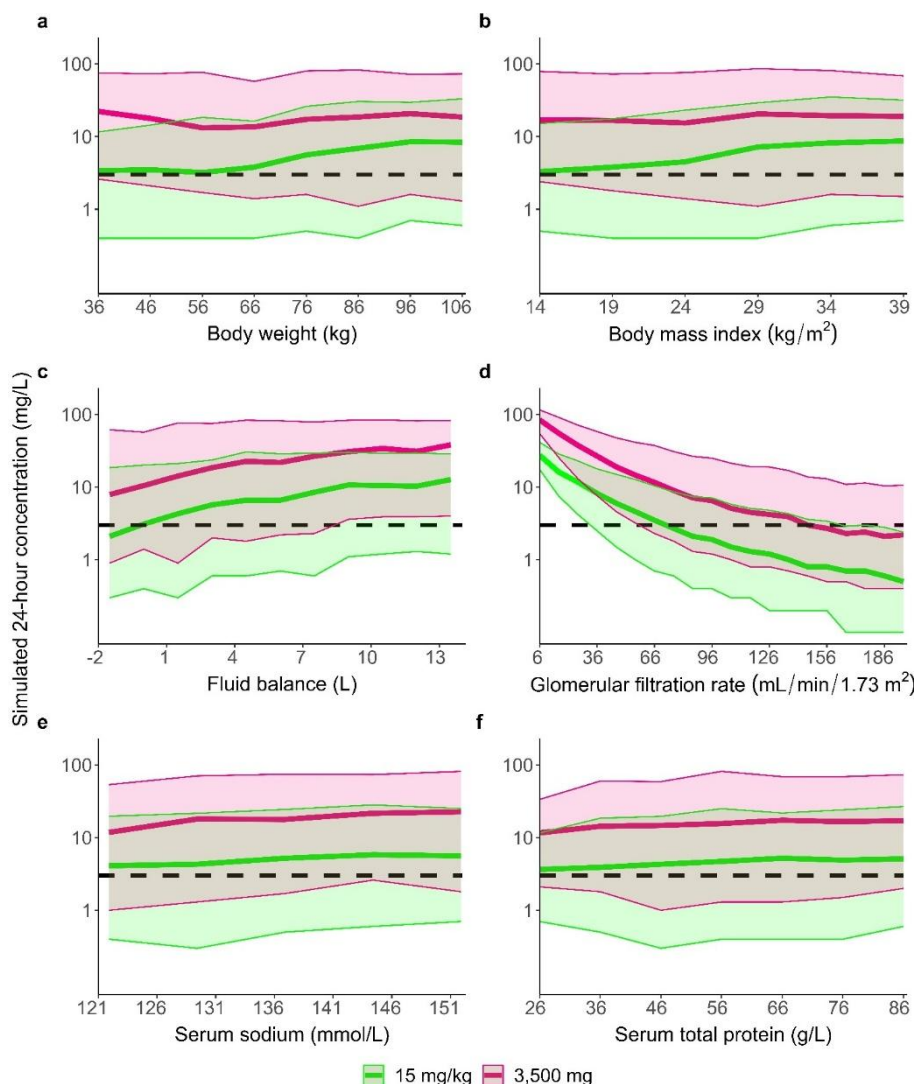


Figure 16 Simulated 24-hour concentration (C_{24h}) over the range of (a) body weight, (b) body mass index, (c) fluid balance, (d) glomerular filtration rate, (e) serum sodium, and (f) serum total protein at the EUCAST clinical breakpoint of 8 mg/L for the dosing strategies of 15 mg/kg and 3,500 mg. Covariate effects are visualized univariately for the ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line, the solid lines, and the shaded areas represent a C_{24h} of 3 mg/L, the median C_{24h} , and 95% prediction interval, respectively

3.4. Evaluation of the dosing strategy at the population level

Considering the MIC₉₀ of 2 mg/L, the predicted population rate of AUC_{24h} target attainment was 94% (95% prediction interval [PI] 78%–100%) under 15 mg/kg dosing and 100% (95% PI 97%–100%) under 1,500 mg flat dosing (p-value >0.05). Similarly, the predicted population rate of C_{max} target attainment was 100% (95% PI 100%–100%) under both 15 mg/kg dosing and 1,500 mg dosing (p-value >0.05).

However, when considering the EUCAST breakpoint of 8 mg/L, the predicted population rate of AUC_{24h} target attainment was only 11% (95% PI 0%–84%) under 15 mg/kg dosing and 96% (95% PI 63%–100%) under 3,500 mg flat dosing (p-value ≤0.05). Similarly, the predicted population rate of C_{max} target attainment was 32% (95% PI 0%–91%) under 15 mg/kg dosing and 100% (95% PI 100%–100%) under 3,500 mg flat dosing (p-value ≤0.05).

4. Discussion

In our simulation study, patients presenting with ARC, that is, with an eGFR_{CKD-EPI,24h} ≥126 mL/min/1.73 m², were vulnerable to subtherapeutic exposure and hence therapeutic failure, following a 15 mg/kg amikacin dose at an MIC of 2 mg/L. However, a flat 1,500 mg dose provided adequate PTA based on AUC_{24h} and C_{max} targets at an MIC of 2 mg/L in all patients, including those with an eGFR_{CKD-EPI,24h} ≥126 mL/min/1.73 m². Yet, its efficacy may be compromised in settings with reduced amikacin susceptibility. The weight-based 25–30 mg/kg dose recommended by EUCAST at the breakpoint for

Enterobacterales of 8 mg/L failed to provide adequate PTA based on AUC_{24h} nor C_{max} . A flat 3,500 mg dose was determined for patients with $eGFR_{CKD-EPI,24h} < 96 \text{ mL/min/1.73 m}^2$ at the EUCAST breakpoint of 8 mg/L, considering adequate PKPD target attainment based on both AUC_{24h} and C_{max} exposure metrics, albeit with safety concerns.

Simulations used the popPK model established by De Winter *et al.* which describes the PK of amikacin in adults admitted to the ED with sepsis or septic shock [21]. Although Boidin *et al.* developed an amikacin popPK model based on ICU patient data, it was deemed unsuitable for our study due to the unique pathophysiological dynamics observed in ED patients [335]. Factors such as edema occurrence, uncommon in emergency settings, and the likelihood of dehydration leading to negative fluid balance distinguish the determinants of amikacin exposure in the ED from those in the ICU. The De Winter *et al.* model, consistent with several amikacin popPK models developed across various patient cohorts, adopts a two-compartment structure [283,336–343]. It included five covariates, with $eGFR_{CKD-EPI,24h}$ explaining variability in amikacin CL. This finding aligns with literature where renal estimators were retained as a covariate impacting amikacin CL [283,336–342,344]. Given amikacin's primarily renal elimination, any alteration in kidney function naturally influences its CL [333]. The $eGFR_{CKD-EPI,24h}$ was not only statistically significant but also clinically impactful under 15 mg/kg dosing on PTA based on AUC_{24h} at the MIC_{90} of 2 mg/L and the EUCAST breakpoint of 8 mg/L, as well as PTA based on C_{max} at the 8 mg/L breakpoint. Furthermore, the

15 mg/kg weight-based dosing implies that body weight is a clinically relevant covariate for amikacin dosing [328,339,345–347]. The popPK model included BMI (rather than body weight). This is not the first amikacin popPK model where a weight-related factor other than body weight was identified as a covariate. In the De Winter *et al.* model, $eGFR_{CKD-EPI,24h}$ and BMI together explained 32% of the variability in amikacin CL, leaving 39% unexplained. A total of 8% of the variability in V_c was explained by BMI, serum total protein, serum sodium, and fluid balance, leaving 18% unexplained. Despite RUV, our simulation study showed that a single flat dose ensures adequate PTA.

Amikacin is notorious for its nephrotoxicity and ototoxicity, with increased death rates and longer hospital stays attributed to amikacin-induced nephrotoxicity [348]. Therefore, TDM has been recommended, typically targeting an amikacin minimum concentration (C_{min}) below 3 mg/L to avoid toxicity in clinical settings where amikacin is administered repeatedly, such as the ICU [335,349]. The French guideline recommends a C_{min} below 2.5 mg/L, the Japanese guideline reports a C_{min} below 4 mg/L, and Yamada *et al.* proposed a C_{min} below 10 mg/L [325,345,348]. Our study revealed that most patients receiving the 15 mg/kg infusion had a C_{24h} above 3 mg/L. However, although nephrotoxicity is associated with increased infusion frequency and prolonged therapy, little information exists on nephrotoxicity in the context of a single amikacin dose [350–356]. Incidences as low as 0.5% was reported in non-septic patients with a lower urinary tract infection treated [357]. Unlike nephrotoxicity, ototoxicity is irreversible, even with a single dose of

aminoglycoside, leading to potential hearing loss [358]. Nevertheless, the recommended 1,500 mg dose at an MIC of 2 mg/L does not exceed the recommended daily dose of 1,500 mg, and hence, it is considered safe [359]. Moreover, the range of $\text{Time}_{3 \text{ mg/L}}$ was similar across the investigated dosing strategies. Nonetheless, further investigation is warranted to assess the potential toxicity of high-dose amikacin [360].

We investigated PTA using two MICs. In clinical practice, the MIC is available within 3 days of bacterial culture. Therefore, the target MIC is often replaced by a locally reported MIC_{90} or the EUCAST breakpoint for Enterobacterales. However, these two MICs are not always interchangeable. The local MIC_{90} we found (2 mg/L) deviated considerably from the EUCAST breakpoint (8 mg/L). The local MIC_{90} is based on the MIC distribution of bacteria cultured at a hospital, whereas the EUCAST breakpoint represents international epidemiology. The local MIC_{90} values are generally lower in northern European countries compared with southern and eastern European countries [361]. This is due to a regional difference in the wild-type distribution of Enterobacterales species and a variation in susceptibility testing methods [362]. Because the selection of a dosing strategy, that affects efficacy, relies on a reliable MIC, EUCAST set a clinical breakpoint of 8 mg/L. It includes the entire wild-type distribution of Enterobacterales species determined based on several million MIC values obtained from standardized antimicrobial testing across Europe [362]. However, this breakpoint corresponds to systemic infection treated with a combination therapy and urinary tract infection treated

by monotherapy [326]. Adjunctive measures are recommended in the case of monotherapy targeting infections other than urinary tract infections [363]. Nonetheless, the investigation of two values of MIC in our study indicated that the selection of an amikacin dose based on MIC represents a dilemma, where selecting an MIC lower than the EUCAST breakpoint skews the efficacy results; however, favoring the EUCAST breakpoint may compromise safety. Moreover, the PKPD target is a surrogate for bacterial eradication with limited evidence demonstrating its clinical efficacy in clinical trials [364,365]. Therefore, further research is needed to identify the amikacin dose with an optimal balance between safety and clinical efficacy.

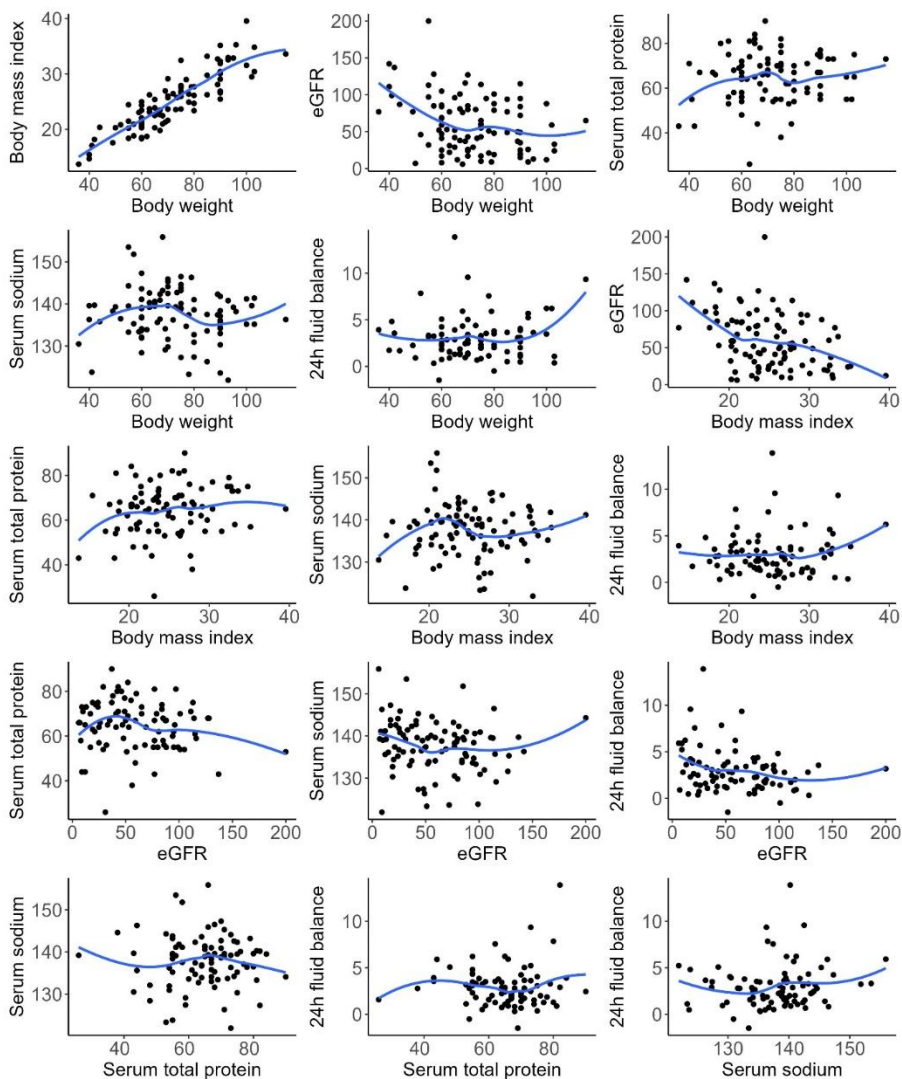
Debate is ongoing on which dosing strategy to apply for amikacin. TDM-guided dosing has been recommended but is not feasible in the ED because amikacin is typically administered as a single dose [325]. Weight-based dosing of amikacin is currently recommended [328,339,345–347]. However, in our simulation study, body weight had no clinically relevant impact on PTA. Moreover, estimating body weight in the ED can be challenging due to the lack of available weighing equipment. Simulations showed that only $eGFR_{CKD-EPI,24h}$ had a clinically relevant impact on PTA under 15 mg/kg dosing. However, SCr, required for estimated GFR calculation, is not available within the first hour of admission to the ED when amikacin administration is crucial to ensure a high survival rate [312]. Therefore, in the absence of a point-of-care GFR test, flat dosing is the only solution. Flat amikacin dosing is more practical than infusing a weight-based dose because it is easier and faster to

administer. In addition, the flat dose has a lower risk of dosing errors and less medication waste. In the absence of prospective confirmatory evidence for flat dosing, we recommend careful monitoring of amikacin concentrations. Furthermore, amikacin concentration monitoring will be essential to evaluate the performance of the proposed dosing strategy in the eagerly anticipated prospective confirmatory clinical trial.

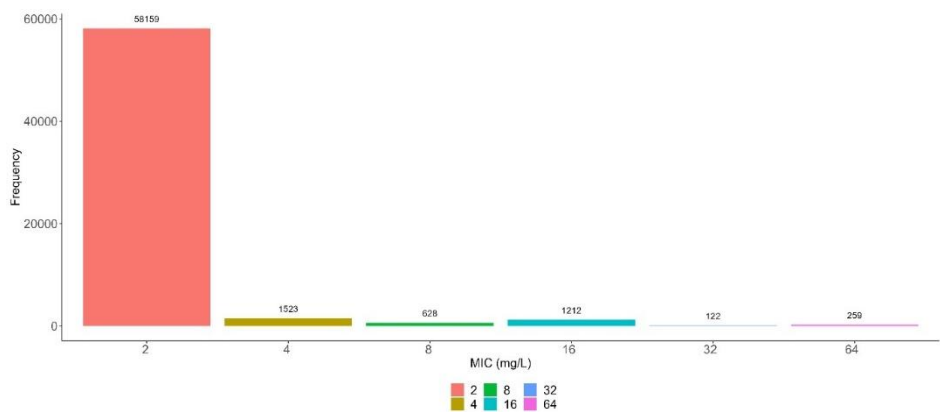
This study has several strengths. Our *in silico* analysis allows the evaluation of an unlimited number of dosing strategies and the identification of the dosing strategy with an optimized PTA, directing research toward a more efficient clinical trial design. Moreover, the patient data originate from a RCT and are of high quality. Our study also has limitations. First, simulations were performed with a popPK model based on data from a single center and therefore may not represent patients admitted to other hospitals, potentially with covariate values going beyond the range covered in our data set. Nevertheless, the sample size was large, and blood sampling was extensive. Second, our simulation study focused only on two MICs. However, these two MICs are for a commonly identified bacteria (Enterobacterales species) on the national and international levels (UZ Leuven and EUCAST). Third, our analysis focused on a safety target of 3 mg/L, although other safety targets (*i.e.*, 5 mg/L) have been reported [331,332,366]. Although further research on the safety of amikacin in this clinical setting is needed, we chose to rely on the slightly more stringent target of 3 mg/L [331,332].

In conclusion, our popPK simulation study advocates for a single 1,500 mg flat dose amikacin infusion, taking into account an MIC₉₀ of 2 mg/L in adults admitted to the ED with sepsis or septic shock. Considering the EUCAST clinical breakpoint of 8 mg/L, a flat amikacin dose of 3,500 mg is proposed when the eGFR_{CKD-EPI,24h} is below 96 mL/min/1.73 m². Selection between the two dosing strategies compromises either efficacy or safety, presenting a Catch-22. Nonetheless, a flat dose can be easily prepared from commercially available vials and ensures early and prompt management.

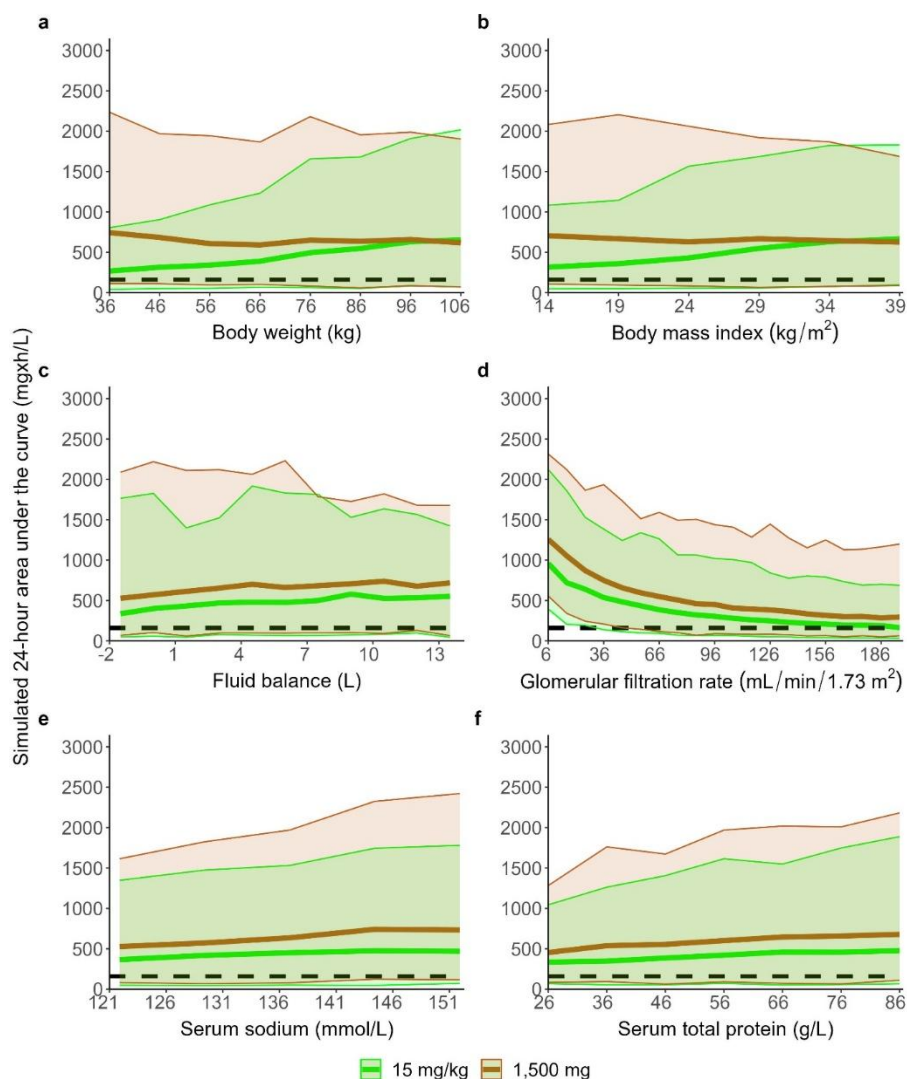
Appendix D



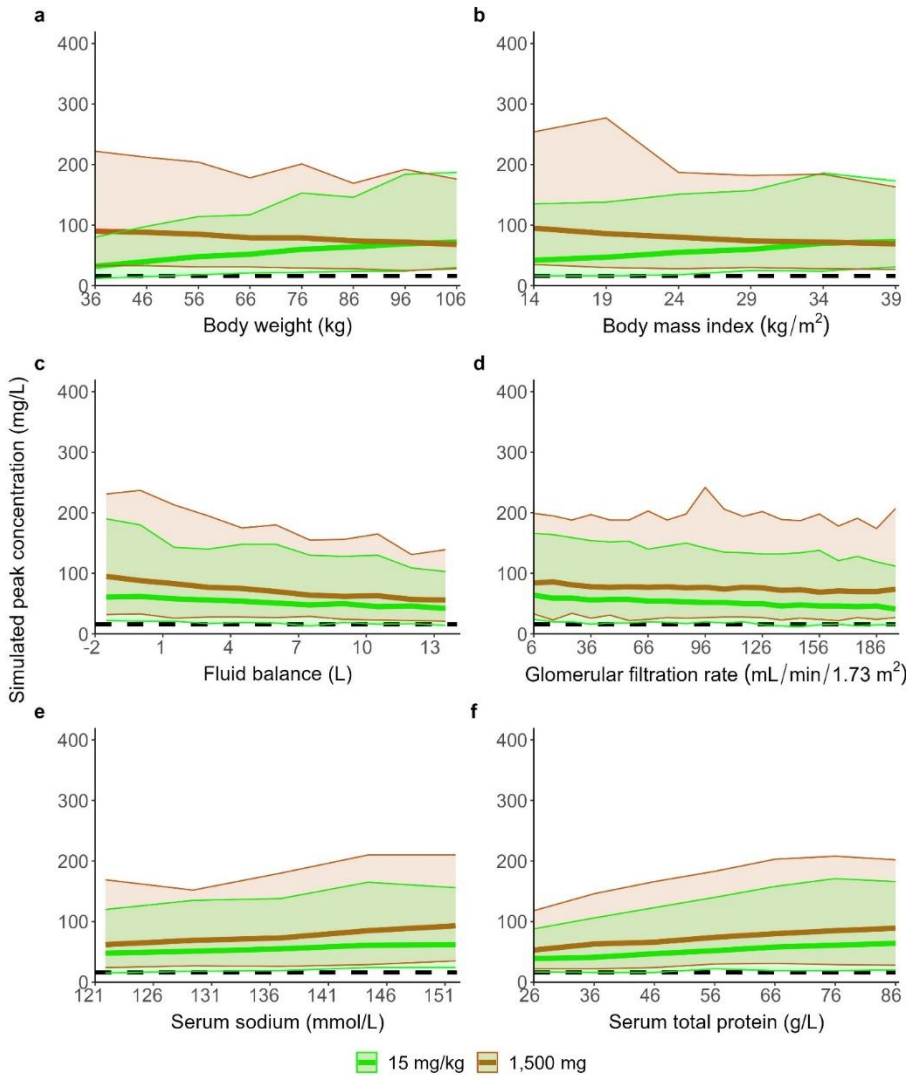
Supplementary Figure 28 Correlation plots of a covariate with either another covariate or body weight. The blue line represents the regression line. eGFR, estimated glomerular filtration rate using the Chronic Kidney Disease Epidemiology Collaboration equation at 24 hours after inclusion; h, hours



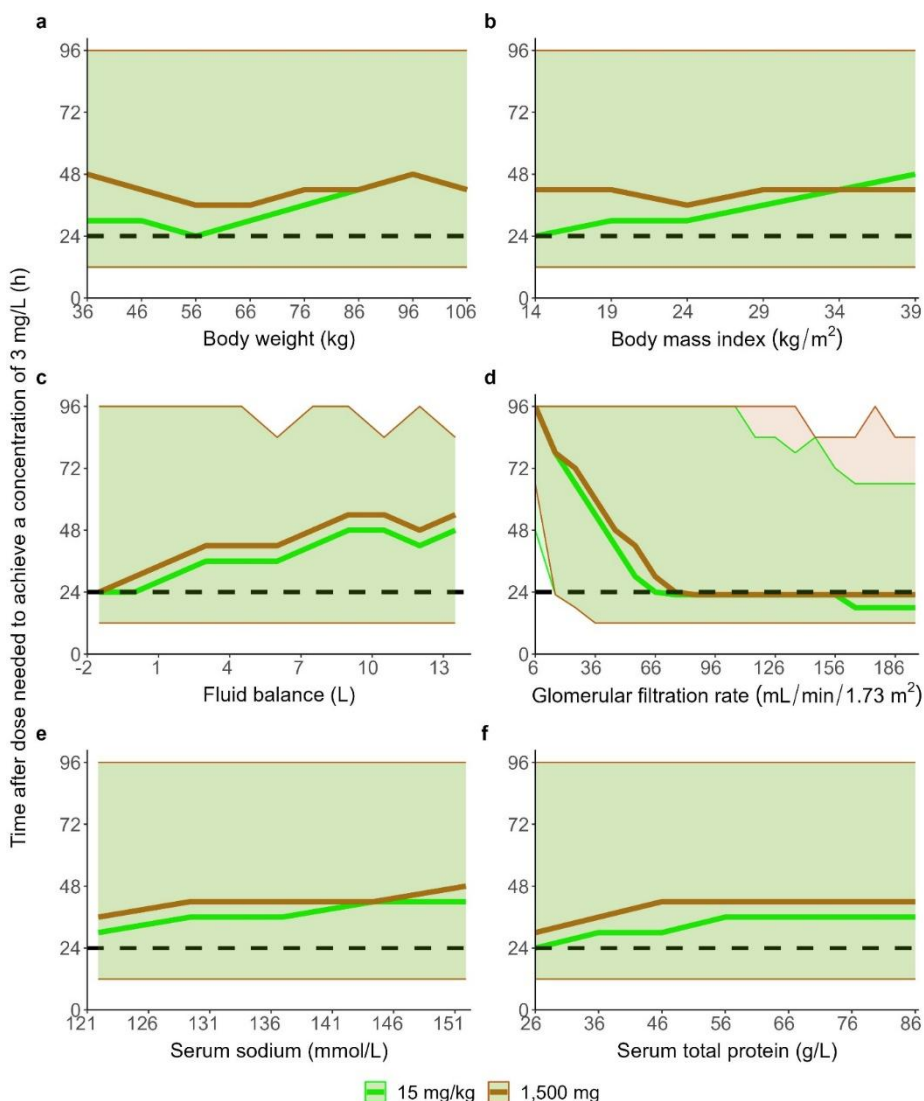
Supplementary Figure 29 MIC for Enterobacterales isolated from blood, urine, lung, or wound cultures obtained from patients on amikacin admitted to UZ Leuven



Supplementary Figure 30 Simulated 24-hour area under the concentration–time curve (AUC_{24h}) over the range of (a) body weight, (b) body mass index (c) fluid balance (d) glomerular filtration rate (e) serum sodium, and (f) serum total protein at an MIC₉₀ of 2 mg/L for dosing scenarios of 15 mg/kg and 1,500 mg, respectively. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line, the solid lines, and the shaded areas represent the 160 mg·h/L AUC_{24h} based on an MIC₉₀ of 2 mg/L, the median AUC_{24h}, and the AUC_{24h} range, respectively

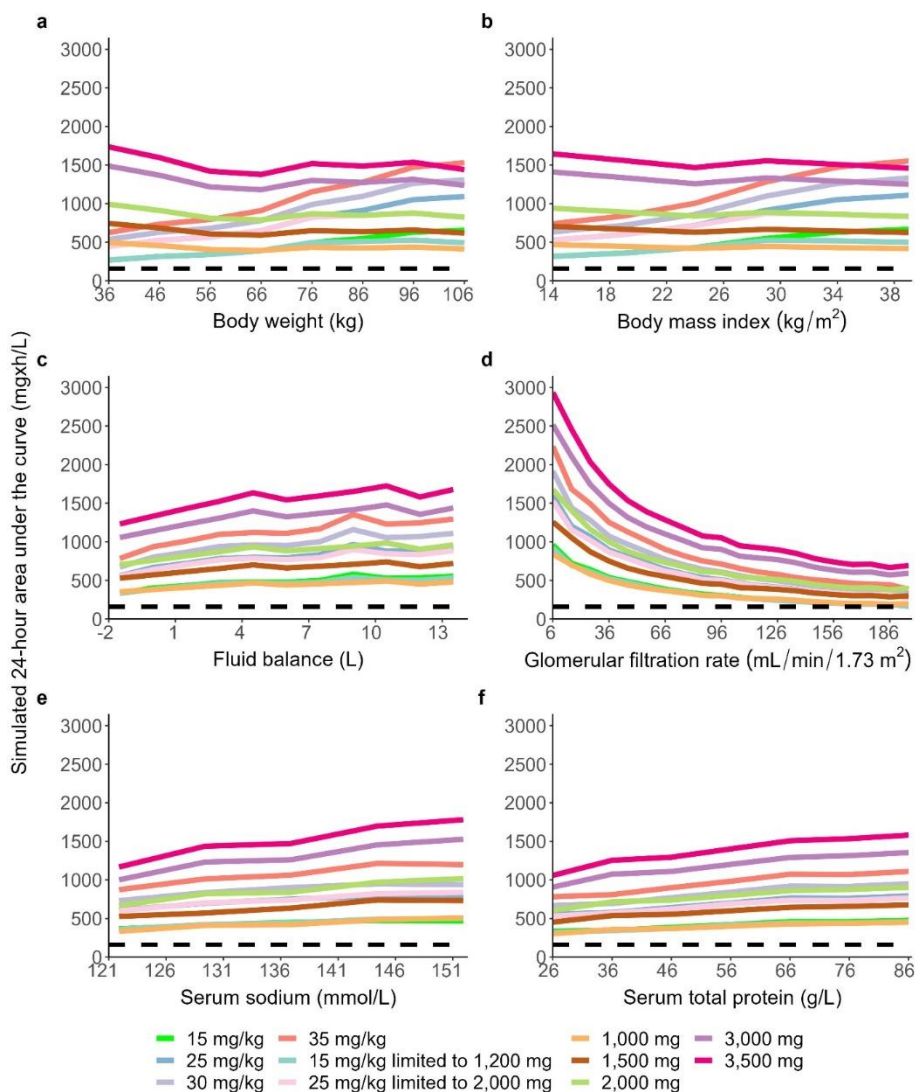


Supplementary Figure 31 Simulated peak concentration (C_{max}) over the range of (a) body weight, (b) body mass index (c) fluid balance (d) glomerular filtration rate (e) serum sodium, and (f) serum total protein at an MIC_{90} of 2 mg/L for the dosing scenarios of 15 mg/kg and 1,500 mg, respectively. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line, the solid lines, and the shaded areas represent the 16 mg/L C_{max} based on an MIC_{90} of 2 mg/L, the median C_{max} , and the C_{max} range, respectively

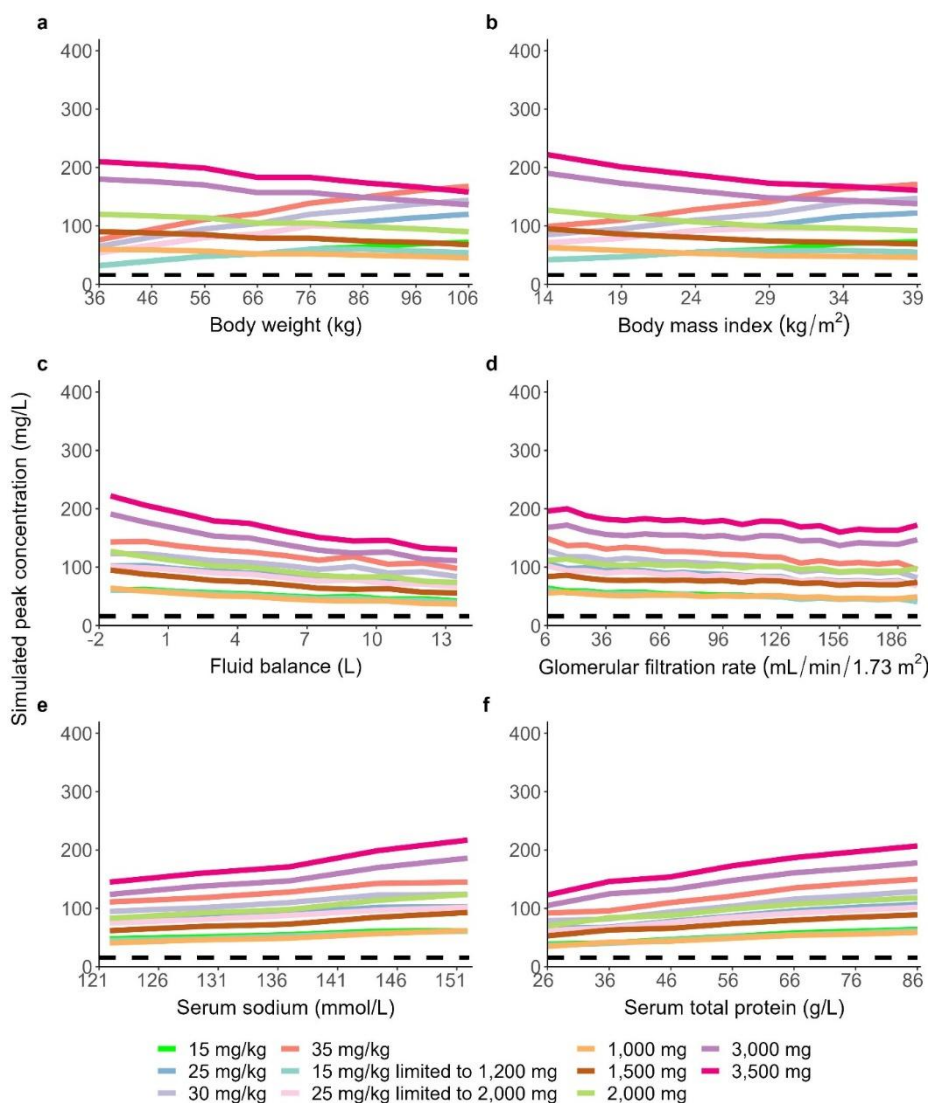


Supplementary Figure 32 Simulated time after dose to achieve a concentration of 3 mg/L ($\text{Time}_{3\text{mg/L}}$) over the range of (a) body weight, (b) body mass index (c) fluid balance (d) glomerular filtration rate (e) serum sodium, and (f) serum total protein at an MIC_{90} of 2 mg/L for the dosing strategies of 15 mg/kg and 1,500 mg, respectively. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line, the

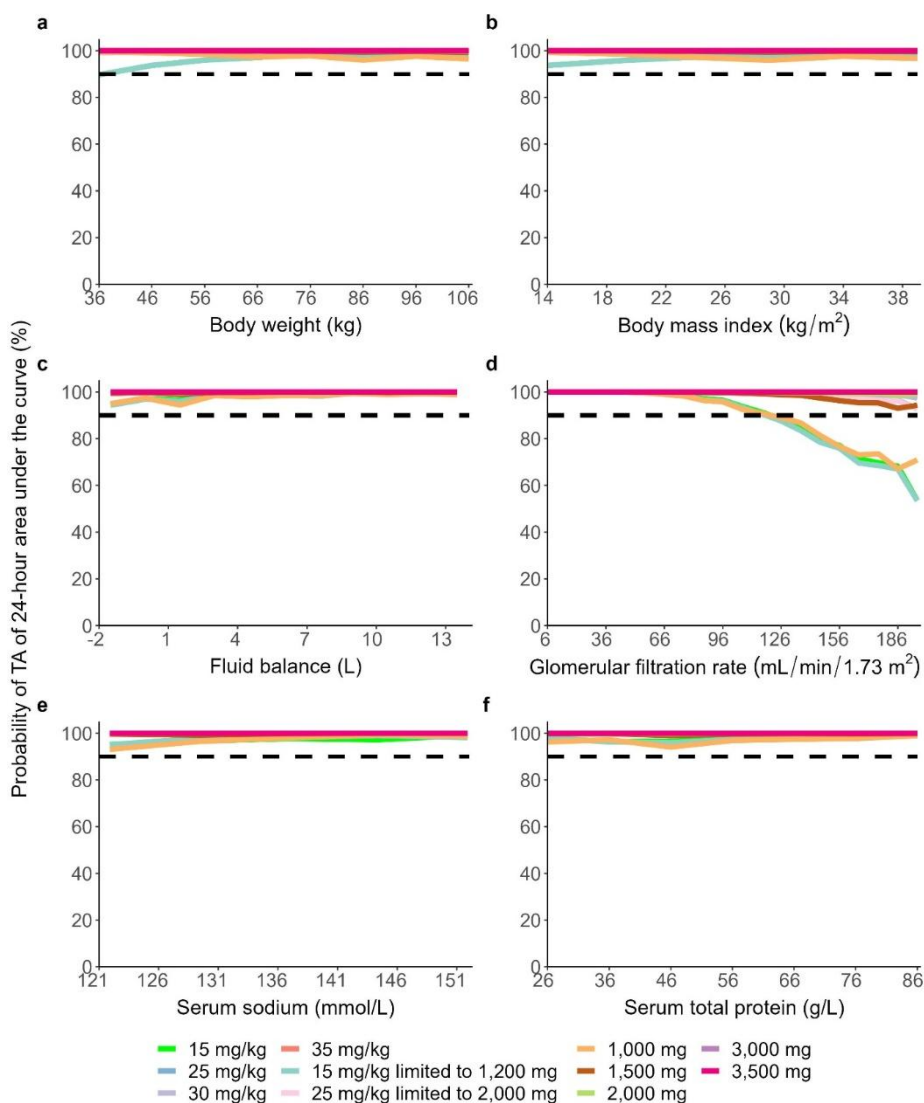
solid lines, and the shaded areas represent $\text{Time}_{3\text{mg/L}}$ of 24 h, the median $\text{Time}_{3\text{mg/L}}$, and the $\text{Time}_{3\text{mg/L}}$ range, respectively



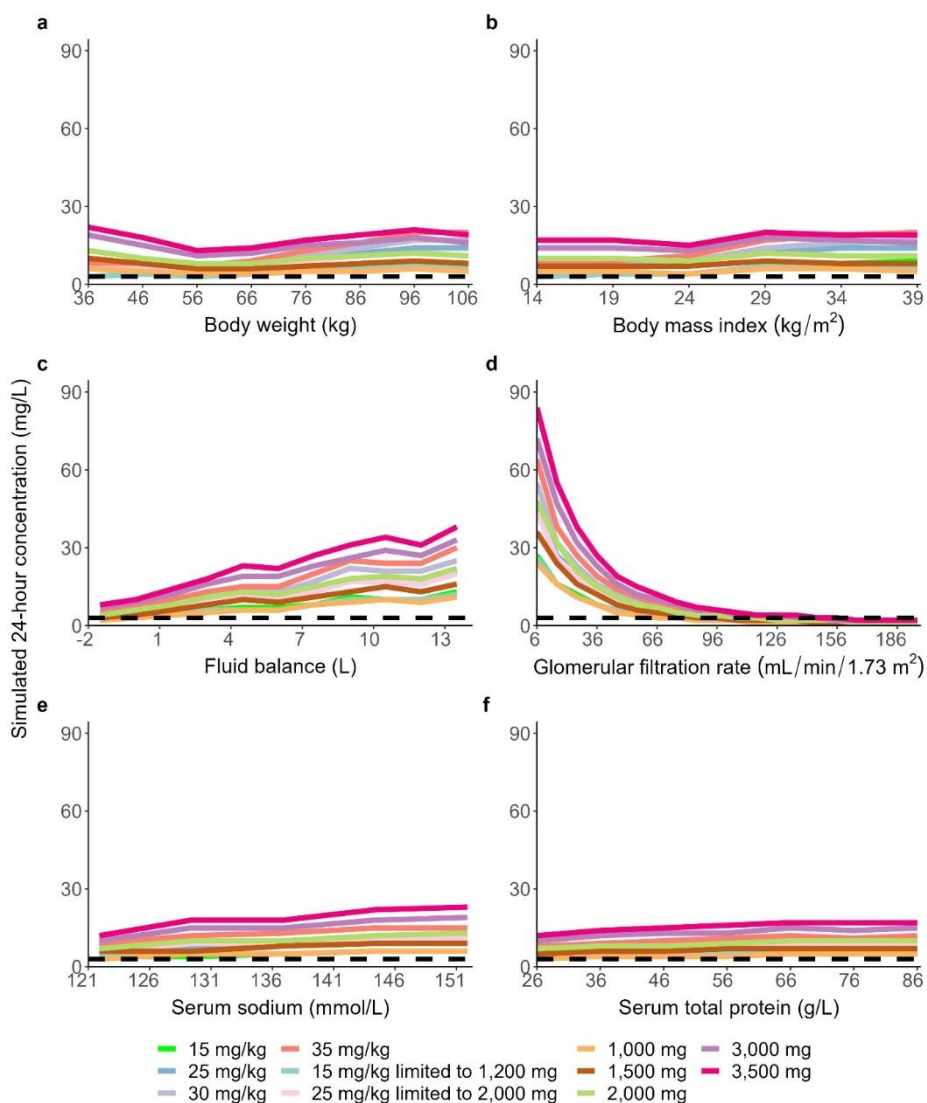
Supplementary Figure 33 Median values of the simulated 24-hour area under the time concentration–time curve (AUC_{24h}) over the range of (a) body weight, (b) body mass index (c) fluid balance (d) glomerular filtration rate (e) serum sodium, and (f) serum total protein at an MIC_{90} of 2 mg/L for eleven single dosing strategies. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line represents the 160 mgxh/L AUC_{24h} based on an MIC_{90} of 2 mg/L



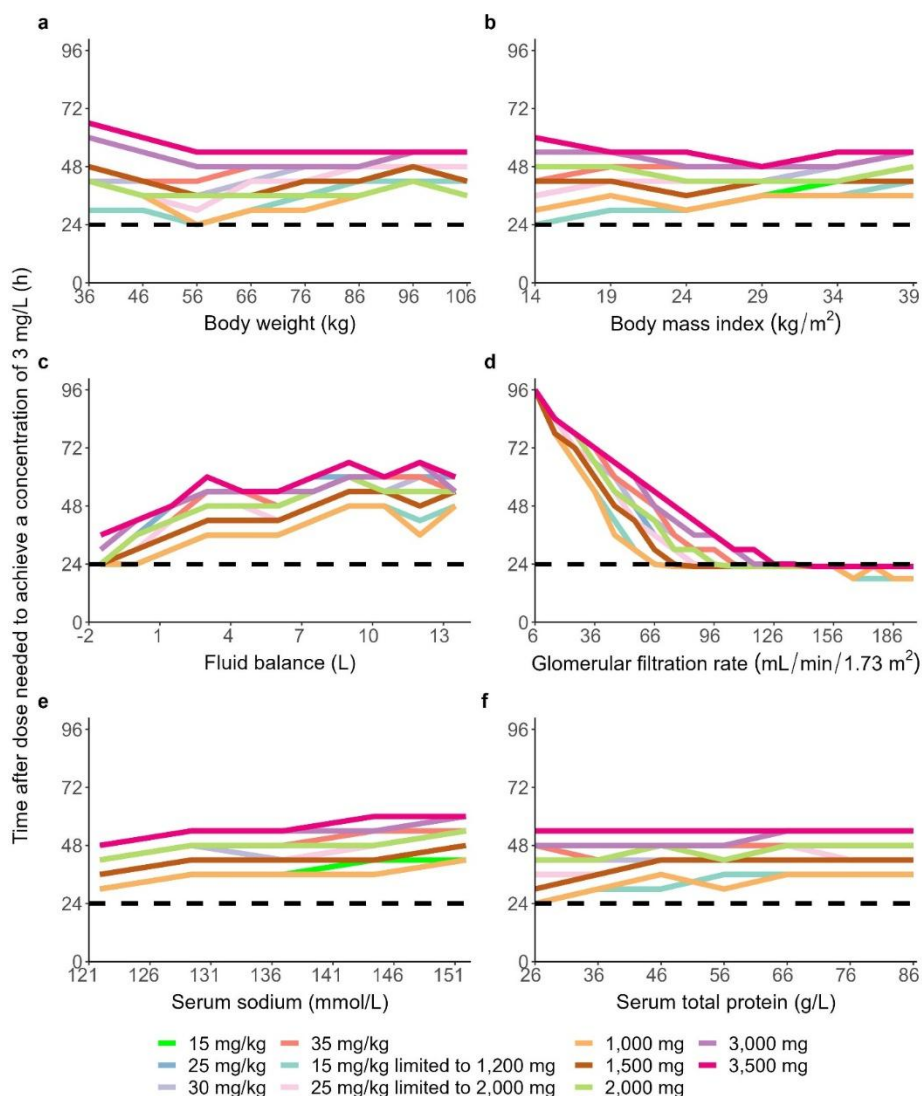
Supplementary Figure 34 Median values of the simulated peak concentration (C_{max}) over the range of (a) body weight, (b) body mass index (c) fluid balance (d) glomerular filtration rate (e) serum sodium, and (f) serum total protein at an MIC_{90} of 2 mg/L for eleven single dosing strategies. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line represents the 16 mg/L C_{max} based on an MIC_{90} of 2 mg/L



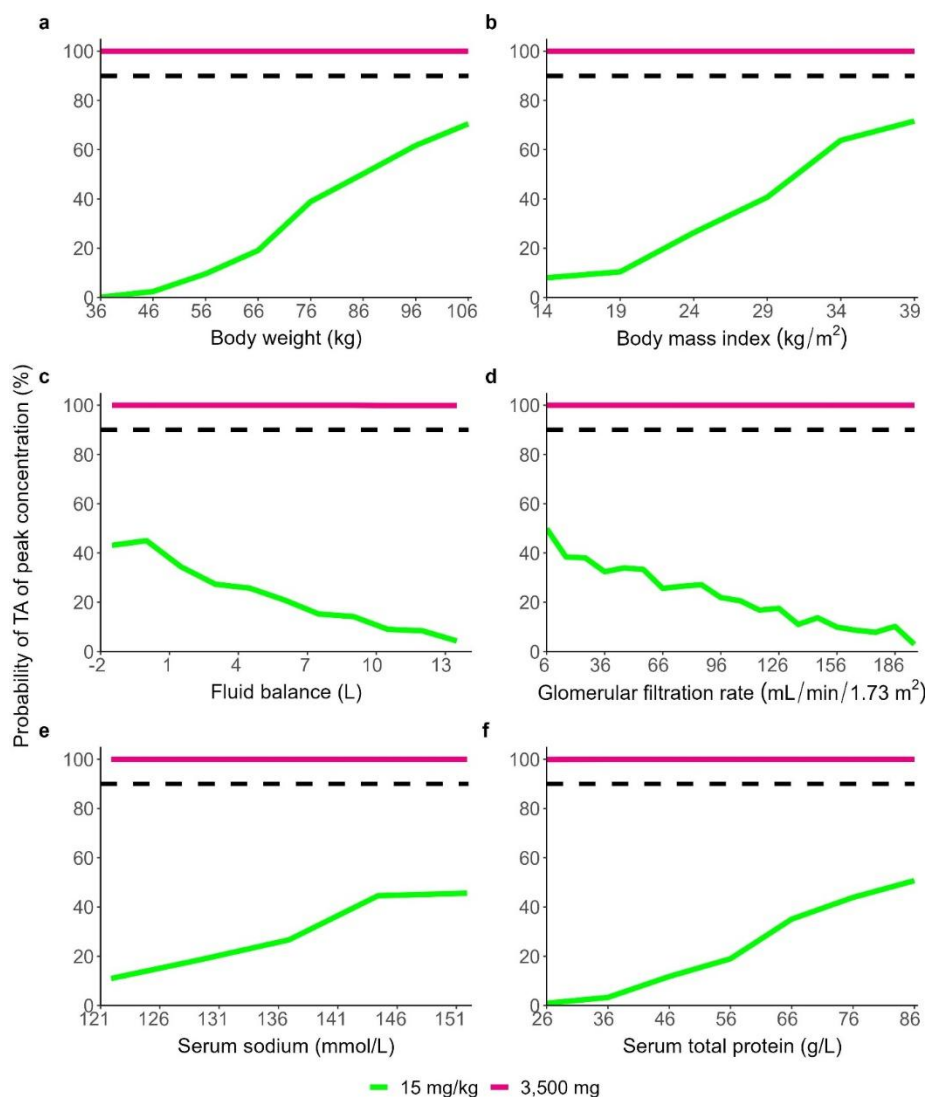
Supplementary Figure 35 Probability of target attainment in terms of 24-hour area under the time concentration–time curve over the range of (a) body weight, (b) body mass index (c) fluid balance (d) glomerular filtration rate (e) serum sodium, and (f) serum total protein at an MIC₉₀ of 2 mg/L for eleven single dosing strategies. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line represents a probability of target attainment of 90%



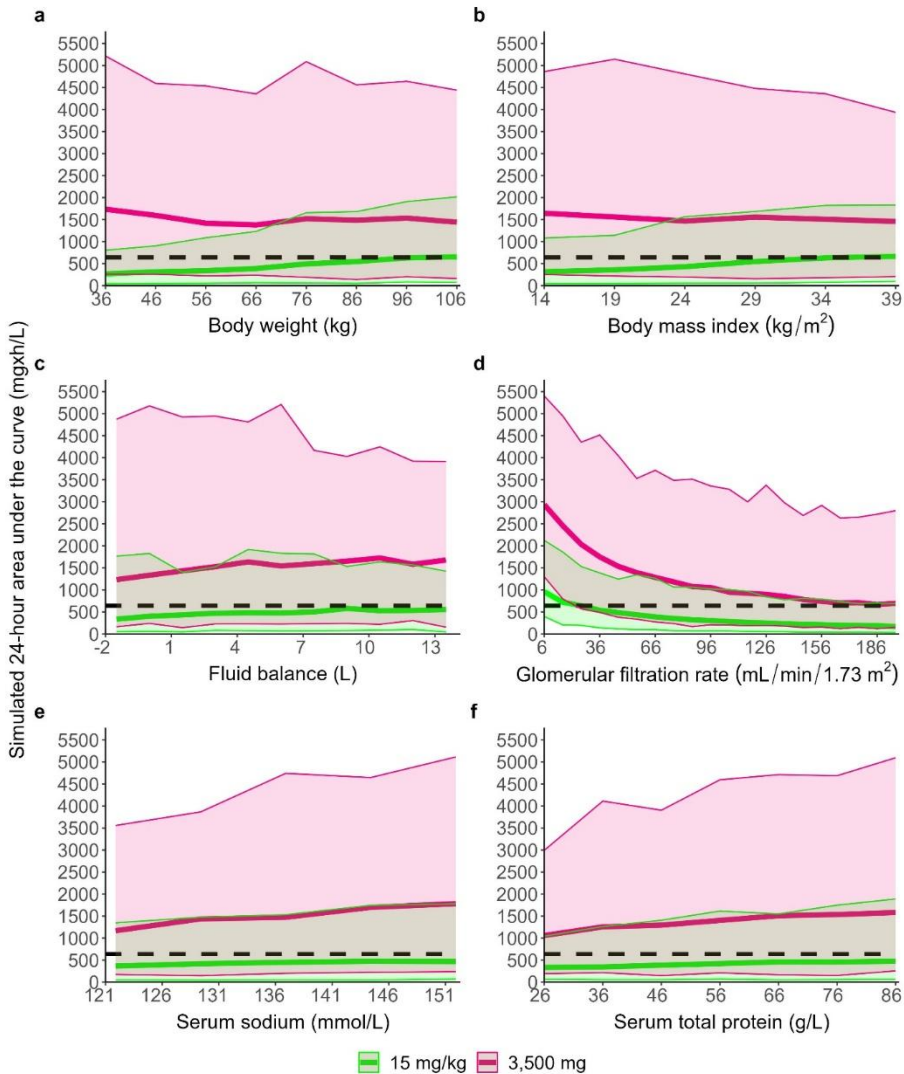
Supplementary Figure 36 Median values of the simulated 24 hours concentration (C_{24h}) across the range of (a) body weight, (b) body mass index (c) fluid balance (d) glomerular filtration rate (e) serum sodium, and (f) serum total protein at an MIC_{90} of 2 mg/L for eleven single dosing strategies. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line represents C_{24h} of 3 mg/L



Supplementary Figure 37 Median values of simulated time after dose to achieve a concentration of 3 mg/L ($\text{Time}_{3\text{mg/L}}$) over the range of (a) body weight, (b) body mass index (c) fluid balance (d) glomerular filtration rate (e) serum sodium, and (f) serum total protein at an MIC_{90} of 2 mg/L for eleven single dosing strategies. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line represents a $\text{Time}_{3\text{mg/L}}$ of 24 h

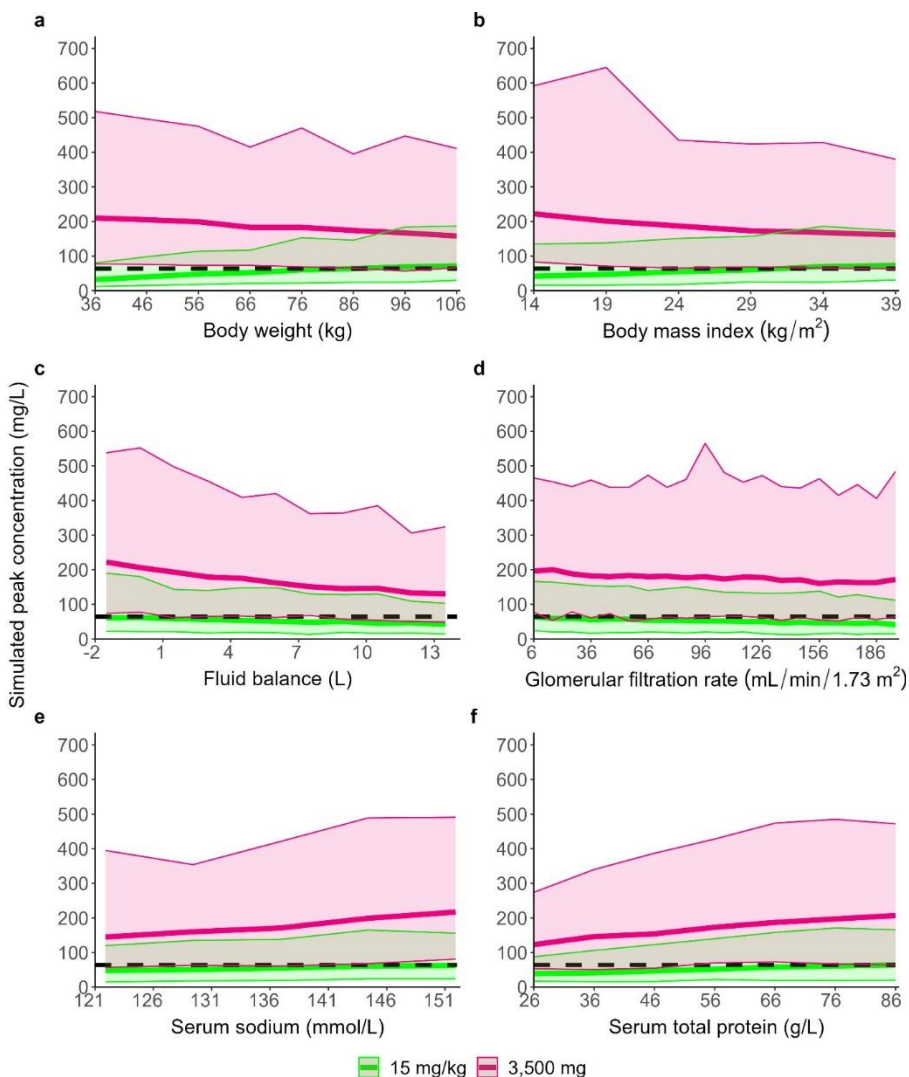


Supplementary Figure 38 Probability of target attainment in terms of peak concentration over the range of (a) body weight, (b) body mass index (c) fluid balance (d) glomerular filtration rate (e) serum sodium, and (f) serum total protein at a EUCAST clinical breakpoint of 8 mg/L for the dosing scenarios of 15 mg/kg and 3,500 mg, respectively. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line represents a probability of target attainment of 90%. TA, target attainment

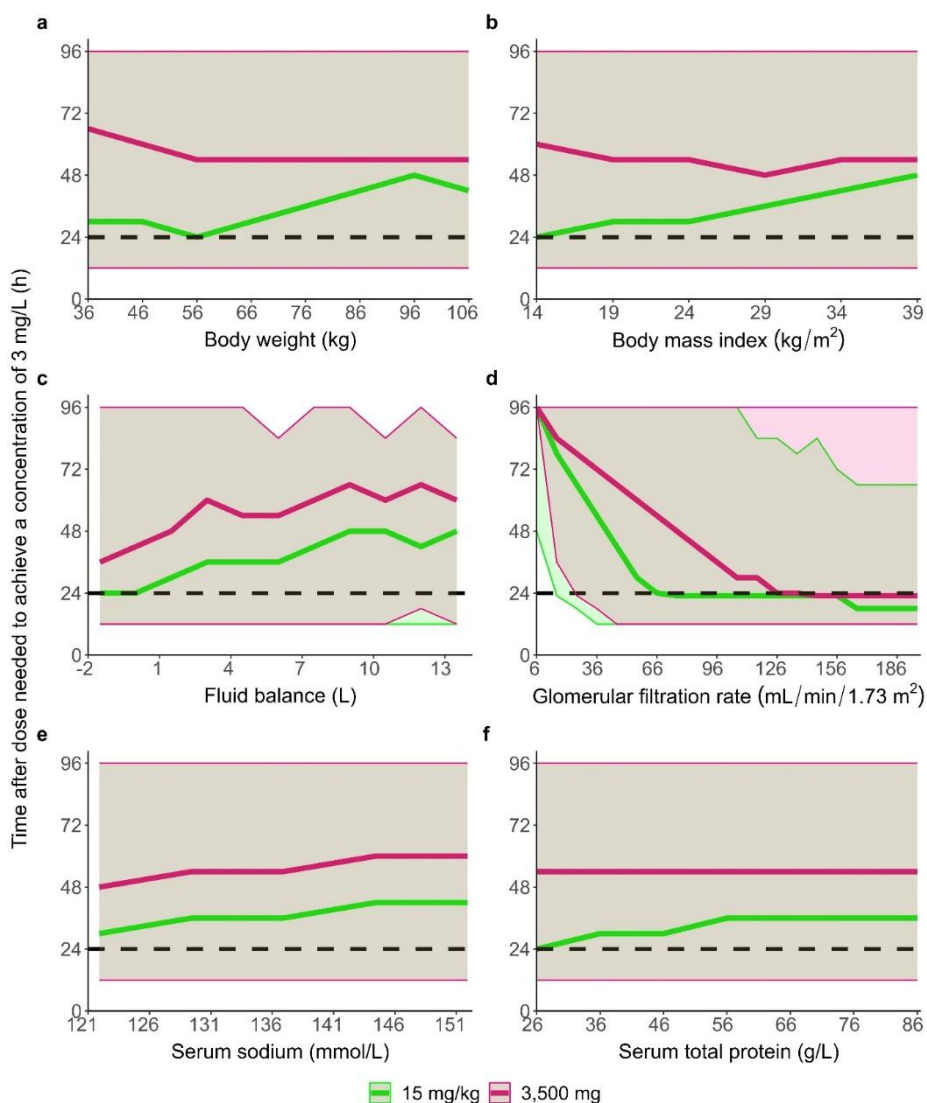


Supplementary Figure 39 Simulated 24-hour area under the concentration–time curve (AUC_{24h}) over the range of (a) body weight, (b) body mass index (c) fluid balance (d) glomerular filtration rate (e) serum sodium, and (f) serum total protein at a EUCAST clinical breakpoint of 8 mg/L for dosing scenarios of 15 mg/kg and 3,500 mg, respectively. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line, the solid lines, and the shaded areas represent the 640 mgxh/L AUC_{24h} based on

a EUCAST clinical breakpoint of 8 mg/L, the median AUC_{24h}, and the AUC_{24h} range, respectively

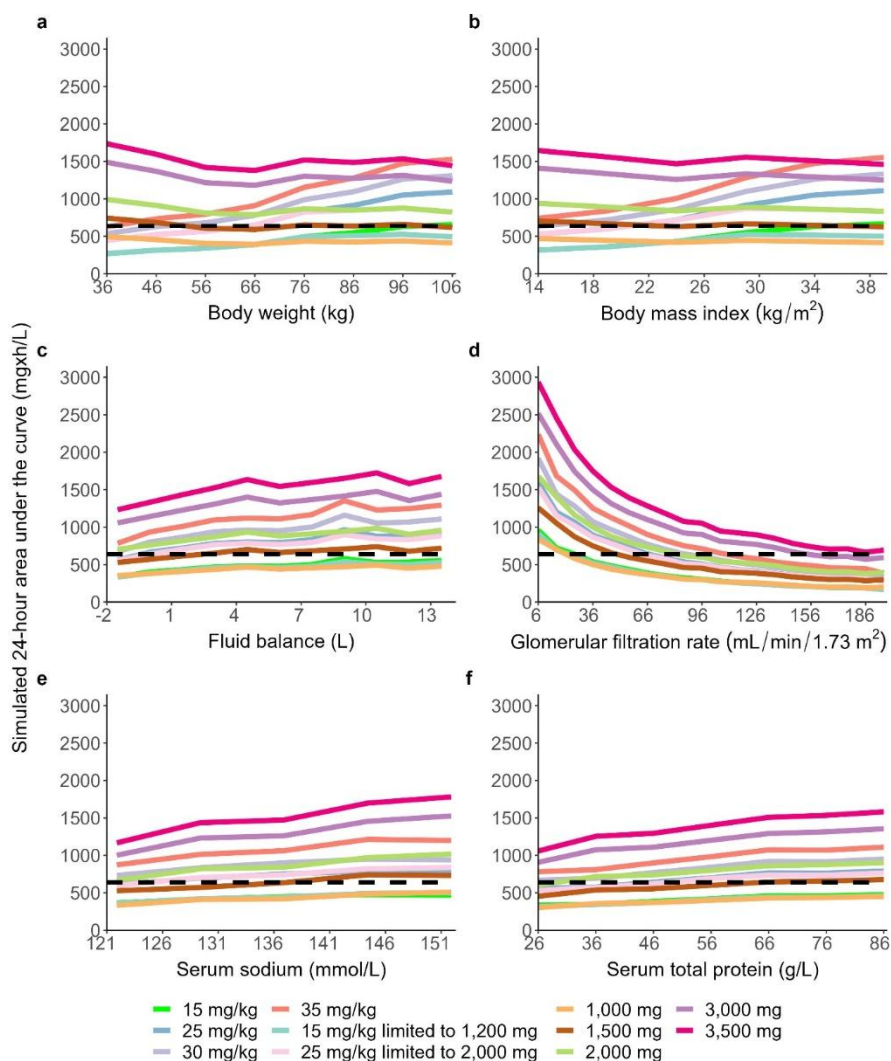


Supplementary Figure 40 Simulated peak concentration ($C_{\max}$) over the range of (a) body weight, (b) body mass index (c) fluid balance (d) glomerular filtration rate (e) serum sodium, and (f) serum total protein at a EUCAST clinical breakpoint of 8 mg/L for dosing scenarios of 15 mg/kg and 3,500 mg, respectively. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line, the solid lines, and the shaded areas represent the 64 mg/L $C_{\max}$ based on a EUCAST clinical breakpoint of 8 mg/L, the median $C_{\max}$, and the $C_{\max}$ range, respectively

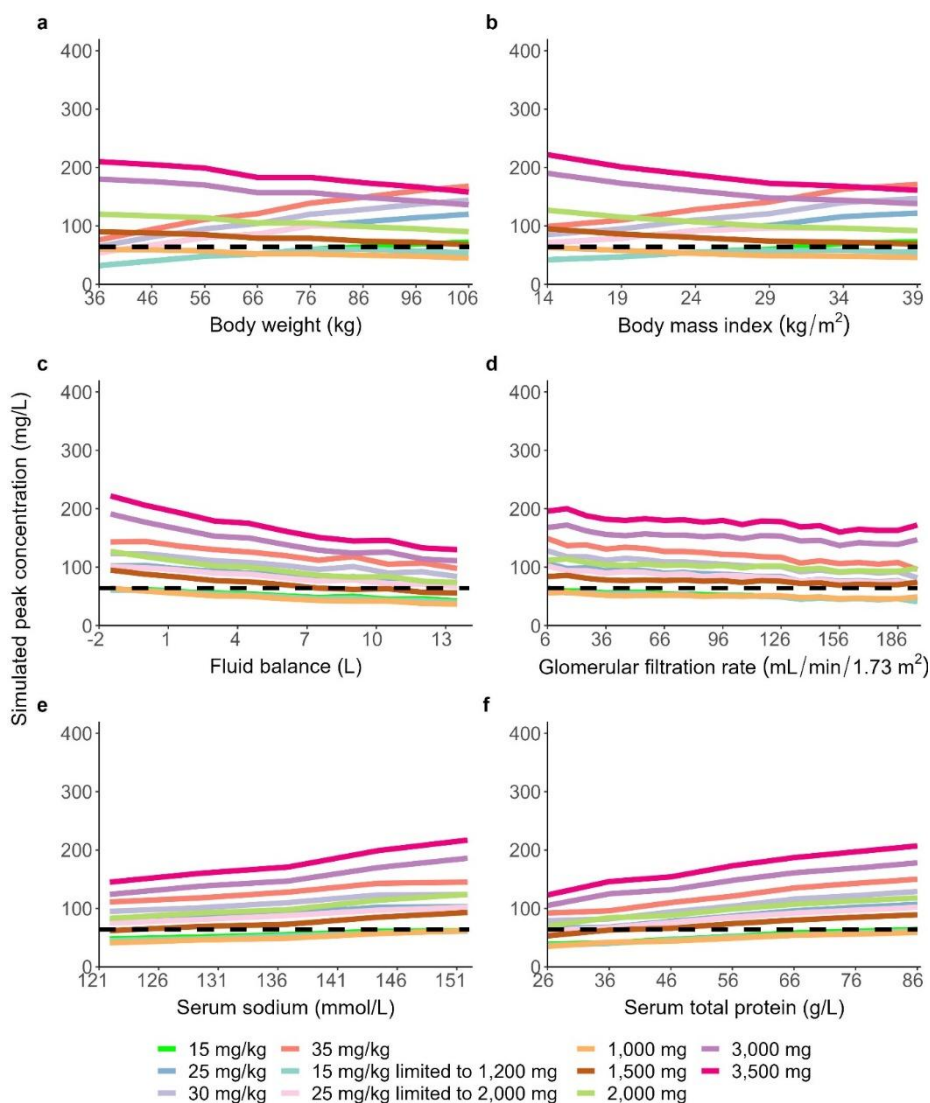


Supplementary Figure 41 Simulated time after dose to achieve a concentration of 3 mg/L (Time_{3 mg/L}) over the range of (a) body weight, (b) body mass index (c) fluid balance (d) glomerular filtration rate (e) serum sodium, and (f) serum total protein at a EUCAST clinical breakpoint of 8 mg/L for the dosing strategies of 15 mg/kg and 3,500 mg. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line, the

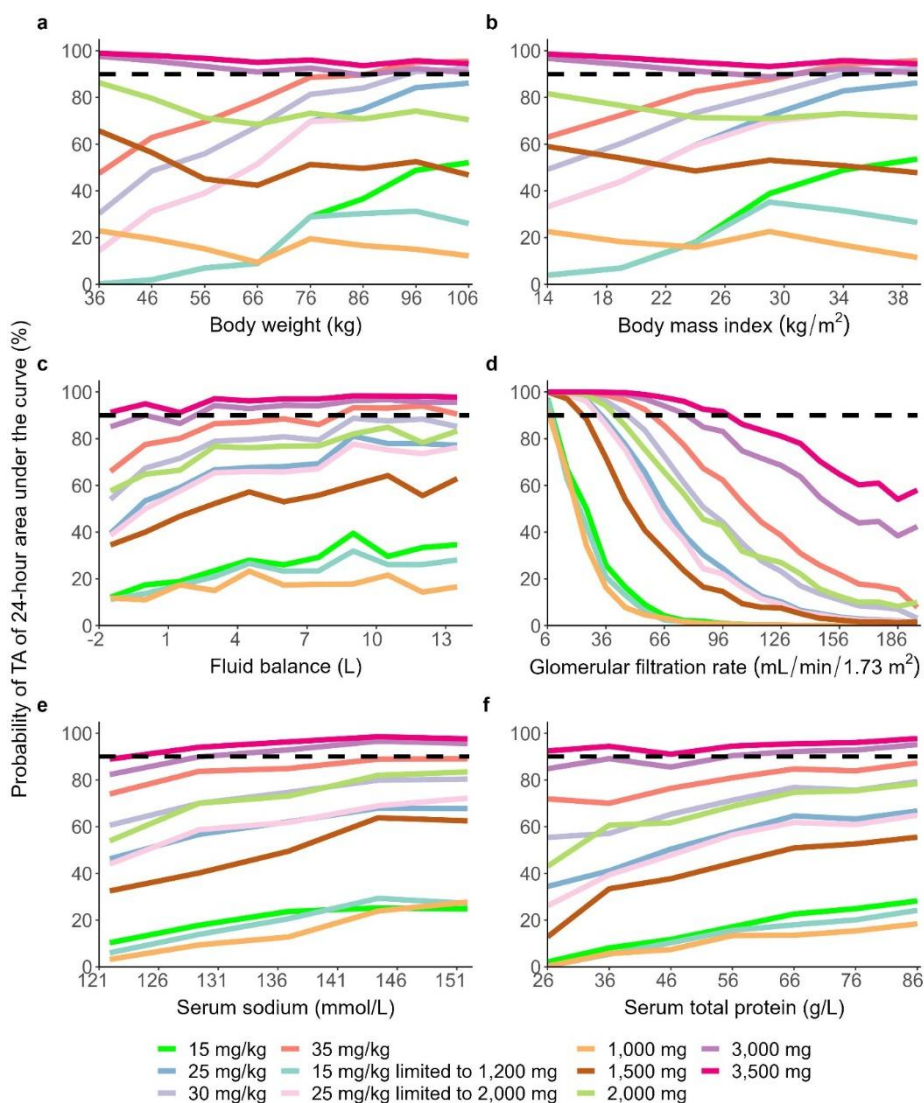
solid lines, and the shaded areas represent $\text{Time}_{3\text{ mg/L}}$, the median $\text{Time}_{3\text{ mg/L}}$, and the $\text{Time}_{3\text{ mg/L}}$ range, respectively



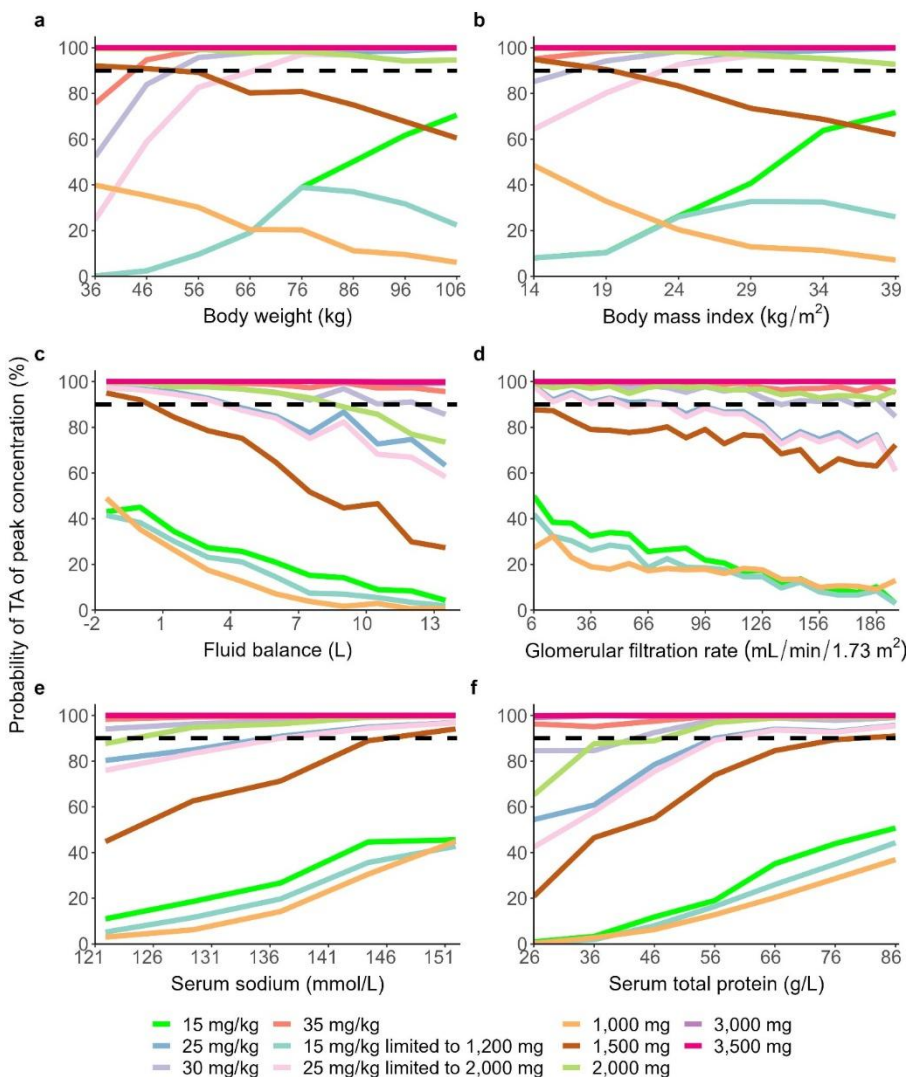
Supplementary Figure 42 Median values of the simulated 24-hour area under the time concentration–time curve (AUC_{24h}) over the range of (a) body weight, (b) body mass index (c) fluid balance (d) glomerular filtration rate (e) serum sodium, and (f) serum total protein at a EUCAST clinical breakpoint of 8 mg/L for eleven single dosing strategies. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line represents the 640 mg×h/L AUC_{24h} based on a EUCAST clinical breakpoint of 8 mg/L



Supplementary Figure 43 Median values of the simulated peak concentration (C_{max}) over the range of (a) body weight, (b) body mass index (c) fluid balance (d) glomerular filtration rate (e) serum sodium, and (f) serum total protein at a EUCAST clinical breakpoint of 8 mg/L for eleven single dosing strategies. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line represents the 64 mg/L C_{max} corresponding to a EUCAST clinical breakpoint of 8 mg/L



Supplementary Figure 44 Probability of target attainment in terms of 24-hour area under the time concentration–time curve over the range of (a) body weight, (b) body mass index (c) fluid balance (d) glomerular filtration rate (e) serum sodium, and (f) serum total protein at a EUCAST clinical breakpoint of 8 mg/L for eleven single dosing strategies. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line represents a probability of target attainment of 90%. TA, target attainment



Supplementary Figure 45 Probability of target attainment in terms of peak concentration over the range of (a) body weight, (b) body mass index (c) fluid balance (d) glomerular filtration rate (e) serum sodium, and (f) serum total protein at a EUCAST clinical breakpoint of 8 mg/L for eleven single dosing strategies. Covariate effects are visualized univariately for ease of interpretation. Correlation between covariates is accounted for by sampling each virtual patient's other covariates from the covariance matrix (*cf.* Materials and methods). The dashed horizontal line represents a probability of target attainment of 90%. TA, target attainment

Supplementary Table 20 Baseline characteristics of patients on a single standard dose of amikacin study (n=97) of the uniformly distributed sample

	Unit	Median	Minimum	Maximum	Stepsize	Number of scenarios	Total number of patients ^a
Body weight	kg	70.0	36.0	115.0	10	8	400
Body mass index	kg/m ²	24.7	13.7	39.6	5.0	6	300
24h fluid balance	L	2.8	-1.5	13.9	1.5	11	550
eGFR _{CKD-EPI,24h}	mL/min/1.73 m ²	53.0	6.0	200.0	10	20	1,000
Serum sodium	mmol/L	138.4	121.9	155.9	7.5	5	250
Serum total protein	g/L	66.0	26.0	90.0	10	7	350
Total						57	2,850

^a 50 patients per scenario. eGFR, estimated glomerular filtration rate using the Chronic Kidney Disease Epidemiology Collaboration equation at 24 hours after inclusion; h, hours

Supplementary Table 21 Variance–covariance matrix of body weight and covariates

	Units	Body weight	Body mass index	24h fluid balance	GFR _{CKD-EPI, 24h}	Serum sodium	Serum total protein
Body weight	kg	256.9	69.4	4.3	-153.7	-9.3	24.8
Body mass index	kg/m ²	69.4	25.1	1.1	-60.7	-1.3	11.1
24h fluid balance	L	4.3	1.1	5.1	-18.8	0.9	0.9
eGFR _C KD-EPI,24h	mL/min/1.73 m ²	-153.7	-60.7	-18.8	1375.8	-31.4	-98.4
Serum sodium	mmol/L	-9.3	-1.3	0.9	-31.4	40.5	-1.4
Serum total protein	g/L	24.8	11.1	0.9	-98.4	-1.4	123.2

eGFR, estimated glomerular filtration rate using the Chronic Kidney Disease Epidemiology Collaboration equation at 24 hours after inclusion; h, hours

NONMEM control stream

```
$PROBLEM AMIKACIN SIMULATION
$INPUT ID TIME AMT RATE DV MDV EVID GROU BMI WT EGFR
TOPR SODI FB FLG
$DATA Input_data_Amikacin.csv ACCEPT= (FLG EQ.1)
IGNORE=C
$SUBROUTINE ADVAN13 TOL=9
$MODEL NCOMPARTMENTS=3
COMP= (CENTRAL DEFOBS)
COMP=(PERIPH)
COMP=(AUC)
$PK TVCL =THETA(1) *(EGFR/52.0)
**THETA(2)*(BMI/24.0)**THETA (3)
CL =TVCL*EXP(ETA(1))
TVV1 =THETA(4)*(BMI/24.0)**THETA(5)*(TOPR/6
6.0)** THETA(6)*
(SODI/138.25)**THETA(7)*(1+THETA(8)*(FB
-2796.5)*THETA(9))* THETA(10)**(1-
THETA(9))
V1 =TVV1*EXP(ETA(2))
TVQ =THETA(11)
Q =TVQ
TVV2 =THETA(12)
V2 =TVV2
S1 =V1
S2 =V2
KE =CL/V1
K12 =Q/V1
K21 =Q/V2
$DES DADT(1) =K21*A(2) - K12*A(1) - KE*A(1)
DADT(2) =K12*A(1) - K21*A(2)
DADT(3) =A(1)/V1
AUC =A(3)
$ERROR IPRED =F
IRES =DV-IPRED
DEL =0
IF (F.EQ.0) DEL=1
W =IPRED+DEL
Y =IPRED+W*ERR(1)
IWRES =IRES/W
$THETA 1.8 ; TVCL
0.86 ; Estimated glomerular filtration rate (EGFR)
on TVCL
```

```

0.86 ; Body mass index (BMI) on TVCL
16 ; TVV1
0.42 ; Body mass index (BMI) on TVV1
-0.53 ; Total protein (TOPR) on TVV1
-2.1 ; Sodium (SODI) on TVV1
0.00005 ; Fluid balance (FB) on TVV1
1 FIX ; FLG1
0.9 ; Correction factor for fluid balance (FB)
0.77 ; TVQ
13 ; TVV2
$OMEGA BLOCK(
2) ; CL
0.14 ; V1
0.04 0.03
$SIGMA 0.04 ; Proportional error
$SIMULATIO (9784275) ONLYSIM SUBPROBLEMS=1000
N
$TABLE ID TIME AMT GROU RATE MDV PRED IPRED AUC BMI
WT EGFR TOPR SODI FB
NOPRINT ONEHEADER FILE=Output_data.csv

```

General discussion and perspectives

In this thesis, we proposed pharmacometric approaches in which popPK models were applied and, where necessary, further developed to support MIPD and simulation to optimize dosing strategies for three intravenous antibiotics mainly eliminated by the kidneys, in selected special patient populations (older adults, critically ill adult patients with preserved renal function or ARC admitted to the ICU, and critically ill adult patients admitted to the ED), while minimizing the risk of toxicity. In these clinical settings, a narrow therapeutic window and high PK variability explain why standard dosing strategies resulted in suboptimal exposure, indicating that a dosing strategy is inappropriate [211,278,328]. MIPD was applied for antibiotics with a narrow therapeutic window where the efficacy-based PKPD target is assessed using concentrations measured at steady state [78]. Otherwise, simulation-based dose optimization using fixed or covariate-based dosing strategies was investigated. The recommended dosing strategy was associated with attainment of the efficacy-based PKPD targets, and a decreased risk of side effects based on safety metrics. Importantly, the contribution of this thesis extends to the identification of clinically feasible dosing strategies that can be implemented in real-world clinical care.

Pharmacometrics is a quantitative discipline that not only quantifies antibiotic PK but also predicts drug exposure, PKPD target attainment and safety metrics in (special) patient populations. Approaches such as MIPD, where TDM is coupled with a popPK model, provide individualized predictions that surpass traditional dosing approaches [78]. PopPK modeling and simulation

also enable systematic evaluation of correlated covariates by incorporating each covariate in parallel models, allowing assessment of their clinical impact through a reproducible and ethically sound scientific framework. In simulations, clinically and practically plausible dosing strategies scenarios are explored to prioritize the most promising dosing strategies, thereby accelerating the identification of dosing strategies that achieve adequate PKPD target attainment and minimal risk of toxicity.

Pharmacometrics can be summarized as a four-step cycle, based on the “learn and confirm” cycle, introduced by Sheiner, the founder of this discipline [367]. In clinical research, this cycle includes (1) data collection, (2) population PK(–PD) modeling, (3) pharmacometrics-based dosing decision approaches (*i.e.*, MIPD or dosing simulations), and (4) prospective clinical validation (**Figure 17**). The first three steps constitute the “learn” phase, while the fourth step represents the “confirm” phase [367]. In the context of this thesis, the focus was on the “learn” phase particularly steps 2 and 3 using available data (step 1). The “confirm” phase falls outside the scope of this thesis, as it requires prospective studies to demonstrate whether the optimized dosing strategies improve PKPD target attainment while maintaining a similar or decreased risk of side effects in real-world clinical practice. This distinction is important from a translational perspective, as the dosing strategies identified in this thesis were selected not only for their efficacy and safety, but also for their realistic feasibility in clinical practice and their suitability for prospective validation.

The following sections include an explanation of each step and proposed improvements to refine the use of pharmacometrics in clinical research and drug development.

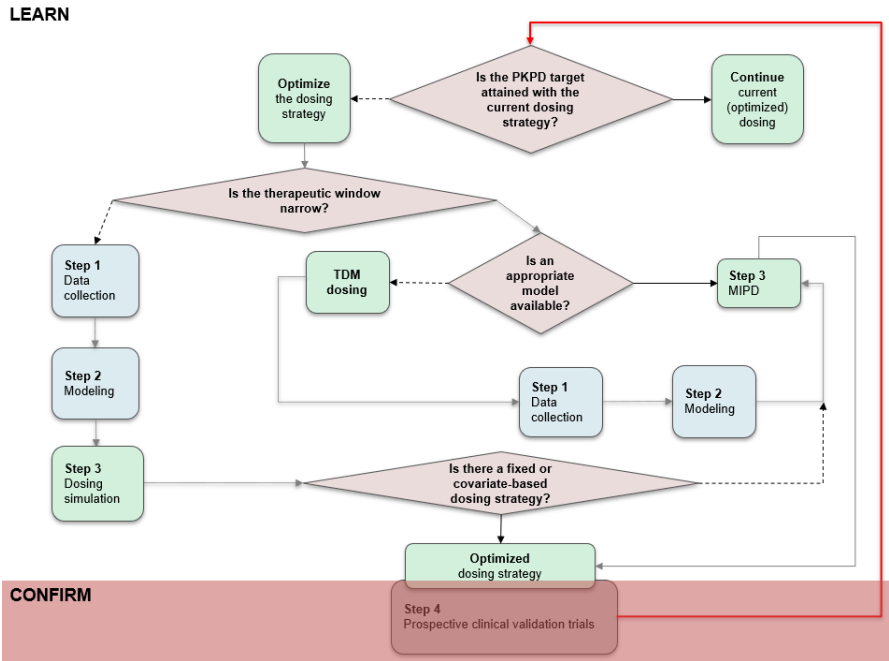


Figure 17. Diagram of the four-step “learn and confirm” cycle for pharmacometrics-based dose optimization in special patient populations [367]. Solid black arrows indicate a “yes” answer, while dashed black arrows indicate a “no” answer. Green arrows represent the sequence of steps followed within each scenario, and the red arrow illustrates the transition from the confirm phase back to the learn phase. The learn phase includes data collection, modeling, and pharmacometrics-based dosing decision approaches such as simulation or model-informed precision dosing (MIPD). The confirm phase consists of prospective clinical validation trials. PKPD, pharmacokinetic–pharmacodynamic; TDM, therapeutic drug monitoring

Step 1. Data collection

Data collection forms a fundamental component of pharmacometrics, as it provides the clinical, dosing, and concentration information required to

perform TDM or MIPD. These data, combined with prior biological and pharmacological knowledge in a (hybrid or/and continuous) modeling approach, are used to develop popPK(–PD) models, and generate virtual patient datasets for simulation-based dosing optimization [70,78]. Data used in pharmacometrics may originate from different sources, including nonclinical datasets (*in vitro* and *in vivo* experiments), RCT, routine clinical TDM, and observational PK(–PD) studies [70]. The latter two formed the basis of this thesis and are therefore discussed further in the subsections below.

For studies based on human data, every dataset must be supported by clearly documented study protocols, and ethics approval to ensure consistency, reproducibility, and regulatory compliance. Informed consent may be waived under specific conditions, but ethics approval remains required. Detailed and complete reporting of sampling times, dosing history, assay characteristics, and (potential) covariates is essential, as inaccuracies in these elements can compromise the subsequent pharmacometric analyses [87,208]. Additionally, standardized reporting frameworks would improve reproducibility and the utility of datasets for MIPD and virtual patient generation [208].

Data can be collected from single-center or multi-center studies located in one or multiple countries. When data are obtained from multiple centers, differences in laboratory assays, sample storage conditions, and other characteristics that may influence drug exposure or response should be

clearly reported. Such reporting facilitates appropriate harmonization, interpretation, and integration of the data. These efforts must also be accompanied by systematic reporting of center-level variability to enable meaningful pooling and comparison. In both cases, representation of diverse patient populations with broader, inclusive criteria is favorable.

There remains considerable room for improvement in how routinely collected (potential) covariates are reported. Definitions must be clear, measurable, standardized and clinically meaningful [87,298]. The overlapping use of ethnicity and race is a good example of two terms that lack clear definition in the field of pharmacometrics [297]. In fact, race is a social construct, whereas ethnicity often reflects cultural background [297]. Body weight provides an example of poor measurement in clinical practice. Although it is frequently used to stratify dosing strategies, it is in many hospital settings estimated rather than measured. SCr measurements also require standardization, as different assay methods can produce inconsistent values [368]. mCrCL_{24h} is a good example of a clinically unmeaningful covariate as it is clinically informative covariate that is not available at early decision times (*e.g.*, on day 1), because it requires a day of urine collection, which limits its use for early dose selection. Improvements in data infrastructure are also required. Integration of EHRs capable of exporting data in standardized, analysis-ready formats, reduces manual extraction time and errors. As hospitals move toward greater digitalization, automated data extraction and standardized biomarker

measurement could further reduce these limitations, enhancing both the quality and usability of real-world data for pharmacometrics research.

The following two subsections describe the applicability, strengths, limitations, and opportunities of refinement based on lessons learned during this PhD, concerning data collection and analysis.

Routine therapeutic drug monitoring data

Routine TDM data are commonly used for both traditional TDM and MIPD. Although TDM data are collected prospectively during clinical care, these measurements are often used retrospectively in research. This retrospective use allows the inclusion of large patient populations without the logistical effort required for prospective studies. However, this advantage comes with limitations. Clinical records may be incomplete, sampling times may be inaccurately reported, and assay techniques may differ over time even within a single center.

From an efficacy perspective, TDM data often reflected the PKPD indices that were considered relevant at the time of data collection. For many antibiotics, this meant sampling $C_{ss, trough}$, which were believed to correlate with exposure and were easy to obtain [64,78]. However, PKPD indices can evolve as new evidence emerges. Vancomycin is a well-known example: earlier TDM practice focused on C_{trough} but accumulating data demonstrated that dosing should shift towards AUC. While C_{trough} remains informative for antibiotics whose PKPD index is $\%fT > MIC$, it is no longer sufficient when the relevant

PKPD index is AUC/MIC [64,78]. In such cases, at least two concentrations are required to estimate AUC using MIPD.

Rapid turnaround time is especially important for antibiotics, as timely TDM may influence clinical outcomes (e.g., survival rates) [312]. This requires not only rapid measurement, but also timely processing of the entire TDM workflow (*i.e.*, sample collection, transport to the laboratory, analysis, validation, and reporting). Automated assays with high precision and minimal sample preparation would support assessment of TDM at the appropriate clinical time points. When participating centers lack on-site assay to quantify drug concentrations, blood samples may be transported to centralized laboratories equipped with validated TDM assays. In this context, emerging logistics solutions — such as specialized medical shipping companies or medical use of drones — offer rapid transport options [369]. These solutions are particularly valuable in rural regions, geographically distant hospitals, and low-resource healthcare centers where laboratory infrastructure is limited [369].

Observational pharmacokinetic (pharmacodynamic) studies

Data collected through observational PK(–PD) studies can serve multiple purposes in pharmacometrics: developing popPK(–PD) models, characterizing the statistical significance and clinical relevance of covariates, and generating virtual patient datasets for simulation-based dosing optimization. These studies may be specifically designed for modeling and

simulation purposes, or they may originate from clinical research and later be repurposed for pharmacometric analysis.

The strengths and limitations of observational studies depend heavily on the study design. Most observational PK datasets are generated from single-center studies with a limited number of participants. While these studies provide controlled data collection, the lack of diversity limits generalizability to subsequent pharmacometric results. Multi-center PK studies provide broader representation, but they are conducted less frequently due to logistical challenges (*e.g.*, differences in assays). A major advantage of observational PK studies is the possibility of rich sampling, in which each patient contributes multiple drug concentration measurements at pre-specified time points after drug administration. Rich sampling enhances the characterization of the concentration–time profile.

Although pharmacometric models developed from observational datasets may allow extrapolation to patient populations not fully represented in the study, extrapolation outside the distributions captured in the data must be approached with caution [298]. For this reason, transparent and detailed reporting of population characteristics — including covariate types and assay methodology (specificity, sensitivity) — is essential to guide the appropriate design and interpretation of subsequent pharmacometric analyses [87,208]. Such comprehensive reporting is also necessary for generating physiologically plausible virtual patient datasets and evaluating clinical

decision-making through simulation studies. For example, if a published popPK model does not report sufficient information on the assay used (e.g., lower limit of quantification), applying that model to simulate dosing in patients expected to have low concentrations — such as those with ARC — may yield unreliable predictions. Without knowing the assay sensitivity, low measured concentrations in the original dataset may reflect assay limitations rather than true drug elimination, making extrapolation questionable.

FUTURE PERSPECTIVES

- Promote diversity and inclusiveness in patient populations by recruiting (special) patient populations traditionally underrepresented in clinical research and drug development.
- Align biomarker collection schedules with clinical workflow to ensure their availability at the time of dosing decisions.
- Explore physiologically plausible biomarkers that are measurable at, or close to, the moment of clinical decision-making, and can support pharmacometric approaches.
- Develop detailed standardized reporting guidelines to ensure use of clinical dataset in MIPD implementation and dose optimization simulation.

Step 2. Modeling

For antibiotics that have been on the market for decades, numerous popPK models have already been published. As a result, developing new models was

not necessary for this thesis. Instead, the availability of these models created an opportunity to use them to address broader challenges in clinical research. In this PhD, published models were leveraged to evaluate the existing evidence and develop quantitative solutions to real-world clinical problems.

The first application involved a structured, evidence-based comparison of published PK models for vancomycin (*cf.* Chapter 1). This comparison revealed where evidence was strong, where important knowledge gaps remained, and what each model could support in subsequent pharmacometric analyses. Beginning with alignment of model-development dataset characteristics with those of the available clinical populations enabled a transparent selection of models driven by clinical context and data compatibility [78,79]. However, a standardized guide for model selection is still needed when multiple models exist.

A second application demonstrated that modeling could address clinical challenges that extend well beyond PK quantification (*cf.* Chapters 3 and 4). One prominent challenge arises when key biomarkers required for dosing decisions are not available at, or close to, the time clinical decisions must be made, making them impractical for routine use (*cf.* Chapters 3 and 4). Although alternative biomarkers may exist, they are correlated (*i.e.*, often reflect partially overlapping physiological information), and therefore cannot be included simultaneously in a traditional covariate model (*cf.* Chapter 3) [87]. Furthermore, no consensus typically exists on interchangeable thresholds

between biomarkers. To overcome this, we used a parallel modeling framework in which multiple models were developed from a single structural and statistical published base model; each incorporating one correlated biomarker at a time. This allowed direct comparison of how each biomarker affected predicted exposure and optimized dosing strategies in subsequent pharmacometric analysis. By doing so, we showed how modeling can identify clinically feasible alternative biomarkers that can support dosing decisions at the required moment, even when the primary biomarker is not available at the time of decision-making. Another approach is to neutralize the biomarker's impact altogether by recommending a fixed dosing strategy in a special patient population (*cf.* Chapter 4).

Together, these applications illustrate that modeling is not limited to quantifying PK. It can systematically compare existing evidence, select appropriate models for specific patient populations, address gaps imposed by biomarker availability or operational constraints, and propose practical solutions to real-world dosing challenges. More broadly, this underscores the role of modeling as an integrative decision-support framework that does not only quantify clinical datasets but also provides a solution to clinically relevant problems in subsequent pharmacometric analysis.

While model selection avoids redundant model development, it introduces several limitations. First, published models are often based on assumptions that are not fully reported (*e.g.*, handling below quantification data) [87].

Second, most models are developed in narrowly defined patient populations whose characteristics may not match those of the new dataset [298]. Third, several models were developed to describe PK rather than predict PK in the subsequent pharmacometric analyses [298]. Together, these limitations increase the risk of misusing the model, thus limiting the generalizability of published models outside their original context. Therefore, when several published models exist for the same drug in the same special patient population, the first step should not automatically be to develop another model. Instead, priority should be given to: (i) systematically comparing their assumptions, and (ii) rigorously validating existing models using external datasets. This supports more efficient use of existing evidence.

Nevertheless, new model development for long-established antibiotics (*e.g.*, vancomycin) remains necessary in specific circumstances particularly when addressing questions that existing popPK(–PD) cannot capture. One such scenario is the evaluation of novel biomarkers that have not yet been tested as covariate. For example, if a newly identified marker of ARC was shown to correlate with enhanced antibiotic CL, a new popPK model would be required to determine its statistical significance and clinical relevance. Another scenario is an evaluation of clinical outcomes beyond exposure, such as evaluating how drug exposure influences transitions between clinical states (*e.g.*, discharge, therapy modification, or death) using multi-state survival model [370]. In such settings, new model development is justified.

FUTURE PERSPECTIVES

- Share popPK model code and all relevant modeling details including all the assumptions made.
- Rationalize the choice between using a published model and developing a new one.
- Call for the development of a standardized guideline on model selection.
- Validate any developed or existent model before its use in subsequent pharmacometric analyses.

Step 3. Pharmacometrics-based dosing decision approaches

PopPK(–PD) models often form the core of published research and are sometimes presented as major scientific outputs that provide directly dose optimization. However, in some cases these models are not followed by any subsequent pharmacometric analyses (e.g., MIPD, or dosing simulations) that are necessary to translate them into dosing recommendation. PopPK(–PD) models value lies in serving as a bridge towards an optimal dosing strategy that improves attainment of efficacy-based PKPD targets while minimizing the risk of toxicity. Only through subsequent pharmacometric analyses can these models contribute to dosing optimization. Pharmacometrics-based dosing decision approaches are required to translate PK(–PD) models into optimized dosing strategies. Additionally, even these approaches must be validated in prospective clinical studies to ensure that *in silico* predictions translate into

real-world clinical benefit. Accordingly, one of the main translational contributions of this thesis was the identification of dosing recommendations developed with real-world feasibility and clinical implementation in mind.

Selection of the MIC used to calculate the efficacy-based PKPD target is subject to debate. In clinical practice, MIC results become available only after three days of bacterial culture. Therefore, dosing optimization often relies on the EUCAST breakpoint or the locally reported MIC₉₀ [28]. However, these values are not interchangeable as shown in the result section of Chapter 4. The EUCAST breakpoint is recommended because it is based on several million MIC values obtained from standardized antimicrobial testing across Europe, thus minimizing regional differences in wild-type distributions and laboratory assays [361,362]. However, relying on the breakpoint can lead to potentially toxic dosing strategies. In all cases, MIC remains based on an *in vitro* measurement, which may not always correlate with antibiotic efficacy, particularly in cases involving high bacterial load, bacteriostatic antibiotics dependent on host immunity, time-varying bacterial susceptibility, or antibiotic synergy. Despite these limitations, MIC is widely used because it is simple, reproducible, and a reasonable approximation of antimicrobial activity at the site of infection. The ongoing debate on which MIC value to use falls outside the pharmacometrician's hands. These uncertainties can only be resolved through prospective clinical trials.

The following two subsections describe the applicability, strengths, limitations, and opportunities of refinement, based on lessons learned during this PhD, for pharmacometrics-based dosing decision approaches.

Model-informed precision dosing

MIPD is applicable to antibiotics with a narrow therapeutic window when a popPK model is available. Its strength lies in combining prior knowledge with patient-specific data to individualize dosing strategy [78,88]. When several popPK models are available, which is common for drugs in the market for decades, matching model development populations to targeted patient population for MIPD is widely considered ideal to select models but not always feasible, limiting MIPD applicability [78,88].

To overcome this limitation, several strategies have been proposed, and many of them have already been explored in the context of antibiotic dosing. A possible strategy is to integrate all available models to reduce dependency on a single model assumption [79]. For vancomycin, for example, multi-model algorithms combining models developed in critically ill patients, obese patients, and patients with sepsis have been evaluated [79]. However, as demonstrated in Chapter 2, a single model performed better than multi-model algorithms, because PK changes rapidly in this population of older patients. This highlights the need to characterize age-related PK dynamics [78]. A second strategy involves flattened priors combined with ML, which aims to modify the weight of prior information relative to patient data, capturing IIV

while retaining the interpretability of the models [198]. In vancomycin dosing, this strategy has been used to allow greater flexibility in individual PK parameter estimation at the time of clinical decision-making in adult patients [198]. A third strategy lies in continuous learning approach, in which a published model is updated with a continuous flow of data (*i.e.*, full re-estimation) [207]. This updates the model parameters to site-specific covariates and clinical practices (*e.g.*, dosing strategies, laboratory assays). Continuous learning has been applied to vancomycin in pediatric ICU, where routinely collected TDM data were incorporated to refine model parameters [207]. A fourth strategy couples physiologically based PK (PBPK) to MIPD, where “virtual twins” are created to mimic the physiology of individual patients and enable individualized dosing [371]. Although most published applications involved non-antibiotic drugs (*e.g.*, apixaban, clozapine), this strategy has also been evaluated for amikacin in critically ill adults with cancer [372]. In all cases, the goal is not to identify a best strategy but to improve the predictive performance of MIPD [373].

Several MIPD software tools are already used in routine practice to support individualized dosing, particularly for drugs such as vancomycin. These tools are primarily intended for use by clinicians and clinical pharmacists as decision-support systems that assist dose selection and dose adjustment. However, their adoption remains limited. As shown in Chapter 1, most tools still lack regulatory certification and prospective clinical validation, restraining their use in routine clinical practice. Additionally, heterogeneous EHR systems

often store laboratory values, drug-administration records, and patient characteristics in separate or incompatible formats. Without standardized data structures, MIPD tools cannot automatically retrieve the information needed for forecasting, limiting the feasibility of real-time individualized dosing. As a result, the impact of these tools on workflow may vary; they have the potential to reduce repetitive calculations and support more consistent dose individualization, but they may also introduce additional workload if data integration and local implementation are insufficient.

Demonstrating clinical benefit is a prerequisite for broader MIPD adoption. RCTs are needed to confirm improvements in exposure, safety, and cost-effectiveness and to strengthen confidence among clinicians and hospital administrators. Once sufficient evidence is established, technical and organizational barriers must be addressed. Harmonizing data structures and promoting interoperability standards such as Fast Healthcare Interoperability Resources would enable automated data exchange between EHR systems and MIPD tools, reducing manual workload. Regulatory clarity and streamlined pathways for software-as-a-medical-device certifications would further support market entry. Finally, successful implementation requires local change management, including multidisciplinary teams, concise onboarding, and integration of MIPD into standard operating procedures. In this context, the added value of MIPD lies in its potential to support safer and more precise dosing, while sustainable implementation will also depend on organizational priorities and the ability of healthcare systems to accommodate the required

infrastructure and resources. Together, these steps can facilitate widespread clinical adoption.

Overall, the lessons learned during this PhD — precisely those demonstrated in Chapter 1 — highlight that while MIPD is theoretically strong, its practical limitations and lack of validation indicate a clear need for methodological refinement and infrastructural investment before widespread clinical adoption becomes feasible.

Dosing simulations

Dosing simulations are applied for drugs without a narrow therapeutic window. They rely on virtual patient populations whose covariates should be close to those of the real-world patient populations. When models include several covariates, it becomes essential to reproduce both the dependency structure (*i.e.*, relationships among covariates such as correlations) and the marginal distribution (*i.e.*, the distribution of each covariate considered independently) [86].

Several methods exist to generate virtual patient populations, falling into two categories: data-driven, or distribution-based methods [86,374]. Data-driven methods require access to individual-level covariate data. These include bootstrapping, which preserves the dependency structure. However, these methods have limitations, notably a dependence on sample size and potential restrictions due to data-protection regulations, which may limit availability and representativeness [374]. This has driven the development of

distribution-based methods, which do not require individual-level data. Here, knowledge of covariate means, and ranges enables the creation of virtual patient datasets. In univariate simulations, random samples are drawn independently from the marginal distribution of each covariate, thereby ignoring correlations. This can result in physiologically implausible patient profiles (e.g., 32 years old adult weighing 20 kg but with an SCr of 1.5 mg/dL). Using a multivariate normal distribution addresses this limitation by incorporating

variance–covariance matrix and therefore preserving correlation structure, as described in Chapter 4 [329,330]. However, this approach assumes normality and may not capture skewed or multimodal real-world distributions. Transformations and truncations, as applied in Chapter 4, can help address these issues [329,330]. More recently, copula-based methods have emerged, allowing simultaneous preservation of dependency structures and marginal distributions [86]. Copulas have shown promising results particularly for high-dimensional covariate spaces, when compared with transformed multivariate normal approaches [86]. Still, the number of covariates, the method's impact on PTA and classification accuracy are important to consider when determining which method to use and how much effort should be invested in generating a virtual patient dataset.

Once the model and virtual patient dataset are selected, the clinical relevance of statistically significant covariates must be evaluated for each dosing strategy. As shown in Chapters 3 and 4, this step is crucial as it determines

whether a covariate meaningfully changes the PTA of a dosing strategy [375]. Clinical relevance can be visualized through PTA covariate plots, receiver operating characteristic curves, and forest plots [376,377].

Selection of a PTA cutoff is subject to debate. PTA of 90%, 95% and 99% appear in the literature indicating that these cutoffs are not harmonized and may be somewhat arbitrary [378]. The EMA's 90% PTA is commonly used in dose optimization studies probably because it is required for regulatory purposes [303]. Under this cutoff, a dosing strategy ensures that 90% of the targeted (special) patient population with a bacterial infection at the MIC of the causative bacteria achieves efficacious therapeutic exposure. However, PKPD targets themselves are surrogate markers for bacterial eradication, and evidence linking them to clinical outcomes in RCTs is limited [364,365,378]. This highlights the need for greater validation of PKPD targets and PTA-based decisions, as emphasized throughout this thesis.

Building on concepts introduced earlier in Section 5.3 of the introduction, this thesis highlights that dosing strategies should not only be stratified based on statistical significance but should also consider clinical relevance and the timing at which biomarkers become available relative to clinical decision-making. While a one-size-fits-all dosing may be inappropriate across all patient populations, a fixed dosing strategy may be still recommended for a special patient population in two scenarios: when no clinically relevant (statistically significant) covariates exist, or when clinically relevant covariates

become available only after the time at which the dosing decision is made. For example, if kidney function markers are delayed or updated infrequently, stratification based on kidney function markers becomes impractical, making a fixed dose more appropriate. When clinically relevant covariates are available at or close to the time of decision-making, a stratified covariate-based dosing strategy may be appropriate. When neither fixed nor stratified dosing is feasible — such as when patient physiology evolves rapidly, a challenge illustrated in older adults in Chapter 2 — MIPD becomes the most suitable solution [96].

Overall, these lessons highlight that generating realistic virtual patient populations, identifying clinically relevant covariates, selecting a context-specific dosing strategy, and evaluating the timing of covariate availability — as demonstrated in Chapters 3 and 4 — are prerequisites for reliable dosing simulations. They further demonstrate that simulation-based optimization can contribute directly to clinical practice when it yields dosing recommendations that are both clinically relevant for the target patient population and feasible within the operational constraints of the intended care setting.

FUTURE PERSPECTIVES

- Establish consensus guidelines on model selection for MIPD.
- Integrate MIPD software tools into hospital EHRs.
- Promote international collaborations to define a scientifically justified, feasible, and continuously updated PKPD cutoff values.
- Create virtual patient population closest to the real-world.
- Recommend a dosing strategy based on clinical relevance, biomarker availability at the decision-making time, and feasibility in real-world clinical practice.
- Conduct prospective clinical validation studies to evaluate the optimized dosing strategy and its safety.

Step 4. Prospective clinical validation

Pharmacometrics is an *in silico* discipline. Its aim is to shorten the road towards identification of a dosing strategy that achieves appropriate therapeutic exposure while minimizing toxicity. However, a recommended dosing strategy can only be considered valid after prospective clinical validation, which provides evidence for clinical utility.

PopPK–PD models play a key role in designing a prospective clinical validation trial. They are the basis for model-informed clinical trial design, where *in silico* trials using virtual patients help determine the required sample size and predict the clinical outcomes [379]. An *in silico* trial reduces real-world

trial inefficiencies and ensures that the dosing strategies evaluated are grounded in plausible clinical scenarios. Effective communication among pharmacists, pharmacologists, pharmacometricians and physicians is essential at this point for successful translation into practice [380].

Despite their importance, prospective validation trials in special patient populations are the least investigated step of the four-step “learn and confirm” cycle. Only a small number of *in silico*-based dosing strategies (e.g., fixed, stratified, or individualized) have been confirmed in such trials. Currently, several MIPD trials are ongoing, illustrating growing momentum toward model-informed dosing [202,203]. These include RCTs evaluating AUC-guided vancomycin dosing supported by MIPD software — one in non-critically ill adults and another in critically ill children — compared with standard dosing. In continuity with this PhD, the recommended dosing strategies of meropenem and amikacin in the specific patient populations investigated in this thesis will be prospectively validated at UZ Leuven, contributing to the evidence base for model-informed therapy.

FUTURE PERSPECTIVES

- Conduct *in silico* trial design to determine the sample size and predict the clinical outcomes.
- Conduct prospective validation trials to confirm the clinical utility and the safety of an optimized dosing strategy.

FINAL REFLECTION

In this thesis, we used pharmacometrics to prove that a standard dosing strategy does not hold for all patient populations. Pharmacometrics-based dosing decision approaches (*i.e.*, MIPD and dosing simulations) were applied to identify a dosing strategy that achieves therapeutic exposure while minimizing toxicity. A particular focus of this work was on special patient populations — such as older adults, critically ill adult patients with preserved renal function or ARC admitted to the ICU, and critically ill adult patients admitted to the ED — who often experience altered PK and are therefore at higher risk of under- or over-exposure when standard dosing is applied. These populations exemplify the clinical need for dosing strategies tailored to specific patient populations and grounded in quantitative evidence. Although this work focused primarily on the learning phases of the four-step “learn and confirm” cycle (*i.e.*, data collection and analysis, modeling, and pharmacometrics-based dosing decision approaches), the fourth step (*i.e.*, prospective clinical validation) remains important. It is during this confirmatory step that the recommended dosing strategy for special patient populations is validated in a clinical trial, indicating that *in silico* results translate to real-world patient population. Every patient deserves a dose – guided by a dosing strategy – that achieves therapeutic exposure safely.

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Curriculum Vitae

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Education

Jul 2021–present	Ph.D. in Pharmaceutical Sciences KU Leuven, Belgium
Jun 2020	Doctor of Pharmacy (Summa Cum Laude) Lebanese International University, Lebanon
Aug 2018	Master of Biomedical Sciences Radboud University, the Netherlands
Jun 2016	Bachelor of Pharmacy (Cum Laude) Lebanese International University, Lebanon
Feb 2011	Bachelor of Science in Medical Laboratory Sciences Balamand University, Lebanon

Honors, Awards, Licenses, and Achievements

Apr 2025	Grant for participation in a conference abroad by FWO
Jun 2024	Grant for participation in a summer school abroad by FWO
Jun 2024	Grant for participation in a conference abroad by FWO
Jan 2023	Certified member of the Belgium Order of Pharmacists
Sep 2019	Academic Excellence Award by Lebanese International University
Jun 2017	Radboud Scholarship Program by Radboud University
Dec 2016	Certified member of the Lebanese Order of Pharmacists
Jun 2016	Radboud Scholarship Program by Radboud University
May 2014	Best poster at the annual Pharmacy Day by Lebanese International University

Teaching experience

Nov–Dec 2025	Department of Pharmaceutical and Pharmacological Sciences, KU Leuven, Belgium Material preparation and assistance in the Master of Pharmacometrics lead by Prof. Erwin Dreesen
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Feb–May 2023	Department of Pharmaceutical and Pharmacological Sciences, KU Leuven, Belgium Material preparation and assistance with the Research skills and science communication course taught by Prof. Erwin Dreesen
Mar–Oct 2023	Department of Pharmaceutical and Pharmacological Sciences, KU Leuven, Belgium Supervision of a master student
Jan–Jul 2021	Department of Clinical Pharmacy, Lebanese International University, Lebanon Volunteer in the clinical training of Pharmacy students in the class of Dr. Diana Malaeb

Professional work experience

Mar 2011–Aug 2012 Laboratory assistant in a medical laboratory, Lebanon

Publications part of the doctoral thesis

Dia N, Hoffert Y, Edwina A E, Tournoy J, Spriet I, Dreesen E. Precision dosing of vancomycin: a systematic review of population pharmacokinetic models and comparative evaluation of software tools. (submitted)

Edwina A E *, Dia N *, Gijsen M, Van der Linden L, Spriet I, Tournoy J, Dreesen E. Optimizing vancomycin dosing in older patients: predictive performance of pharmacometric models. (in preparation; * shared first author)

Dia N, Blaine Z, Gijsen M, Wauters J, Spriet I, Dreesen E. Model-based evaluation of an optimized meropenem dosing strategy for critically ill patients with preserved or augmented renal clearance. (in preparation)

Dia N, De winter S, Gijsen M, Desmet S, Vanbrabant P, Peetermans W, Spriet I, Dreesen E. Amikacin dose optimization in the emergency department: a population pharmacokinetic simulation study. *Therapeutic drug monitoring*. 2024. DOI: 10.1097/FTD.0000000000001279

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Malaeb D, Mansour S, Dia N, Kassem N M, Haddad C, Dabbous M, Ismail O, Adel F, Gamal M, Lucca J M, El Khatib S, Salameh P, Hallit S, Hosseini H. Scoping review about pathogenesis, risk factors, and treatment of venous and arterial thrombosis in coronavirus infection. *Frontiers in Cardiovascular Medicine*. 2025;12. DOI: 10.3389/fcvm.2025.1688115

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software tools. *Clin Pharmacokinet.* 2024; 63 (10). DOI: 10.1007/s40262-024-01414-y

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Edwina AE, Dia N, Dreesen E, Vanasche T, Verhamme P, Spriet I, Van der Linden L, Tournoy J. Insights into the Pharmacokinetics and Pharmacodynamics of Direct Oral Anticoagulants in Older Adults with Atrial Fibrillation: A Structured Narrative Review. *Clinical Pharmacokinetics.* 2023; 62. DOI: 10.1007/s40262-023-01222-w

Dakroub D, Sakr F, Dabbous M, Dia N, Hammoud J, Rida A, Ibrahim A, Fahs H, Obeid S, Hallit S, Malaeb D. The sociodemographic and lifestyle characteristics associated with quality of life among diabetic patients in Lebanon: a cross-sectional study. *Pharmacy Practice.* 2023; 21. DOI: 10.18549/PharmPract.2023.1.2775

Gastmans H, Dreesen E, Wicha S, Dia N, Spreuwers E, Dompas A, Allegaert K, Desmet S, Lagrou K, Peetermans W, Debaveye Y, Spriet I, Gijzen M. Systematic comparison of hospital-wide standard and model-based therapeutic drug monitoring of vancomycin in adults. *Pharmaceutics.* 2022; 14. DOI: 10.3390/pharmaceutics14071459

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Haidar L, AlHarfany H, Cherri S, Malaeb D, Dia N, Salameh P, Hosseini H. Evaluation of hypertension treatment in acute ischemic stroke. *Clinical Epidemiology and Global Health.* 2021; 11. DOI: 10.1016/j.cegh.2021.100776

Hartman SJF, Bruggemans RJ, Orriëns L, Dia N, Schreuder MF, de Wildt SN. Pharmacokinetics and Target Attainment of Antibiotics in Critically Ill Children: A Systematic Review of Current Literature. *Clinical Pharmacokinetics.* 2020; 59. DOI:10.1007/s40262-019-00813-w

Oral presentations

Dia N, Blaine Z, Gijssen M, Wauters J, Spriet I, Dreesen E. (2025, April), "Model-based evaluation of an optimized meropenem dosing strategy in the intensive care unit". ISAP-EPASG joint anti-infective PKPD symposium, Vienna, Austria

Dia N, De Winter S, Elkayal O, Mertens B, Vanbrabant P, Peetermans W, Spriet I, Dreesen E. (2023, April), "Amikacin dose optimization in the emergency department: A population simulation study". 18th Pharmacometrics Network Benelux Spring meeting, Antwerp, Belgium

Dia N, De Winter S, Elkayal O, Mertens B, Vanbrabant P, Peetermans W, Spriet I, Dreesen E. (2022, June), "Amikacin dose optimization in the emergency department: A population simulation study". Pre-symposium Population Approach Group in Europe, Ljubljana, Slovenia

Poster presentations

Dia N, Blaine Z, Gijssen M, Wauters J, Spriet I, Dreesen E. (2025, April), "Model-based evaluation of an optimized meropenem dosing strategy in the intensive care unit". European Congress of Clinical Microbiology and Infectious Diseases, Vienna, Austria

Dia N, Blaine Z, Gijssen M, Wauters J, Spriet I, Dreesen E. (2024, June), "Model-informed comparison of the clinical relevance of four renal function markers on meropenem target attainment in critically ill septic patients". Population Approach Group in Europe, Rome, Italy

Dia N, De Winter S, Elkayal O, Mertens B, Vanbrabant P, Peetermans W, Spriet I, Dreesen E. (2022, June), "Amikacin dose optimization in the emergency department: A population simulation study". Population Approach Group in Europe, Ljubljana, Slovenia

Dia N, De Winter S, Elkayal O, Mertens B, Vanbrabant P, Peetermans W, Spriet I, Dreesen E. (2022, April), "Amikacin dose optimization in the emergency department: Does one dose fit all?". European Congress of Clinical Microbiology and Infectious Diseases, Lisbon, Portugal

Dia N, De Winter S, Elkayal O, Mertens B, Vanbrabant P, Peetermans W, Spriet I, Dreesen E. (2021, November), "Amikacin dose optimization in the emergency department: Does one dose fit all?". Asian Pharmacometrics Network, online

Dia N, Mansour M, Aridi R, Cherfan M. (2014, May). "Thalassemia Awareness: It is not the years in your life that count. It is the life in your years". Annual Pharmacy Day organized by the Lebanese International University, Bekaa, Lebanon

Acknowledgments, personal contribution, conflict of interest, and use of generative AI

Acknowledgments

Chapter 1

Nada Dia and Erwin Dreesen conceptualized the study and developed the methodology. Nada Dia conducted the investigation, curated the data, performed the formal analysis, visualized the results, and wrote the original manuscript draft. Erwin Dreesen supervised the project. Erwin Dreesen and Isabel Spriet provided the resources needed to conduct the study. Yannick Hoffert, Angela Elma Edwina, Jos Tournoy, Isabel Spriet, and Erwin Dreesen critically revised the manuscript. We thank all the pharmacometricians who developed the popPK models referred to in the study and all the software tools providers who participated in our survey.

Chapter 2

Isabel Spriet, Jos Tournoy, and Erwin Dreesen conceptualized the study and its design. Jos Tournoy provided the data. Angela Elma Edwina and Nada Dia conducted the investigation, curated the data, performed the formal analysis, visualized the results, and wrote the original manuscript draft. Erwin Dreesen supervised the project. Matthias Gijzen, Lorenz van der Linden, Isabel Spriet, Jos Tournoy, and Erwin Dreesen critically revised the manuscript. We thank all the patients whose data was retrospectively used in the study.

Chapter 3

Isabel Spriet and Erwin Dreesen conceptualized the study and its design. Nada Dia, Zakaria Blaine, and Isabel Spriet provided and curated the data. Nada Dia conducted the investigation, performed the formal analysis, visualized the results, and wrote the original manuscript draft. Erwin Dreesen supervised the project. Zakaria Blaine, Matthias Gijssen, Joost Wauters, Isabel Spriet, and Erwin Dreesen critically revised the manuscript. We thank all the patients who participated in the study.

Chapter 4

Isabel Spriet and Erwin Dreesen conceptualized the study and its design. Nada Dia, Sabrina de Winter, and Isabel Spriet provided and curated the data. Nada Dia conducted the investigation, performed the formal analysis, visualized the results, and wrote the original manuscript draft. Erwin Dreesen supervised the project. Sabrina de Winter, Matthias Gijssen, Stefanie Desmet, Peter Vanbrabant, Willy Peetermans, Isabel Spriet, and Erwin Dreesen critically revised the manuscript. We thank all the patients who participated in the study.

Personal contribution

For **Chapter 1**, the author of this thesis was entirely responsible for the investigation, data curation, formal analysis, visualization, and writing the

manuscript. The author was partially responsible, together with the promoter Erwin Dreesen, for the study conceptualization, and methodology.

For **Chapter 2**, the author of the thesis and Angela Elma Edwina carried out the investigation, data curation, formal analysis, visualization, and wrote the manuscript.

For **Chapter 3**, the author of this thesis was entirely responsible for the investigation, formal analysis, visualization, and writing the manuscript. The author was partially responsible, together with Zakaria Blaine, and the co-promoter Isabel Spriet, for the data collection and curation.

For **Chapter 4**, the author of this thesis was entirely responsible for the investigation, data curation, formal analysis, visualization, and writing the manuscript.

Conflict of interest

Angela Elma Edwina was supported by AGePOP, a Horizon 2020 research and innovation program (Marie Skłodowska–Curie grant number 956146).

Erwin Dreesen received consultancy fees from Alimentiv, lecture fees from Celltrion and Galapagos, and financial support from Celltrion, Janssen, Pfizer, Prometheus, R–Biopharm, and Sandoz, with all honoraria/fees being paid to KU Leuven and not to any personal account of Erwin Dreesen.

Isabel Spriet received consultancy support and lecture fees from Argenx, Pfizer, Merck, and Cidara. She also had a grant from the Clinical Research Fund of UZ Leuven.

Lorenz van der Linden had a grant from the Clinical Research Fund of UZ Leuven.

Nada Dia was supported by a Starting Grant of KU Leuven assigned to Erwin Dreesen (grant number KGG-E1099-STG/21/005).

Yannick Hoffert (1SHAA24N) was supported by a doctoral research grant from the Research Foundation – Flanders, Belgium.

Zakaria Baline was supported by an Erasmus mobility grant (grant number 84475).

Other authors declared no conflict of interest.

Use of generative AI

Generative AI assistance tools were not used for data analysis or scientific interpretation. They were used solely for language editing. Specifically, M365 Copilot operating under KU Leuven Enterprise Data Protection was applied to support linguistic refinement. The use of generative AI adhered to the principles outlined in the *Living Guidelines on the Responsible Use of Generative AI in Research Published*, including transparency, validation, source verification, and acknowledgement of limitations. The author retains full responsibility and accountability for the scientific content, interpretation,

and integrity of this thesis, in compliance with relevant ethical and legal standards.